SSC JE

THERMAL ENGINEERING

Topper-Style Concept Notes

SSC JE Mechanical • Paper-I & Paper-II • Concept + Formula + PYQ-Trend + Solved Numericals

Thermodynamics • Pure Substances • Gas Power Cycles • IC Engines • Vapour Power Cycles • Boilers • Nozzles & Steam Turbines • Condensers • Power Plants • Air Compressors • Gas Turbines • Refrigeration • Psychrometry & AC • Heat Transfer

Concept-based notes – no MCQ practice. "PYQ-Trend" boxes list concepts SSC JE has repeatedly tested (based on recurring topics, not copied questions). Cross-check values with a standard textbook.

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INTRODUCTION

How to Use These Notes

Why Thermal Engineering Is the Biggest Block

Thermal engineering is the largest block of SSC JE Mechanical. Across Paper-I it typically contributes 25–30 of the 100 technical questions, and in Paper-II it forms the biggest share of numericals. The same ideas repeat year after year: zeroth/first/second law statements, Carnot efficiency and COP, entropy change, Otto/Diesel/Dual comparison, octane and cetane numbers, Rankine improvements, boiler mountings and accessories, critical pressure ratio of nozzles, Parsons turbine, isothermal compression and intercooling, refrigerant numbering and 1 TR, psychrometric processes, critical radius of insulation, dimensionless numbers and LMTD.

Area	Typical questions (Mech Paper-I)	Priority
Basic thermodynamics (laws, entropy, properties)	6–8	Very high
Gas power cycles + IC engines	5–7	Very high
Steam, boilers, turbines, nozzles, power plants	4–6	High
Air compressors, gas turbines	2–3	Medium
Refrigeration & air conditioning	3–5	Very high
Heat transfer	3–5	High

How Each Chapter Is Built

- Concept points → yellow formula box → blue PYQ-TREND box (concepts SSC asks repeatedly, from recurring topics – not copied questions) → green TOPPER TRICK → red COMMON MISTAKE → purple Paper-II question with a model answer.
- Chapters 24–25 hold solved numericals; Chapter 26 is a concept-recall bank; Chapter 27 has Paper-II answers; Chapter 28 is the last-minute sheet. Appendix A collects every formula; Appendix B collects every PYQ-trend box.
- Standard data used: air R = 0.287 kJ/kg·K, cp = 1.005, cv = 0.718 kJ/kg·K, γ = 1.4; universal gas constant R̄ = 8.314 kJ/kmol·K; water cp = 4.187 kJ/kg·K; hfg at 1 atm = 2257 kJ/kg; 1 TR = 3.5167 kW.

✔ **TOPPER TRICK:** Always convert temperatures to kelvin before using any ratio formula (Carnot, isentropic relations, entropy). This single habit prevents most numerical errors.

✖ **COMMON MISTAKE:** Sign conventions differ between books. This book uses: heat added to the system positive, work done by the system positive ($Q = \Delta U + W$).

CHAPTER 1

Basic Concepts of Thermodynamics

1.1 System, Surroundings and Boundary

- Thermodynamics deals with energy, its forms (heat and work) and its interaction with matter. Classical (macroscopic) thermodynamics treats matter as a continuum; statistical (microscopic) thermodynamics works with molecules.
- System: the quantity of matter or region of space chosen for study. Surroundings: everything outside it. Boundary: real or imaginary surface separating them; it can be fixed or moving.

System	Mass transfer	Energy transfer	Examples
Closed (control mass)	No	Yes	Gas in a piston–cylinder, sealed tank, pressure cooker (before whistle)
Open (control volume)	Yes	Yes	Turbine, compressor, nozzle, pump, boiler, IC engine (during intake/exhaust), heat exchanger
Isolated	No	No	Universe, perfectly insulated thermos flask (ideal)

- Adiabatic system: no heat transfer across the boundary (work may cross). Diathermic wall: allows heat transfer.
- Homogeneous system: single phase (water, air). Heterogeneous: more than one phase (ice + water).

★ **PYQ-TREND:** Classification of systems with examples (turbine – open; piston–cylinder – closed; universe – isolated). Adiabatic vs isolated. Control mass vs control volume.

1.2 Properties, State, Process, Cycle

- Property: a measurable characteristic that defines the state (p, V, T, U, H, S). It depends only on the state, not on the path – a point function with an exact differential.
- Intensive properties do not depend on mass: pressure, temperature, density, specific volume, specific enthalpy, viscosity, velocity, specific heat. Extensive properties depend on mass: volume, mass, energy, enthalpy, entropy, internal energy. Specific (per unit mass) values of extensive properties are intensive. The ratio of two extensive properties is intensive.
- State postulate: the state of a simple compressible system is fixed by two independent intensive properties.
- Process: change of state. Path: series of states passed through. Cycle: process(es) returning to the initial state – net change of every property is zero ($\oint dX = 0$).

- Quasi-static (quasi-equilibrium) process: infinitely slow, so the system is always infinitely close to equilibrium; it can be drawn as a continuous line on a property diagram. Only quasi-static processes have a definable path and work ∫p dV.
- Thermodynamic equilibrium requires simultaneous mechanical (no unbalanced forces – uniform pressure), thermal (uniform temperature) and chemical (no chemical reaction/mass transfer) equilibrium; phase equilibrium is part of chemical equilibrium.
- Heat and work are not properties – they are path functions (inexact differentials δQ, δW); they appear only during a process, at the boundary.

Point functions (properties)	Path functions
p, V, T, U, H, S, G, F, ρ	Heat Q, Work W

`Property test: ∮dX = 0 for any cycle; path function: ∮δQ ≠ 0 in general`

★ **PYQ-TREND:** Intensive vs extensive identification (very frequent); heat and work as path functions; quasi-static process definition; thermodynamic equilibrium components; state postulate (two independent properties).

✔ **TOPPER TRICK:** Divide the system in two equal halves: if the property stays the same in each half it is intensive (T, p, ρ); if it halves it is extensive (V, m, U).

1.3 Zeroth Law and Temperature

- Zeroth law (R.H. Fowler, 1931): if two bodies are each in thermal equilibrium with a third body, they are in thermal equilibrium with each other. It is the basis of temperature measurement – the thermometer is the 'third body'.
- Temperature: property that determines whether a body is in thermal equilibrium with another.

```
Kelvin: T(K) = t(°C) + 273.15; Fahrenheit: t(°F) = 1.8 t(°C) + 32; Rankine: T(R) = t(°F) + 459.67 = 1.8 T(K)
°C and °F read the same at −40; °F and K read the same at 574.59
Linear thermometer scale: (t − ice point)/(steam point − ice point) is the same on all linear scales
```

- Fixed points: old two-point scale – ice point (0 °C) and steam point (100 °C) at 1 atm. Since 1954 a single fixed point – the triple point of water, 273.16 K (0.01 °C, 0.6117 kPa) – defines the kelvin; ITS-90 uses several fixed points. Since 2019 the kelvin is defined via the Boltzmann constant.
- Thermometric properties: length of liquid column (mercury-in-glass), pressure at constant volume (constant-volume gas thermometer – most accurate, independent of the gas at low pressures; used as the standard), volume at constant pressure, electrical resistance (RTD, thermistor), thermo-emf (thermocouple – Seebeck effect), radiation intensity (pyrometer).

Thermometer	Range / note
Mercury-in-glass	−39 to 357 °C
Alcohol	Low temperatures (to −115 °C)
Constant-volume gas	Standard, very accurate
Platinum resistance (Pt-100)	−200 to 850 °C, accurate, linear
Thermocouple	−200 to 1800 °C (type-dependent), fast response
Optical/radiation pyrometer	> 700 °C, non-contact

★ **PYQ-TREND:** Zeroth law is the basis of temperature measurement (very frequent); temperature equal on °C and °F (−40); triple point of water 273.16 K; constant-volume gas thermometer as standard; thermocouple works on Seebeck effect.

✖ **COMMON MISTAKE:** The triple point of water is 0.01 °C (273.16 K), not 0 °C. The ice point (0 °C) is at 1 atm; the triple point is at 0.611 kPa.

1.4 Pressure, Energy and Units

```
1 bar = 10⁵ Pa; 1 atm = 1.01325 bar = 760 mm Hg = 10.33 m H₂O; 1 torr = 1 mm Hg
p_abs = p_atm + p_gauge; vacuum = p_atm − p_abs
1 kJ = 1000 N·m; 1 kWh = 3600 kJ; 1 cal = 4.187 J; 1 hp (metric) = 735.5 W
```

- Energy forms: stored (internal, kinetic, potential, chemical, nuclear) and in transit (heat, work). Internal energy U: sum of microscopic energies (molecular translation, rotation, vibration, binding).
- Specific heat: energy to raise unit mass by one degree. For gases two principal values: cp (constant pressure) > cv (constant volume) because at constant pressure the gas also does expansion work.

★ **PYQ-TREND:** Unit conversions; absolute vs gauge pressure; internal energy is a property; cp > cv for gases.

✎ **DESCRIPTIVE Q:** Explain the concept of thermodynamic equilibrium and quasi-static process.

A system is in thermodynamic equilibrium when no spontaneous change occurs in it without an external influence. It requires mechanical equilibrium (no unbalanced forces, so uniform pressure), thermal equilibrium (uniform temperature equal to that of the surroundings across diathermic walls) and chemical equilibrium (no chemical reaction or mass transfer between parts, including phase equilibrium). Only equilibrium states can be described by properties. A quasi-static process is one carried out so slowly that the system passes through a continuous succession of equilibrium states; each intermediate state can be plotted on a p–V diagram, so the path is a continuous line and work can be calculated as $\int p\,dV$. Real processes are not quasi-static, but slow piston motion approximates them; for example, removing tiny weights one at a time from a loaded piston.

CHAPTER 2

Work, Heat and the First Law (Closed Systems)

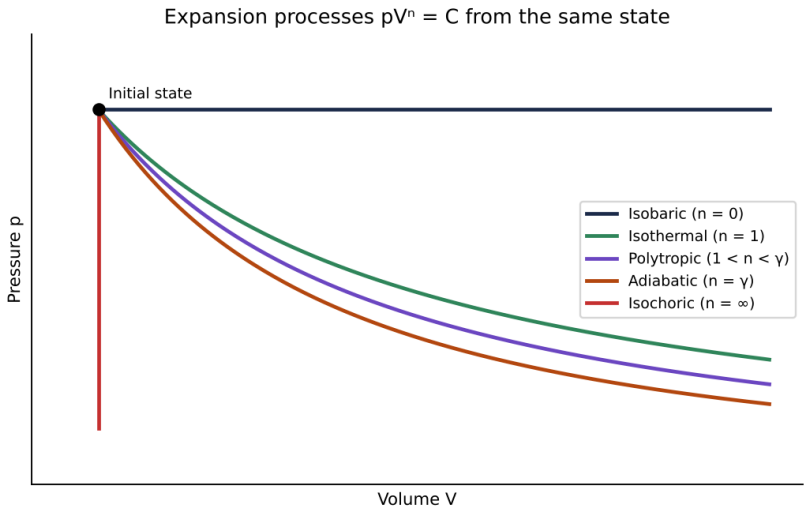
2.1 Work and Heat

- Work: energy interaction at the boundary that could be reduced to raising a weight. Displacement (p dV) work: $W = \int p\,dV$ for a quasi-static process – area under the curve on a p–V diagram.
- Other work modes: shaft work (T·θ), electrical work (V·I·t), stretching a wire (F dL), surface tension (σ dA), magnetisation.
- Heat: energy transfer due to temperature difference. Heat and work are both boundary phenomena and path functions.
- Free expansion (expansion against vacuum): W = 0 because no external resistance; irreversible; not quasi-static.
- Sign convention used here: heat into the system +, work done by the system +.

```
Work done in different quasi-static processes (closed system, ideal gas)
Constant volume (isochoric): W = 0
Constant pressure (isobaric): W = p(V₂ − V₁) = mR(T₂ − T₁)
Isothermal (pV = C): W = p₁V₁ ln(V₂/V₁) = mRT ln(p₁/p₂)
Adiabatic reversible (pV^γ = C): W = (p₁V₁ − p₂V₂)/(γ − 1) = mR(T₁ − T₂)/(γ − 1)
Polytropic (pV^n = C): W = (p₁V₁ − p₂V₂)/(n − 1)
```

- Polytropic index n: n = 0 → isobaric; n = 1 → isothermal; n = γ → reversible adiabatic (isentropic); n = ∞ → isochoric. On a p–V diagram the curves get steeper as n increases.
- Slope of adiabatic on p–V = γ × slope of isothermal through the same point (adiabatic is steeper).

```
Isentropic relations: T₂/T₁ = (V₁/V₂)^(γ−1) = (p₂/p₁)^((γ−1)/γ)
Polytropic: T₂/T₁ = (V₁/V₂)^(n−1) = (p₂/p₁)^((n−1)/n)
```



★ **PYQ-TREND:** Work in free expansion = 0 (very frequent); n values for each process; adiabatic steeper than isothermal on p–V; work formulas; area under p–V = work.

✔ **TOPPER TRICK:** For compression from the same state to the same final pressure, work input is least for isothermal and greatest for adiabatic; for expansion, isothermal gives the most work out.

2.2 First Law of Thermodynamics

- Joule's experiment (paddle wheel stirring water in an insulated vessel): for a cycle, net heat equals net work. $\oint \delta Q = \oint \delta W$ (with Q and W in the same units; earlier J = 4.186 J/cal).
- For a process: Q = ΔU + W (energy conservation). Internal energy is a property (a corollary of the first law).
- Isolated system: Q = W = 0 → ΔU = 0 (energy of an isolated system is constant).
- PMM-1 (perpetual motion machine of the first kind): a device producing work continuously without any energy input – impossible by the first law.

Cycle: $\oint \delta Q = \oint \delta W$

Process: $Q_{1-2} = (U_2 - U_1) + W_{1-2}$; per unit mass $q = \Delta u + w$

Enthalpy $H = U + pV$; $h = u + pv$

Ideal gas: $du = cv\ dT$, $dh = cp\ dT$, $cp - cv = R$, $\gamma = cp/cv$

$cv = R/(\gamma - 1)$; $cp = \gamma R/(\gamma - 1)$; molar: $\bar{C}p - \bar{C}v = \bar{R} = 8.314$ kJ/kmol·K

Polytropic heat: $Q = W\ (\gamma - n)/(\gamma - 1)$; polytropic specific heat $c_n = cv\ (n - \gamma)/(n - 1)$ (negative for $1 < n < \gamma$)

- Joule's law: internal energy (and enthalpy) of an ideal gas is a function of temperature only.
- γ values: monatomic (He, Ar) 1.67; diatomic (air, N_2, O_2, H_2) 1.4; triatomic (CO_2, H_2O vapour) ≈ 1.3.
- Constant-volume process: Q = ΔU = mcvΔT. Constant-pressure: Q = ΔH = mcpΔT. Isothermal (ideal gas): ΔU = 0, Q = W. Adiabatic: Q = 0, W = −ΔU.

Process	Q	W	ΔU
Isochoric	m cv ΔT	0	m cv ΔT
Isobaric	m cp ΔT	p ΔV	m cv ΔT
Isothermal	W	$p_1V_1 \ln(V_2/V_1)$	0
Adiabatic (reversible)	0	$(p_1V_1 - p_2V_2)/(\gamma-1)$	−W
Polytropic	$W(\gamma - n)/(\gamma - 1)$	$(p_1V_1 - p_2V_2)/(n-1)$	m cv ΔT
Free expansion (ideal gas)	0	0	0 (T constant)
Throttling	0	0	h constant

★ **PYQ-TREND:** First law for cycle and process; internal energy as a property; PMM-1; U of ideal gas depends only on T (Joule's law); cp − cv = R; γ values; heat in each process.

✖ **COMMON MISTAKE:** In a polytropic expansion with $1 < n < \gamma$, heat is ADDED even though temperature falls – the gas does more work than the heat supplied.

2.3 Real Gas Behaviour (Overview)

Ideal gas: $pv = RT$; $pV = mRT = n\bar{R}T$; $R = \bar{R}/M$

```
Compressibility factor Z = pv/RT (Z = 1 ideal; Z < 1 attractive forces dominate; Z > 1 repulsive)
Van der Waals: (p + a/v̄²)(v̄ − b) = R̄T; a — intermolecular attraction, b — molecular volume
Critical constants (vdW): a = 27R̄²Tc²/(64pc); b = R̄Tc/(8pc); Zc = 3/8 = 0.375 (actual gases ≈ 0.27–0.29)
```

- Gases behave ideally at low pressure and high temperature (relative to critical values). Law of corresponding states: all gases have nearly the same Z at the same reduced pressure pr = p/pc and reduced temperature Tr = T/Tc (generalised compressibility chart).
- Other equations of state: Beattie–Bridgeman, Berthelot, Dieterici, Redlich–Kwong, virial equation.
- Gas mixtures: Dalton's law – total pressure = sum of partial pressures; Amagat's law – total volume = sum of partial volumes; mole fraction = pressure fraction = volume fraction.

★ **PYQ-TREND:** Ideal behaviour at low p and high T; meaning of vdW constants a and b; Zc = 0.375 from vdW; Dalton's and Amagat's laws; R̄ = 8.314 kJ/kmol·K and R_air = 0.287.

2.4 Specific Heats of Solids and Liquids

```
Incompressible substances: cp ≈ cv = c; du ≈ dh = c dT
```

- Water c ≈ 4.187 kJ/kg·K; ice ≈ 2.1; steel ≈ 0.46; aluminium ≈ 0.9; copper ≈ 0.385.
- Dulong–Petit law: molar heat capacity of many solids ≈ 3R̄ ≈ 25 kJ/kmol·K at room temperature.

★ **PYQ-TREND:** cp = cv for incompressible substances; specific heat of water.

✎ **DESCRIPTIVE Q:** State the first law of thermodynamics for a closed system undergoing a cycle and a process, and prove that internal energy is a property.

For a cycle, the first law states that the net heat transfer equals the net work transfer: $\oint \delta Q = \oint \delta W$. Consider a system taken from state 1 to 2 along path A and returned along path C, and also along path B and back along C. For cycle 1-A-2-C-1: $Q_A + Q_C = W_A + W_C$, so $Q_A - W_A = -(Q_C - W_C)$. For cycle 1-B-2-C-1: $Q_B - W_B = -(Q_C - W_C)$. Hence $Q_A - W_A = Q_B - W_B$: the quantity $(Q - W)$ is the same for every path between the two states. It therefore depends only on the end states and is the change of a property – the internal energy: $Q - W = \Delta U$, i.e., $Q = \Delta U + W$ for a process. Since dU is an exact differential, U is a point function (property), even though Q and W individually are path functions.

CHAPTER 3

First Law for Open Systems (Control Volume)

3.1 Flow Work and the SFEE

- In an open system, mass crossing the boundary carries energy. Flow work (flow energy) is the work needed to push a unit mass into or out of the control volume: pv per unit mass. This is why enthalpy h = u + pv appears in flow processes.
- Steady flow: properties at any point do not change with time; mass flow in = mass flow out; rates of heat and work are constant.

```
Mass balance: ṁ = ρ₁A₁V₁ = ρ₂A₂V₂ = A V/v
Steady Flow Energy Equation (per kg): h₁ + V₁²/2 + g z₁ + q = h₂ + V₂²/2 + g z₂ + w
Rate form: ṁ(h₁ + V₁²/2 + gz₁) + Q̇ = ṁ(h₂ + V₂²/2 + gz₂) + Ẇ
Units check: V²/2 in J/kg − divide by 1000 to add to h in kJ/kg
Shaft work (reversible, negligible KE/PE): w = −∫v dp
```

- For a steady-flow device the reversible work is −∫v dp (area to the left of the curve on p–v), whereas closed-system work is ∫p dv (area under the curve).

★ **PYQ-TREND:** Flow work = pv; SFEE terms; reversible steady-flow work −∫v dp vs closed-system ∫p dv; enthalpy as the natural property for flow processes.

3.2 SFEE Applied to Devices

Device	Assumptions	Result
Nozzle	q = 0, w = 0, Δz ≈ 0	$V_2 = \sqrt{2(h_1 - h_2) + V_1^2} \approx 44.72\sqrt{(\Delta h \text{ in kJ/kg})}$
Diffuser	q = 0, w = 0	$h_2 - h_1 = (V_1^2 - V_2^2)/2$ – velocity converted to pressure
Turbine (adiabatic)	q ≈ 0, ΔKE, ΔPE ≈ 0	$w = h_1 - h_2$ (work output)
Compressor (adiabatic)	q ≈ 0	$w = -(h_2 - h_1)$ (work input)
Pump	Incompressible	$w = -v(p_2 - p_1)$
Boiler / condenser / evaporator	w = 0, ΔKE ≈ 0	$q = h_2 - h_1$
Heat exchanger (no loss)	Adiabatic overall	ṁ_h(h_h1 − h_h2) = ṁ_c(h_c2 − h_c1)
Throttling valve	q = 0, w = 0, ΔKE ≈ 0	$h_1 = h_2$ (isenthalpic)
Mixing chamber	Adiabatic	Σṁh in = Σṁh out

```
Nozzle exit velocity (V₁ negligible): V₂ = 44.72 √(h₁ − h₂) m/s with Δh in kJ/kg
```

★ **PYQ-TREND:** Throttling is isenthalpic (very frequent); nozzle velocity 44.72√Δh; turbine work = $h_1 - h_2$; boiler heat = $h_2 - h_1$; diffuser converts KE to pressure.

✔ **TOPPER TRICK:** 44.72 = √2000. For Δh = 100 kJ/kg → V = 447 m/s; Δh = 400 → 894 m/s.

3.3 Throttling and the Joule–Thomson Effect

- Throttling: flow through a restriction (partially open valve, porous plug, capillary tube) with a pressure drop and no work or heat transfer. It is irreversible; enthalpy before and after is equal ($h_1 = h_2$), but enthalpy is not constant in between.

```
Joule–Thomson coefficient μ_JT = (∂T/∂p)_h
```

- $\mu_{JT} > 0$: gas cools on throttling (below the inversion temperature – most gases at room temperature). $\mu_{JT} < 0$: heats up (hydrogen, helium at room temperature; above inversion). $\mu_{JT} = 0$ for an ideal gas (T unchanged since h = f(T)).
- Inversion temperature: where μ_{JT} changes sign. Maximum inversion temperature: $N_2 \approx 621$ K, $H_2 \approx 202$ K, He ≈ 40 K – hence hydrogen and helium must be precooled before Linde liquefaction.
- Applications: refrigeration expansion valves and capillary tubes, gas liquefaction (Linde–Hampson), throttling calorimeter, pressure control in steam lines.

★ **PYQ-TREND:** Throttling isenthalpic and irreversible; $\mu_{JT} = 0$ for ideal gas; cooling only below inversion temperature; hydrogen heats on throttling at room temperature.

3.4 Unsteady Flow Processes

```
Filling an evacuated rigid tank from a line at constant h_line (adiabatic): u₂ = h_line → T₂ = γ T_line (ideal gas)
General: m₂u₂ − m₁u₁ = (m₂ − m₁)h_line + Q − W
```

- When an empty tank is filled from a supply line, the gas in the tank ends up hotter than the line gas ($T_2 = 1.4\ T_{line}$ for air), because flow work is converted into internal energy.
- Discharging a tank: the remaining gas cools (approximately isentropic expansion of the gas left behind).

★ **PYQ-TREND:** Final temperature when filling an evacuated tank = γ T_line; filling and emptying processes are unsteady.

✎ **DESCRIPTIVE Q:** Apply the steady flow energy equation to a steam turbine and a nozzle and obtain the expressions for work and exit velocity.

For a turbine, heat loss is small (insulated), the inlet and exit velocities and heights differ little, so the SFEE $h_1 + V_1^2/2 + gz_1 + q = h_2 + V_2^2/2 + gz_2 + w$ reduces to $w = h_1 - h_2$: the work output equals the drop in enthalpy of the steam; power $= \dot{m}(h_1 - h_2)$. For a nozzle there is no work and negligible heat transfer and height change, so $h_1 + V_1^2/2 = h_2 + V_2^2/2$, giving $V_2 = \sqrt{(2(h_1 - h_2) + V_1^2)}$. With Δh in kJ/kg and negligible inlet velocity, $V_2 = \sqrt{(2000\ \Delta h)} = 44.72\sqrt{\Delta h}$ m/s. The nozzle converts enthalpy into kinetic energy, which the turbine blades then convert into shaft work.

CHAPTER 4

Second Law of Thermodynamics

4.1 Why a Second Law?

- The first law allows any process that conserves energy, but nature imposes a direction: heat flows by itself only from hot to cold; work can be fully converted to heat, but heat cannot be fully converted to work in a cycle. The second law states these limits and defines entropy and the quality of energy.
- Thermal energy reservoir: a body with such large heat capacity that its temperature does not change when heat is added or removed (atmosphere, ocean, large furnace). Source – supplies heat; sink – receives heat.

★ **PYQ-TREND:** Direction of heat flow; reservoirs (source and sink).

4.2 Heat Engine, Refrigerator, Heat Pump

Heat engine efficiency: $\eta = W/Q_1 = 1 - Q_2/Q_1$

Refrigerator: $COP_R = Q_2/W = Q_2/(Q_1 - Q_2)$

Heat pump: $COP_{HP} = Q_1/W = Q_1/(Q_1 - Q_2)$

$COP_{HP} = COP_R + 1$ (for the same Q_1, Q_2)

- A heat engine takes Q_1 from a high-temperature source, converts part into work W and rejects Q_2 to a low-temperature sink. A refrigerator uses work to remove Q_2 from a cold space and rejects Q_1 to the surroundings; a heat pump does the same but its useful output is Q_1 (heating).
- COP can exceed 1 (it is not an efficiency); COP_{HP} is always > 1.

★ **PYQ-TREND:** $COP_{HP} - COP_R = 1$ (very frequent); COP can exceed unity; definitions of efficiency and COP.

4.3 Statements of the Second Law

- Kelvin–Planck statement: it is impossible to construct a device that, operating in a cycle, produces no effect other than the raising of a weight (work) and the exchange of heat with a single reservoir. ⇒ No heat engine can have 100% efficiency; a sink is always needed. A machine violating it is a PMM-2 (perpetual motion machine of the second kind).
- Clausius statement: it is impossible to construct a device that, operating in a cycle, produces no effect other than the transfer of heat from a cooler body to a hotter body. ⇒ A refrigerator needs work input.
- The two statements are equivalent: violation of one implies violation of the other.

★ **PYQ-TREND:** Kelvin–Planck vs Clausius statements (very frequent); PMM-2 violates Kelvin–Planck; equivalence of the two statements.

4.4 Reversible and Irreversible Processes

- Reversible process: one that can be reversed so that both the system and the surroundings return to their original states with no trace left. It is an idealisation: quasi-static and without dissipative effects.
- Causes of irreversibility: friction (mechanical and fluid), heat transfer across a finite temperature difference, unrestrained (free) expansion, mixing of different substances or fluids at different states, throttling, inelastic deformation, electrical resistance (I^2R), hysteresis, chemical reactions, combustion.
- Internally reversible: no irreversibility inside the system; externally reversible: heat transfer with surroundings across infinitesimal ΔT.

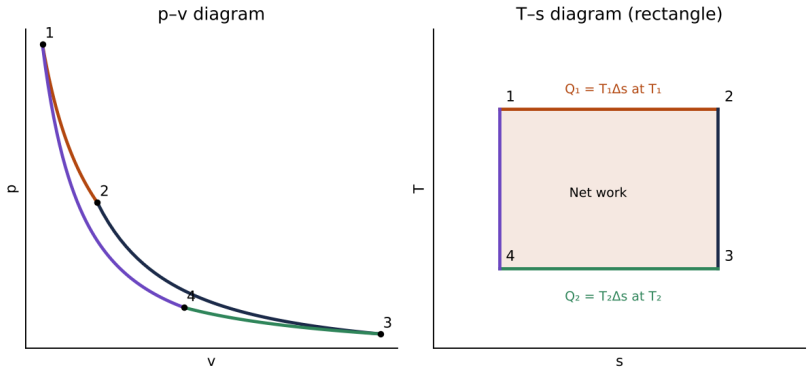
★ **PYQ-TREND:** Causes of irreversibility (friction, finite ΔT heat transfer, free expansion, mixing, throttling) – frequent; reversible process is an idealisation.

4.5 Carnot Cycle and Carnot Theorem

- Carnot cycle (Sadi Carnot, 1824): two reversible isothermal processes and two reversible adiabatic (isentropic) processes. 1–2 isothermal heat addition at T_1, 2–3 adiabatic expansion, 3–4 isothermal heat rejection at T_2, 4–1 adiabatic compression.

```
Carnot efficiency: η_C = 1 − T₂/T₁ (temperatures in kelvin)
Carnot refrigerator: COP_R = T₂/(T₁ − T₂); Carnot heat pump: COP_HP = T₁/(T₁ − T₂)
```

- Carnot theorem: no heat engine operating between two reservoirs can be more efficient than a reversible engine operating between the same reservoirs. Corollary: all reversible engines operating between the same two reservoirs have the same efficiency, independent of the working substance.
- Carnot efficiency increases if T_1 increases or T_2 decreases. Lowering T_2 by ΔT increases efficiency more than raising T_1 by the same ΔT ($\partial\eta/\partial T_2 = -1/T_1 > \partial\eta/\partial T_1 = T_2/T_1^2$ in magnitude). But T_2 cannot practically go below ambient.
- Carnot cycle is impractical: isothermal heat transfer is extremely slow; adiabatic processes must be very fast; the work ratio is small (for gases); with steam in the wet region, compressing a wet mixture is difficult.
- 100% efficiency would require $T_2 = 0$ K, which the third law says is unattainable.




```
Engines in series (same heat flow through): equal work → T = (T₁ + T₂)/2; equal
efficiency → T = √(T₁T₂)
```

★ **PYQ-TREND:** Carnot efficiency formula and dependence only on temperatures (very frequent); Carnot cycle processes; Carnot COP formulas; series engines ($\sqrt{T_1T_2}$ for equal efficiency); impracticality of Carnot cycle.

✔ **TOPPER TRICK:** For COP problems, write COP_R = T_L/ΔT and COP_HP = T_H/ΔT – the denominator is always the temperature difference in K.

✖ **COMMON MISTAKE:** Never use °C in Carnot formulas. η = 1 − 27/327 is wrong; correct is 1 − 300/600 = 0.5.

4.6 Thermodynamic Temperature Scale

- Kelvin showed that for reversible engines Q_1/Q_2 depends only on the reservoir temperatures; choosing $Q_1/Q_2 = T_1/T_2$ defines the absolute (Kelvin) thermodynamic scale, independent of any substance. It coincides with the ideal-gas temperature scale.

```
Q₁/Q₂ = T₁/T₂ (reversible engine)
```

★ **PYQ-TREND:** Thermodynamic temperature scale is independent of the working substance.

✎ **DESCRIPTIVE Q:** Prove that the efficiency of a reversible engine is greater than that of an irreversible engine working between the same reservoirs (Carnot's theorem).

Assume an irreversible engine I is more efficient than a reversible engine R between the same source and sink. Let both receive Q_1 from the source; I produces $W_I > W_R$ and rejects less heat. Since R is reversible, run it backward as a heat pump driven by part of W_I: it takes W_R, extracts heat from the sink and delivers Q_1 back to the source. The source then has no net exchange; the combined device takes net heat $(W_I - W_R)$ from the sink alone and converts it entirely into work $W_I - W_R$. This is a PMM-2, violating the Kelvin–Planck statement. Hence the assumption is false: $\eta_I \le \eta_R$. By similar reasoning, two reversible engines between the same reservoirs must have equal efficiency, so the reversible efficiency depends only on the reservoir temperatures: $\eta = 1 - T_2/T_1$.

CHAPTER 5

Entropy

5.1 Clausius Inequality and Entropy

Clausius inequality: ∮ δQ/T ≤ 0 (= 0 for reversible cycles, < 0 for irreversible cycles)
Definition: dS = (δQ/T)_rev; $S_2 - S_1 = \int(\delta Q/T)_{rev}$ (kJ/K; specific s in kJ/kg·K)
For any process: dS ≥ δQ/T; $S_2 - S_1 = \int \delta Q/T + S_{gen}$ ($S_{gen} \geq 0$)

- Entropy is a property (∮dS = 0 for a reversible cycle), a measure of molecular disorder / randomness and of the unavailability of energy.
- On a T–s diagram the area under a reversible process curve equals the heat transfer.
- Entropy can be transferred by heat (and by mass in open systems) – never by work. It is generated by irreversibilities.

★ **PYQ-TREND:** Clausius inequality; dS = δQ_rev/T; entropy as a property; area under T–s = heat; work does not transfer entropy.

5.2 Principle of Increase of Entropy

Isolated system: $\Delta S_{isolated} \geq 0$; $\Delta S_{universe} = \Delta S_{system} + \Delta S_{surroundings} \geq 0$

- Entropy of an isolated system (or the universe) never decreases. ΔS > 0 for real (irreversible) processes, = 0 for reversible ones, < 0 impossible.
- Entropy of a system can decrease (e.g., when it is cooled) provided the surroundings' entropy increases by more.

Heat transfer Q from T_H to T_L: $\Delta S_{gen} = Q(1/T_L - 1/T_H) > 0$

★ **PYQ-TREND:** Entropy of isolated system never decreases (very frequent); entropy generated in heat transfer across finite ΔT; entropy of a system may decrease.

5.3 Entropy Change of Ideal Gases, Solids and Liquids

Tds relations: T ds = du + p dv; T ds = dh − v dp
Ideal gas: $s_2 - s_1 = c_v \ln(T_2/T_1) + R \ln(v_2/v_1) = c_p \ln(T_2/T_1) - R \ln(p_2/p_1) = c_v \ln(p_2/p_1) + c_p \ln(v_2/v_1)$
Isothermal: $\Delta s = R \ln(v_2/v_1) = R \ln(p_1/p_2)$
Isochoric: $\Delta s = c_v \ln(T_2/T_1)$; Isobaric: $\Delta s = c_p \ln(T_2/T_1)$
Polytropic: $\Delta s = c_v (n - \gamma)/(n - 1) \ln(T_2/T_1)$
Reversible adiabatic: Δs = 0 (isentropic)
Solids/liquids: $\Delta s = c \ln(T_2/T_1)$
Phase change at constant T: $\Delta s = h_{fg}/T_{sat}$ (water at 100 °C: 2257/373.15 = 6.05 kJ/kg·K)
Mixing of two equal masses at T_1 and T_2: $\Delta s = 2mc \ln[(T_1 + T_2)/(2\sqrt{(T_1 T_2)})] > 0$ (AM > GM)

- On a T–s diagram, constant-volume lines are steeper than constant-pressure lines (slope T/cv > T/cp).
- On an h–s (Mollier) diagram for an ideal gas, isobars diverge to the right.

- Free expansion of ideal gas into vacuum (volume doubles): $\Delta s = R \ln 2 > 0$ although $Q = 0$ – because the process is irreversible.

★ **PYQ-TREND:** Entropy change formulas for each process; reversible adiabatic = isentropic (very frequent); isochore steeper than isobar on T–s; entropy change in free expansion; entropy change of vaporisation.

✖ **COMMON MISTAKE:** Adiabatic does not always mean isentropic. An irreversible adiabatic process (throttling, adiabatic friction) has $\Delta s > 0$. Only reversible adiabatic = isentropic.

5.4 Third Law and Absolute Entropy

- Third law (Nernst heat theorem / Planck statement): the entropy of a pure, perfectly crystalline substance at absolute zero is zero. Consequence: absolute zero cannot be reached in a finite number of steps.
- Allows absolute entropies to be tabulated (useful in chemical thermodynamics).

★ **PYQ-TREND:** Third law statement; unattainability of absolute zero.

5.5 Isentropic Efficiencies

Turbine: η_T = actual work/isentropic work = $(h_1 - h_2)/(h_1 - h_{2s})$

Compressor/pump: η_C = isentropic work/actual work = $(h_{2s} - h_1)/(h_2 - h_1)$

Nozzle: η_N = actual KE at exit / isentropic KE = $(h_1 - h_2)/(h_1 - h_{2s})$

- Irreversibilities make the actual exit state lie to the right of the isentropic state on h–s/T–s diagrams (higher entropy).

★ **PYQ-TREND:** Definitions of isentropic efficiency of turbine and compressor (note the inverse ratio for compressors).

✎ **DESCRIPTIVE Q:** State the principle of increase of entropy and prove that entropy of an isolated system increases in any irreversible process.

For any process, the Clausius inequality gives $dS \geq \delta Q/T$, with equality for reversible processes. An isolated system exchanges no heat ($\delta Q = 0$), so $dS_{isolated} \geq 0$: its entropy either increases (irreversible process) or remains constant (reversible process); it can never decrease. Since the universe (system + surroundings) is an isolated system, $\Delta S_{universe} = \Delta S_{sys} + \Delta S_{surr} \geq 0$. Example: heat Q flowing from a hot body at T_H to a cold body at T_L: $\Delta S = -Q/T_H + Q/T_L = Q(T_H - T_L)/(T_H T_L) > 0$ since $T_H > T_L$. The increase measures the irreversibility and the degradation of energy; this principle is why natural processes have a direction.

CHAPTER 6

Availability, Thermodynamic Relations and Mixtures

6.1 Available and Unavailable Energy

- Available energy (exergy): the maximum useful work obtainable from a system or energy as it is brought into equilibrium with the environment (dead state, T_0, p_0).

Available energy from heat Q at constant T: $AE = Q(1 - T_0/T)$; unavailable energy $UE = T_0 \Delta S$

Heat from a body cooled from T_1 to T_0 (variable T): $AE = Q - T_0 \Delta s = mc[(T_1 - T_0) - T_0 \ln(T_1/T_0)]$

Loss of available energy due to heat transfer through finite ΔT: $\Delta AE = T_0 \Delta S_{gen}$

Closed-system availability: $\varphi = (u - u_0) + p_0(v - v_0) - T_0(s - s_0) + V^2/2 + gz$

Flow availability (exergy): $\psi = (h - h_0) - T_0(s - s_0) + V^2/2 + gz$

Irreversibility (Gouy–Stodola theorem): $I = W_{rev} - W_{actual} = T_0 S_{gen}$

Second-law efficiency: $\eta_{II} = \eta_{th}/\eta_{rev}$ (engines) = COP/COP_rev (refrigerators)

- Energy is conserved but exergy is destroyed by irreversibility. High-temperature heat has more available energy per kJ than low-temperature heat ('quality of energy').

★ **PYQ-TREND:** Unavailable energy = $T_0\Delta S$ (frequent); Gouy–Stodola theorem; available energy decreases when heat flows to a lower temperature; second-law efficiency.

6.2 Thermodynamic Potentials and Maxwell Relations

Helmholtz function $F = U - TS$; Gibbs function $G = H - TS$

$dU = T\,dS - p\,dV$; $dH = T\,dS + V\,dp$; $dF = -S\,dT - p\,dV$; $dG = -S\,dT + V\,dp$

Maxwell: $(\partial T/\partial v)_s = -(\partial p/\partial s)_v$; $(\partial T/\partial p)_s = (\partial v/\partial s)_p$; $(\partial p/\partial T)_v = (\partial s/\partial v)_T$; $(\partial v/\partial T)_p = -(\partial s/\partial p)_T$

Clausius–Clapeyron: $dp/dT = h_{fg}/(T v_{fg})$ ($\approx h_{fg}\, p/(R T^2)$ for vapour as ideal gas, $v_f \ll v_g$)

$c_p - c_v = -T (\partial v/\partial T)_p^2 (\partial p/\partial v)_T = T v \beta^2/\kappa_T \geq 0$

Joule–Thomson coefficient: $\mu = (\partial T/\partial p)_h = [T(\partial v/\partial T)_p - v]/c_p$

- At constant T and p, a system reaches equilibrium when G is minimum (phase change, chemical equilibrium: dG = 0). At constant T and V, F is minimum.
- $c_p = c_v$ for incompressible substances and at the density maximum of water (4 °C, where $\beta = 0$).
- Volume expansivity $\beta = (1/v)(\partial v/\partial T)_p$; isothermal compressibility $\kappa_T = -(1/v)(\partial v/\partial p)_T$.

★ **PYQ-TREND:** Gibbs and Helmholtz definitions; Maxwell relations; Clausius–Clapeyron relates saturation pressure slope to latent heat; $c_p \geq c_v$ always; $c_p = c_v$ for water at 4 °C; G minimum at equilibrium at constant T, p.

6.3 Gas Mixtures

Dalton's law: $p = \Sigma p_i$; $p_i = y_i\, p$ (y_i = mole fraction)

Amagat's law: $V = \Sigma V_i$; volume fraction = mole fraction

Mixture properties: $R_{mix} = \bar{R}/M_{mix}$; $M_{mix} = \Sigma y_i M_i$; $c_{p,mix} = \Sigma(\text{mass fraction} \times c_{p_i})$

Entropy of mixing ideal gases: $\Delta S = -\bar{R}\, \Sigma\, n_i \ln y_i > 0$

- Air composition (dry, by volume): N_2 78.08%, O_2 20.95%, Ar 0.93%, $CO_2 \approx$ 0.04%; by mass $O_2 \approx$ 23.2%; molecular mass 28.97 kg/kmol.

★ **PYQ-TREND:** Dalton's law of partial pressures; mole fraction = volume fraction; composition of air; entropy of mixing positive.

CHAPTER 7

Properties of Pure Substances and Steam

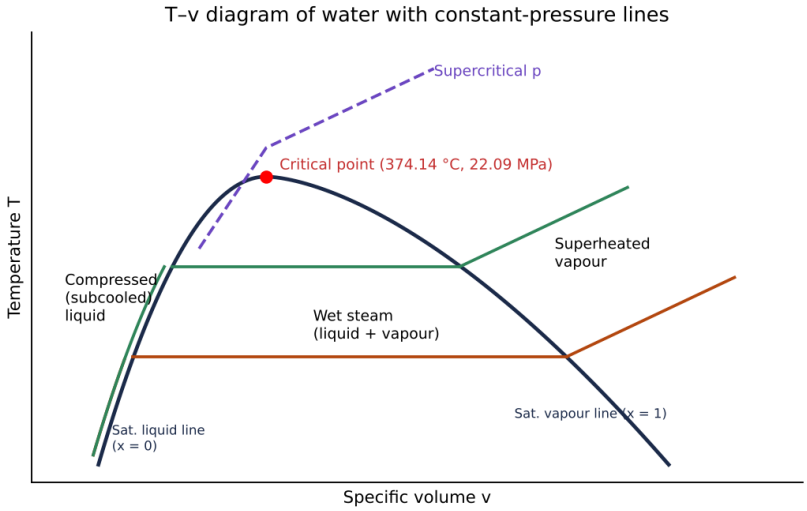
7.1 Pure Substance and Phase Change

- Pure substance: homogeneous and invariable in chemical composition throughout (it may exist in more than one phase). Water, steam, ice–water mixture, nitrogen, air (as long as it does not condense partially) are pure substances; a mixture of liquid air and gaseous air is not (different compositions).
- Heating water at constant pressure: compressed (subcooled) liquid → saturated liquid → wet steam (liquid + vapour) → dry saturated vapour → superheated vapour.
- Saturation temperature: temperature at which a substance boils at a given pressure (100 °C at 1.01325 bar; ≈ 179.9 °C at 10 bar; ≈ 99.6 °C at 1 bar).
- Sensible heat: changes temperature. Latent heat: changes phase at constant temperature. Latent heat of vaporisation of water at 1 atm ≈ 2257 kJ/kg; it decreases with pressure and becomes zero at the critical point.

Point	Water value
Critical point	22.064 MPa (≈ 221.2 bar), 373.95 °C (≈ 374.15 °C in older tables), v_c = 0.003106 m³/kg
Triple point	0.01 °C (273.16 K), 0.6117 kPa

- Above the critical point there is no distinct phase change (supercritical fluid); latent heat is zero.
- At the triple point solid, liquid and vapour coexist. Below the triple-point pressure, solid sublimates directly to vapour (dry ice sublimes at 1 atm because CO_2's triple point is at 5.1 atm).
- Water expands on freezing, so its solid–liquid line on a p–T diagram has a negative slope (ice melts under pressure – regelation, ice skating). Most substances have a positive slope.

```
Gibbs phase rule: F = C − P + 2 (pure substance at triple point: F = 1 − 3 + 2 = 0)
```



★ **PYQ-TREND:** Critical point of water (221.2 bar, 374 °C) and latent heat zero there (very frequent); triple point 0.01 °C, 0.611 kPa; degrees of freedom at triple point = 0; negative slope of fusion curve for water; pure substance definition; sublimation of dry ice.

7.2 Properties of Wet and Superheated Steam

```
Dryness fraction x = m_vapour/(m_vapour + m_liquid) (0 ≤ x ≤ 1)
Wetness = 1 − x; quality as percentage = 100x
Wet steam: v = v_f + x v_fg ≈ x v_g; h = h_f + x h_fg; s = s_f + x s_fg; u = u_f + x u_fg
Superheated: h = h_g + c_ps(T_sup − T_sat) (c_ps ≈ 2.1 kJ/kg·K); degree of superheat = T_sup − T_sat
Internal energy: u = h − pv
```

- Steam tables: saturation (temperature-based and pressure-based), superheated, compressed liquid. Reference: u and s of saturated liquid at the triple point = 0.
- Dry saturated steam: x = 1; on the saturated vapour line. Superheated steam behaves more like a gas the higher the superheat.
- Advantages of superheated steam: higher work per kg, less condensation and erosion in turbines, higher cycle efficiency.

★ **PYQ-TREND:** Dryness fraction definition; h = h_f + x h_fg; degree of superheat; reference state of steam tables.

7.3 Measurement of Dryness Fraction

Calorimeter	Principle / suitability
Barrel (bucket) calorimeter	Steam condensed in cold water – heat balance; least accurate
Separating calorimeter	Mechanically separates water droplets; gives x_1 (approximate, for very wet steam)
Throttling calorimeter	Steam throttled to superheated state at low pressure; h before = h after; works only if steam is fairly dry (x > ≈ 0.93–0.95)
Combined separating & throttling	Separator first, then throttling; for very wet steam; $x = x_1 \times x_2$
Electrical calorimeter	Electric heating to superheat a sample

```
Throttling calorimeter: h_f1 + x h_fg1 = h_g2 + c_ps(T₂ − T_sat2) → x
Combined: x = x_separating × x_throttling
```

★ **PYQ-TREND:** Throttling calorimeter uses isenthalpic throttling and needs the steam to become superheated; combined calorimeter for very wet steam; x = x_1x_2 (very frequent).

7.4 Property Diagrams

- p–v, T–v, T–s, h–s (Mollier) and p–h diagrams. Saturation dome: left side = saturated liquid line, right side = saturated vapour line, meeting at the critical point.
- On T–s: constant-pressure lines in the wet region are horizontal (coincide with constant temperature); in the superheated region they rise steeply; constant-volume lines are steeper than constant-pressure lines.
- Mollier chart (h–s): used for steam turbines and nozzles; isentropic process is a vertical line, throttling is a horizontal line; constant-pressure lines diverge upward; constant-dryness lines below the saturation line; constant-temperature lines become nearly horizontal far into the superheated region (ideal-gas behaviour, h = f(T)).
- p–h chart: used for refrigerants (VCR cycle); throttling is a vertical line.

★ **PYQ-TREND:** Mollier diagram is h–s (very frequent); isentropic = vertical on h–s; throttling = horizontal on h–s, vertical on p–h; isotherms nearly horizontal in superheated region far from saturation.

✔ **TOPPER TRICK:** Throttling → 'horizontal on Mollier, vertical on p–h' – because enthalpy is the x-axis of p–h and the y-axis of h–s.

✎ **DESCRIPTIVE Q:** Explain the throttling calorimeter and its limitation.

A sample of wet steam from the main is passed through a small orifice (throttle) into a chamber at a lower pressure, close to atmospheric. Throttling is isenthalpic: $h_1 = h_2$. If the throttled steam becomes superheated, its pressure and temperature fix $h_2 = h_{g2} + c_{ps}(T_2 - T_{sat2})$, and the dryness of the original steam follows from $h_{f1} + x\,h_{fg1} = h_2$. The method requires that the steam be superheated after throttling – which happens only if the original steam is fairly dry (dryness above about 0.93–0.95 at moderate pressures). For wetter steam the throttled state remains wet and cannot be fixed by p and T alone. In that case a separating calorimeter is placed ahead to remove most water first, and the overall dryness is $x = x_{separator} \times x_{throttling}$.

CHAPTER 8

Gas Power Cycles (Air-Standard Cycles)

8.1 Air-Standard Assumptions and Terms

- Working fluid is air, behaving as an ideal gas with constant specific heats; cycle is closed (exhaust and intake replaced by heat rejection); combustion replaced by heat addition from an external source; all processes internally reversible. The 'air-standard efficiency' is an upper limit for real engines.

```
Compression ratio r = V₁/V₂ = (V_s + V_c)/V_c; clearance ratio c = V_c/V_s → r = 1 + 1/c
Cut-off ratio ρ = V₃/V₂ (Diesel); pressure (explosion) ratio α = p₃/p₂ (Otto, Dual)
Mean effective pressure: MEP = net work per cycle / swept volume
Work ratio = net work / gross (positive) work
```

- MEP is the constant pressure that, acting over the full stroke, would give the same net work – used to compare engines of different sizes.

★ **PYQ-TREND:** Air-standard assumptions; compression ratio and clearance ratio relation; MEP definition (very frequent).

8.2 Otto Cycle (Constant-Volume Cycle) – SI Engines

- Processes: 1–2 isentropic compression; 2–3 constant-volume heat addition; 3–4 isentropic expansion; 4–1 constant-volume heat rejection.

```
η_Otto = 1 − 1/r^(γ−1)
Heat added Q₁ = cv(T₃ − T₂); rejected Q₂ = cv(T₄ − T₁); T₂/T₁ = T₃/T₄ = r^(γ−1)
MEP_Otto = p₁ r [(α − 1)(r^(γ−1) − 1)] / [(γ − 1)(r − 1)]
For maximum work between T_min and T_max: optimum r^(γ−1) = √(T₃/T₁) → T₂ = T₄ = √(T₁T₃)
```

- Efficiency depends only on r (and γ), not on heat added. Increasing r is limited by detonation (knocking) in SI engines; practical r = 6–10 (up to ~12–13 with direct injection).

r (γ = 1.4)	η_Otto
4	42.6%
6	51.2%
8	56.5%
10	60.2%
12	63.0%

★ **PYQ-TREND:** Otto efficiency formula and dependence only on r (very frequent); limit of r by knocking; $T_2 = T_4 = \sqrt{(T_1T_3)}$ for maximum work.

✔ **TOPPER TRICK:** Quick values: 2^0.4 = 1.32, 4^0.4 = 1.74, 8^0.4 = 2.30, 16^0.4 = 3.03, 10^0.4 = 2.51.

8.3 Diesel Cycle (Constant-Pressure Cycle) – CI Engines

- Processes: 1–2 isentropic compression; 2–3 constant-pressure heat addition; 3–4 isentropic expansion; 4–1 constant-volume heat rejection.

```
η_Diesel = 1 − (1/r^(γ−1)) × [(ρ^γ − 1)/(γ(ρ − 1))]
Q₁ = cp(T₃ − T₂); Q₂ = cv(T₄ − T₁); T₃/T₂ = ρ; T₄/T₁ = ρ^γ
MEP_Diesel = p₁ r^γ [γ(ρ − 1) − r^(1−γ)(ρ^γ − 1)] / [(γ − 1)(r − 1)]
```

- The bracketed term is > 1, so for the same r, Diesel efficiency < Otto efficiency. Diesel efficiency falls as ρ (load) increases; η_Diesel → η_Otto as ρ → 1.
- CI engines use r = 14–22, so in practice they are more efficient than SI engines.

```
Cut-off as % of stroke: (V₃ − V₂)/(V₁ − V₂) = (ρ − 1)/(r − 1)
```

★ **PYQ-TREND:** Diesel efficiency formula; efficiency decreases with increasing cut-off ratio; Diesel → Otto when ρ = 1; cut-off percentage of stroke.

8.4 Dual Cycle (Limited-Pressure / Mixed Cycle)

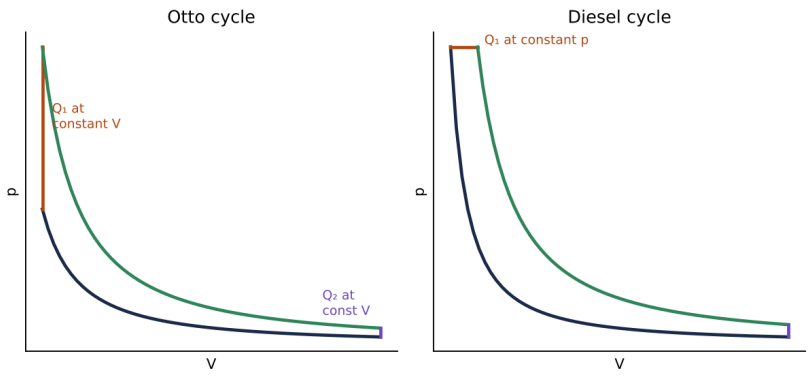
- Heat added partly at constant volume (2–3) and partly at constant pressure (3–4); closer to real high-speed CI engines.

```
η_Dual = 1 − (1/r^(γ−1)) × [(αρ^γ − 1)/((α − 1) + γα(ρ − 1))]
```

- Reduces to Otto when ρ = 1, to Diesel when α = 1.

8.5 Comparison of Otto, Diesel and Dual Cycles

Condition	Order of efficiency
Same r and same heat input	Otto > Dual > Diesel
Same maximum pressure and temperature (same heat rejection)	Diesel > Dual > Otto
Same maximum pressure and same heat input	Diesel > Dual > Otto
Same r and same heat rejection	Otto > Dual > Diesel



★ **PYQ-TREND:** Comparison orders (asked almost every year). Remember: same r → Otto best; same peak pressure → Diesel best.

8.6 Other Gas Cycles

Cycle	Processes	Remarks
Stirling	2 isothermal + 2 constant-volume, with perfect regenerator	η = Carnot; external combustion (Stirling engine)
Ericsson	2 isothermal + 2 constant-pressure, with regenerator	η = Carnot
Atkinson	2 isentropic + constant-volume addition + constant-pressure rejection	Expansion ratio > compression ratio; more efficient than Otto; used in hybrid cars (via late intake-valve closing)
Lenoir	Constant-volume heat addition, isentropic expansion, constant-pressure rejection (no compression)	Pulse-jet; low efficiency
Brayton (Joule)	2 isentropic + 2 constant-pressure	Gas turbines
Reversed Brayton (Bell–Coleman)	–	Air refrigeration

- Carnot, Stirling and Ericsson (with ideal regeneration) all have the maximum possible efficiency between two temperatures.

★ **PYQ-TREND:** Stirling and Ericsson efficiencies equal Carnot (with regenerator); Atkinson cycle efficiency higher than Otto; Lenoir cycle has no compression; Bell–Coleman for air refrigeration.

8.7 Brayton (Joule) Cycle – Gas Turbines

- Processes: 1–2 isentropic compression (compressor); 2–3 constant-pressure heat addition (combustion chamber); 3–4 isentropic expansion (turbine); 4–1 constant-pressure heat rejection (exhaust/cooler).

```
η_Brayton = 1 − 1/r_p^((γ−1)/γ) (r_p = pressure ratio)
Net work w = cp[(T₃ − T₄) − (T₂ − T₁)]
Optimum pressure ratio for maximum net work: r_p,opt = (T₃/T₁)^(γ/(2(γ−1))) → T₂ = T₄ = √(T₁T₃)
Maximum net work: w_max = cp(√T₃ − √T₁)²
Back-work ratio = compressor work / turbine work (40–80% in gas turbines — very high)
With actual efficiencies: η_c = (T₂s − T₁)/(T₂ − T₁); η_t = (T₃ − T₄)/(T₃ − T₄s)
```

- Efficiency depends only on pressure ratio; net work depends on both r_p and T_3/T_1 (turbine inlet temperature limited by blade materials ≈ 1100–1600 °C with cooling).

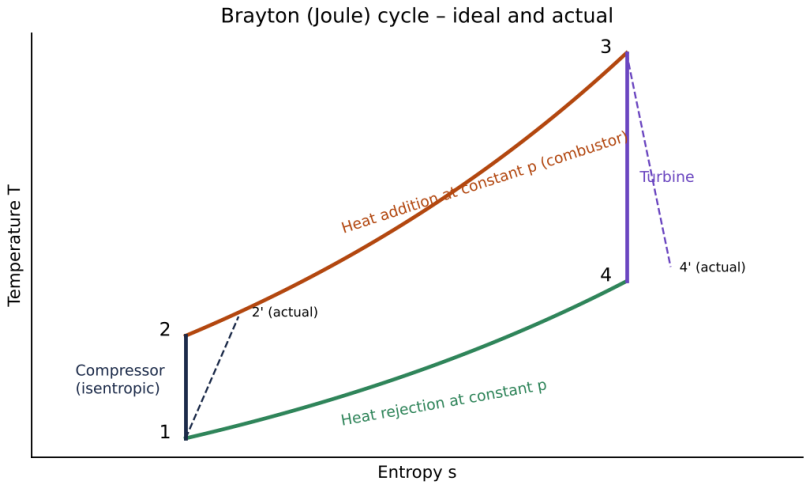
Modifications of the gas turbine cycle

Modification	Effect
Regeneration (heat exchanger using exhaust to preheat compressed air)	Increases efficiency, same net work; effective only when $T_4 > T_2$ (low r_p); ideal regenerator: $\eta = 1 - (T_1/T_3)\, r_p^{((\gamma-1)/\gamma)}$
Intercooling (between compressor stages)	Reduces compressor work → increases net work; alone, reduces efficiency (lower T_2 needs more fuel), but with regeneration increases it
Reheating (between turbine stages)	Increases turbine work → net work; alone reduces efficiency; with regeneration increases it

Modification	Effect
Intercooling + reheating + regeneration	Approaches Ericsson cycle → approaches Carnot efficiency

`Regenerator effectiveness ε = (T_x − T₂)/(T₄ − T₂)`

- Open-cycle gas turbine: air from atmosphere, combustion products exhausted. Closed cycle: working fluid recirculated (air, helium), heat added in a heat exchanger – allows any fuel (nuclear, coal) and higher pressures.
- Gas turbines: compact, light, quick start, low cooling-water need; used for aircraft propulsion, peak-load power plants, combined cycles (Brayton topping + Rankine bottoming, efficiency 55–62%).



★ **PYQ-TREND:** Brayton efficiency depends only on r_p (very frequent); optimum r_p for maximum work; back-work ratio high; regeneration increases efficiency (not work); intercooling and reheating increase work; closed vs open cycle; combined cycle efficiency.

✖ **COMMON MISTAKE:** Regeneration helps only if turbine exhaust temperature exceeds compressor delivery temperature. At high pressure ratios ($T_2 > T_4$) a regenerator would reduce efficiency.

✎ **DESCRIPTIVE Q:** Derive the expression for the air-standard efficiency of the Diesel cycle.

Let the compression ratio be $r = V_1/V_2$ and cut-off ratio $\rho = V_3/V_2$. Heat added at constant pressure $Q_1 = cp(T_3 - T_2)$; heat rejected at constant volume $Q_2 = cv(T_4 - T_1)$. $\eta = 1 - cv(T_4 - T_1)/(cp(T_3 - T_2)) = 1 - (T_4 - T_1)/(\gamma(T_3 - T_2))$. From isentropic compression $T_2 = T_1 r^{(\gamma-1)}$. At constant pressure $T_3 = T_2\rho = T_1\rho\, r^{(\gamma-1)}$. Expansion ratio $V_4/V_3 = r/\rho$, so $T_4 = T_3(\rho/r)^{(\gamma-1)} = T_1\rho^{\gamma}$. Substituting: $\eta = 1 - T_1(\rho^{\gamma} - 1)/(\gamma T_1 r^{(\gamma-1)}(\rho - 1)) = 1 - (1/r^{(\gamma-1)})(\rho^{\gamma} - 1)/(\gamma(\rho - 1))$. Since $(\rho^{\gamma} - 1)/(\gamma(\rho - 1)) > 1$ for $\rho > 1$, the Diesel efficiency is lower than Otto's for the same compression ratio, and it decreases as the cut-off ratio (load) increases.

CHAPTER 9

IC Engines – Classification, Working and Components

9.1 Heat Engines and Classification of IC Engines

- External combustion (EC) engines: combustion outside the working cylinder (steam engine, steam turbine, Stirling engine, closed gas turbine). Internal combustion (IC) engines: combustion inside the engine; working fluid is the air–fuel mixture/combustion products.
- Advantages of IC engines: compact, high power-to-weight ratio, quick starting, high efficiency at small sizes. Disadvantages: need high-quality fuels, vibration (reciprocating), limited sizes compared with steam plants.

Basis	Types
Ignition	Spark ignition (SI) / compression ignition (CI)
Cycle of operation	Otto, Diesel, Dual
Strokes per cycle	Four-stroke / two-stroke
Fuel	Petrol, diesel, gas (CNG, LPG, biogas), alcohol, bi-fuel, dual-fuel
Cooling	Air-cooled / water-cooled
Cylinder arrangement	In-line, V, opposed (flat/boxer), radial (aircraft), W, opposed-piston
Valve location	Overhead (I-head), side valve (L-head), T-head, F-head
Speed	Low (< 350 rpm), medium (350–1200), high (> 1200)
Charging	Naturally aspirated, supercharged, turbocharged
Load control (governing)	Quantity (SI), quality (CI), hit-and-miss (gas engines)
Application	Automotive, locomotive, marine, stationary, aircraft

★ **PYQ-TREND:** EC vs IC engine examples; classification bases; radial engines in aircraft; quantity vs quality governing.

9.2 Engine Terminology

Bore D; stroke $L = 2 \times$ crank radius; swept (displacement) volume $V_s = (\pi/4)D^2L$

Total cylinder volume $V = V_s + V_c$; compression ratio $r = (V_s + V_c)/V_c$

Mean piston speed $V_p = 2LN/60$ (m/s)

Engine displacement = $V_s \times$ number of cylinders

Stroke-to-bore ratio L/D: square engine $L = D$; over-square (short stroke) $L < D$ — high-speed engines; under-square (long stroke) $L > D$

- TDC (top dead centre) / IDC: piston at its top, minimum volume. BDC (bottom dead centre) / ODC: piston at its bottom, maximum volume.

- Clearance volume: volume above the piston at TDC.

★ **PYQ-TREND:** Swept volume and compression ratio calculations; mean piston speed; definition of TDC/BDC; square and over-square engines.

9.3 Four-Stroke Engines

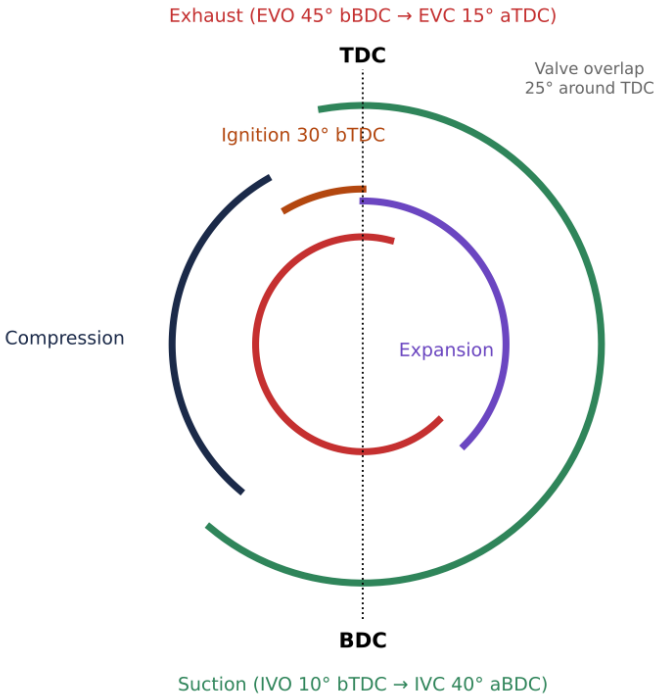
- Four strokes per cycle = two crankshaft revolutions per cycle: suction (intake), compression, power (expansion), exhaust. One power stroke every two revolutions → heavier flywheel needed.
- SI four-stroke: air–fuel mixture drawn in; compressed ($r \approx 6$–10); ignited by spark near TDC. CI four-stroke: only air drawn in; compressed ($r \approx 14$–22) to 500–700 °C; fuel injected near TDC, ignites by itself.
- Camshaft rotates at half the crankshaft speed in a four-stroke engine (one valve opening per cycle).

Valve timing (typical, four-stroke)

Event	SI engine	CI engine
Inlet valve opens (IVO)	10–20° before TDC	10–25° before TDC
Inlet valve closes (IVC)	30–50° after BDC	25–50° after BDC
Exhaust valve opens (EVO)	40–50° before BDC	30–50° before BDC
Exhaust valve closes (EVC)	10–15° after TDC	10–15° after TDC
Fuel injection / ignition	Spark 10–40° before TDC (advanced with speed)	Injection 10–20° before TDC

- Valve overlap: period near TDC when both inlet and exhaust valves are open – uses the inertia of exhaust gases to scavenge and draw in fresh charge; larger in high-speed and CI engines.
- IVC after BDC uses the ram (inertia) effect of the incoming charge to improve volumetric efficiency. EVO before BDC (blow-down) reduces back pressure during the exhaust stroke.

Typical valve timing diagram – four-stroke SI engine



★ **PYQ-TREND:** Valve overlap definition and purpose (very frequent); camshaft speed half of crankshaft; EVO before BDC (blow-down); IVC after BDC (ram effect); spark advance increases with speed.

9.4 Two-Stroke Engines

- Complete cycle in two strokes (one crankshaft revolution). Ports in the cylinder wall (inlet/transfer/exhaust) controlled by the piston; crankcase compression in small engines; no valves (or exhaust valves in large uniflow diesels).
- Scavenging: removal of exhaust gases by fresh charge. Types: cross-flow (deflector piston), loop (reverse) scavenging (Schnurle), uniflow (most efficient – air enters at one end, exhaust leaves at the other; used in large marine diesels).

```
Scavenging ratio = mass of delivered air / (swept volume × ambient density); scavenging efficiency = mass of fresh charge retained / total charge in cylinder
```

Feature	Four-stroke	Two-stroke
Power strokes	One per 2 revolutions	One per revolution
Theoretical power (same size, speed)	1	≈ 2 (actually 1.7–1.8)
Flywheel	Heavier	Lighter (more uniform torque)
Valves	Poppet valves, valve gear	Ports (simple, cheap)
Thermal efficiency	Higher	Lower (charge loss, incomplete scavenging)
Volumetric efficiency	Higher	Lower
Fuel consumption	Lower	Higher (fresh charge short-circuits to exhaust in SI)

Feature	Four-stroke	Two-stroke
Lubricating oil consumption	Lower	Higher (petroil lubrication in small SI)
Emissions (HC)	Lower	Higher
Wear, cooling	Less	More (more frequent heat input)
Uses	Cars, trucks, most engines	Mopeds, scooters (older), chainsaws, outboards, large marine diesels

★ **PYQ-TREND:** Two-stroke completes cycle in one revolution; uniflow scavenging most efficient; lighter flywheel for two-stroke; higher HC and lube-oil consumption; comparison table points.

9.5 Engine Components and Materials

Component	Function / material
Cylinder block	Houses cylinders; grey cast iron or aluminium alloy
Cylinder liner (wet/dry)	Replaceable wearing surface; alloy CI
Cylinder head	Closes cylinder; contains valves, combustion chamber; CI/Al alloy
Piston	Transmits gas force to connecting rod; aluminium alloy (light, good conductivity) or CI
Piston rings	Compression rings (seal gas) at top, oil (scraper) ring at bottom; alloy CI
Gudgeon (wrist) pin	Connects piston and small end of connecting rod; case-hardened steel
Connecting rod	Converts reciprocating to rotary motion; I-section forged steel
Crankshaft	Converts reciprocating motion to rotary; forged steel / SG iron
Camshaft	Operates valves; runs at half crank speed (4-stroke); driven by timing chain/belt/gears
Flywheel	Smooths torque fluctuations
Valves	Poppet valves; exhaust valve of heat-resistant steel (sometimes sodium-cooled)
Crankcase / oil sump	Encloses crank; stores oil
Gasket	Seals head and block
Spark plug / injector	Ignition / fuel injection

★ **PYQ-TREND:** Number and function of piston rings (compression vs oil rings); connecting rod I-section; camshaft speed; materials of piston and liner; sodium-cooled exhaust valves.

✎ **DESCRIPTIVE Q:** Compare four-stroke and two-stroke engines.

A four-stroke engine completes its cycle in four piston strokes (two revolutions): suction, compression, power and exhaust, with poppet valves operated by a camshaft. A two-stroke engine completes the cycle in two strokes (one revolution) using ports uncovered by the piston, with crankcase compression and scavenging. For the same size and speed the two-stroke engine theoretically gives double the power (in practice about 1.7–1.8 times), has a more uniform torque and a lighter flywheel, and is simpler, lighter and cheaper because it has no

valve gear. However, it has lower thermal and volumetric efficiency because some fresh charge escapes with the exhaust and scavenging is incomplete; fuel and lubricating-oil consumption and HC emissions are higher, and it runs hotter. Four-stroke engines are therefore used in cars, trucks and most stationary engines, while two-stroke engines are used where light weight and simplicity matter (small tools, outboard motors) and in very large uniflow marine diesels where they are highly efficient.

CHAPTER 10

IC Engines – Combustion, Knocking and Fuels

10.1 Combustion in SI Engines

- Stages (Ricardo): (1) ignition lag/preparation phase – chemical process, a flame nucleus forms after the spark (≈ 0.0015 s); (2) flame propagation – rapid, turbulent flame front crosses the chamber, sharp pressure rise; (3) after-burning – combustion of remaining charge during expansion.
- Factors increasing flame speed: turbulence (most important), higher compression ratio, slightly rich mixture (A/F ≈ 12.5–13), higher intake temperature/pressure, higher engine speed (more turbulence).
- Air–fuel ratio requirements: stoichiometric ≈ 14.7:1 (petrol); maximum power ≈ 12.5:1 (rich); maximum economy ≈ 16–17:1 (lean); cold start and idling need rich mixtures (choke).

```
Equivalence ratio φ = (F/A)_actual / (F/A)_stoich; φ > 1 rich, φ < 1 lean; λ = 1/φ
```

★ **PYQ-TREND:** Stages of SI combustion; stoichiometric A/F 14.7; rich mixture for maximum power and cold start; turbulence increases flame speed.

10.2 Detonation (Knocking) in SI Engines

- After the spark, the advancing flame compresses and heats the unburnt end gas. If the end gas reaches its self-ignition temperature and the ignition delay elapses before the flame arrives, it auto-ignites spontaneously, producing a sudden pressure rise and pressure waves that hit the cylinder walls – the metallic 'pinking/pinging' sound. This is detonation (knock).
- Effects: noise, loss of power and efficiency, overheating of piston crown and spark plug, mechanical damage (piston crown erosion, ring breakage), pre-ignition.
- Pre-ignition: ignition of the charge by a hot spot (glowing carbon deposit, overheated spark plug or exhaust valve) before the spark – different from knock but can be caused by it.

Factor	To reduce knock in SI engines
Compression ratio	Decrease
Intake temperature	Decrease
Intake pressure (supercharging)	Decrease
Spark timing	Retard
Engine speed	Increase (more turbulence, less time)
Turbulence	Increase
Cylinder size / flame travel distance	Smaller; central spark plug
Fuel octane number	Increase
Mixture	Very rich or very lean
Coolant temperature / end-gas temperature	Decrease
Combustion chamber	Compact, with squish; end gas in cool region

Factor	To reduce knock in SI engines
EGR	Increase (dilutes charge)

★ **PYQ-TREND:** Detonation due to auto-ignition of end gas (very frequent); factors reducing SI knock (lower r, retard spark, higher octane, more turbulence, higher speed); pre-ignition vs detonation.

10.3 Combustion and Knock in CI Engines

- Stages: (1) ignition delay – physical delay (atomisation, vaporisation, mixing) + chemical delay (pre-flame reactions); (2) rapid (uncontrolled) combustion – fuel accumulated during the delay burns almost at once; (3) controlled combustion – fuel burns as it is injected; (4) after-burning.
- Diesel knock: if the ignition delay is long, much fuel accumulates and burns suddenly, causing a very high rate of pressure rise and a knocking sound. It occurs at the start of combustion (unlike SI knock, which occurs at the end).

Factor	To reduce knock in CI engines
Compression ratio	Increase
Intake temperature	Increase
Intake pressure (supercharging)	Increase
Injection timing	Slightly retard / optimise (inject at higher temperature)
Engine speed	Decrease (in time terms)
Fuel cetane number	Increase
Cylinder size	Increase
Atomisation/turbulence	Improve
Coolant temperature	Increase

- Rule: the factors that reduce knock in SI engines (lower temperature, lower r) increase knock in CI engines, because SI knock needs a long delay of the end gas whereas CI knock comes from a long delay at the start.

★ **PYQ-TREND:** Diesel knock due to long ignition delay; CI knock at start vs SI knock at end of combustion; opposite effect of factors in SI and CI engines (very frequent).

✔ **TOPPER TRICK:** SI knock is fought with 'cool and slow end gas' (low r, low T, high octane); CI knock is fought with 'hot and quick ignition' (high r, high T, high cetane).

10.4 Combustion Chambers

- SI chambers: T-head (old, knock-prone), L-head/side valve (low r), I-head/overhead valve (most common – compact, high r), F-head; wedge, hemispherical (pent-roof, multi-valve), bath-tub. Good chamber: compact (low surface/volume), central spark plug, squish areas for turbulence, exhaust valve near the plug, cool end-gas region.
- CI chambers: (1) open/direct injection (quiescent; multi-hole nozzle, high injection pressure; large slow engines; best fuel economy, cold-starts well); (2) swirl chamber (Ricardo Comet); (3) pre-combustion chamber; (4) air-cell/energy cell (Lanova). Indirect-injection (IDI) chambers use single-hole/pintle nozzles at lower pressure, run smoother and quieter but have higher heat loss and need glow plugs for cold starting.

- Swirl (rotation about cylinder axis) and squish (radial inward motion as piston approaches TDC) improve air–fuel mixing.

★ **PYQ-TREND:** I-head most efficient SI chamber; open chamber vs swirl/pre-combustion chambers; glow plugs in IDI engines; squish and swirl.

10.5 Fuels and Their Rating

- SI fuel (petrol/gasoline): high volatility, high self-ignition temperature (≈ 370 °C+) and high octane number so that it does not auto-ignite. CI fuel (diesel): low volatility, low self-ignition temperature (≈ 250 °C) and high cetane number so that it ignites quickly.
- Octane number: percentage by volume of iso-octane (2,2,4-trimethylpentane, ON = 100) in a mixture with normal heptane (ON = 0) that gives the same knock intensity as the fuel in a CFR engine. Methods: Research ON (RON, mild) and Motor ON (MON, severe); anti-knock index (RON + MON)/2. Indian petrol: regular ≈ 91 RON; premium 95–100.
- Performance number for fuels above 100 ON; anti-knock agents: tetraethyl lead (TEL – banned; poisons catalytic converters), MTBE, ethanol, aromatics, MMT.
- Cetane number: percentage of cetane (n-hexadecane, CN = 100) in a mixture with α-methyl naphthalene (CN = 0) that matches the ignition delay of the fuel. Diesel ≈ 40–55 (BS-VI ≥ 51). Cetane improvers: amyl nitrate, ethyl nitrate, ether. Diesel index = aniline point (°F) × API gravity/100.
- Hydrocarbon families: paraffins (C_nH_{2n+2} – straight-chain: good CI, poor SI; iso-paraffins: good SI), olefins, naphthenes, aromatics (high ON – good SI fuels, poor CI fuels). Long straight chains = high CN, low ON; compact/branched/ring molecules = high ON, low CN.
- Calorific value: higher (HCV) includes latent heat of water vapour formed; lower (LCV) excludes it. LCV ≈ 43–44 MJ/kg for petrol and diesel; LCV = HCV − 9H × 2442 kJ/kg (H = hydrogen mass fraction). Measured with a bomb calorimeter (solids, liquids) or Junkers gas calorimeter (gases).

Alternative fuel	Key facts
CNG	Mainly methane; ON ≈ 120–130; SI engines, clean
LPG	Propane + butane; ON ≈ 105–110
Ethanol	ON ≈ 108; E20 blending; oxygenated fuel
Methanol	ON ≈ 110; toxic
Biodiesel	Methyl/ethyl esters from vegetable oils via transesterification; CN ≈ 50–60; B20 blends
Hydrogen	Highest calorific value per kg (≈ 120 MJ/kg LCV); wide flammability; NOx only
Biogas	55–70% methane + CO_2

Stoichiometric A/F for a hydrocarbon C_xH_y: $(A/F)_s = 34.56\,(4x + y)/(12x + y)$ → octane C_8H_{18} ≈ 15:1

★ **PYQ-TREND:** Octane and cetane reference fuels (very frequent); good SI fuel = high ON, good CI fuel = high CN; TEL banned; aromatics high ON; paraffins high CN; HCV vs LCV; bomb calorimeter; biodiesel by transesterification; CNG main constituent methane.

✎ **DESCRIPTIVE Q:** Explain the phenomenon of detonation in SI engines and the factors affecting it.

In a spark-ignition engine the flame started by the spark travels across the chamber. Ahead of it, the unburnt end gas is compressed and heated by the expanding burnt gases and by radiation. If the end gas reaches its self-ignition temperature and remains there longer than its ignition delay before the flame arrives, it ignites spontaneously at many points. The almost instantaneous release of energy creates a very rapid pressure rise and pressure waves that reflect from the walls, causing the characteristic knocking sound, loss of power, overheating and damage to pistons, rings and plugs. Detonation is favoured by anything that raises the end-gas temperature or density or increases the time available for it to auto-ignite: high compression ratio, supercharging, high intake temperature, advanced spark, low engine speed, large cylinder bore with long flame travel, poor turbulence, low-octane fuel and hot spots. It is controlled by using high-octane fuel, retarding the spark (knock sensors), reducing compression ratio, compact chambers with central plugs and squish, better cooling of the end-gas region and EGR.

CHAPTER 11

IC Engines – Fuel Supply, Ignition, Cooling, Lubrication

11.1 Carburettor and Petrol Injection (SI Engines)

- Carburettor: prepares a combustible air–fuel mixture of the right ratio and atomises/vaporises the fuel. Simple carburettor parts: float chamber (constant fuel level by float and needle valve), venturi (throat creates suction), fuel nozzle/jet at the throat, throttle valve (butterfly – controls quantity of mixture = load), choke valve (upstream – restricts air for a rich cold-start mixture).

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Air velocity at throat: V = √(2 Δp/ρ_a); fuel flow ṁ_f = C_df A_f √(2ρ_f(Δp − ρ_f g z))
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- A simple carburettor gives a progressively richer mixture as airflow increases (fuel flow rises faster than air flow because air density falls at the throat). Compensating devices: auxiliary air valve/port, air-bleed jet, compensating jet, emulsion tube, back-suction control.
- Additional circuits: idling (slow-running) system, accelerating pump (extra fuel on sudden throttle opening), economiser/power enrichment, choke for cold start.
- Petrol injection replaced carburettors: TBI (throttle-body, single point), MPFI (multi-point port injection – one injector per cylinder, ECU-controlled), GDI (gasoline direct injection – into cylinder; allows higher r and stratified charge). Benefits: precise A/F control, better economy, lower emissions (essential for catalytic converters with closed-loop λ-sensor control), better cold starting and response.

★ **PYQ-TREND:** Functions of float chamber, venturi, throttle and choke (very frequent); simple carburettor richens mixture at high speeds; compensating jet; MPFI advantages; choke gives rich mixture.

11.2 Fuel Injection in CI Engines

- Requirements: meter the exact quantity per cycle, inject at the correct time and rate, atomise finely, distribute and penetrate properly, cut off sharply (no dribbling).
- Systems: air injection (obsolete – fuel blown in by compressed air); solid (airless) injection: individual pump system (one pump per cylinder), unit injector (pump and nozzle combined, cam-driven), distributor (rotary) pump, common-rail direct injection (CRDI – high-pressure pump fills a common rail at 1,500–2,500+ bar; electronically controlled injectors allow multiple injections per cycle; low noise and emissions).
- Fuel injection pump: Bosch-type jerk pump with helix-grooved plunger; quantity varied by rotating the plunger (rack and pinion) – quality governing.
- Nozzles: pintle (IDI; self-cleaning), pintaux, single-hole, multi-hole (DI; 3–10 holes). Injection pressure 100–300 bar (older systems).
- Fuel feed pump lifts fuel from the tank to the injection pump through filters.

★ **PYQ-TREND:** CRDI full form and features; jerk pump with helical plunger; multi-hole nozzle for DI, pintle for IDI; air injection obsolete; unit injector.

11.3 Ignition Systems (SI Engines)

- Battery (coil) ignition: battery (12 V) → ignition switch → primary winding of ignition coil (few turns, thick wire) → contact breaker (cam-operated) → condenser (capacitor) across the breaker. When the breaker opens, primary current collapses rapidly, inducing 10,000–25,000 V in the secondary winding (many turns), which the distributor sends to the correct spark plug.
- Condenser: prevents arcing across the contact-breaker points (protects them from burning) and speeds up the collapse of the magnetic field for a stronger spark.
- Magneto ignition: generates its own current (permanent magnets), no battery; spark intensity improves with speed – used in motorcycles, racing cars, aircraft piston engines; poor starting at low speed.
- Electronic ignition: transistorised coil ignition (TCI), capacitor discharge ignition (CDI – two-wheelers), distributorless/coil-on-plug systems with ECU control.
- Spark advance: centrifugal advance (with speed) and vacuum advance (with load/manifold vacuum – more advance at part load). Ignition timing is measured with a timing light (stroboscope).
- Spark plug: centre electrode, ground electrode, ceramic insulator; gap ≈ 0.6–1.0 mm. Hot plug (long insulator nose – retains heat; low-speed, low-compression engines, prevents fouling); cold plug (short nose – dissipates heat; high-speed, high-compression engines, prevents pre-ignition).
- Firing order: chosen to balance crankshaft loads and avoid adjacent cylinders firing successively. 4-cylinder in-line: 1-3-4-2 (or 1-2-4-3); 6-cylinder in-line: 1-5-3-6-2-4; 8-cylinder V: 1-5-4-8-6-3-7-2 (typical).

★ **PYQ-TREND:** Function of condenser in battery ignition (very frequent); magneto better at high speed; cold plug for high-compression engines; centrifugal and vacuum advance; firing order 1-3-4-2; secondary voltage range.

11.4 Cooling Systems

- Purpose: about 25–35% of fuel energy goes to the coolant; cooling prevents overheating of cylinder walls, pistons, valves; maintains lubricating oil film (oil breaks down above ≈ 200–250 °C); prevents pre-ignition/detonation. Over-cooling reduces efficiency and increases wear and sludge – so a thermostat maintains the optimum temperature (≈ 80–90 °C).
- Air cooling: fins on cylinder and head, natural (motorcycles) or forced by a fan (aircraft radial engines, some cars). Simple and light; no leakage or freezing; but less uniform cooling, noisier, limited to smaller engines.
- Water (liquid) cooling: thermosyphon (natural circulation by density difference, no pump – low capacity), pump circulation with radiator and fan (most cars), pressurised system (radiator cap at ≈ 1–1.5 bar raises boiling point to ≈ 110–120 °C), evaporative cooling.
- Thermostat: wax-pellet or bellows type; stays closed when the engine is cold (coolant bypasses radiator for quick warm-up), opens at ≈ 70–85 °C.
- Radiator: tube-and-fin cross-flow/down-flow heat exchanger made of copper/brass or aluminium. Coolant: water + ethylene glycol (antifreeze; lowers freezing point, raises boiling point, inhibits corrosion).

★ **PYQ-TREND:** Function of thermostat (very frequent); thermosyphon has no pump; pressurised radiator cap raises boiling point; antifreeze (ethylene glycol); percentage of heat to coolant.

11.5 Lubrication Systems

- Functions of lubricant: reduce friction and wear, cool moving parts, seal piston rings, clean (detergent action), prevent corrosion, cushion shocks.
- Systems: petroil (mist) lubrication – 2–5% oil mixed with petrol in small two-strokes; splash lubrication – dippers on connecting-rod caps splash oil (small four-strokes); pressure-feed (forced) – gear pump forces oil through drilled passages to main and big-end bearings, camshaft, valve gear (most automotive engines); semi-pressure; dry sump (oil stored in a separate tank, scavenge pumps – racing cars, aircraft) vs wet sump (oil in the crankcase sump).
- Oil filters (full-flow, by-pass), oil cooler, pressure relief valve, oil pressure gauge.
- Lubricating oil properties: viscosity, viscosity index (VI – resistance of viscosity change with temperature; higher is better; multigrade oils have high VI using VI improvers), flash and fire points, pour point and cloud point (low-temperature fluidity), oiliness, carbon residue, total base number, detergency.
- SAE grades: e.g., SAE 20W-40 – 'W' means winter grade (low-temperature viscosity, ≈ SAE 20 cold) and 40 is the viscosity at 100 °C; multigrade oils serve both.
- Crankcase ventilation: PCV (positive crankcase ventilation) valve routes blow-by gases to the intake – emission control.

★ **PYQ-TREND:** Petroil lubrication in two-strokes; splash vs pressure-feed; dry sump in racing/aircraft; viscosity index; SAE 20W-40 meaning; flash and fire points; PCV valve.

11.6 Governing and Starting

- Quantity governing (SI): throttle varies the quantity of mixture, keeping A/F nearly constant – throttling losses at part load. Quality governing (CI): quantity of fuel injected per cycle varied, air quantity nearly constant – A/F changes with load (lean at part load), no throttling loss. Hit-and-miss governing: fuel cut off for some cycles (small gas engines). Combination governing.
- Governors: mechanical (centrifugal), pneumatic, hydraulic, electronic (ECU).
- Starting: electric starter motor (DC series motor – high starting torque) with Bendix drive engaging the flywheel ring gear; kick start (motorcycles); compressed air starting (large diesels); hand cranking (small engines).

★ **PYQ-TREND:** Quantity governing in SI and quality governing in CI (very frequent); starter is a DC series motor; Bendix drive.

✎ **DESCRIPTIVE Q:** Explain the working of a simple carburettor and its limitations.

Fuel from the tank is kept at a constant level in the float chamber by a float-operated needle valve; the level is slightly below the tip of the fuel jet to prevent spilling when the engine is stopped. During the suction stroke air flows through the venturi; its velocity rises and pressure falls at the throat. The pressure difference between the float chamber (vented to atmosphere) and the throat forces fuel through the metering jet into the air stream as fine droplets, which partly vaporise. The throttle valve downstream of the venturi controls the quantity of mixture entering the cylinders, and so the engine power, while a choke valve upstream can restrict air to give a rich mixture for cold starting. Limitations: the simple carburettor delivers a progressively richer mixture as speed increases because the fuel flow increases faster than the air flow; it gives a very lean mixture at idling; it cannot provide extra fuel for acceleration; it is affected by altitude and icing. Compensating jets, air-bleed systems, idling circuits and

accelerating pumps were added, and modern engines replace carburettors by electronic fuel injection.

CHAPTER 12

IC Engines – Performance, Testing, Supercharging, Emissions

12.1 Performance Parameters

Indicated power IP = p_mi L A n k / 60 (W, with p_mi in Pa) — n = N/2 (four-stroke), n = N (two-stroke); k = number of cylinders

Brake power BP = 2πNT/60 = torque × angular speed

Friction power FP = IP − BP

Mechanical efficiency η_m = BP/IP (typically 75–90%)

Indicated thermal efficiency η_ith = IP/(ṁ_f × CV); brake thermal efficiency η_bth = BP/(ṁ_f × CV) = η_ith × η_m

Relative efficiency η_rel = η_ith / air-standard efficiency

Brake mean effective pressure: bmep = BP × 60/(L A n k)

Specific fuel consumption: bsfc = ṁ_f/BP (kg/kWh); bsfc = 3600/(η_bth × CV) with CV in kJ/kg

Volumetric efficiency η_v = actual mass (or volume at intake conditions) of air inducted / swept volume × ρ_air

Air–fuel ratio = ṁ_air/ṁ_fuel

Torque and bmep: T = bmep × V_s × k/(2π × n_R) (n_R = 2 for four-stroke, 1 for two-stroke)

- Typical values: η_bth – SI 25–33%, CI 30–40% (large diesels up to ~50%); bsfc – SI 0.25–0.35 kg/kWh, CI 0.20–0.25; η_v – NA SI 75–90%.
- Indicator diagram (p–V) from an engine indicator gives indicated work: area × spring scale. Imep = area of indicator diagram × spring constant / length of diagram.

Imep from indicator card: p_mi = (area of diagram / length of diagram) × spring stiffness

★ **PYQ-TREND:** IP formula with n = N/2 for four-stroke (very frequent); BP = 2πNT/60; mechanical efficiency; bsfc units; volumetric efficiency definition; relation η_bth = η_ith × η_m.

✔ **TOPPER TRICK:** Most IP mistakes come from using N instead of N/2 for a four-stroke engine. Write 'n = power strokes per minute' and decide first.

12.2 Measurement and Testing

- Brake power: dynamometers – absorption (Prony brake, rope brake, hydraulic (Froude), eddy-current, electric swinging-field) and transmission (torsion meter).
- Friction power measurement:

Method	Applicable to	Principle
Willan's line (fuel-rate extrapolation)	CI engines only (at constant speed)	Plot fuel consumption vs BP; extrapolate to zero fuel → FP

Method	Applicable to	Principle
Morse test	Multi-cylinder engines (SI mainly)	Cut out each cylinder in turn, restore speed, measure drop in BP; IP of that cylinder = BP_all − BP_(n−1)
Motoring test	Any	Drive the engine with an electric motor at the same speed with ignition/fuel off
Retardation test	Any	Rate of fall of speed after cutting fuel
Indicator diagram	Any	IP − BP

- Air consumption: air box with orifice and manometer (with damping), viscous-flow (Alcock) meter, positive-displacement meters. Fuel consumption: burette and stopwatch, gravimetric, flow meters.
- Exhaust analysis: Orsat apparatus (CO_2, O_2, CO by volume – dry basis), NDIR analysers (CO, CO_2), FID (HC), chemiluminescence (NOx), smoke meters (Bosch, Hartridge) for diesel smoke.
- Heat balance sheet (per minute or per hour): heat supplied = BP + heat to cooling water + heat to exhaust gases + unaccounted losses (radiation, friction). Typical: BP 25–35%, coolant 25–35%, exhaust 30–40%, unaccounted 5–10%.

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Heat to coolant = ṁ_w c_pw (t_out − t_in); heat to exhaust = ṁ_g c_pg (t_g − t_ambient)
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- Performance curves: at constant speed vs load (bsfc minimum near 75–90% load); at full throttle vs speed (torque maximum at medium speed; BP rises then falls; bsfc minimum at medium speed).

★ **PYQ-TREND:** Willan's line for CI engines, Morse test for multi-cylinder SI engines (very frequent); heat balance components and percentages; Orsat apparatus; Prony brake; motoring test.

12.3 Supercharging and Turbocharging

- Supercharging: supplying air (or mixture) at a pressure above atmospheric to increase the mass of charge per cycle → higher power from the same engine size. Uses: aircraft engines at altitude (compensate thin air), racing cars, high-output diesels.
- Supercharger types: positive displacement (Roots blower – no internal compression; vane; screw), centrifugal; driven mechanically from the crankshaft (absorbs some power).
- Turbocharger: an exhaust-gas turbine drives a centrifugal compressor on a common shaft – uses waste exhaust energy, improving overall efficiency. Issue: turbo lag (delayed response at low speed). Waste-gate valve limits boost pressure; variable-geometry turbines reduce lag.
- Intercooler (charge air cooler): cools the compressed air → higher density, more power, lower knock risk (SI) and lower NOx.
- Supercharging is more suitable for CI engines – it reduces ignition delay (less knock) and improves combustion; in SI engines it raises knock tendency, so compression ratio must be reduced or higher-octane fuel used.

★ **PYQ-TREND:** Purpose of supercharging; turbocharger driven by exhaust gases (very frequent); Roots blower; intercooler; supercharging better for CI engines; altitude compensation in aircraft.

12.4 Emissions and Their Control

Pollutant	Source/cause	Main effect
CO	Incomplete combustion, rich mixture (SI idling, deceleration)	Combines with haemoglobin – toxic
HC (unburnt hydrocarbons)	Flame quenching at cold walls, crevice volumes, misfire, scavenging loss in 2-strokes	Smog formation, carcinogenic
NOx (NO + NO_2)	High peak combustion temperature with excess O_2 (CI, lean SI)	Photochemical smog, acid rain
Particulate matter (soot)	Diesel – locally rich zones, incomplete combustion	Respiratory diseases
SOx	Sulphur in fuel	Acid rain
CO_2	Complete combustion of carbon	Greenhouse gas

- Control methods: three-way catalytic converter (SI) – platinum/palladium oxidise CO and HC, rhodium reduces NOx; needs stoichiometric mixture ($\lambda = 1$) maintained by an oxygen (λ) sensor in closed loop; poisoned by lead. Oxidation catalyst (DOC) and diesel particulate filter (DPF) for diesels; selective catalytic reduction (SCR) with urea (AdBlue) for NOx; lean NOx traps.
- In-engine methods: EGR (exhaust gas recirculation – dilutes charge, lowers peak temperature → less NOx), retarded injection/ignition timing, better mixing, high injection pressure (CRDI), combustion chamber design, PCV valve (crankcase HC), evaporative emission control (charcoal canister).
- Norms in India: Bharat Stage (BS) emission standards based on Euro norms; BS-IV nationwide from 2017; BS-VI from 1 April 2020 (BS-V skipped).
- There is a trade-off in diesels: measures that reduce NOx (lower temperature) tend to increase soot/PM, and vice versa.

★ **PYQ-TREND:** EGR reduces NOx (very frequent); three-way catalyst metals and $\lambda = 1$ requirement; CO from rich mixture; HC from quenching; DPF and SCR for diesels; BS-VI from 2020; NOx–PM trade-off.

✎ **DESCRIPTIVE Q:** Explain the Morse test and state its limitations.

The Morse test determines the indicated power, and hence friction power and mechanical efficiency, of a multi-cylinder engine. The engine runs at a fixed speed and load, and its brake power BP is measured with a dynamometer. One cylinder is then cut out (spark shorted in an SI engine or fuel cut off in a CI engine); the speed drops, the dynamometer load is reduced to restore the original speed, and the new brake power BP_1 is recorded. The indicated power of the cut-out cylinder is $IP_1 = BP - BP_1$, assuming friction power is unchanged. Repeating for every cylinder, total $IP = \Sigma IP_i = nBP - \Sigma BP_i$; then $FP = IP - BP$ and $\eta_m = BP/IP$. Limitations: it applies only to multi-cylinder engines; it assumes the friction and pumping losses stay the same when a cylinder is cut out, which is not strictly true (exhaust pressure and pumping change); cutting out cylinders can alter the mixture distribution in carburetted engines; and readings must be taken quickly to keep conditions steady.

CHAPTER 12B

IC Engines – Modern Technologies, Emission Norms and Special Engines

12B.1 Fuel Injection and Engine Management

System	Description	Benefit
Throttle-body injection (TBI, single point)	One or two injectors above the throttle	Better metering than carburettor; simple
Multi-point fuel injection (MPFI, port injection)	One injector per cylinder spraying at the intake port; timed sequentially by ECU	Uniform distribution, better economy and emissions, quick response
Gasoline direct injection (GDI)	Fuel injected directly into the cylinder at 50–200 bar; homogeneous or stratified charge	Charge cooling allows higher r, better economy; but more particulate emissions
Common rail direct injection (CRDI)	High-pressure pump keeps fuel in a common rail at ≈ 1,600–2,500 bar; ECU-controlled solenoid or piezo injectors	Precise timing, multiple injections (pilot, main, post) – quieter combustion, lower NOx and smoke, better economy
Unit injector	Pump and injector combined for each cylinder, cam-driven	Very high pressure, used in heavy engines

- The engine control unit (ECU) reads sensors and controls injection quantity and timing, ignition timing, idle speed, EGR and turbo boost.

Sensor	Function
MAF (mass air flow, hot-wire/hot-film) / MAP (manifold absolute pressure)	Air quantity for fuel calculation
Oxygen (lambda) sensor (zirconia)	Exhaust O_2 – closed-loop control to keep $\lambda \approx 1$ for the three-way catalyst
Throttle position sensor	Driver demand, acceleration enrichment
Crankshaft/camshaft position sensors	Engine speed and piston position for injection and ignition timing
Coolant temperature sensor	Cold-start enrichment, fan control
Knock sensor (piezoelectric)	Detects detonation vibrations; ECU retards spark

- Ignition: conventional battery/coil with contact breaker → electronic (transistorised) ignition → distributor-less ignition (DIS) → coil-on-plug; magneto ignition in two-wheelers and small engines without battery.
- Variable valve timing (VVT) and lift (e.g. VTEC, VVT-i): adjusts valve events with speed and load for better torque, power and economy. Cylinder deactivation shuts off cylinders at light load.
- Downsizing: smaller turbocharged GDI engines replace larger naturally aspirated engines with equal power and lower fuel use; intercooler cools the compressed air.

★ **PYQ-TREND:** MPFI vs carburettor; CRDI common rail pressure and multiple injections (very frequent); lambda sensor for closed-loop control; knock sensor piezoelectric; GDI.

12B.2 Emission Control Technologies

Device	Purpose
Three-way catalytic converter (Pt, Pd, Rh on ceramic honeycomb)	Oxidises CO and HC, reduces NOx – effective only near stoichiometric ($\lambda \approx 1$) in SI engines
Diesel oxidation catalyst (DOC)	Oxidises CO, HC and soluble organic part of PM in diesel exhaust
Diesel particulate filter (DPF)	Wall-flow ceramic filter traps soot; periodically regenerated by burning the soot
Selective catalytic reduction (SCR)	Urea solution (AdBlue/DEF) injected, forms NH_3 which reduces NOx to N_2 and H_2O
Lean NOx trap (LNT)	Stores NOx during lean operation, releases and reduces it during brief rich periods
Exhaust gas recirculation (EGR)	Recirculated exhaust lowers peak combustion temperature – reduces NOx
Positive crankcase ventilation (PCV)	Blow-by gases returned to the intake – reduces HC
Evaporative emission control (charcoal canister)	Traps fuel vapour from the tank
On-board diagnostics (OBD)	Monitors emission-related components and warns the driver

- Trade-off in diesel engines: measures that reduce NOx (lower combustion temperature, retarded injection) tend to increase PM and fuel consumption, and vice versa – the 'NOx–PM trade-off'; after-treatment (DPF + SCR) breaks it.
- Indian norms (Bharat Stage, based on Euro norms): BS-IV nationwide from April 2017; India leapfrogged BS-V and implemented BS-VI from 1 April 2020 with 10 ppm sulphur fuel; BS-VI Phase 2 (2023) added real driving emission (RDE) testing and stricter OBD.
- Alternative fuels and blends: ethanol blending (E20 petrol targeted for 2025), flex-fuel vehicles, CNG (high octane, low PM), LPG, biodiesel blends (B20), hydrogen ICE and fuel-cell vehicles.

★ **PYQ-TREND:** Three-way catalyst needs stoichiometric mixture; SCR uses urea (very frequent); DPF traps soot; EGR reduces NOx; BS-VI from 2020 skipping BS-V; NOx–PM trade-off; PCV and charcoal canister.

12B.3 Hybrid and Advanced Combustion Concepts

Concept	Description
Series hybrid	Engine drives only a generator; wheels driven by electric motor
Parallel hybrid	Engine and motor can both drive the wheels
Series–parallel (power-split) hybrid	Planetary gear splits engine power between wheels and generator (e.g. Toyota hybrid system)
Mild hybrid / start–stop	Small motor-generator assists the engine, stops it at idle and recovers braking energy
Plug-in hybrid (PHEV)	Larger battery charged from the grid; electric-only range

Concept	Description
HCCI (homogeneous charge compression ignition)	Premixed lean charge autoignites by compression – very low NOx and soot, high efficiency, but difficult to control combustion timing
Stratified-charge engine	Rich mixture near the spark plug, lean elsewhere – overall lean operation
Dual-fuel engine	Gaseous fuel (CNG/biogas) premixed with air, ignited by a pilot diesel injection

- Regenerative braking converts kinetic energy into electrical energy stored in the battery, improving city fuel economy.

12B.4 Special Engines

Engine	Key features
Wankel (rotary) engine	Triangular rotor in an epitrochoidal housing; three power impulses per rotor revolution (one per output-shaft revolution); compact, light, smooth, few parts; drawbacks – apex-seal wear and leakage, high HC emissions and fuel consumption, low compression ratio
Stirling engine	External combustion, closed regenerative cycle with working gas (air, He, H_2); quiet, multi-fuel (solar, biomass), theoretically Carnot efficiency; low power density, high cost
Free-piston engine	No crankshaft; pistons drive a compressor or linear generator
Gas engines	Run on natural gas, biogas, producer gas; spark ignited or dual fuel
Opposed-piston engine	Two pistons per cylinder moving toward each other; no cylinder head; used in some two-stroke diesels
Variable compression ratio engine	Changes r with load for efficiency and knock control

★ **PYQ-TREND:** Wankel engine rotor shape and apex seals (frequent); Stirling engine is external combustion with regenerator; dual-fuel engine with pilot diesel.

12B.5 Engine Balancing and Flywheel (Quick Facts)

- Primary unbalanced force of a reciprocating mass: $F_p = m\, r\, \omega^2 \cos\theta$; secondary force: $F_s = m\, r\, \omega^2 \cos 2\theta / n$ ($n = l/r$, ratio of connecting rod length to crank radius). Secondary force varies at twice crank speed.
- Single-cylinder engines are partially balanced by rotating counterweights (balancing usually 2/3 to 3/4 of reciprocating mass), which introduces an unbalanced force perpendicular to the line of stroke.
- In-line four-cylinder engine (cranks at 180°) is balanced for primary forces and couples but has unbalanced secondary forces; six-cylinder in-line engine is fully balanced for primary and secondary forces and couples.
- Flywheel stores energy during the power stroke and releases it during other strokes, reducing speed fluctuation. Coefficient of fluctuation of speed $C_s = (N_{max} - N_{min})/N_{mean}$; energy fluctuation $\Delta E = I\, \omega^2\, C_s$. Two-stroke and multi-cylinder engines need smaller flywheels than single-cylinder four-stroke engines because torque is more

uniform.

$\Delta E = I\,\omega^2\,C_s = m\,k^2\,\omega^2\,C_s;\ C_s = (\omega_{max} - \omega_{min})/\omega_{mean}$

★ **PYQ-TREND:** Six-cylinder in-line engine fully balanced; four-stroke single-cylinder engine needs the heaviest flywheel; secondary force at twice crank speed.

✎ **DESCRIPTIVE Q:** Explain the working of a common rail direct injection (CRDI) system and its advantages.

In a CRDI system a high-pressure pump, driven by the engine, supplies diesel continuously to a thick-walled common rail (accumulator) that is kept at a very high, controlled pressure – typically 1,600 to 2,500 bar – independent of engine speed. Short high-pressure pipes connect the rail to electronically controlled injectors (solenoid or piezoelectric) in each cylinder. The ECU, using inputs from crank-position, accelerator, air-flow, rail-pressure and temperature sensors, decides when and for how long each injector opens, so the injection timing, quantity and pressure are precisely controlled. It can split injection into pilot, main and post injections: the small pilot injection shortens the ignition delay of the main injection, reducing diesel knock and noise; post injections help burn soot and regenerate after-treatment devices. Advantages: finer atomisation and better mixing due to high pressure, lower smoke and NOx, better fuel economy, higher power and torque, quieter and smoother running, and flexibility to meet BS-VI norms with after-treatment. Disadvantages include high cost, sensitivity to fuel quality and the need for clean fuel filtration and electronic diagnostics.

CHAPTER 13

Vapour Power Cycles (Rankine and Modifications)

13.1 Why Not the Carnot Cycle with Steam?

- A Carnot cycle using wet steam would need (1) compression of a wet mixture (liquid + vapour) to saturated liquid – difficult to control, needs a large compressor and the two-phase flow damages it; (2) heat addition only in the wet region, limiting the maximum temperature to the saturation temperature (low efficiency); (3) the turbine would handle very wet steam (blade erosion); (4) very low work ratio. If superheat is used, isothermal heat addition at varying pressure is impractical.
- The Rankine cycle condenses the steam completely, so a small pump handles only liquid; the work ratio is close to 1, and superheating raises the mean temperature of heat addition.

★ **PYQ-TREND:** Reasons the Carnot cycle is not used in steam plants; Rankine work ratio close to 1.

13.2 Ideal Rankine Cycle

- Processes: 1–2 isentropic compression of liquid in the pump; 2–3 constant-pressure heat addition in the boiler (subcooled liquid → saturated → superheated); 3–4 isentropic expansion in the turbine; 4–1 constant-pressure heat rejection in the condenser.

Pump work $w_p = v_f (p_2 - p_1) = h_2 - h_1$ ($\approx 0.001\ m^3/kg \times \Delta p$)

Turbine work $w_t = h_3 - h_4$

Heat supplied $q_1 = h_3 - h_2$; heat rejected $q_2 = h_4 - h_1$

$\eta_{Rankine} = (w_t - w_p)/q_1 = [(h_3 - h_4) - (h_2 - h_1)]/(h_3 - h_2) \approx (h_3 - h_4)/(h_3 - h_1)$ if pump work is neglected

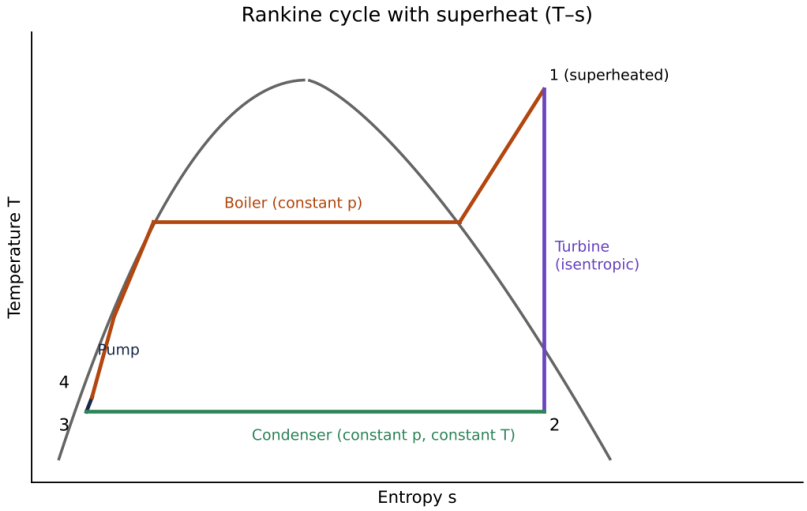
Specific steam consumption SSC = 3600/w_net (kg/kWh, w_net in kJ/kg)

Heat rate = $3600/\eta$ (kJ/kWh)

Work ratio = w_net/w_t

Mean temperature of heat addition $T_{m1} = (h_3 - h_2)/(s_3 - s_2)$; $\eta = 1 - T_2/T_{m1}$ (T_2 = condenser temperature)

- Pump work is tiny (≈ 1% of turbine work) because the specific volume of water is very small – so it is often neglected in quick calculations.



★ **PYQ-TREND:** Rankine efficiency expression; pump work small due to low specific volume of water (very frequent); specific steam consumption 3600/w_net; mean temperature of heat addition concept; heat rate.

13.3 Improving Rankine Efficiency

Method	Effect on efficiency	Other effects
Increase boiler pressure (same max temperature)	Increases (higher T_m1)	Exhaust dryness decreases (more moisture) – needs reheat
Superheating	Increases	Exhaust dryness increases; higher specific work; needs heat-resistant materials
Lower condenser pressure (better vacuum)	Increases (lower T_2)	Exhaust moisture increases; more air leakage; limited by cooling-water temperature
Reheating	Small increase (or none) depending on pressure	Main purpose: increase exhaust dryness; more work per kg
Regeneration (feedwater heating with bled steam)	Increases significantly	Less work per kg of steam, more steam flow, smaller condenser
Supercritical pressures	Increases (40–46%)	Once-through (Benson) boilers

- Turbine exhaust dryness should not drop below about 0.88–0.90 to avoid blade erosion.

★ **PYQ-TREND:** Effect of boiler pressure, superheat, condenser pressure, reheat and regeneration (very frequent). Main purpose of reheat = dryness; regeneration raises efficiency by raising feedwater temperature; exhaust dryness ≥ 0.88.

13.4 Reheat Cycle

Turbine work = $(h_3 - h_4) + (h_5 - h_6)$ (HP + LP turbines); heat supplied = $(h_3 - h_2) + (h_5 - h_4)$

$\eta_{reheat} = [(h_3 - h_4) + (h_5 - h_6) - w_p] / [(h_3 - h_2) + (h_5 - h_4)]$

- Steam partly expanded in the HP turbine is returned to the boiler reheater, heated (usually back to the initial temperature) at constant pressure, then expanded in the LP turbine. Optimum reheat pressure ≈ 20–25% of the initial pressure.
- Modern plants use single or double reheat.

★ **PYQ-TREND:** Reheat mainly improves exhaust dryness; optimum reheat pressure ≈ 1/4 of boiler pressure.

13.5 Regenerative Cycle and Feedwater Heaters

- Ideal regenerative cycle: feedwater passes around the turbine casing and is heated by the expanding steam (impractical). Practical: steam is bled (extracted) from intermediate turbine stages to heat the feedwater in feedwater heaters (FWH).
- Open (direct-contact) FWH: bled steam mixes with feedwater; exit is saturated liquid at heater pressure; simple, good heat transfer, also removes dissolved gases (deaerator); needs a pump after each heater.
- Closed (surface/shell-and-tube) FWH: no mixing; drains cascaded backward or pumped forward; more expensive but needs fewer pumps.

```
Open heater mass balance: m(h_bled) + (1 − m)h_feed = h_sat,out → m = (h_out − h_feed)/(h_bled − h_feed)
```

- Effects: raises mean temperature of heat addition → higher efficiency; reduces mass of steam to condenser (smaller condenser, less heat rejected); increases total steam flow per kW; reduces thermal stresses in the boiler (warmer feedwater). Number of heaters in large plants: 6–8.

★ **PYQ-TREND:** Regeneration increases efficiency and reduces condenser size; open heater = deaerator; closed heater = no mixing; bleeding fraction calculation.

13.6 Binary, Cogeneration and Combined Cycles

- Binary vapour cycle: two working fluids – mercury (topping; high temperature at low pressure) and water/steam (bottoming); mercury condenser acts as the steam boiler. Ideal working fluid: high critical temperature with low saturation pressure at high temperature, high latent heat, low specific heat of liquid, saturation vapour line close to vertical, non-toxic, cheap, stable.
- Cogeneration (combined heat and power, CHP): simultaneous generation of electricity and useful process heat (back-pressure or pass-out/extraction turbines) – overall energy utilisation 70–90%. Used in sugar mills, paper, textile and chemical industries.
- Combined cycle power plant (CCPP): gas turbine (Brayton – topping) exhaust (≈ 500–600 °C) heats a heat-recovery steam generator (HRSG) feeding a steam turbine (Rankine – bottoming). Efficiency 55–62%.

```
Combined efficiency (ideal coupling): η_cc = η_GT + η_ST − η_GT η_ST
Utilisation factor (cogeneration) = (net work + process heat)/heat supplied
```

★ **PYQ-TREND:** Combined cycle uses HRSG and gas turbine as topping cycle; $\eta = \eta_1 + \eta_2 - \eta_1\eta_2$; mercury in binary vapour cycles; cogeneration definition and back-pressure turbines.

✎ **DESCRIPTIVE Q:** Explain the effect of reheating and regeneration on the Rankine cycle.

Reheating: after partial expansion in the high-pressure turbine, steam is sent back to the boiler's reheater and heated at constant pressure, normally to the original temperature, then expanded in the low-pressure turbine. The expansion line moves to the right on the T–s diagram, so the turbine exhaust is much drier, reducing blade erosion and allowing higher boiler pressures. The work per kg of steam also increases. The effect on efficiency is small – it increases only if the mean temperature of the additional heat is higher than the cycle's mean

temperature of heat addition. Regeneration: steam bled from intermediate stages heats the feedwater in open or closed heaters before it enters the boiler. Heat that would have been added at low temperature in the boiler is now supplied internally, so the mean temperature of external heat addition rises and the cycle efficiency increases appreciably (approaching Carnot in the ideal limit). It also reduces the steam reaching the condenser and the heat rejected, at the cost of slightly less work per kg of steam generated.

CHAPTER 14

Steam Generators (Boilers)

14.1 Introduction and Classification

- Boiler (steam generator): a closed vessel in which water is heated by combustion (or hot gases/electricity/nuclear heat) to produce steam at the required pressure and temperature. Indian Boilers Act 1923 – a vessel above 22.75 litres capacity used for steam generation is a 'boiler' under the Act; Indian Boiler Regulations (IBR) 1950.
- Requirements of a good boiler: maximum steam with minimum fuel; quick starting; light, compact; easily accessible for cleaning and inspection; meets fluctuating demand; safe; low maintenance.

Basis	Classification
Contents of tubes	Fire-tube (hot gases inside tubes, water outside) / water-tube (water inside tubes, gases outside)
Position of furnace	Internally fired / externally fired
Axis of shell	Vertical / horizontal / inclined
Number of tubes	Single-tube (Cornish, simple vertical) / multi-tubular (Lancashire has two flues; Cochran, locomotive, B&W)
Circulation	Natural (thermosyphon, by density difference) / forced (pump)
Pressure	Low (< 10 bar), medium (10–60), high (> 60), supercritical (> 221.2 bar)
Use	Stationary / mobile (locomotive, marine)
Heat source	Fuel, waste heat, electrical, nuclear, solar

Fire-tube boilers	Water-tube boilers
Hot gases in tubes	Water in tubes
Low pressure (up to ≈ 25 bar), small capacity	High pressure (up to 300+ bar), large capacity
Large water volume – slow to raise steam, but steady pressure	Small water volume – rapid steaming, responds to load
Shell bursting is very dangerous (large mass of water at saturation)	Tube failure has only a local effect – safer at high pressure
Cochran, Cornish, Lancashire, locomotive, Scotch marine, simple vertical	Babcock & Wilcox, Stirling, La Mont, Benson, Loeffler, Velox, Schmidt–Hartmann
Used in small industries	Used in power plants

★ **PYQ-TREND:** Fire-tube vs water-tube classification with examples (very frequent); Lancashire has two flue tubes; Cornish single flue; water-tube for high pressure; Indian Boilers Act.

14.2 Important Boilers

Boiler	Type & key features
Simple vertical	Fire-tube, vertical, small capacity; cross tubes
Cochran	Vertical, multi-tubular fire-tube, internally fired; hemispherical crown; compact, small floor area
Cornish	Horizontal, single flue tube, internally fired fire-tube
Lancashire	Horizontal, two flue tubes, internally fired fire-tube; ≈ 15–20 bar; uses brickwork flues around shell (side and bottom flues); reliable, large water space
Locomotive	Horizontal, multi-tubular fire-tube, internally fired; high steaming rate per unit size; draught by exhaust steam blast through a blast pipe; mobile
Scotch marine	Horizontal multi-tubular fire-tube; ships
Babcock & Wilcox	Water-tube, horizontal drum with inclined straight tubes (15°), externally fired, natural circulation; power stations (medium)
Stirling	Water-tube with bent tubes and several drums
La Mont	Forced circulation water-tube (centrifugal pump circulates 8–10 times the evaporation) – high pressure
Benson	Supercritical, once-through, drumless (above critical pressure ≈ 225 bar; no distinct evaporation); forced flow
Loeffler	Forced circulation; superheated steam from superheater is used to evaporate feedwater in an evaporating drum – avoids salt deposition in tubes
Velox	Gas velocities above sonic speed (pressurised furnace with gas turbine–driven compressor) – very high heat transfer rates, compact
Schmidt–Hartmann	Two-circuit: primary circuit of distilled water heats secondary water – can use impure feedwater
Once-through (supercritical, ultra-supercritical)	Modern large power stations (> 600 MW): 250–300 bar, 600 °C+

★ **PYQ-TREND:** Boiler–feature matching (Benson supercritical drumless; Loeffler avoids salt deposits; Velox supersonic gases; La Mont forced circulation; B&W externally fired water-tube; Cochran vertical fire-tube; locomotive draught by exhaust steam) – very frequent.

14.3 Boiler Mountings and Accessories

- Mountings: fittings mounted on the boiler for safety and complete control of the steam generation process; a boiler cannot be operated safely without them (required by IBR).

Mounting	Function
Water level indicator (gauge glass)	Shows water level; two per boiler
Pressure gauge (Bourdon)	Indicates steam pressure
Safety valves (two)	Release steam if pressure exceeds the working limit – dead-weight, lever (deadweight-lever), spring-loaded (Ramsbottom – locomotive, marine), high-steam low-water safety valve

Mounting	Function
Fusible plug	Melts (lead/tin alloy core) when water level falls too low and the plug is uncovered and overheated; steam/water rushes in and puts out the fire
Steam stop valve (junction valve)	Regulates/shuts off steam flow from boiler to the main
Feed check valve	Non-return valve that admits feedwater and prevents back-flow
Blow-off cock	Removes sludge and sediment from the bottom; empties the boiler
Manhole / mud box	Access for inspection and cleaning

- Accessories: auxiliary equipment that increases efficiency and helps operation (boiler can work without them).

Accessory	Function
Economiser	Heats feedwater using flue gases before it enters the boiler (Green's economiser) – saves 15–20% fuel
Air preheater	Heats combustion air using flue gases (after economiser) – better combustion, saves fuel
Superheater	Superheats saturated steam (convective, radiant types)
Reheater	Reheats partially expanded steam from HP turbine
Feed pump	Supplies water to the boiler at boiler pressure
Injector	Feeds water using a steam jet (no moving parts) – locomotives, small boilers
Steam separator	Removes water particles from steam
Steam trap	Drains condensate from steam pipes without letting steam escape
Pressure reducing valve	Maintains constant lower pressure
Soot blower	Removes soot from tube surfaces

- Typical gas path in a large plant: furnace → superheater → reheater → economiser → air preheater → electrostatic precipitator → ID fan → chimney.

★ **PYQ-TREND:** Mountings vs accessories (economiser is an accessory; safety valve and fusible plug are mountings) – very frequent; function of fusible plug; economiser heats feedwater; air preheater heats air; order of heat-recovery equipment in gas path; injector principle; number of safety valves.

✔ **TOPPER TRICK:** If it keeps the boiler safe or controllable, it is a mounting. If it saves fuel or improves performance, it is an accessory.

14.4 Draught

- Draught: pressure difference that causes the flow of air into the furnace and flue gases through the boiler to the chimney. Needed to supply combustion air and remove products. Expressed in mm of water column (typically 10–30 mm).
- Natural draught: produced by a chimney due to density difference between hot flue gases inside and cold outside air. Free of running cost, but low and weather-dependent; needs a

tall chimney.

Natural draught (mm of water): $h_w = 353\ H\ [1/T_a - ((m+1)/m)(1/T_g)]$ (H = chimney height in m, m = air–fuel ratio by mass, temperatures in K)

Draught as column of hot gas: $H_1 = H[(m/(m+1))(T_g/T_a) - 1]$

Condition for maximum discharge of gases through a chimney: $T_g/T_a = 2(m+1)/m$ (≈ 2 → flue gases about twice ambient absolute temperature)

Chimney efficiency = (useful work equivalent of draught) / (extra heat carried by flue gases) — very low, ≈ 0.25–0.5%

- Artificial draught: (1) mechanical – forced draught (FD fan at the base pushes air through the furnace; positive pressure; handles cold air – smaller fan power), induced draught (ID fan at the chimney base sucks gases; negative pressure; handles hot gases – larger fan), balanced draught (FD + ID; furnace pressure slightly below atmospheric – most power stations); (2) steam-jet draught (induced in locomotives; forced in some).

★ **PYQ-TREND:** Natural draught formula and dependence on chimney height and temperature; condition for maximum discharge ($T_g \approx 2T_a$); FD vs ID vs balanced draught (very frequent); steam-jet draught in locomotives; chimney efficiency very low.

14.5 Boiler Performance

Equivalent evaporation (from and at 100 °C): $m_e = m_a\ (h - h_{fw})/2257$ kg of steam per kg of fuel

Boiler efficiency $\eta_b = m_a\ (h - h_{fw})/CV$ (m_a = actual steam per kg fuel)

Factor of evaporation $F_e = (h - h_{fw})/2257$

Boiler horsepower (old unit) = equivalent evaporation of 15.65 kg/h from and at 100 °C

- Heat losses in a boiler: dry flue gases (largest), moisture in fuel and from hydrogen combustion, unburnt carbon in ash, incomplete combustion (CO), radiation and convection, blowdown. Typical efficiency: 75–92%.
- Boiler trial: determines evaporation rate, efficiency and heat balance.
- Water treatment: hardness causes scale (reduces heat transfer, overheats tubes), corrosion (O_2, CO_2), priming and foaming (carry-over of water with steam), caustic embrittlement. External treatment: softening (lime-soda, zeolite, ion exchange, demineralisation); internal treatment: phosphate dosing, deaeration, sodium sulphite/hydrazine (oxygen scavengers).

★ **PYQ-TREND:** Equivalent evaporation from and at 100 °C with 2257 kJ/kg (very frequent); boiler efficiency formula; largest loss is dry flue gas; scale, priming, foaming, caustic embrittlement; deaeration.

✎ **DESCRIPTIVE Q:** Distinguish between boiler mountings and accessories with two examples each, and explain the working of a fusible plug.

Mountings are fittings mounted on the boiler shell for safety and for complete control of steam generation; the boiler cannot be operated without them under the Boiler Regulations. Examples: safety valves (release steam when pressure exceeds the set limit), water level indicator, pressure gauge, fusible plug, steam stop valve, feed check valve, blow-off cock. Accessories are auxiliary devices that improve the efficiency and convenience of the plant; the boiler can work without them. Examples: economiser (preheats feedwater with flue gases), air preheater, superheater, feed pump, injector, steam separator. A fusible plug is fitted on the furnace crown, which is normally covered by water. It is a gun-metal body with a copper plug held in place by a low-melting alloy (lead–tin). If the water level falls dangerously low, the

crown and the plug are exposed to fire and overheat; the alloy melts, the copper plug drops, and steam and water rush into the furnace, extinguishing the fire and warning the operator before the shell is damaged.

CHAPTER 15

Steam Nozzles and Diffusers

15.1 Flow Through Nozzles

- Nozzle: a duct of varying cross-section that converts enthalpy (pressure energy) into kinetic energy. Diffuser: converts kinetic energy into pressure (reverse).
- Types: convergent (exit at most sonic; used when exit pressure ≥ critical pressure), divergent (diffuser for subsonic; nozzle for supersonic), convergent–divergent (de Laval) – accelerates to supersonic when exit pressure < critical pressure.

Exit velocity (adiabatic, inlet velocity negligible): $V_2 = \sqrt{2(h_1 - h_2)} = 44.72\sqrt{\Delta h}$, kJ/kg)

With inlet velocity: $V_2 = \sqrt{2000\,\Delta h + V_1^2}$

Isentropic (pv^n = const): $V_2 = \sqrt{\left[\frac{2n}{n-1}\right] p_1 v_1 \left(1 - (p_2/p_1)^{(n-1)/n}\right)}$

Mass flow per unit area: $\dot{m}/A = V/v$; continuity $A_1V_1/v_1 = A_2V_2/v_2$

Area–velocity relation: $dA/A = (M^2 - 1)\,dV/V$ → subsonic: area decreases as velocity increases; supersonic: area increases

★ **PYQ-TREND:** Nozzle converts enthalpy to KE; exit velocity $44.72\sqrt{\Delta h}$ (very frequent); convergent vs convergent–divergent nozzles; area–velocity relation.

15.2 Critical Pressure Ratio and Choking

Critical pressure ratio: $p_c/p_1 = \left(\frac{2}{n+1}\right)^{n/(n-1)}$

Fluid	Index n	p_c/p_1
Superheated steam	1.3	0.546
Dry saturated steam	1.135	0.577
Wet steam (Zeuner: n = 1.035 + 0.1x)	≈ 1.1–1.13	≈ 0.58
Air (γ = 1.4)	1.4	0.528

Critical (throat) velocity: $V_c = \sqrt{\frac{2n}{n+1} \times p_1 v_1} = \sqrt{n\,p_c\,v_c}$ = speed of sound at throat

Maximum mass flow (choked): $\dot{m}_{max}/A_t = \sqrt{n\,(p_1/v_1)\left(\frac{2}{n+1}\right)^{(n+1)/(n-1)}}$

- When the throat reaches the critical pressure, the flow is choked: Mach number at the throat = 1 and mass flow is maximum; lowering the back pressure further does not increase the mass flow (it only changes the flow downstream of the throat).
- In a convergent–divergent nozzle with exit pressure higher than design, shocks form in the divergent part (over-expansion / under-expansion cases).

★ **PYQ-TREND:** Critical pressure ratio values (0.546 superheated, 0.577 saturated, 0.528 air) – very frequent; M = 1 at the throat; choked flow and maximum discharge.

✔ **TOPPER TRICK:** '546 superheated, 577 saturated, 528 air' – superheated steam behaves closest to air.

15.3 Supersaturated (Metastable) Flow and Nozzle Efficiency

- In a rapidly expanding nozzle, steam does not condense when it crosses the saturation line; it remains as a supersaturated (supercooled) vapour in a metastable state until the Wilson line (≈ 96–97% dryness on the Mollier chart), after which condensation suddenly occurs.
- Effects of supersaturation: the steam is denser than in equilibrium → mass flow increases by 2–5%; the enthalpy drop is slightly reduced; entropy increases (irreversible condensation); dryness fraction at exit increases slightly.

```
Nozzle efficiency η_n = actual enthalpy drop / isentropic enthalpy drop (0.90–0.98)
Velocity coefficient C_v = √η_n; coefficient of discharge = actual mass flow / theoretical mass flow
```

- Friction in the divergent portion causes most losses; divergent cone half-angle about 5–7° (too large → separation; too small → long nozzle, more friction).

★ **PYQ-TREND:** Supersaturated flow increases discharge (very frequent); Wilson line; nozzle efficiency definition; friction effect.

✎ **DESCRIPTIVE Q:** What is supersaturated flow in a steam nozzle? State its effects.

When steam expands rapidly through a nozzle, it passes through the saturation line so quickly that there is not enough time for condensation nuclei to form. The steam therefore continues to expand as a dry, supercooled vapour – in a metastable, supersaturated state – below the saturation temperature corresponding to its pressure. Condensation occurs suddenly only when the degree of supersaturation reaches a limit, roughly at the Wilson line (about 96–97% dryness), after which the steam returns to equilibrium. Effects: the supersaturated steam has a higher density than equilibrium wet steam, so the discharge through the nozzle increases by about 2–5%; the available enthalpy drop is slightly reduced, so the exit velocity is slightly lower; the sudden condensation is irreversible, increasing entropy; and the exit dryness is slightly higher. The supersaturation ratio is the ratio of the actual pressure to the saturation pressure at the supercooled temperature.

CHAPTER 16

Steam Turbines

16.1 Principle and Classification

- A steam turbine converts the enthalpy of steam into kinetic energy in nozzles/fixed blades, and then into shaft work by changing the momentum of the steam on moving blades. It is a rotary, continuous-flow machine – smooth torque, high speed, large unit sizes (up to 1,000+ MW), high efficiency.

Basis	Types
Action of steam	Impulse, reaction, impulse–reaction (combination)
Direction of flow	Axial (most), radial (Ljungström), tangential
Number of stages	Single-stage, multi-stage
Exhaust condition	Condensing, back-pressure (non-condensing), pass-out/extraction, mixed-pressure
Steam pressure	Low, medium, high, supercritical
Compounding	Velocity, pressure, pressure–velocity

Impulse turbine	Reaction turbine
Complete pressure drop in nozzles only; pressure constant across moving blades	Pressure drops in both fixed blades (acting as nozzles) and moving blades
Blades symmetrical (profile) – equal inlet and outlet angles	Aerofoil-shaped asymmetric blades
Relative velocity unchanged (ideal): $V_{r2} = V_{r1}$ (or $k \cdot V_{r1}$ with friction)	Relative velocity increases across moving blades
Partial admission possible (steam only through some nozzles)	Full admission necessary
Fewer stages for the same pressure drop; small size	More stages; larger size
Lower efficiency; lower blade leakage effect	Higher efficiency at design; tip leakage matters (needs balance piston/dummy piston for axial thrust)
De Laval, Curtis, Rateau, Zoelly	Parsons

★ **PYQ-TREND:** Impulse vs reaction turbine differences (very frequent); De Laval simple impulse; Parsons reaction; pressure constant on impulse moving blades; partial admission in impulse; axial thrust balanced by dummy piston in reaction turbines.

16.2 Velocity Triangles and Work (Impulse Stage)

- Inlet: nozzle angle α (steam absolute velocity V_1), blade speed u, relative velocity V_{r1} at blade inlet angle θ; outlet: blade angle φ, relative velocity V_{r2}, absolute velocity V_2 at angle β.

```
Whirl components: V_w1 = V₁ cosα; V_w2 = V₂ cosβ
Work per kg = u (V_w1 ± V_w2) (+ when V_w2 is opposite to V_w1 – usual)
```

Tangential force (driving thrust) per kg/s = $V_{w1} \pm V_{w2}$; axial thrust per kg/s = $V_{f1} - V_{f2}$

Blade (diagram) efficiency $\eta_b = 2u(V_{w1} \pm V_{w2})/V_1^2$

Stage efficiency = blade efficiency × nozzle efficiency = work/(enthalpy drop in stage)

Impulse blade efficiency (symmetrical blades, no friction, k = 1): $\eta_b = 4\rho(\cos\alpha - \rho)$, where $\rho = u/V_1$

Optimum blade speed ratio (single-stage impulse): $u/V_1 = \cos\alpha/2$; $\eta_{b,max} = \cos^2\alpha$

With blade friction (k = V_{r2}/V_{r1}) and blade angles: $\eta_{b,max} = (\cos^2\alpha/2)(1 + k\cos\varphi/\cos\theta)$

- Typical nozzle angle α = 15–20°; blade speed limited by stress (≈ 400 m/s).

★ **PYQ-TREND:** Optimum speed ratio cosα/2 and $\eta_{max} = \cos^2\alpha$ for impulse turbine (very frequent); work = $u(V_{w1} + V_{w2})$; axial thrust from difference in flow components.

✖ **COMMON MISTAKE:** For an impulse turbine use $u/V_1 = \cos\alpha/2$; for a 50% reaction (Parsons) turbine use $u/V_1 = \cos\alpha$. Mixing them up is the most common error.

16.3 Compounding of Impulse Turbines

- Purpose: in a single-stage impulse (De Laval) turbine the whole pressure drop occurs in one set of nozzles, giving very high steam velocity (≈ 1,000+ m/s) and very high rotor speed (≈ 30,000 rpm) – needs reduction gearing; large leaving loss. Compounding reduces rotor speed by absorbing the energy in several steps.

Compounding	How	Example	Remarks
Velocity compounding	One set of nozzles (full pressure drop), then several rows of moving blades with fixed guide blades between them (no pressure drop in guide blades)	Curtis turbine	Velocity absorbed in steps; fewer stages, cheap, but low efficiency; optimum $u/V_1 = \cos\alpha/(2n)$ for n rows
Pressure compounding	Several stages, each with nozzles + one row of moving blades; pressure dropped in steps	Rateau turbine (also Zoelly)	Lower steam velocity, higher efficiency, but more stages/longer
Pressure–velocity compounding	Pressure dropped in stages; each stage velocity-compounded	Curtis stages in series	Combines both

Velocity-compounded (2 rows): optimum $u/V_1 = \cos\alpha/4$; work in rows ratio 3:1 (first row does 3/4 of total work)

For n rows: optimum $u/V_1 = \cos\alpha/(2n)$; work ratio of rows: (2n−1):(2n−3):…:1

★ **PYQ-TREND:** Curtis = velocity compounding; Rateau = pressure compounding (very frequent); purpose of compounding (reduce rotor speed); work ratio 3:1 in two-row Curtis wheel; De Laval needs gearing.

16.4 Reaction Turbines and Degree of Reaction

Degree of reaction R = enthalpy drop in moving blades / enthalpy drop in the stage = $\Delta h_{moving}/(\Delta h_{fixed} + \Delta h_{moving})$

- R = 0 → pure impulse; R = 0.5 (50% reaction) → Parsons turbine: fixed and moving blades are identical in shape (symmetrical velocity triangles), α = φ and θ = β.

```
Parsons (50% reaction): optimum u/V₁ = cosα; η_b,max = 2cos²α/(1 + cos²α)
Height of blades from continuity: ṁ v = π D_m h V_f (D_m = mean diameter, h = blade height)
```

- Reaction turbines need many stages; blade height increases toward the LP end as specific volume increases (exhaust ends use double-flow LP turbines).
- Losses in steam turbines: nozzle friction, blade friction, leakage (tip, gland), disc friction, partial admission, residual (leaving) velocity, wetness (moisture), shock (incidence) losses, mechanical friction, radiation.

★ **PYQ-TREND:** Degree of reaction definition; Parsons turbine is 50% reaction with identical fixed and moving blades (very frequent); optimum ratio cosα for Parsons; losses list.

16.5 Governing and Other Points

- Governing (keeping speed constant as load changes): throttle governing (reduce steam pressure by throttling – simple but causes loss), nozzle-control governing (open/close groups of nozzles – efficient at part load, used in impulse turbines), by-pass governing (admit extra steam to later stages at overload), combination. Emergency (overspeed) governor trips the turbine at about 110% speed.
- Reheat factor (multistage): ratio of the cumulative stage isentropic heat drops to the overall isentropic heat drop (≈ 1.03–1.05) – because friction reheats steam, later stages get a little more available drop.

```
Reheat factor RF = Σ(Δh_stage,isentropic) / (Δh_overall,isentropic); internal efficiency η_int = η_stage × RF
Willans line: steam consumption varies linearly with load for throttle-governed turbines
```

- Back-pressure turbine: exhausts at above-atmospheric pressure for process heating (cogeneration). Pass-out (extraction) turbine: part of the steam extracted at an intermediate pressure for process use, the rest condensed.
- Turbine blade materials: stainless steel (12% Cr), Inconel, titanium for last LP blades; stellite shields against erosion.

★ **PYQ-TREND:** Throttle vs nozzle-control governing; Willans line; reheat factor > 1; back-pressure and pass-out turbines for cogeneration; overspeed trip.

✎ **DESCRIPTIVE Q:** Explain the need for compounding in steam turbines and describe velocity compounding.

In a simple impulse (De Laval) turbine, steam expands from boiler pressure to condenser pressure in a single set of nozzles, so its velocity becomes very high (1,000 m/s or more). For good blade efficiency the blade speed must be about half the steam velocity, giving rotor speeds near 30,000 rpm. Such speeds need heavy reduction gearing, impose high centrifugal stresses, and leave a large exit velocity loss. Compounding divides the energy absorption into steps so the rotor runs at a practical speed. In velocity compounding (Curtis), the whole pressure drop occurs in one set of nozzles, producing a high-velocity jet. This jet passes through a first row of moving blades, where part of its velocity is absorbed; it then passes through a row of fixed guide blades that only redirect the steam (no pressure drop) onto a second row of moving blades, which absorbs more velocity, and so on. The pressure remains

constant across all rows. For two rows the optimum blade speed is a quarter of the jet velocity, and the first row does three-quarters of the work. It is compact and cheap but less efficient because of friction losses at the high velocities in the later rows.

CHAPTER 17

Condensers and Cooling Towers

17.1 Steam Condensers

- A condenser condenses the exhaust steam from the turbine at a pressure well below atmospheric. Functions: (1) lower back pressure → greater enthalpy drop and more work, higher cycle efficiency; (2) recover pure condensate for reuse as boiler feedwater (saves water treatment).
- Elements of a condensing plant: condenser, cooling-water pump, condensate (extraction) pump, air extraction pump (or steam-jet air ejector), hot well, cooling tower/pond, make-up water.

Type	Description
Jet (mixing) condensers	Steam and cooling water mix directly; condensate is lost (cannot be reused unless cooling water is pure); simpler, cheaper, less cooling water; parallel-flow, counter-flow, ejector, barometric (tall pipe – condensate drains by gravity, ≈ 10.3 m) types
Surface (non-mixing) condensers	Steam condenses on the outside of tubes carrying cooling water; condensate recovered pure; used in all large power plants; shell-and-tube; down-flow, central-flow, evaporative, regenerative types
Evaporative condenser	Cooling water sprayed over tubes, evaporates in air stream – for places with water scarcity

- Air leakage into the condenser (through glands and joints, and dissolved in feedwater) raises the total pressure (Dalton's law: p_total = p_steam + p_air), reduces vacuum and the heat transfer coefficient, and causes corrosion. Air is removed by air pumps (Edwards air pump) or steam-jet air ejectors.

```
Vacuum = barometric reading − absolute condenser pressure (both in mm Hg)
Vacuum efficiency = actual vacuum / ideal vacuum; ideal vacuum = barometer reading − saturation pressure of steam at condensate temperature
Condenser efficiency = (t_w,out − t_w,in) / (t_sat − t_w,in) (actual rise of cooling water / maximum possible rise)
Cooling water required: ṁ_w c_pw (t_w,out − t_w,in) = ṁ_s (h_exhaust − h_condensate)
Dalton's law in condenser: p_condenser = p_s + p_a; mass of air ṁ_a = p_a V/(R T)
```

- Typical condenser pressure ≈ 0.05–0.1 bar (vacuum ≈ 710–735 mm Hg); cooling water temperature rise ≈ 8–10 °C; about 50–60% of the heat supplied in a thermal plant is rejected in the condenser.

★ **PYQ-TREND:** Purpose of condenser; jet vs surface condensers (surface used in power plants, condensate reused) – very frequent; effects of air leakage and Dalton's law; vacuum efficiency and condenser efficiency formulas; barometric condenser height ≈ 10.3 m.

17.2 Cooling Towers and Ponds

- Cooling tower: cools the warm condenser water by evaporating a small part of it into air (evaporative cooling – mainly latent heat transfer). Water is sprayed or splashed over fill (packing) while air flows through.

Type	Feature
Natural draught (hyperbolic)	Tall reinforced-concrete shell of hyperbolic shape; air flows by density difference (chimney effect); no fan power; large power plants
Mechanical draught – forced	Fan at the base pushes air in
Mechanical draught – induced	Fan at the top draws air out (less recirculation, more common)
Cross-flow / counter-flow	Air horizontal across falling water / air upward against falling water
Dry (closed) cooling tower	No direct contact; water in finned tubes; no evaporation loss – water-scarce areas; less effective
Spray pond / cooling pond	Simple, large area

```
Range = water inlet temperature − water outlet temperature
Approach = water outlet temperature − wet bulb temperature of entering air (the outlet water can never be cooled below WBT)
Cooling tower efficiency = range/(range + approach) = (t_in − t_out)/(t_in − t_wb)
```

- Losses: evaporation (≈ 1–2% of circulating water per 6–7 °C range), drift (carried-off droplets, ≈ 0.1–0.2%), blowdown (to control dissolved solids). Make-up water replaces all three.
- Hyperbolic shape: structural strength with less material and good natural draught; the narrowing then widening shape accelerates the air.

★ **PYQ-TREND:** Approach and range definitions (very frequent); water cannot be cooled below WBT; natural draught towers are hyperbolic; induced vs forced draught towers; losses and make-up.

✎ **DESCRIPTIVE Q:** Compare jet and surface condensers.

In a jet condenser the exhaust steam and cooling water mix directly; heat transfer is very effective, so less cooling water is needed and the condenser is small, simple and cheap. However, the condensate is mixed with cooling water and cannot be reused as boiler feedwater unless the cooling water itself is pure, more air enters with the cooling water (needing a larger air pump), and a higher power is needed for pumping. In a surface condenser the steam condenses on the outer surface of tubes through which cooling water flows, so the two never mix. The pure condensate is recovered and returned to the boiler, impure cooling water (river, sea) can be used, and a higher vacuum is obtainable. It is, however, larger, costlier, needs more cooling water and more maintenance (tube cleaning, leak prevention). Surface condensers are therefore used in all modern steam power stations, while jet condensers suit small plants where feedwater is plentiful and cheap.

CHAPTER 18

Power Plant Engineering

18.1 Load Curves and Economics

- Load curve: plot of load (MW) vs time; load duration curve: loads arranged in descending order of magnitude.

```
Load factor = average load / maximum demand (over the period)
Demand factor = maximum demand / connected load (≤ 1)
Diversity factor = sum of individual maximum demands / maximum demand of the whole system (≥ 1)
Plant capacity factor = average load / rated plant capacity = actual energy produced / (capacity × hours)
Plant use factor = actual energy produced / (capacity × hours of actual operation)
Reserve capacity = plant capacity − maximum demand
```

- High load factor and diversity factor reduce the cost per kWh. Base-load plants (run continuously): large thermal (coal), nuclear, run-of-river hydro. Peak-load plants (quick start): gas turbines, diesel, pumped-storage hydro, storage hydro.
- Tariffs: flat-rate, block-rate, two-part (fixed + energy charge), maximum-demand, time-of-day.
- Cost of power: fixed costs (interest, depreciation, insurance, taxes – depend on capacity) + operating costs (fuel, labour, maintenance – depend on energy). Depreciation methods: straight-line, sinking-fund, diminishing balance.

★ **PYQ-TREND:** Load factor, diversity factor (≥ 1), demand factor, capacity factor definitions (very frequent); base vs peak load plants; two-part tariff.

18.2 Thermal (Coal-Fired) Power Plant

- Circuits: (1) coal and ash circuit; (2) air and flue-gas circuit; (3) feedwater and steam circuit; (4) cooling-water circuit.
- Coal handling: unloading → crushing → storage → pulverising (ball mill, bowl mill) → burners (pulverised-fuel firing, 70–80% passing 200 mesh) – or stoker firing (travelling grate, spreader) for small units, or fluidised-bed combustion (FBC – burns low-grade coal at 800–900 °C with limestone to capture SO_2; low NOx).
- Ash handling: hydraulic, pneumatic, mechanical; fly ash collected by electrostatic precipitators (ESP – > 99% efficiency), cyclone separators, bag filters. Flue gas desulphurisation (FGD) with limestone slurry scrubbers removes SO_2.
- Coal analysis: proximate analysis – moisture, volatile matter, fixed carbon, ash (by heating); ultimate analysis – C, H, O, N, S, ash (elemental). Ranks (increasing carbon and calorific value): peat → lignite → sub-bituminous → bituminous → anthracite (highest fixed carbon, burns with short smokeless flame, hard to ignite).
- Overall efficiency of thermal plants ≈ 33–40% (subcritical); supercritical/ultra-supercritical ≈ 40–46%. Major losses: condenser (≈ 50%), boiler flue gases.

- Major Indian thermal plants: Vindhyachal (MP, NTPC – largest in India), Mundra (Gujarat), Talcher, Singrauli, Korba, Ramagundam.

★ **PYQ-TREND:** Four circuits of thermal plant; ESP removes fly ash; proximate vs ultimate analysis; anthracite highest carbon; FBC and limestone; pulverised coal firing; largest loss in condenser.

18.3 Hydroelectric Power Plant

- Components: dam/reservoir, spillway, intake with trash rack, penstock, surge tank (protects penstock from water hammer), turbine, draft tube, tail race, powerhouse.

```
Power P = η ρ g Q H (W); energy = P × time
```

- Classification by head: low (< 30 m – Kaplan/propeller), medium (30–300 m – Francis), high (> 300 m – Pelton). By storage: run-of-river (with/without pondage), storage, pumped storage (water pumped up at off-peak hours using surplus power, released at peak).
- Advantages: renewable, no fuel cost, quick start, long life, low running cost, multipurpose (irrigation, flood control). Disadvantages: high capital cost, long construction, site-dependent, transmission distance, ecological and resettlement issues.

★ **PYQ-TREND:** Surge tank function; pumped storage for peak load; turbine selection by head; P = ηρgQH.

18.4 Nuclear Power Plant

- Fission: U-235 (or Pu-239, U-233) nucleus absorbs a slow (thermal) neutron and splits into two fragments, releasing ≈ 200 MeV energy and 2–3 fast neutrons → chain reaction. Mass defect converted to energy ($E = mc^2$). 1 kg of U-235 ≈ 8×10^{13} J (≈ 3,000 tonnes of coal).
- Fertile materials (U-238, Th-232) are converted into fissile Pu-239 and U-233 by neutron capture (breeding).

Component	Function / materials
Fuel	Natural uranium (0.7% U-235), enriched uranium (3–5%), plutonium, thorium (via U-233)
Moderator	Slows fast neutrons to thermal speeds by elastic collisions – heavy water (D_2O, best, allows natural U), ordinary water, graphite, beryllium
Control rods	Absorb neutrons to control the reaction rate – cadmium, boron (boron carbide), hafnium
Coolant	Removes heat from the core – ordinary water, heavy water, CO_2, helium, liquid sodium (fast reactors)
Reflector	Reflects escaping neutrons back into the core – graphite, beryllium
Shielding	Concrete, lead, water – protects against radiation
Reactor vessel, containment	Strong pressure vessel; containment building

Reactor	Fuel / moderator / coolant
PWR (pressurised water)	Enriched U / light water / light water at ≈ 150 bar (no boiling in core); steam generator
BWR (boiling water)	Enriched U / light water / light water boils in core – direct cycle

Reactor	Fuel / moderator / coolant
PHWR / CANDU	Natural U / heavy water / heavy water – India's mainstay (Kaiga, Kakrapar, Rawatbhata, Narora, Tarapur 3&4)
Gas-cooled (GCR, AGR)	Natural/enriched U / graphite / CO_2
Fast breeder (FBR)	Pu–U mixed oxide / no moderator / liquid sodium – breeds more fuel; PFBR Kalpakkam

- India's three-stage nuclear programme (Homi Bhabha): PHWRs (natural U) → fast breeders (Pu) → thorium-based reactors (U-233).
- Indian plants: Tarapur (first, 1969, BWR), Rawatbhata, Kalpakkam, Narora, Kakrapar, Kaiga, Kudankulam (VVER – largest, Russian).

★ **PYQ-TREND:** Moderator and control rod materials (very frequent); heavy water allows natural uranium; FBR has no moderator and uses liquid sodium; PHWR in India; fertile vs fissile; Tarapur first; three-stage programme.

18.5 Diesel and Gas Turbine Plants

- Diesel power plant: diesel engine + generator; quick start, compact, simple, small capacity (up to ≈ 50 MW); used for standby, peak load, remote areas, emergency (hospitals). Systems: fuel, air intake, exhaust, cooling, lubrication, starting.
- Gas turbine plant: compressor + combustion chamber + turbine; quick start (minutes), low water requirement, peak-load duty or combined cycle base load. Efficiency of simple open cycle ≈ 30–40%; combined cycle 55–62%.

★ **PYQ-TREND:** Diesel plants for standby and peak loads; gas turbines quick start; combined cycle efficiency.

18.6 Non-Conventional Energy Sources

Source	Key facts
Solar	Flat-plate collectors (water heating, < 100 °C), concentrating collectors (parabolic trough, dish, central tower/heliostats – power), photovoltaic cells (silicon; efficiency 15–22%); solar constant ≈ 1,367 W/m²
Wind	Power ∝ V^3 ($P = \frac{1}{2}\rho A V^3 \times Cp$); Betz limit $Cp,max = 16/27 = 59.3\%$; HAWT and VAWT (Darrieus, Savonius); Muppandal (TN) wind farm
Geothermal	Heat from the earth's interior; Puga valley (Ladakh), Manikaran (HP) potential sites
Tidal	Gulf of Kutch, Gulf of Khambhat, Sundarbans potential sites
OTEC (ocean thermal)	Uses temperature difference ≈ 20 °C between surface and deep water; Claude (open) and Anderson (closed) cycles
Biomass / biogas	Biogas (55–70% CH_4) from anaerobic digestion; KVIC floating-dome and Janata fixed-dome (Deenbandhu) digesters
Fuel cells	Direct conversion of chemical energy ($H_2 + O_2$) to electricity; hydrogen–oxygen; efficiency 40–60%

Source	Key facts
MHD (magnetohydrodynamics)	Hot ionised gas through a magnetic field generates DC directly
Thermoelectric / thermionic	Seebeck effect / electron emission from hot cathode

★ **PYQ-TREND:** Wind power ∝ V^3 and Betz limit 59.3% (very frequent); solar constant; PV cell material; biogas composition; OTEC temperature difference; fuel cell principle; MHD generates DC.

✎ **DESCRIPTIVE Q:** Explain the working of a nuclear power plant with the functions of the moderator, control rods and coolant.

In a nuclear power plant heat produced by controlled fission in the reactor core replaces the furnace of a thermal plant. Fuel rods of uranium oxide sit in the core; each fission of a U-235 nucleus by a slow neutron releases about 200 MeV and 2–3 fast neutrons. The moderator (heavy water, light water or graphite) surrounds the fuel and slows these fast neutrons by elastic collisions to thermal speeds, at which they are most likely to cause further fission, sustaining the chain reaction. Control rods of cadmium or boron, inserted to a varying depth, absorb excess neutrons and keep the multiplication factor exactly 1 for steady power; full insertion shuts the reactor down. The coolant (pressurised water, heavy water, CO_2 or liquid sodium) circulates through the core, removes the heat and transfers it in a steam generator to a secondary water circuit. The steam drives a turbine and generator exactly as in a conventional plant, and is condensed and returned. A reflector reduces neutron leakage, and thick concrete and lead shielding with a containment building protect people from radiation.

CHAPTER 19

Air Compressors

19.1 Uses and Classification

- Uses: pneumatic tools (drills, hammers, riveters), spray painting, sand blasting, braking of trains and trucks, supercharging, gas turbines, refrigeration, blast furnaces, pneumatic conveying, starting diesel engines, tyre inflation.

Basis	Types
Principle	Positive displacement (reciprocating; rotary – Roots blower, vane, screw, scroll) / dynamic (centrifugal, axial)
Stages	Single-stage, multi-stage
Action	Single-acting, double-acting
Pressure	Fans (< 1.1 pressure ratio), blowers (1.1–2.5), compressors (> 2.5)

Reciprocating	Rotary/dynamic
Low flow, high pressure (up to 1,000+ bar multistage)	High flow, lower pressure per stage
Pulsating delivery; needs receiver	Continuous, smooth delivery
Slow speed; heavy foundations; more vibration	High speed; compact; balanced
Lubrication contaminates air	Oil-free air possible (centrifugal)
Higher isothermal efficiency at high pressure ratios	Higher efficiency at large flow rates

★ **PYQ-TREND:** Classification; reciprocating for high pressure/low flow; centrifugal and axial for large flow; uses of compressed air; fan/blower/compressor pressure ratios.

19.2 Work of a Single-Stage Reciprocating Compressor

Work per cycle (no clearance), polytropic $pV^n = C$: $W = \frac{n}{(n-1)} p_1V_1 \left[\left(\frac{p_2}{p_1}\right)^{\frac{(n-1)}{n}} - 1\right]$

Isentropic ($n = \gamma$): $W = \frac{\gamma}{(\gamma-1)} p_1V_1 \left[\left(\frac{p_2}{p_1}\right)^{\frac{(\gamma-1)}{\gamma}} - 1\right] = \dot{m}\, c_p (T_2 - T_1)$

Isothermal ($n = 1$): $W = p_1V_1 \ln\left(\frac{p_2}{p_1}\right)$ — minimum work

With clearance, the work formula is the same with V_1 replaced by the effective swept (induced) volume $V_1 - V_4$

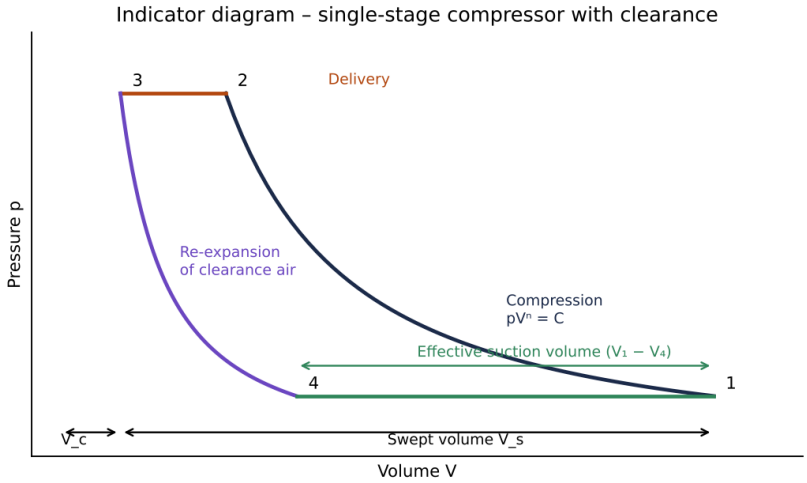
Isothermal efficiency η_{iso} = isothermal work / actual work

Delivery temperature: $T_2 = T_1 \left(\frac{p_2}{p_1}\right)^{\frac{(n-1)}{n}}$

Power = $W \times N_{working\ strokes\ per\ second}$ (single-acting: $N/60$; double-acting: $2N/60$)

- Work input is the area of the p–V diagram enclosed between suction, compression and delivery lines. Isothermal compression needs the least work (smallest area); adiabatic the most. Practical compressors have $1.2 < n < 1.35$ with cooling (water jackets, fins); the aim is to approach isothermal compression.

- Methods to approach isothermal compression: cooling the cylinder (water jackets/fins), spraying water, multistaging with intercooling (most effective).



★ **PYQ-TREND:** Isothermal compression requires minimum work (very frequent); work formulas; isothermal efficiency definition; delivery temperature; methods to reduce work.

19.3 Clearance and Volumetric Efficiency

Clearance ratio C = V_c/V_s (typically 4–10%)

Volumetric efficiency η_v = (V₁ − V₄)/V_s = 1 + C − C (p₂/p₁)^(1/n)

Maximum pressure ratio (η_v = 0): (p₂/p₁)_max = ((1 + C)/C)^n

Free air delivery (FAD): volume of air delivered reduced to intake (free-air) conditions − p_a V_a/T_a = p₁V₁/T₁

Overall volumetric efficiency referred to FAD: η_v' = (FAD per cycle)/V_s

- Clearance air re-expands during the suction stroke, reducing the induced volume. Volumetric efficiency decreases as clearance or pressure ratio increases. Clearance does not change the work per kg of air (ideal) but reduces capacity.
- Clearance is necessary to prevent the piston from striking the cylinder head and to accommodate valves.

★ **PYQ-TREND:** Volumetric efficiency formula and effect of clearance and pressure ratio (very frequent); FAD definition; maximum pressure ratio for zero delivery; work per kg independent of clearance (ideal).

19.4 Multistage Compression and Intercooling

- Compression in two or more stages with intercoolers between them (air cooled back to intake temperature = perfect intercooling).

Two-stage, perfect intercooling, equal n: optimum intermediate pressure p₂ = √(p₁p₃)

Pressure ratio per stage for z stages: (p_final/p₁)^(1/z) (equal in each stage)

Minimum total work (z stages): W = z (n/(n−1)) p₁V₁ [(p_{z+1}/p₁)^((n−1)/(zn)) − 1]

Under these conditions: work in each stage is equal, delivery temperature from each stage is equal, heat rejected in each intercooler is equal

Heat rejected in intercooler = ṁ cp (T₂ − T₁); heat rejected during polytropic compression = W (γ − n)/(γ − 1)

- Advantages of multistaging: less work (closer to isothermal – saving shown as area between the curves), higher volumetric efficiency (lower pressure ratio per stage reduces clearance effect), lower delivery temperatures (better lubrication, no oil decomposition/explosion risk), lighter LP cylinders, better mechanical balance, less leakage.
- Usual number of stages: up to 5 bar – one; 5–35 bar – two; 35–85 bar – three; above – four or more.

★ **PYQ-TREND:** Optimum intermediate pressure $\sqrt{(p_1 p_3)}$ (very frequent); equal work and equal pressure ratio per stage; advantages of multistaging; intercooler heat.

✔ **TOPPER TRICK:** For a three-stage compressor from 1 bar to 27 bar, the stage pressures are 3 bar and 9 bar (ratio 3 each).

19.5 Rotary Compressors

- Roots blower: two lobed rotors trap air and carry it from inlet to outlet; no internal compression – compression occurs irreversibly when the pocket opens to the higher-pressure delivery (back-flow). Efficient only at low pressure ratios (≈ 1.2–2); used for supercharging and scavenging two-stroke diesels.

Roots blower work = $(p_2 - p_1)V$; Roots efficiency = isentropic work / actual work

- Vane blower: eccentric rotor with sliding vanes; partial internal compression (then back-flow compression) → more efficient than Roots at the same ratio.
- Screw (Lysholm) compressor: two meshing helical rotors; internal compression; oil-flooded or oil-free; widely used in industry and refrigeration (medium flows, 7–15 bar).
- Scroll compressor: orbiting spiral – domestic AC and refrigeration (quiet, efficient).

Centrifugal compressor

- Air enters the eye of the impeller axially, is accelerated radially outward by the rotating vanes (kinetic energy and some pressure rise), then decelerated in a diffuser (vaned or vaneless) and volute casing to raise the pressure. Pressure ratio per stage ≈ 4–5 (up to ≈ 10 in special designs).

Work input per kg (radial vanes, no prewhirl) = $\sigma \psi u_2^2$ (σ = slip factor ≈ 0.9, ψ = power input factor ≈ 1.03–1.05)

Pressure coefficient = isentropic work / u_2^2; temperature rise of air $\Delta T = \sigma\psi u_2^2/cp$

- Slip: the air leaves the impeller with less whirl than the blade angle implies (finite blades, relative eddy).
- Surging: unstable operation at low mass flow when the pressure ratio peak is passed – flow reversal and violent pulsations. Choking (stonewalling): maximum flow when the velocity at some section reaches sonic speed – flow cannot increase further. Stalling: separation of flow from blades (rotating stall).

Axial-flow compressor

- Many stages of rotating (rotor) and stationary (stator) blade rows; each stage gives a small pressure rise (≈ 1.1–1.3), with 50% degree of reaction common; high efficiency (85–90%) and very large flow; used in jet engines and large gas turbines. Inlet guide vanes, variable stators.

Centrifugal	Axial
Radial flow; high pressure ratio per stage (4–5)	Axial flow; low pressure ratio per stage (1.1–1.3)

Centrifugal	Axial
Robust, wider stable range, less sensitive to dirt	Higher efficiency, larger mass flow, smaller frontal area
Larger frontal area – early jet engines, turbochargers	Modern jet engines and large gas turbines

★ **PYQ-TREND:** Roots blower has no internal compression; surging at low flow and choking at maximum flow (very frequent); slip factor; axial compressors for jet engines (high efficiency, large flow, 50% reaction); centrifugal for turbochargers; screw compressor.

✎ **DESCRIPTIVE Q:** Explain the advantages of multistage compression with intercooling and derive the condition for minimum work in a two-stage compressor.

Air compressed in a single stage to a high pressure becomes very hot, needs much more work than isothermal compression, and the large pressure ratio makes clearance air re-expand over most of the suction stroke, lowering volumetric efficiency. Dividing compression into stages with intercooling brings the air back to near intake temperature between stages, so the process approaches isothermal compression. This saves work, gives lower delivery temperatures (safer lubrication), better volumetric efficiency, lighter cylinders and better balance. For two stages with perfect intercooling (T at the start of each stage = T_1) and the same index n, total work $W = (n/(n-1))mRT_1[(p_2/p_1)^k + (p_3/p_2)^k - 2]$, with $k = (n-1)/n$. Setting $dW/dp_2 = 0$: $k\, p_2^{(k-1)}/p_1^k - k\, p_3^k/p_2^{(k+1)} = 0$, giving $p_2^2 = p_1 p_3$, i.e., $p_2 = \sqrt{(p_1 p_3)}$. Then the pressure ratio and work are equal in both stages, and so are the delivery temperatures.

CHAPTER 20

Gas Turbines and Jet Propulsion

20.1 Gas Turbine Plants

- Simple open-cycle gas turbine: atmospheric air → compressor → combustion chamber (fuel burnt at nearly constant pressure) → turbine → exhaust. About half to two-thirds of the turbine work drives the compressor (high back-work ratio). Ideal cycle: Brayton (Chapter 8).
- Closed cycle: same gas (air, helium, CO_2) circulated; heat added in an external heat exchanger and rejected in a cooler. Advantages: higher pressure (smaller size), clean working fluid (no erosion), any fuel (coal, nuclear), constant efficiency at part load. Disadvantages: large heat exchangers, cooling water needed.
- Components: axial/centrifugal compressor; combustion chambers (can, annular, can-annular); turbine (axial, cooled blades of nickel superalloys); regenerator; intercooler; reheat combustor.
- Combustion chamber: primary zone (≈ 20–30% of air, near-stoichiometric burning), secondary/dilution zone (rest of the air to cool gases to turbine inlet temperature); overall A/F ≈ 60–100:1 because the turbine inlet temperature is limited by blade materials.
- Turbine inlet temperature (TIT): 900–1,100 °C in industrial units; up to ≈ 1,500–1,600 °C in modern aero/advanced units with blade cooling (internal air passages, film cooling) and thermal barrier coatings.

Simple cycle with component efficiencies: $w_{net} = c_p[\eta_t T_3(1 - 1/r_p^k) - T_1(r_p^k - 1)/\eta_c]$, $k = (\gamma-1)/\gamma$

Thermal efficiency $\eta = w_{net} / [c_p(T_3 - T_2)]$

- Advantages of gas turbines: high power-to-weight ratio, compact, quick start, little water, low vibration, multi-fuel. Disadvantages: lower efficiency than diesels at small sizes, high compressor work, expensive high-temperature materials, poor part-load efficiency.
- Uses: aircraft propulsion, peak-load and combined-cycle power plants, ships, gas pipelines (compressor drives), locomotives (rare), emergency power.

★ **PYQ-TREND:** High back-work ratio; overall air–fuel ratio is very lean because of TIT limit; open vs closed cycle; blade cooling; uses of gas turbines.

20.2 Jet Propulsion

- Jet propulsion works on Newton's third law: a high-velocity jet of gas leaving the engine produces an equal and opposite thrust.

Thrust (air-breathing): $F = \dot{m}_a[(1 + f)V_j - V_a] + (p_e - p_a)A_e \approx \dot{m}_a(V_j - V_a)$

Thrust power = $F \times V_a$

Propulsive (Froude) efficiency η_p = thrust power / KE supplied to jet = $2V_a/(V_a + V_j) = 2/(1 + V_j/V_a)$

Thermal efficiency = increase in KE of gases / heat supplied by fuel; overall efficiency = $\eta_p \times \eta_{th} = F\ V_a/(\dot{m}_f\ CV)$

Thrust specific fuel consumption TSFC = $\dot{m}_f / F$

- Propulsive efficiency is highest when the jet speed is close to the flight speed – but then thrust per unit mass flow is small. This is why high-bypass turbofans (large air mass at modest jet velocity) are efficient at subsonic speeds.

Engine	Key features
Turbojet	Compressor, combustor, turbine (drives compressor), nozzle; all thrust from the jet; efficient at high (supersonic) speeds; noisy
Turbofan	Fan driven by turbine; bypass air (bypass ratio 5–12 in airliners) gives most thrust; high propulsive efficiency, quiet; modern airliners
Turboprop	Turbine drives a propeller through gearbox (80–90% thrust from propeller); efficient at low speeds (< 700 km/h); transport aircraft
Turboshaft	Shaft power only – helicopters, ships, generators
Ramjet	No compressor or turbine; air compressed by ram effect of high flight speed (needs initial speed, M ≈ 2–4); cannot start from rest; simple; missiles
Scramjet	Supersonic combustion ramjet (M > 5)
Pulse jet	Intermittent combustion with flapper valves; V-1 flying bomb
Afterburner (reheat)	Extra fuel burnt in the jet pipe to boost thrust (military aircraft)
Rocket	Carries its own oxidiser – works in vacuum/space; liquid (LOX + LH_2/kerosene), solid, cryogenic, hybrid propellants

`Rocket thrust F = ṁ V_e + (p_e − p_a)A_e; specific impulse I_sp = F/(ṁ g) (seconds)`

★ **PYQ-TREND:** Propulsive efficiency 2/(1 + V_j/V_a) (very frequent); ramjet cannot start from rest and has no moving parts; rocket carries oxidiser; turbofan most efficient for airliners; turboprop for low speeds; afterburner increases thrust; specific impulse.

✎ **DESCRIPTIVE Q:** Explain the working of a turbojet engine and why turbofan engines are more efficient for subsonic aircraft.

In a turbojet, air enters through a diffuser intake where its velocity falls and pressure rises (ram compression). An axial or centrifugal compressor further raises its pressure, and fuel is burnt continuously in the combustion chamber at nearly constant pressure. The hot gases expand partly in the turbine, which drives the compressor, and then in the propelling nozzle to a high velocity, producing thrust $F \approx \dot{m}(V_j - V_a)$. Because all the thrust comes from a small mass flow accelerated to a very high jet velocity, the jet carries away a lot of kinetic energy and the propulsive efficiency $2V_a/(V_a + V_j)$ is low at subsonic flight speeds. A turbofan adds a large fan, driven by an extra turbine stage, which moves a much larger mass of air (bypass air) at a moderate velocity around the core. For the same thrust, accelerating more air by a smaller velocity increment wastes less kinetic energy, so propulsive efficiency and fuel economy improve markedly and noise falls. That is why airliners use high-bypass turbofans, while pure turbojets (often with afterburners) suit supersonic flight.

CHAPTER 21

Refrigeration – Fundamentals, Air Refrigeration and VCR

21.1 Basics

- Refrigeration: process of removing heat from a body or space to maintain it at a temperature below that of the surroundings. By the Clausius statement, work (or heat in absorption systems) must be supplied.

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1 tonne of refrigeration (TR) = 211 kJ/min = 3.5167 kW (≈ 3.5 kW) ≈ 50 kcal/min ≈ 12,000 BTU/h
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- Definition: refrigerating effect produced by melting 1 short ton (907 kg / 2000 lb) of ice at 0 °C into water at 0 °C in 24 hours (using latent heat 335 kJ/kg: 907 × 335/1440 ≈ 211 kJ/min).

```
COP = refrigerating effect / work input; relative COP = actual COP / theoretical COP
Carnot (reversed) COP_R = T_L/(T_H − T_L); COP_HP = COP_R + 1
Energy efficiency ratio EER = COP × 3.412 (BTU/Wh)
```

- Applications: food preservation, air conditioning, ice making, cold storage, chemical and pharmaceutical industries, cryogenics, medical.
- Methods of refrigeration: ice, evaporative cooling, dry ice, throttling (Joule–Thomson), vapour compression, vapour absorption, air expansion (Bell–Coleman), steam jet, thermoelectric (Peltier), vortex tube, adiabatic demagnetisation (near 0 K).

★ **PYQ-TREND:** 1 TR = 3.5 kW = 211 kJ/min (very frequent); Carnot COP formula and that COP decreases as the temperature difference increases; COP_HP = COP_R + 1.

✔ **TOPPER TRICK:** For a Carnot refrigerator between −23 °C and 27 °C: COP = 250/50 = 5. Halving the temperature lift doubles the COP.

21.2 Reversed Carnot Cycle and Its Limits

- Reversed Carnot cycle with a vapour: isentropic compression, isothermal condensation, isentropic expansion, isothermal evaporation. Impractical: isentropic expansion of liquid in an expander gives very little work for its cost; wet compression causes damage and poor lubrication ('liquid slugging'); isothermal heat transfer needs infinite area/time. The practical VCR cycle replaces the expander by a throttle valve and uses dry compression.

★ **PYQ-TREND:** Why reversed Carnot is not practical; throttle valve replaces expander.

21.3 Air Refrigeration (Bell–Coleman / Reversed Brayton)

- Air (gas) remains a gas throughout – no phase change. Processes: isentropic compression, constant-pressure cooling in a heat exchanger, isentropic expansion in an expander (turbine) to low temperature, constant-pressure heat absorption in the refrigerated space.

COP = $T_1/(T_2 - T_1) = 1/(r_p^{((\gamma-1)/\gamma)} - 1)$ (r_p = pressure ratio; T_1 = compressor inlet, T_2 = compressor outlet)

Refrigerating effect = $cp(T_1 - T_4)$; work = $cp[(T_2 - T_3) - (T_1 - T_4)]$

- COP is low (≈ 0.5–1) because heat is absorbed and rejected sensibly (not at constant temperature), needing large mass flows. But air is free, safe and non-toxic, and the system is light – used in aircraft cabin cooling (using air bled from the engine compressor).
- Aircraft refrigeration systems: simple, bootstrap (secondary compressor driven by the cooling turbine – for high-speed aircraft), regenerative, reduced-ambient, simple evaporative.

★ **PYQ-TREND:** Air refrigeration used in aircraft because of low weight per TR (very frequent); Bell–Coleman COP formula; low COP; bootstrap system.

21.4 Simple Vapour Compression Refrigeration (VCR) Cycle

- Components and processes (on p–h and T–s): 1–2 compressor – isentropic compression of saturated (or slightly superheated) vapour to condenser pressure (becomes superheated); 2–3 condenser – desuperheating and condensation at constant pressure (heat rejected to air/water); 3–4 expansion device – throttling at constant enthalpy ($h_3 = h_4$; part of the liquid flashes to vapour); 4–1 evaporator – evaporation at constant low pressure and temperature, absorbing heat from the refrigerated space.

Refrigerating effect RE = $h_1 - h_4 = h_1 - h_3$ (kJ/kg)

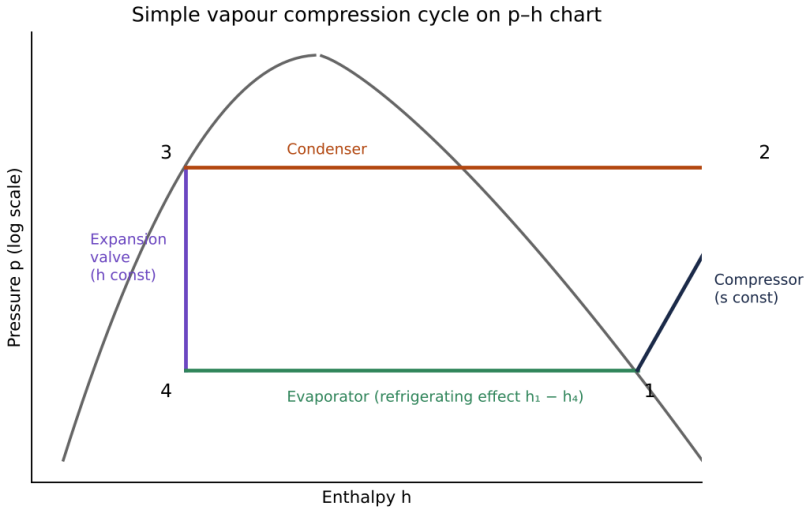
Compressor work W = $h_2 - h_1$; heat rejected $Q_c = h_2 - h_3 = RE + W$

COP = $(h_1 - h_4)/(h_2 - h_1)$

Refrigerant mass flow $\dot{m}$ = 211 × TR / RE (kg/min) or 3.5167 × TR / RE (kg/s)

Compressor power = $\dot{m}$ ($h_2 - h_1$); power per TR = 3.5167/COP kW

Volumetric efficiency of compressor: $\eta_v = 1 + C - C (p_d/p_s)^{(1/n)}$; piston displacement = $\dot{m}\, v_1/\eta_v$



- Dry compression (vapour saturated or superheated at compressor inlet) is used; wet compression risks liquid slugging and valve damage.

Effect of operating conditions

Change	RE	Work	COP
Lower evaporator pressure/temperature	Decreases	Increases	Decreases
Higher condenser pressure/temperature	Decreases	Increases	Decreases
Subcooling the liquid after condenser	Increases	No change	Increases
Superheating vapour (useful, in evaporator)	Increases	Increases	Depends on refrigerant (increases for R-12/R-134a, decreases for NH_3)
Pressure drop in suction line	–	Increases	Decreases
Liquid–vapour (suction-line) heat exchanger	Subcools liquid, superheats vapour – useful for halocarbons	–	Usually increases

★ **PYQ-TREND:** Throttling in expansion valve is isenthalpic (very frequent); COP = $(h_1 - h_4)/(h_2 - h_1)$; subcooling increases COP; lowering evaporator temperature decreases COP; mass flow 211×TR/RE; dry vs wet compression; condenser heat = RE + W.

✖ **COMMON MISTAKE:** In VCR numericals take $h_4 = h_3$ (throttling). Don't use $h_4 = h_f$ at the evaporator pressure.

21.5 Components of VCR Systems

Component	Types / facts
Compressor	Reciprocating (small–medium, up to ~100 TR), rotary (vane, rolling piston – small), scroll (domestic AC – quiet, efficient), screw (medium–large, 50–1,000 TR), centrifugal (large, > 200 TR; low-pressure refrigerants); hermetic (motor + compressor sealed in a shell – domestic refrigerators, AC), semi-hermetic, open type (ammonia plants)
Condenser	Air-cooled (natural convection – domestic fridge, wire-and-tube; forced – window/split AC), water-cooled (shell-and-tube, shell-and-coil, double-pipe), evaporative condenser (water sprayed over coil + air – large plants)
Expansion device	Capillary tube (domestic refrigerators, small AC – fixed restriction, no moving parts, pressures equalise at stop so low starting torque motor), automatic (constant-pressure) expansion valve, thermostatic expansion valve (TEV – maintains constant superheat at evaporator outlet; most common in commercial systems), float valves (high-side and low-side – flooded evaporators), electronic expansion valve
Evaporator	Bare-tube, finned-tube, plate (domestic freezer); flooded (evaporator full of liquid – better heat transfer) vs dry-expansion (direct expansion, DX); natural vs forced convection; chillers (shell-and-tube)

Component	Types / facts
Accessories	Receiver, drier (removes moisture – silica gel, activated alumina, molecular sieves), filter, sight glass, solenoid valve, accumulator, oil separator, pressure switches (HP/LP cut-outs), thermostat

- Domestic refrigerator: hermetic reciprocating/rotary compressor, wire-and-tube or natural-convection condenser at the back, capillary tube, evaporator in the freezer; refrigerant R-134a or R-600a (isobutane).
- Moisture in halocarbon systems freezes at the expansion device (choking) and forms acids; driers are essential.

★ **PYQ-TREND:** Capillary tube in domestic refrigerators (very frequent); TEV maintains constant superheat; hermetic compressors; centrifugal compressors for large capacity; evaporative condenser; purpose of drier.

✎ **DESCRIPTIVE Q:** Explain the working of a simple vapour compression refrigeration system with its p–h diagram and give the expression for COP.

Low-pressure dry saturated vapour from the evaporator (state 1) is compressed isentropically by the compressor to condenser pressure, leaving superheated at state 2 (work $h_2 - h_1$). In the condenser it rejects heat to the cooling air or water, first losing superheat and then condensing at constant pressure to saturated (or subcooled) liquid at state 3 (heat $h_2 - h_3$). The liquid passes through the expansion valve, where it is throttled at constant enthalpy to evaporator pressure ($h_4 = h_3$); its temperature falls and a part flashes into vapour. In the evaporator the low-temperature wet mixture absorbs heat from the refrigerated space and evaporates at constant pressure back to state 1 (refrigerating effect $h_1 - h_4$). On the p–h chart the compression is a curve along a constant-entropy line, condensation and evaporation are horizontal lines, and throttling is a vertical line. COP = refrigerating effect/work = $(h_1 - h_4)/(h_2 - h_1)$. For a load of TR tonnes the refrigerant flow is $211 \times TR/(h_1 - h_4)$ kg/min and the compressor power is $\dot{m}(h_2 - h_1)$.

CHAPTER 22

Refrigerants, Vapour Absorption and Other Systems

22.1 Refrigerants

- Primary refrigerants undergo the cycle (ammonia, halocarbons, CO_2, hydrocarbons). Secondary refrigerants (brines – NaCl or $CaCl_2$ solutions, glycols, water) are cooled by the primary refrigerant and then carry cold to the load without phase change.

Numbering (ASHRAE)

Halocarbons $C_mH_nF_pCl_q$: R-(m−1)(n+1)(p) − hundreds digit = C − 1, tens = H + 1, units = F

Inorganic: R-7XX where XX = molecular mass (NH_3 = 17 → R-717; H_2O = 18 → R-718; CO_2 = 44 → R-744; air = 29 → R-729)

Blends: zeotropic R-4XX (R-404A, R-407C, R-410A); azeotropic R-5XX (R-502); hydrocarbons R-600 (n-butane), R-600a (isobutane), R-290 (propane)

Refrigerant	Chemical	Boiling point (1 atm)	Uses / remarks
R-11	CCl_3F	23.8 °C	Centrifugal chillers (old); CFC – banned
R-12	CCl_2F_2	−29.8 °C	Old domestic refrigerators, car AC; CFC – banned
R-22	$CHClF_2$	−40.8 °C	Window/split AC, cold storage; HCFC – being phased out
R-134a	CH_2FCF_3	−26.1 °C	Domestic refrigerators, car AC; HFC; ODP 0, GWP ≈ 1,430
R-410A	Blend of R-32/R-125	−48.5 °C	Modern ACs; ODP 0, high GWP
R-32	CH_2F_2	−51.7 °C	Modern split ACs; lower GWP (≈ 675)
R-717 (ammonia)	NH_3	−33.3 °C	Cold storage, ice plants, large industrial plants; highest refrigerating effect per kg; toxic, flammable at certain concentrations; attacks copper – use steel; ODP 0, GWP 0
R-744 (CO_2)	CO_2	−78.5 °C (sublimes)	Ships, transcritical systems, supermarkets; very high pressure; ODP 0, GWP 1

Refrigerant	Chemical	Boiling point (1 atm)	Uses / remarks
R-718 (water)	H_2O	100 °C	Absorption (LiBr–water) chillers, steam-jet systems
R-600a (isobutane)	C_4H_{10}	−11.7 °C	Modern domestic refrigerators; natural, flammable
R-290 (propane)	C_3H_8	−42 °C	Natural refrigerant in small AC
R-1234yf	HFO	−29 °C	New car AC refrigerant; very low GWP

- Desirable properties of a refrigerant: thermodynamic – low boiling point, high latent heat of vaporisation, low specific volume of vapour (small compressor), moderate condensing pressure, high critical temperature (well above condensing temperature), low freezing point, high COP; chemical – non-toxic, non-flammable, non-corrosive, chemically stable, inert with oil and materials; physical – good thermal conductivity, low viscosity, easy leak detection, high dielectric strength (hermetic units); environmental – zero ODP, low GWP; economic – cheap and available.
- Leak detection: ammonia – sulphur candle (dense white fumes of ammonium sulphite), litmus/phenolphthalein paper, smell; halocarbons – halide torch (flame turns green), electronic detectors; all – soap solution.
- Environment: CFCs (contain chlorine) deplete stratospheric ozone (Montreal Protocol 1987 – phase-out); HCFCs being phased out; HFCs have zero ODP but high GWP (Kigali Amendment 2016 – phase-down). Trend towards natural refrigerants (NH_3, CO_2, hydrocarbons) and HFOs.

★ **PYQ-TREND:** Refrigerant numbering (R-717 ammonia, R-744 CO_2, R-718 water; R-134a formula) – very frequent; ammonia not used with copper; halide torch for Freon leaks; sulphur candle for ammonia; Montreal (ozone) vs Kigali (HFC) protocols; R-134a has zero ODP; desirable properties; secondary refrigerants (brine).

✔ **TOPPER TRICK:** R-134a: '1' (C = 2), '3' (H = 2), '4' (F = 4) → $C_2H_2F_4$. Add 90 to any halocarbon number to get C-H-F directly: 134 + 90 = 224 → $C_2H_2F_4$; 12 + 90 = 102 → $C_1H_0F_2$ (rest chlorine).

22.2 Vapour Absorption Refrigeration (VAR)

- The compressor of a VCR system is replaced by a 'thermal compressor': absorber + pump + generator (+ heat exchanger). Low-pressure refrigerant vapour from the evaporator is absorbed in a weak solution in the absorber (heat removed by cooling water); the strong solution is pumped (small pump work) to the generator, where heat (steam, waste heat, solar, gas flame) drives the refrigerant vapour out; condenser, expansion valve and evaporator work as in VCR.

System	Refrigerant	Absorbent	Uses / notes
Aqua-ammonia	NH_3	Water	Sub-zero temperatures; needs analyser and rectifier to remove water vapour from ammonia leaving the generator

System	Refrigerant	Absorbent	Uses / notes
Lithium bromide–water	Water	LiBr	Above 0 °C (water freezes) – large AC chillers (single/double effect); operates under vacuum; crystallisation risk
Electrolux (Platen–Munters, three-fluid)	NH_3	Water	Hydrogen as third fluid equalises total pressure throughout – no pump, no moving parts; gravity circulation; small domestic/hotel refrigerators, caravans

Ideal (maximum) COP of VAR: COP = [(T_G − T_0)/T_G] × [T_E/(T_0 − T_E)] (T_G generator, T_0 ambient/condenser, T_E evaporator)

Practical COP: single-effect LiBr ≈ 0.6–0.8; double-effect ≈ 1.0–1.3; aqua-ammonia ≈ 0.5–0.7

- Advantages: uses low-grade heat (waste steam, solar), few moving parts, quiet, no compressor wear, capacity control easy, refrigerant is natural. Disadvantages: low COP, bulky, higher initial cost, longer start-up.

VCR	VAR
Mechanical work (compressor)	Heat energy (generator) + small pump work
High COP (2–5)	Low COP (0.5–1.3)
Noisy, moving parts	Quiet, few moving parts
Performance drops sharply at low evaporator pressures	Less affected by evaporator pressure
Electricity costly	Can use waste/solar heat

★ **PYQ-TREND:** Electrolux uses hydrogen as third fluid with no pump (very frequent); LiBr–water: water is the refrigerant (above 0 °C only); aqua-ammonia: ammonia is refrigerant, needs rectifier and analyser; ideal COP of VAR formula; VAR vs VCR comparison.

22.3 Other Refrigeration Systems

System	Principle / use
Steam-jet (water vapour) refrigeration	High-velocity steam jet (ejector) creates vacuum so water evaporates at low temperature (≈ 4–10 °C); chilled water for AC; no moving parts except pumps
Thermoelectric	Peltier effect – heat absorbed at one junction of dissimilar semiconductors (Bi_2Te_3) when current flows; portable coolers, electronics; low COP
Vortex tube (Ranque–Hilsch)	Compressed air enters tangentially, splits into hot outer and cold inner streams; spot cooling, no moving parts
Evaporative cooling (desert cooler)	Water evaporates into dry air; effective only in hot, dry climates
Adiabatic demagnetisation	Paramagnetic salts – temperatures near 0 K

System	Principle / use
Cryogenics (< −150 °C)	Linde–Hampson (J–T throttling with regenerative heat exchanger), Claude (expansion engine + throttling), cascade systems; liquefaction of air, N_2, O_2, H_2, He
Cascade refrigeration	Two or more VCR circuits in series, each with a suitable refrigerant – very low temperatures (−60 to −100 °C)
Multi-pressure / multi-evaporator systems	Flash intercooling, flash gas removal for large temperature lifts

- Dry ice (solid CO_2) sublimes at −78.5 °C at 1 atm; used for transport of frozen food, stage fog. Liquid nitrogen boils at −196 °C.

★ **PYQ-TREND:** Peltier effect; vortex tube; Linde–Hampson and Claude cycles; desert coolers in dry climates; dry ice −78.5 °C; liquid nitrogen −196 °C; cascade systems for very low temperature.

✎ **DESCRIPTIVE Q:** Explain the Electrolux refrigerator and why it needs no pump.

The Electrolux (three-fluid) absorption refrigerator uses ammonia as refrigerant, water as absorbent and hydrogen as an auxiliary inert gas. Hydrogen fills the evaporator and absorber, so that the total pressure is the same everywhere in the system; in the evaporator the partial pressure of ammonia is low (most of the total pressure is due to hydrogen), so liquid ammonia evaporates at a low temperature by diffusing into the hydrogen, absorbing heat from the cabinet. The heavy ammonia–hydrogen mixture sinks to the absorber, where water absorbs the ammonia; the light hydrogen rises back to the evaporator. The strong solution is heated in the generator by a gas flame or electric heater; vapour bubbles lift the solution up a narrow tube (bubble pump) to a separator, where ammonia vapour goes to the air-cooled condenser and the weak water solution returns by gravity to the absorber. Because the total pressure is uniform, no pump or valve is needed – the system has no moving parts, runs silently and can operate from any heat source, though its COP is low.

CHAPTER 23

Psychrometry and Air Conditioning

23.1 Psychrometric Properties

- Atmospheric air = dry air + water vapour (moist air). Psychrometry studies its thermodynamic properties. Dry air and vapour are treated as ideal gases (Dalton's law: $p_b = p_a + p_v$).

Property	Definition
Dry bulb temperature (DBT)	Temperature measured by an ordinary thermometer
Wet bulb temperature (WBT)	Temperature indicated by a thermometer with a wetted wick in moving air (≈ adiabatic saturation temperature for air–water)
Dew point temperature (DPT)	Temperature at which vapour starts condensing when air is cooled at constant pressure (= saturation temperature at p_v)
Wet bulb depression	DBT − WBT (zero for saturated air)
Dew point depression	DBT − DPT
Specific humidity (humidity ratio) ω	Mass of water vapour per kg of dry air
Relative humidity φ	Ratio of actual vapour pressure to saturation vapour pressure at the same DBT
Degree of saturation μ	ω/ω_s at the same DBT
Absolute humidity	Mass of vapour per m^3 of moist air (vapour density)
Saturated air	Air holding the maximum vapour at its DBT (φ = 100%)

$\omega = 0.622\, p_v/(p_b - p_v)$ kg/kg of dry air

$\varphi = p_v/p_s$; $\mu = \omega/\omega_s = \varphi\,(p_b - p_s)/(p_b - p_v)$ $(\mu \approx \varphi)$

Enthalpy of moist air: $h = 1.005\, t + \omega(2500 + 1.88\, t)$ kJ/kg of dry air (t in °C)

Humid specific heat $c_{pa} = 1.005 + 1.88\omega \approx 1.022$ kJ/kg·K

Specific volume $v = R_a\, T/p_a = 0.287\, T/(p_b - p_v)$ m^3/kg dry air

Carrier's equation: $p_v = p_{vs}' - (p_b - p_{vs}')(t_{db} - t_{wb})/(1544 - 1.44\, t_{wb})$ (p_{vs}' = saturation pressure at WBT)

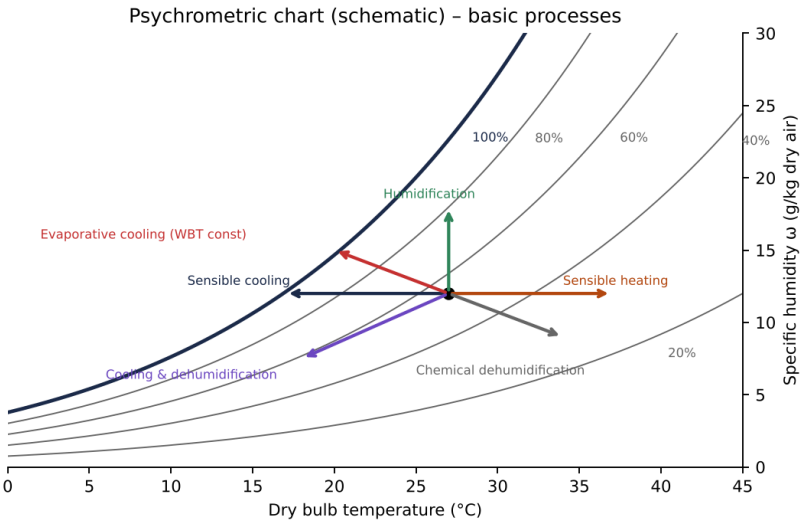
- For unsaturated air DBT > WBT > DPT; for saturated air DBT = WBT = DPT.
- Instruments: sling psychrometer (DBT and WBT, whirled), Assmann aspiration psychrometer, hygrometers (hair, dew-point, electronic).

★ **PYQ-TREND:** Order DBT > WBT > DPT and equality at saturation (very frequent); $\omega = 0.622\, p_v/(p_b - p_v)$; definitions of RH, specific humidity, degree of saturation, dew point; enthalpy of moist air; sling psychrometer.

✖ **COMMON MISTAKE:** Relative humidity and specific humidity are different. Sensible heating keeps ω constant but lowers RH.

23.2 Psychrometric Chart and Processes

- Chart: DBT on the x-axis, specific humidity (and vapour pressure) on the y-axis; saturation curve (100% RH) is the upper boundary; RH curves below it; WBT lines (nearly parallel to constant-enthalpy lines) slope downward to the right; specific-volume lines are steeper; dew point read horizontally to the saturation curve.



Process	Path on chart	Example
Sensible heating	Horizontal to the right (ω constant, RH falls)	Heating coil, winter AC
Sensible cooling	Horizontal to the left (above dew point)	Cooling coil with surface temperature > DPT
Humidification (pure)	Vertical upward	Steam injection at constant DBT
Dehumidification (pure)	Vertical downward	– (theoretical)
Cooling and dehumidification	Down and to the left towards ADP of the coil	Summer AC, cooling coil below DPT
Heating and humidification	Up and to the right	Winter AC
Evaporative (adiabatic) cooling	Along constant WBT line, up and left (DBT falls, ω rises)	Desert cooler, air washer
Chemical dehumidification (adsorption)	Along constant WBT/enthalpy, down and right (DBT rises, ω falls)	Silica gel, activated alumina
Heating with steam injection / hot water spray	Depends on water temperature	–
Adiabatic mixing of two streams	Point on straight line joining the states, divided inversely as the masses	Mixing fresh and recirculated air

Sensible heat: $Q_s = \dot{m}_a c_{pa} \Delta T \approx 1.022\ \dot{m}_a \Delta T$; latent heat: $Q_L = \dot{m}_a \Delta\omega\ h_{fg} \approx 2500\ \dot{m}_a \Delta\omega$

Total heat = $\dot{m}_a (h_2 - h_1) = Q_s + Q_L$

Sensible heat factor SHF = $Q_s/(Q_s + Q_L)$ (slope of the condition line)

Bypass factor BPF = $(t_2 - t_{ADP})/(t_1 - t_{ADP})$; contact (coil) efficiency = 1 − BPF

Mixing: $\dot{m}_1\omega_1 + \dot{m}_2\omega_2 = (\dot{m}_1 + \dot{m}_2)\omega_3$; $\dot{m}_1 h_1 + \dot{m}_2 h_2 = (\dot{m}_1 + \dot{m}_2) h_3$; $t_3 \approx$ weighted average

Humidifying efficiency (air washer) = $(t_1 - t_2)/(t_1 - t_{wb})$

- Apparatus dew point (ADP): effective surface temperature of the cooling coil; leaving air would reach ADP if the coil were 100% effective. Bypass factor accounts for air that does not touch the coil; more rows and lower air velocity reduce BPF.

★ **PYQ-TREND:** Sensible heating is horizontal (very frequent); evaporative cooling follows constant WBT; chemical dehumidification along constant enthalpy with DBT rising; cooling and dehumidification for summer AC; SHF and BPF definitions; adiabatic mixing on a straight line; ADP.

23.3 Human Comfort and Air Conditioning Systems

- Factors of comfort: DBT, RH, air motion, mean radiant temperature, purity of air, activity and clothing. Typical comfort conditions: summer 24–26 °C DBT with 45–55% RH; winter 20–22 °C with 40–50% RH; air velocity 6–9 m/min (0.1–0.15 m/s) in the occupied zone.
- Effective temperature (ET): index combining DBT, humidity and air movement – temperature of saturated still air that gives the same feeling of comfort. Comfort chart (ASHRAE): optimum ET ≈ 20.6 °C (winter), 21.7 °C (summer) in older charts.
- Fresh air (ventilation) requirement ≈ 0.3–0.5 m³/min per person; more in smoking or crowded areas.
- Metabolic heat of an adult seated at rest ≈ 100–120 W total (sensible + latent).

System	Description
Window AC	Self-contained unit; all components in one box through the wall
Split AC	Outdoor unit (compressor + condenser), indoor unit (evaporator + blower); quieter; inverter types vary compressor speed
Packaged unit	Factory-assembled larger unit (5–20 TR) with ducting
Central AC	Chillers + air handling units (AHU) + ducts for large buildings
All-air system	Conditioned air only (single duct CAV/VAV, dual duct)
All-water system	Fan-coil units with chilled water
Air–water system	Induction units, fan coils with primary air
Summer AC	Cooling and dehumidification (+ reheat sometimes)
Winter AC	Heating and humidification
Year-round AC	Both, with controls

Cooling load estimation

- Sensible loads: heat gain through walls, roof, floor ($U\ A\ \Delta T_e$ – equivalent temperature difference considers solar radiation and time lag), solar radiation through glass (largest in many buildings), occupants (sensible part), lights (incandescent fully sensible; fluorescent ×1.25 for ballast), equipment and appliances, infiltration and ventilation air (sensible), duct heat gain, fan work.
- Latent loads: occupants (perspiration, respiration), infiltration/ventilation air moisture, cooking, steam from processes.
- Room sensible heat factor (RSHF), grand SHF (GSHF), effective SHF (ESHF).

★ **PYQ-TREND:** Comfort conditions (24 °C, 50% RH) and effective temperature; sources of sensible and latent load; split AC components; winter AC = heating + humidification; central AC with chillers and AHUs.

✎ **DESCRIPTIVE Q:** Explain the summer air-conditioning system for hot and humid weather with a psychrometric sketch.

In hot and humid weather, outdoor air has high DBT and high humidity. Fresh air is mixed with recirculated room air (state on the straight line between them), filtered, and passed over a cooling coil whose surface temperature (apparatus dew point) is below the dew point of the air. The air is cooled sensibly and, once below its dew point, moisture condenses on the coil, so the air is cooled and dehumidified – on the psychrometric chart it moves down and to the left towards the ADP; the bypass factor determines how close it gets. If the leaving air is too cold or the RH too low for the required sensible heat factor, it may be reheated slightly (horizontal process). The fan then supplies the conditioned air to the room, where it absorbs the room sensible and latent loads and returns to the room condition (≈ 25 °C DBT, 50% RH) along the room SHF line. Condensate is drained, and part of the return air is exhausted to make room for fresh ventilation air.

CHAPTER 23B

Refrigeration and AC – Applications and Equipment

23B.1 Food Preservation and Cold Storage

- Refrigeration slows bacterial growth, enzyme activity and respiration of fresh produce. Chilling: cooling to just above freezing (0–10 °C) – fruits, vegetables, milk, eggs. Freezing: storage at −18 °C or below – meat, fish, frozen foods; ice cream ≈ −25 °C.
- Quick (fast) freezing forms small ice crystals that do not rupture cells, preserving texture and taste; slow freezing forms large crystals and causes 'drip loss' on thawing. Methods: air-blast freezing, plate (contact) freezing, immersion in brine, individually quick frozen (IQF) fluidised-bed freezing, cryogenic freezing with liquid N_2 (−196 °C) or CO_2.
- Chilling injury: some tropical produce (bananas, mangoes) is damaged below about 10–13 °C, so they are stored above that.
- Controlled-atmosphere (CA) storage: reduced O_2 and raised CO_2 with low temperature slow respiration – used for apples.

Load in cold storage	Source
Transmission load	Heat through walls, roof, floor (U A ΔT) – reduced by thick PUF/EPS insulation with vapour barrier
Product load	Sensible heat above freezing + latent heat of freezing + sensible heat below freezing; respiration heat of fruits and vegetables
Air change (infiltration) load	Warm moist air entering through doors – air curtains, strip curtains reduce it
Miscellaneous	Occupants, lights, fork-lifts, fan motors, defrost heaters

Product load (freezing): $Q = m[c_1(T_1 - T_f) + L + c_2(T_f - T_2)]$ (c_1 above freezing, c_2 below freezing, L latent heat of product)

★ **PYQ-TREND:** Quick freezing preserves quality (small crystals); frozen food at −18 °C; cold storage load components; chilling vs freezing; cryogenic freezing with liquid nitrogen.

23B.2 Ice Plant, Water Coolers and Domestic Appliances

- Can-ice plant: galvanised cans filled with water are immersed in a tank of brine (NaCl or $CaCl_2$ solution) cooled to about −10 to −12 °C by an ammonia evaporator coil; an agitator circulates the brine; after freezing (≈ 24–48 h depending on can size) the cans are dipped in warm water to loosen the blocks. Air is blown into the cans to make clear ice. Capacity is expressed in tonnes of ice per day.
- Water coolers: storage type (large tank, intermittent demand – offices) and instantaneous/pressure type (continuous flow); capacity in litres per hour of water cooled, typically from about 30 °C to 10 °C. Bottled water dispensers use small hermetic compressors or thermoelectric modules.

- Domestic refrigerators: direct-cool (natural convection, manual defrost) vs frost-free (forced-air circulation with automatic defrost heater and timer); door gaskets with magnets; refrigerant R-134a or R-600a; star rating by BEE.
- Automobile AC: compressor driven by engine belt via an electromagnetic clutch (swash-plate or variable-displacement types; electric compressors in EVs), condenser in front of the radiator, TXV or orifice tube with accumulator, evaporator in the dashboard; refrigerant R-134a or R-1234yf.
- Deep freezers (−18 to −30 °C), visi-coolers (display), bottle coolers, ice-cream cabinets, cold chains (reefers – refrigerated trucks and containers).

23B.3 Air-Conditioning Equipment

Equipment	Facts
Window AC	Single unit; capacity 0.75–2 TR; rotary/reciprocating hermetic compressor; fan with double-shaft motor drives evaporator blower and condenser fan
Split AC	Indoor (evaporator + blower) and outdoor (compressor + condenser) units joined by refrigerant piping; quieter indoors; inverter models use variable-speed BLDC compressors for better part-load efficiency
Packaged unit	5–20 TR; factory-assembled; ducted air distribution
VRF/VRV system	One outdoor unit serves many indoor units; refrigerant flow varied by inverter compressor and electronic expansion valves; zone control, heat recovery versions
Chillers	Produce chilled water (≈ 7 °C supply, 12 °C return) for AHUs/FCUs; reciprocating, scroll, screw, centrifugal (large) or absorption (waste heat)
Air handling unit (AHU)	Filter, cooling coil, heating coil, humidifier, fan, mixing box; supplies conditioned air through ducts
Fan coil unit (FCU)	Small unit in each room with chilled water coil and fan
Cooling tower	Rejects condenser heat from water-cooled chillers
Thermal (ice) storage	Ice or chilled water made at night (cheaper power) and used during peak hours

```
Energy efficiency ratio EER = cooling capacity (W)/power input (W) (= COP for cooling)
Chiller performance often quoted in kW/TR = 3.517/COP (efficient water-cooled centrifugal ≈ 0.5–0.65 kW/TR)
```

- In India, room ACs carry BEE star labels based on ISEER (Indian Seasonal Energy Efficiency Ratio), which weights performance over the range of outdoor temperatures in a year. Inverter ACs score higher because they run at reduced speed most of the time instead of cycling on and off.

★ **PYQ-TREND:** Split AC components (very frequent); inverter AC variable-speed compressor; chillers and AHUs in central AC; EER/COP definitions; VRF systems; thermal storage for peak shaving.

23B.4 Air Distribution – Ducts, Fans, Filters

Duct design methods

Method	Description
Velocity reduction	Velocities chosen arbitrarily, decreasing along the duct – simple systems
Equal friction	Same friction loss per metre in all sections (≈ 0.8–1 Pa/m) – most common for low-velocity supply, return and exhaust ducts
Static regain	Velocity reduced after each branch so the regained static pressure offsets friction losses – large, high-velocity systems; gives balanced static pressure at outlets

```
Equivalent diameter of rectangular duct (same friction and flow): D_eq = 1.3(ab)^0.625/(a + b)^0.25; for same velocity and friction: D_h = 2ab/(a + b)
```

- Aspect ratio of rectangular ducts kept below about 4:1 (ideally close to 1) to reduce material and friction. Typical main duct velocity: 5–8 m/s in comfort systems (low noise).
- Air outlets: grilles (fixed or adjustable vanes), registers (grilles with dampers), diffusers (ceiling – circular, square, slot, linear). Terms: throw (distance the air jet travels before velocity falls to ≈ 0.25 m/s), drop (vertical fall of the jet), spread (divergence), aspiration/induction (room air entrained by the jet).

Fans

Fan type	Features / uses
Forward-curved centrifugal	Many shallow blades; high flow at low pressure; compact, quiet; overloading power characteristic – AHUs, furnaces, window ACs
Backward-curved centrifugal	Highest efficiency, non-overloading power curve – large HVAC and industrial systems
Radial-blade centrifugal	Self-cleaning, rugged – dusty material handling
Axial (propeller, tube-axial, vane-axial)	High flow at low pressure – exhaust, condenser fans, cooling towers

```
Fan laws (same fan, density constant): Q ∝ N; Δp ∝ N²; P ∝ N³
Geometrically similar fans: Q ∝ N D³; Δp ∝ N² D²; P ∝ N³ D⁵
```

Air filters

Filter	Use
Viscous impingement (oil-coated)	Coarse dust, cheap, washable
Dry (fabric, fibreglass, bag)	General HVAC; MERV rated
HEPA	≥ 99.97% at 0.3 µm – hospitals, clean rooms, pharmaceuticals
Electrostatic precipitator	Charged particles collected on plates – fine particles, smoke
Activated carbon	Odours and gases (adsorption)

★ **PYQ-TREND:** Equal friction method most common (very frequent); static regain for high-velocity systems; fan laws P ∝ N³; HEPA filter 0.3 µm; throw definition; forward-curved fans overloading.

23B.5 Low-Temperature Applications and Cryogenics

Fluid	Normal boiling point
Helium	−269 °C (4.2 K) – lowest
Hydrogen	−253 °C (20 K)
Nitrogen	−196 °C (77 K)
Argon	−186 °C
Oxygen	−183 °C (90 K)
Methane (LNG)	−162 °C
CO_2 (sublimation)	−78.5 °C

- Air liquefaction: Linde–Hampson process (compression, regenerative heat exchanger, Joule–Thomson throttling); Claude process adds an expansion engine/turbine for higher yield. Liquid air is separated into O_2, N_2 and Ar by fractional distillation (double column) because of their different boiling points.
- Gases with low inversion temperature ($H_2 \approx 200$ K, He ≈ 40 K) must be precooled below the inversion temperature before throttling can cool them.
- Uses: liquid N_2 for food freezing, cryosurgery, preservation of biological samples; liquid O_2 in rockets, steel making and hospitals; LNG transport; superconducting magnets (MRI) cooled by liquid helium; cryopumps; Dewar flasks with vacuum and silvered walls (multilayer insulation).

★ **PYQ-TREND:** Liquid nitrogen −196 °C, helium lowest boiling point; Linde vs Claude cycle; J–T precooling of hydrogen and helium; Dewar flask.

✎ **DESCRIPTIVE Q:** Explain the equal-friction method of duct design and state its merits and demerits.

In the equal-friction method, the duct system is designed so that the pressure loss due to friction per unit length is the same in all sections. The designer first selects a velocity for the main duct leaving the fan – based on noise limits and the type of building – and, from the required air flow, determines the main duct size and its friction loss per metre from a friction chart. This friction rate is then used for every branch and sub-branch: knowing the air flow in each section, its diameter is read from the chart at the same friction rate, and rectangular sizes are chosen with an equivalent diameter. Velocities automatically decrease as the flow decreases along the duct. The total pressure required from the fan is the friction loss of the longest (index) run plus losses in fittings. Merits: simple, quick and widely used for supply, return and exhaust ducts in low-velocity systems; it gives reasonable duct sizes and automatic velocity reduction. Demerits: branches close to the fan have excess pressure and need dampers for balancing; it does not equalise static pressure at outlets, which the static-regain method does for large, high-velocity systems.

CHAPTER 24

Heat Transfer I – Conduction, Fins, Transient Conduction

24.1 Modes and Basic Laws

Mode	Mechanism	Law
Conduction	Molecular interaction / free electrons within a medium, no bulk motion	Fourier: $q = -k\,A\,dT/dx$
Convection	Conduction + bulk fluid motion at a surface	Newton's law of cooling: $q = h\,A\,(T_s - T_\infty)$
Radiation	Electromagnetic waves; no medium needed	Stefan–Boltzmann: $q = \varepsilon\sigma A(T_s^4 - T_{sur}^4)$

- Thermal conductivity k (W/m·K): diamond ≈ 2,000+ > silver 429 > copper 386–401 > gold 317 > aluminium 205–237 > brass ≈ 110 > iron/carbon steel 45–80 > stainless steel ≈ 15 > concrete ≈ 1 > glass ≈ 0.8–1 > water 0.6 > brick ≈ 0.7 > wood ≈ 0.15 > glass wool ≈ 0.04 > air 0.026.
- Order: pure metals > alloys > non-metallic solids > liquids > insulators > gases. k of pure metals generally decreases with temperature (impurities/alloys may increase); k of gases increases with temperature ($\propto \sqrt{T}$); k of most liquids decreases (water increases up to ≈ 150 °C).

Thermal diffusivity $\alpha = k/(\rho\, c_p)$ (m^2/s) – how fast temperature changes propagate in transient conduction

Heat conduction equation (Cartesian): $\partial^2T/\partial x^2 + \partial^2T/\partial y^2 + \partial^2T/\partial z^2 + q_g/k = (1/\alpha)\,\partial T/\partial t$

Steady 1-D, no generation: $d^2T/dx^2 = 0 \rightarrow$ linear temperature profile (Laplace equation)

Poisson equation (steady with generation): $\nabla^2T + q_g/k = 0$; Fourier equation (transient, no generation): $\nabla^2T = (1/\alpha)\partial T/\partial t$

★ **PYQ-TREND:** Fourier's law and its sign; order of thermal conductivity (silver highest metal; diamond overall); k of gases increases with temperature; thermal diffusivity definition (very frequent); forms of heat conduction equation.

24.2 Steady Conduction – Walls, Cylinders, Spheres

Plane wall: $Q = kA(T_1 - T_2)/L$; thermal resistance $R = L/(kA)$

Convection resistance $R = 1/(hA)$; radiation resistance $R = 1/(h_r A)$, $h_r = \varepsilon\sigma(T_1^2 + T_2^2)(T_1 + T_2)$

Composite wall (series): $Q = (T_{\infty 1} - T_{\infty 2})/(1/(h_1A) + \Sigma L_i/(k_iA) + 1/(h_2A))$

Parallel layers: $1/R_{eq} = \Sigma 1/R_i$

Overall heat transfer coefficient (plane): $1/U = 1/h_1 + \Sigma L/k + 1/h_2$; $Q = U\,A\,\Delta T$

Hollow cylinder: $Q = 2\pi kL(T_1 - T_2)/\ln(r_2/r_1)$; $R = \ln(r_2/r_1)/(2\pi kL)$

Logarithmic mean area (cylinder): $A_m = (A_2 - A_1)/\ln(A_2/A_1) \rightarrow Q = kA_m\,\Delta T/(r_2 - r_1)$

Hollow sphere: $Q = 4\pi k\, r_1 r_2 (T_1 - T_2)/(r_2 - r_1)$; $R = (r_2 - r_1)/(4\pi k\, r_1 r_2)$; geometric mean area $A_m = \sqrt{(A_1 A_2)}$

Variable conductivity $k = k_0(1 + \beta T)$: use $k_m = k_0(1 + \beta\, T_{mean})$ in the constant-k formula (plane wall)

- Temperature profile: linear in a plane wall (constant k), logarithmic in a cylinder, hyperbolic in a sphere. With k increasing with temperature, the profile in a plane wall is convex upward.
- Contact resistance: imperfect contact at interfaces adds $R_c = 1/(h_c A)$; reduced by pressure, smooth surfaces and thermal grease.

★ **PYQ-TREND:** Electrical analogy of heat flow (series/parallel resistances) – frequent; overall U; cylinder formula with $\ln(r_2/r_1)$; log-mean area for cylinders, geometric mean for spheres; temperature distribution shapes.

✔ **TOPPER TRICK:** Think of temperature difference as voltage, heat flow as current and L/(kA) as resistance; all Ohm's-law tricks then work.

24.3 Critical Radius of Insulation

Cylinder: $r_c = k_{ins}/h_o$; sphere: $r_c = 2k_{ins}/h_o$

- Adding insulation to a small pipe or wire increases the outer surface area. Below r_c, heat loss increases with added insulation (convection resistance falls faster than conduction resistance rises); heat loss is maximum at $r = r_c$; beyond it, insulation reduces heat loss.
- Applications: electric cables – keep insulation radius near r_c so that heat is dissipated (prevents overheating); steam pipes – outer radius must be well above r_c for insulation to be useful. For most pipes r_c is a few mm, smaller than the pipe radius, so insulation always helps.

★ **PYQ-TREND:** $r_c = k/h$ (cylinder) and $2k/h$ (sphere) – very frequent; heat loss maximum at critical radius; electric cables use this.

✖ **COMMON MISTAKE:** Insulating a small-diameter wire can increase heat loss. Insulation helps only when the outer radius is above r_c.

24.4 Conduction with Heat Generation

Plane wall (thickness 2L, both surfaces at T_s): $T_{max} = T_s + q_g L^2/(2k)$ at the centre

Solid cylinder (radius R): $T_{max} = T_s + q_g R^2/(4k)$; $T(r) = T_s + q_g(R^2 - r^2)/(4k)$

Solid sphere: $T_{max} = T_s + q_g R^2/(6k)$

Surface temperature from convection: $T_s = T_\infty + q_g V/(hA)$ (cylinder: $T_\infty + q_g R/(2h)$)

- Examples: electric wires and heating elements (I^2R), nuclear fuel rods, curing concrete, exothermic chemical reactions.

★ **PYQ-TREND:** Maximum temperature formulas for plane wall, cylinder, sphere with generation; location of maximum temperature at the centre.

24.5 Extended Surfaces (Fins)

- Fins increase the surface area to enhance heat transfer from a surface, most useful where h is low (gas side). Examples: motorcycle cylinder fins, car radiators, transformer fins, electronic heat sinks, air-cooled condensers.

Fin parameter $m = \sqrt{(hP/(kA_c))}$ (P = perimeter, A_c = cross-section area)

Infinitely long fin: $Q = \sqrt{(hPkA_c)}\ \theta_0$; $\theta/\theta_0 = e^{(-mx)}$

Fin with insulated tip: $Q = \sqrt{(hPkA_c)}\ \theta_0 \tanh(mL)$; $\theta/\theta_0 = \cosh m(L - x)/\cosh mL$

Fin with convective tip: $Q = \sqrt{(hPkA_c)}\ \theta_0$ [sinh mL + (h/mk) cosh mL]/[cosh mL + (h/mk) sinh mL]; approximate with corrected length $L_c = L + t/2$ (rectangular), $L + D/4$ (pin)

Fin efficiency η_f = actual heat transfer / heat transfer if the whole fin were at base temperature; insulated tip: $\eta_f = \tanh(mL)/(mL)$; long fin $\eta \approx 1/(mL)$

Fin effectiveness ε_f = heat transfer with fin / without fin = $Q_{fin}/(h\ A_c\ \theta_0) = \sqrt{(kP/(hA_c))}$ (long fin)

- Fins are justified only if effectiveness > 2. Effectiveness is high when k is high (copper, aluminium), h is low (gases, natural convection), and the fin is thin (high P/A_c). Fins should be placed on the side with the lower heat-transfer coefficient.
- $\tanh(mL) \approx 1$ when mL > 2.5–3: making a fin longer than this adds almost nothing.

★ **PYQ-TREND:** Fins used where h is low and placed on that side (very frequent); effectiveness formula and > 2 condition; fin efficiency definition; long-fin heat transfer; corrected length; thin closely spaced fins of high-k material.

24.6 Transient (Unsteady) Conduction

Biot number $Bi = h\ L_c / k_{solid}$ (ratio of internal conduction resistance to surface convection resistance)

Characteristic length $L_c = V/A_s$: plane wall (thickness 2L) = L; long cylinder = R/2; sphere = R/3; cube of side a = a/6

Lumped-capacitance analysis (valid if Bi < 0.1): $(T - T_\infty)/(T_i - T_\infty) = \exp(-hA_s t/(\rho Vc)) = \exp(-Bi \cdot Fo)$

Fourier number $Fo = \alpha\ t/L_c^2$ (dimensionless time)

Time constant $\tau = \rho Vc/(hA_s)$ — time to reach 63.2% of the total temperature change

Instantaneous heat transfer rate $q(t) = hA_s(T_i - T_\infty)\ e^{(-t/\tau)}$; total $Q = \rho Vc(T_i - T_\infty)(1 - e^{(-t/\tau)})$

- For Bi > 0.1, internal temperature gradients are significant – use Heisler charts (one-term series solutions) or numerical methods. Semi-infinite solids use the error-function solution.
- Thermocouple response: a small bead with high h has a small time constant (fast response).

★ **PYQ-TREND:** Lumped analysis valid for Bi < 0.1 (very frequent); Biot vs Nusselt (k of solid vs fluid); characteristic lengths (sphere R/3); time constant and 63.2%; Heisler charts; Fourier number.

✎ **DESCRIPTIVE Q:** What is the critical radius of insulation? Derive its expression for a cylinder.

Consider a pipe of outer radius r_1 at temperature T_1 covered by insulation of conductivity k to an outer radius r, losing heat to surroundings at T_∞ with coefficient h. Heat loss $Q = 2\pi L(T_1 - T_\infty)/[\ln(r/r_1)/k + 1/(hr)]$. As r increases, the conduction resistance $\ln(r/r_1)/k$ increases while the convection resistance 1/(hr) decreases because the outer area grows. Q is maximum when the total resistance is minimum: $d/dr[\ln(r/r_1)/k + 1/(hr)] = 1/(kr) - 1/(hr^2) = 0$, giving $r = k/h$. The second derivative is positive, so the resistance is a minimum and the heat loss a maximum at $r_c = k/h$. If the pipe's radius is below r_c, adding insulation up to r_c increases heat loss; beyond r_c insulation reduces it. For a sphere the same procedure gives $r_c = 2k/h$. The concept is used to choose insulation for electric cables (where heat dissipation is desirable) and to check that insulation on small pipes is effective.

CHAPTER 25

Heat Transfer II – Convection, Boiling and Condensation

25.1 Convection Fundamentals

- Forced convection: fluid motion created by external means (fan, pump, wind). Free (natural) convection: motion caused by buoyancy from density differences due to temperature. Mixed convection: both comparable.
- Thermal boundary layer: region near the surface where temperature changes from T_s to T_∞; velocity boundary layer – velocity changes. Their relative thickness depends on Prandtl number (δ_t/δ ≈ Pr^(−1/3) for laminar flow over a plate).
- At the wall (no-slip), heat transfer is by conduction through the stagnant fluid layer: h = −k_f (∂T/∂y)_wall/(T_s − T_∞).

Situation	Typical h (W/m²·K)
Free convection – gases	2–25
Free convection – liquids	10–1,000
Forced convection – gases	25–250
Forced convection – liquids	50–20,000
Boiling and condensation	2,500–100,000

★ **PYQ-TREND:** Forced vs free convection; boiling/condensation give the highest h; h is lowest for free convection in gases.

25.2 Dimensionless Numbers

Number	Expression	Physical meaning	Used in
Reynolds Re	ρVL/μ = VL/ν	Inertia / viscous forces	Forced convection – laminar/turbulent
Prandtl Pr	μ c_p/k = ν/α	Momentum diffusivity / thermal diffusivity (relative thickness of velocity and thermal BL)	All convection
Nusselt Nu	hL/k_fluid	Convective / conductive heat transfer across the fluid layer (dimensionless h)	All convection
Grashof Gr	gβΔT L³/ν²	Buoyancy / viscous forces	Free convection (role of Re)
Rayleigh Ra	Gr × Pr	Onset/strength of free convection	Free convection
Stanton St	Nu/(Re Pr) = h/(ρVc_p)	Heat transferred / thermal capacity of fluid flow	Forced convection, analogies

Number	Expression	Physical meaning	Used in
Peclet Pe	Re × Pr = VL/α	Advection / diffusion of heat	Liquid metals, forced convection
Biot Bi	hL/k_solid	Internal conduction / external convection resistance	Transient conduction
Fourier Fo	αt/L²	Dimensionless time	Transient conduction
Graetz Gz	Re Pr (D/L)	Entry-length heat transfer	Laminar duct flow
Lewis Le	α/D_AB = Sc/Pr	Heat / mass diffusivity	Simultaneous heat & mass transfer
Schmidt Sc	ν/D_AB	Momentum / mass diffusivity	Mass transfer
Sherwood Sh	h_m L/D_AB	Dimensionless mass-transfer coefficient	Mass transfer
Eckert Ec	V²/(c_p ΔT)	Kinetic energy / enthalpy difference	High-speed flow

- Prandtl number values: liquid metals 0.004–0.03 (thermal BL ≫ velocity BL); gases ≈ 0.7–1.0 (air 0.71); water ≈ 7 at 20 °C (1.75 at 100 °C); light oils 100–1,000; heavy oils up to 10^5.
- Functional relationships: forced convection Nu = f(Re, Pr); free convection Nu = f(Gr, Pr) = f(Ra).

★ **PYQ-TREND:** Meaning of Nu, Pr, Gr, St, Bi (very frequent); Pr of air ≈ 0.7, liquid metals very low; Nu = f(Re, Pr) for forced and f(Gr, Pr) for free convection; Biot uses k of solid, Nusselt uses k of fluid.

✔ **TOPPER TRICK:** Biot vs Nusselt look identical (hL/k) – the difference is whose k: solid → Biot (transient conduction), fluid → Nusselt (convection).

25.3 Forced Convection Correlations

```
Flat plate, laminar (Re_x < 5 × 10⁵): local Nu_x = 0.332 Re_x^0.5 Pr^(1/3); average Nu_L = 0.664 Re_L^0.5 Pr^(1/3)
Flat plate, turbulent: Nu_x = 0.0296 Re_x^0.8 Pr^(1/3); average Nu_L = 0.037 Re_L^0.8 Pr^(1/3)
Laminar fully developed tube flow: Nu = 3.66 (constant wall temperature), Nu = 4.36 (constant heat flux)
Turbulent tube flow (Dittus–Boelter): Nu = 0.023 Re^0.8 Pr^n; n = 0.4 (fluid being heated), 0.3 (cooled)
Sieder–Tate (large viscosity variation): Nu = 0.027 Re^0.8 Pr^(1/3) (μ/μ_s)^0.14
Reynolds analogy (Pr = 1): St = C_f/2; Colburn analogy: St Pr^(2/3) = C_f/2 (j-factor)
Hydraulic diameter for non-circular ducts: D_h = 4A/P
Bulk (mixing-cup) temperature and LMTD for tubes: q = h A ΔT_lm
```

- In fully developed laminar tube flow h is constant (Nu const) – h ∝ 1/D; in turbulent flow h ∝ V^0.8/D^0.2.
- Laminar entry region has higher h than the fully developed region.

★ **PYQ-TREND:** Nu = 3.66 and 4.36 (very frequent); Dittus–Boelter exponents 0.8 and 0.4/0.3; Reynolds/Colburn analogy; flat plate laminar Nu ∝ Re^0.5; hydraulic diameter 4A/P.

25.4 Free (Natural) Convection

```
Nu = C (Gr Pr)^n = C Ra^n; laminar (10⁴ < Ra < 10⁹): n = 1/4; turbulent (Ra > 10⁹): n = 1/3 (h independent of length)
Volume expansion coefficient for ideal gas: β = 1/T (absolute film temperature)
Film temperature T_f = (T_s + T_∞)/2 — properties evaluated at it
```

- Transition from laminar to turbulent free convection on a vertical plate at Ra ≈ 10^9.
- Examples: cooling of transformers, radiators (domestic heaters), electronic components, hot pipes in still air, room heating, atmospheric and oceanic circulation.

★ **PYQ-TREND:** Grashof number governs free convection; β = 1/T for gases; turbulent free convection exponent 1/3; film temperature for properties.

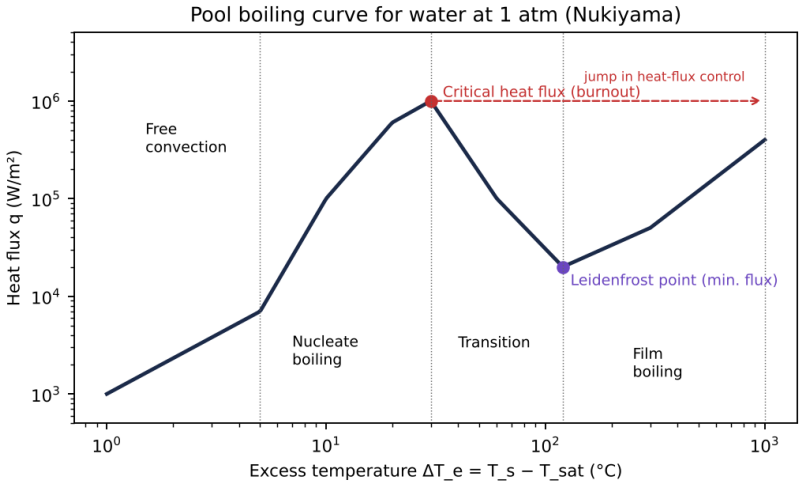
25.5 Boiling

- Boiling occurs when a surface temperature exceeds the saturation temperature of the liquid (excess temperature $\Delta T_e = T_s - T_{sat}$). Pool boiling (stagnant liquid) vs flow (forced-convection) boiling; subcooled vs saturated boiling.

Pool boiling curve (Nukiyama, water at 1 atm)

Regime	ΔT_e (approx.)	Features
Free-convection boiling	< 5 °C	Liquid superheated, evaporates at free surface
Nucleate boiling	5–30 °C	Bubbles form at nucleation sites; isolated bubbles then jets/columns; very high h; ends at the critical heat flux (CHF, peak heat flux, burnout point ≈ 1 MW/m² for water at 1 atm)
Transition (unstable film) boiling	30–120 °C	Partial vapour film, unstable; heat flux falls as ΔT rises
Film boiling	> 120 °C	Stable vapour blanket; heat transfer by conduction and radiation through vapour; minimum heat flux at the Leidenfrost point

- Burnout: in heat-flux-controlled systems (electric heaters, nuclear fuel), exceeding CHF causes a jump to film boiling with a very large temperature rise that can melt the surface. Systems are designed to operate safely below CHF in nucleate boiling.
- Leidenfrost effect: water drops dance on a very hot pan because a vapour film insulates them.
- Factors increasing boiling heat transfer: rough surfaces (more nucleation sites), higher pressure (for nucleate boiling up to a point), agitation.



★ **PYQ-TREND:** Nucleate boiling gives maximum h (very frequent); critical heat flux / burnout point; Leidenfrost point marks film boiling minimum; regimes order; film boiling insulates by vapour.

25.6 Condensation

- Filmwise condensation: condensate wets the surface and forms a continuous film that adds thermal resistance (Nusselt's analysis). Dropwise condensation: condensate forms droplets (non-wetting surfaces, promoters like oleic acid, Teflon, gold coatings); much bare surface exposed – h is 5–10 times higher than filmwise. Because dropwise condensation is hard to sustain, condensers are designed for filmwise condensation.

```
Nusselt: h_film (vertical plate) = 0.943 [ρ_l(ρ_l − ρ_v)g h_fg k_l³/(μ_l L ΔT)]^(1/4);
horizontal tube uses 0.725 with D in place of L
```

- Horizontal tubes give better condensation than vertical tubes of the same size (thinner film); in a bank of tubes, inundation from upper tubes reduces h of lower tubes.
- Presence of non-condensable gases (air) drastically reduces condensation heat transfer – hence air extraction in steam condensers.

★ **PYQ-TREND:** Dropwise condensation gives higher h but filmwise is used for design (very frequent); effect of non-condensable gases; horizontal tubes better than vertical.

✎ **DESCRIPTIVE Q:** Explain the physical significance of Nusselt, Prandtl and Grashof numbers.

The Nusselt number Nu = hL/k_f is the ratio of convective heat transfer to the heat that would be conducted through a stationary fluid layer of thickness L under the same temperature difference; it is the dimensionless heat-transfer coefficient (Nu = 1 means pure conduction). The Prandtl number Pr = ν/α = μc_p/k is a fluid property comparing the diffusion of momentum to the diffusion of heat; it fixes the relative thickness of the velocity and thermal boundary layers (Pr ≈ 1 for gases – similar thicknesses; Pr ≪ 1 for liquid metals – thick thermal layer; Pr ≫ 1 for oils – thin thermal layer). The Grashof number Gr = gβΔTL³/ν² is the ratio of buoyancy forces to viscous forces in natural convection; it plays the role that the Reynolds number plays in forced convection, determining whether free-convection flow is laminar or turbulent (with Ra = Gr Pr ≈ 10^9 as the transition value on vertical plates).

CHAPTER 26

Heat Transfer III – Radiation and Heat Exchangers

26.1 Radiation Fundamentals

- Thermal radiation: electromagnetic waves emitted by all bodies above 0 K because of their temperature; wavelengths ≈ 0.1–100 μm (part of UV, visible and infrared). Travels at the speed of light, needs no medium – fastest mode; it is most significant at high temperatures and in vacuum (space).

Absorptivity α + reflectivity ρ + transmissivity τ = 1; opaque body τ = 0 → α + ρ = 1

Body	Property
Black body	α = 1 (absorbs all radiation); ideal emitter (ε = 1); perfectly diffuse
White body	ρ = 1 (reflects all)
Transparent (diathermanous) body	τ = 1
Grey body	ε (= α) < 1 and constant over all wavelengths
Opaque body	τ = 0
Specular reflection	Angle of reflection = angle of incidence (polished surfaces)
Diffuse reflection	Reflected equally in all directions (rough surfaces)

- Real surfaces: polished metals have low emissivity (0.02–0.1), oxidised metals higher; non-metals (paint, brick, skin, water) high (0.8–0.95). Snow is nearly black to long-wave (infrared) radiation though white to visible light. Emissivity of most non-metals is high regardless of colour at infrared wavelengths.

Stefan–Boltzmann: $E_b = \sigma T^4$; $\sigma = 5.67 \times 10^{-8}$ W/m²·K⁴; real surface $E = \varepsilon\sigma T^4$

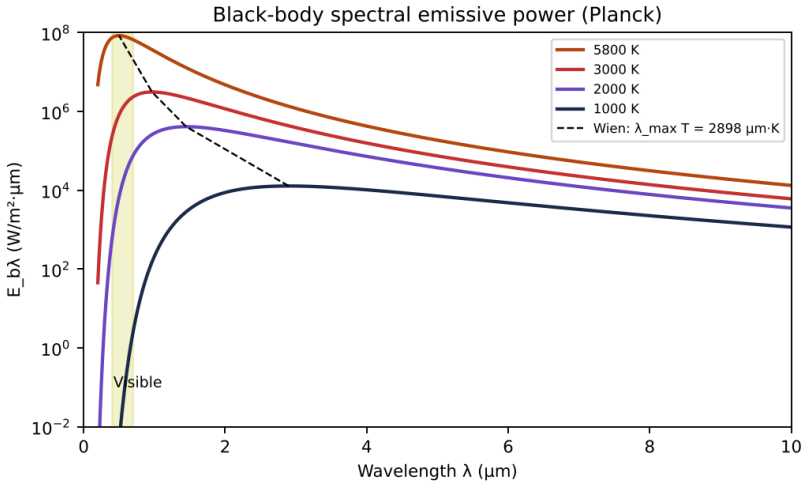
Wien's displacement law: $\lambda_{max} T = 2898$ μm·K (≈ 2900)

Planck's law: $E_{b\lambda} = C_1\lambda^{-5}/(e^{(C_2/\lambda T)} - 1)$; $C_1 = 3.742 \times 10^8$ W·μm⁴/m², $C_2 = 1.4388 \times 10^4$ μm·K

Kirchhoff's law: at thermal equilibrium ε = α (for a grey body at all temperatures)

Lambert's cosine law: intensity of a diffuse emitter I is independent of direction; emissive power E = πI

- Sun (≈ 5,800 K): λ_{max} ≈ 0.5 μm (visible). Bodies at room temperature (≈ 300 K): λ_{max} ≈ 10 μm (infrared). Glass transmits short-wave solar radiation but is opaque to long-wave radiation from objects inside – greenhouse effect.
- Gases: monatomic and symmetric diatomic gases (N_2, O_2, H_2, dry air) are transparent to thermal radiation; CO_2, H_2O vapour, CO, SO_2, NH_3, CH_4 absorb and emit in bands (important in furnaces and the atmosphere).



★ **PYQ-TREND:** Black body α = ε = 1 (very frequent); Kirchhoff's law α = ε; Wien's law constant 2898 μm·K; Stefan–Boltzmann constant; grey body definition; greenhouse effect; N_2/O_2 transparent, CO_2/H_2O participating; α + ρ + τ = 1.

26.2 Radiation Exchange Between Surfaces

Shape (view/configuration) factor F_{12}: fraction of radiation leaving surface 1 that reaches surface 2 directly

Reciprocity: $A_1F_{12} = A_2F_{21}$; summation: $\Sigma_j F_{1j} = 1$ (enclosure)

Flat or convex surface: $F_{11} = 0$; concave surface: $F_{11} > 0$

Small body (1) completely enclosed by a large enclosure (2): $F_{12} = 1$; $F_{21} = A_1/A_2$

Two large parallel plates: $q = \sigma(T_1^4 - T_2^4)/(1/\varepsilon_1 + 1/\varepsilon_2 - 1)$ per unit area

Concentric cylinders/spheres: $Q = \sigma A_1(T_1^4 - T_2^4)/[1/\varepsilon_1 + (A_1/A_2)(1/\varepsilon_2 - 1)]$

Small body in large enclosure: $Q = \varepsilon_1\sigma A_1(T_1^4 - T_2^4)$

Black surfaces: $Q_{12} = A_1F_{12}\sigma(T_1^4 - T_2^4)$

Surface resistance $(1 - \varepsilon)/(\varepsilon A)$; space (geometric) resistance $1/(A_1F_{12})$ – radiation network

Hemisphere over its base: F(base → hemisphere) = 1; F(hemisphere → base) = 0.5; F(hemisphere → itself) = 0.5

Radiation shields

n shields between two parallel plates (all emissivities equal): heat transfer reduced to 1/(n + 1) of the unshielded value

One shield with ε_s between plates with ε_1, ε_2: resistance = $(1/\varepsilon_1 + 1/\varepsilon_2 - 1) + (2/\varepsilon_s - 1)$ per unit area

- Shields of low-emissivity (polished) material are used in thermos flasks, cryogenic storage (multi-layer insulation), thermocouple shields (to reduce radiation error) and furnace walls.

★ **PYQ-TREND:** Shape factor reciprocity and summation rules (very frequent); $F_{11} = 0$ for flat/convex surfaces; small body in large enclosure; parallel plates formula; n shields → 1/(n+1); hemisphere view factors.

✔ **TOPPER TRICK:** Radiation shields: 1 shield halves the heat flow, 2 shields reduce it to one-third, 3 to one-fourth (equal emissivities).

26.3 Heat Exchangers

Classification	Types
Nature of contact	Direct contact (cooling towers, jet condensers), indirect – recuperators (separated by wall: shell-and-tube, plate), regenerators (matrix alternately heated and cooled: rotary air preheater, Cowper stoves)
Flow arrangement	Parallel flow, counter flow, cross flow (mixed/unmixed), multi-pass
Construction	Double pipe, shell-and-tube, plate, plate-fin, finned-tube, spiral
Phase	Condensers, evaporators, boilers, radiators
Compactness	Compact HX: area density > 700 m^2/m^3 (car radiator ≈ 1,000; human lung ≈ 20,000)

Heat transfer: $Q = U A \Delta T_{lm}$

$LMTD = (\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$

Parallel flow: $\Delta T_1 = T_{h,in} - T_{c,in}$, $\Delta T_2 = T_{h,out} - T_{c,out}$; counter flow: $\Delta T_1 = T_{h,in} - T_{c,out}$, $\Delta T_2 = T_{h,out} - T_{c,in}$

When $\Delta T_1 = \Delta T_2$ (counter flow, equal capacity rates): $LMTD = \Delta T$ (constant along length)

Cross-flow and multipass: $Q = U A F \Delta T_{lm,counter}$ (F = correction factor ≤ 1)

Overall U (tube, based on outer area): $1/U_o = 1/h_o + R_{fo} + (r_o \ln(r_o/r_i))/k + (r_o/r_i) R_{fi} + r_o/(r_i h_i)$

Fouling factor (scale resistance) $R_f = 1/U_{dirty} - 1/U_{clean}$

Effectiveness–NTU method

Capacity rate $C = \dot{m} c_p$; C_{min}, C_{max}; capacity ratio $C_r = C_{min}/C_{max}$

Effectiveness $\varepsilon = Q_{actual}/Q_{max}$; $Q_{max} = C_{min} (T_{h,in} - T_{c,in})$

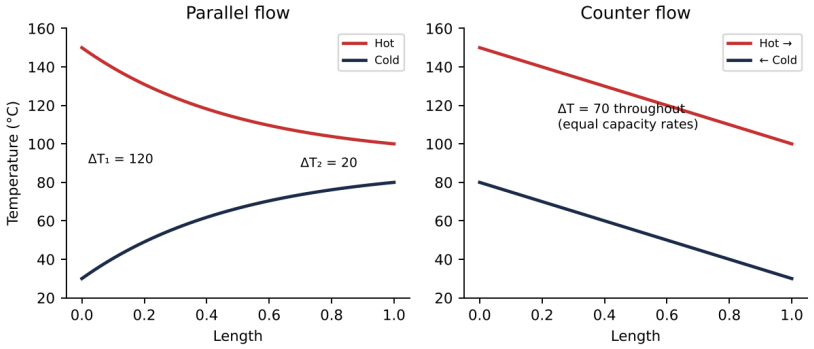
Number of transfer units $NTU = U A / C_{min}$

Parallel flow: $\varepsilon = [1 - \exp(-NTU(1 + C_r))]/(1 + C_r)$

Counter flow: $\varepsilon = [1 - \exp(-NTU(1 - C_r))]/[1 - C_r \exp(-NTU(1 - C_r))]$; $C_r = 1$: $\varepsilon = NTU/(1 + NTU)$

Phase change (condenser/evaporator, $C_r = 0$): $\varepsilon = 1 - \exp(-NTU)$ (same for all arrangements)

- Counter flow gives the larger LMTD and therefore needs less area for the same duty; in counter flow the cold-fluid outlet temperature can exceed the hot-fluid outlet temperature, which is impossible in parallel flow. Maximum effectiveness of parallel flow ($C_r = 1$) is 50%.
- When one fluid changes phase (condenser, evaporator), its temperature is constant and the LMTD is the same for parallel and counter flow.
- LMTD method is convenient when all inlet and outlet temperatures are known (sizing); ε–NTU when outlet temperatures are unknown (rating).



★ **PYQ-TREND:** Counter flow more effective than parallel flow (very frequent); LMTD formula; ε = 1 − e^(−NTU) for condensers/evaporators; NTU definition; fouling factor; car radiator is cross-flow compact HX; regenerators vs recuperators; LMTD = ΔT when $\Delta T_1 = \Delta T_2$.

✖ **COMMON MISTAKE:** In LMTD, ΔT_1 and ΔT_2 are the end temperature differences, defined differently for parallel and counter flow. Using the parallel-flow definitions for a counter-flow exchanger is the most common error.

✎ **DESCRIPTIVE Q:** Compare parallel-flow and counter-flow heat exchangers.

In a parallel-flow exchanger both fluids enter at the same end and flow in the same direction. The temperature difference is large at the inlet and falls steadily along the length, so the LMTD is relatively small; the outlet temperature of the cold fluid always remains below the outlet temperature of the hot fluid, and the maximum effectiveness (for equal capacity rates) is only 50%. In a counter-flow exchanger the fluids enter at opposite ends and flow in opposite directions; the temperature difference is more uniform along the length, the LMTD is higher for the same terminal temperatures, so a smaller area is needed for the same heat duty, and the cold-fluid outlet can be heated above the hot-fluid outlet temperature; effectiveness can approach 100% with large area. Counter flow also gives lower thermal stresses because the temperature difference across the wall is more uniform. Parallel flow is used when a rapid initial heating/cooling is needed or when the wall temperature must be limited. When one fluid changes phase at constant temperature, both arrangements give the same LMTD and effectiveness.

CHAPTER 26B

Fuels and Combustion

26B.1 Classification of Fuels

Type	Natural (primary)	Prepared (secondary)
Solid	Wood, peat, lignite, bituminous coal, anthracite	Coke, charcoal, briquettes, pulverised coal
Liquid	Crude petroleum	Petrol, kerosene, diesel, furnace oil, alcohols, biodiesel
Gaseous	Natural gas	LPG, CNG, producer gas, water gas, coal gas, biogas, blast-furnace gas

- Rank of coal (increasing age and carbon): wood → peat → lignite (brown coal) → sub-bituminous → bituminous → semi-anthracite → anthracite. With rank, fixed carbon and calorific value generally rise; moisture and volatile matter fall. Anthracite has the highest carbon (≈ 90–95%), burns with a short smokeless flame and is hard to ignite; bituminous coal (most common steam coal) has high volatile matter and burns with a long smoky flame.
- Indian coals are mostly bituminous/sub-bituminous with high ash (30–45%), which affects boiler design (large ESPs, erosion of tubes).
- Coking coal: bituminous coal that softens and fuses when heated without air to give strong, porous coke (for blast furnaces). Carbonisation: low temperature (≈ 500–700 °C) gives soft coke with more volatile matter; high temperature (≈ 1000–1200 °C) gives hard metallurgical coke and coal gas.

★ **PYQ-TREND:** Order of coal rank (anthracite highest carbon/CV; peat lowest); coke made by carbonisation; Indian coal high ash.

26B.2 Analysis of Coal

Analysis	Determines	Method
Proximate	Moisture, volatile matter, ash, fixed carbon (by difference)	Moisture: heat 1 g at 105–110 °C for 1 h; volatile matter: covered crucible at ≈ 950 °C for 7 min; ash: open crucible at ≈ 700–750 °C until constant mass; FC = 100 − (M + VM + A)
Ultimate	C, H, N, S, O (and ash)	Chemical: C and H by combustion (CO_2 absorbed in KOH, H_2O in $CaCl_2$); N by Kjeldahl; S by Eschka/bomb washings; O by difference

- Proximate analysis is simple and used commercially (grading, firing behaviour); ultimate analysis is used for combustion calculations (air required, flue-gas quantity, calorific value).
- High moisture lowers CV and flame temperature; high volatile matter gives easy ignition but long smoky flame; high ash lowers CV, increases handling and disposal and causes

clinkering; sulphur causes corrosion and SO_2 pollution.

★ **PYQ-TREND:** Proximate vs ultimate analysis (very frequent); fixed carbon found by difference; oxygen by difference in ultimate analysis.

26B.3 Calorific Value

- Calorific value: heat released by complete combustion of unit mass (solid/liquid, kJ/kg) or unit volume (gas, kJ/m³ at standard conditions) of fuel.
- Higher (gross) calorific value HCV: products cooled to the reference temperature so that the water vapour formed is condensed – includes its latent heat. Lower (net) calorific value LCV: water remains as vapour. Engines, boilers and furnaces actually obtain LCV, because exhaust leaves hot.

```
LCV = HCV − m_w × h_fg; m_w = 9H (+ moisture M) kg per kg fuel; h_fg ≈ 2442 kJ/kg (at 25 °C)
Dulong's formula: HCV ≈ 33,800 C + 144,000 (H − O/8) + 9,270 S kJ/kg (C, H, O, S as mass fractions)
```

- (H − O/8) is the 'available hydrogen': oxygen in the fuel is assumed to be already combined with one-eighth of its mass of hydrogen as water.

Calorimeter	Fuel	Remarks
Bomb calorimeter	Solid and liquid	Constant volume, burns in pure O_2 at ≈ 25 atm; gives HCV; water equivalent and corrections for fuse wire, acid formation, cooling
Junkers gas calorimeter	Gaseous (and volatile liquids)	Steady-flow, constant pressure; gas burnt continuously, heat absorbed by flowing water; gives HCV, LCV from condensate collected
Boys calorimeter	Gaseous	Similar continuous-flow principle

Fuel	HCV (approx.)
Hydrogen	142 MJ/kg (LCV ≈ 120) – highest per kg
Methane / natural gas	55.5 MJ/kg (LCV ≈ 50); ≈ 37–39 MJ/m³
LPG	≈ 50 MJ/kg
Petrol	≈ 47 MJ/kg (LCV ≈ 44)
Diesel	≈ 45.5 MJ/kg (LCV ≈ 42.8)
Carbon (to CO_2)	33.8 MJ/kg (to CO only 10.2 MJ/kg)
Anthracite / bituminous coal	≈ 30–35 / 25–33 MJ/kg
Indian non-coking coal	≈ 13–20 MJ/kg (high ash)
Wood (dry)	≈ 18–20 MJ/kg
Ethanol / methanol	29.7 / 22.7 MJ/kg
Biogas (≈ 60% CH_4)	≈ 20–24 MJ/m³
Producer gas	≈ 4–6 MJ/m³ (lowest)

★ **PYQ-TREND:** HCV vs LCV difference (latent heat of water vapour from hydrogen) – very frequent; bomb calorimeter for solid/liquid, Junkers for gaseous fuels; hydrogen highest CV per kg; Dulong's formula.

✖ COMMON MISTAKE: In LCV correction, water formed is 9 times the mass of hydrogen in the fuel (2 kg H_2 + 16 kg O_2 → 18 kg H_2O), not equal to it.

26B.4 Combustion Equations and Air Requirement

Reaction	Mass balance (per kg of combustible)
$C + O_2 \rightarrow CO_2$	1 kg C + 8/3 kg O_2 → 11/3 kg CO_2
$2C + O_2 \rightarrow 2CO$ (incomplete)	1 kg C + 4/3 kg O_2 → 7/3 kg CO
$2CO + O_2 \rightarrow 2CO_2$	1 kg CO + 4/7 kg O_2 → 11/7 kg CO_2
$2H_2 + O_2 \rightarrow 2H_2O$	1 kg H_2 + 8 kg O_2 → 9 kg H_2O
$S + O_2 \rightarrow SO_2$	1 kg S + 1 kg O_2 → 2 kg SO_2
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	1 kg CH_4 + 4 kg O_2 → 2.75 kg CO_2 + 2.25 kg H_2O

Minimum O_2 per kg of fuel = (8/3)C + 8H + S − O kg

Theoretical (stoichiometric) air = O_2 required/0.232 ≈ 4.31 × O_2 (mass; air has 23.2% O_2 by mass, 21% by volume)

Hydrocarbon C_xH_y: $C_xH_y + (x + y/4)(O_2 + 3.76\ N_2) \rightarrow x\ CO_2 + (y/2)\ H_2O + 3.76(x + y/4)\ N_2$

Equivalence ratio φ = (F/A)_actual/(F/A)_stoich; φ > 1 rich, φ < 1 lean; λ (excess-air ratio) = 1/φ

Excess air % = (actual air − theoretical air)/theoretical air × 100 ≈ O_2%/(21 − O_2%) × 100 (from dry flue gas, complete combustion)

Fuel	Stoichiometric A/F (by mass)
Hydrogen	34.3
Methane	17.2
Petrol (≈ C_8H_{18})	14.7–15.1
Diesel	≈ 14.5
Carbon	11.5
Ethanol / methanol	9.0 / 6.4

Typical excess air	Value
Gas burners	5–10%
Oil burners	10–20%
Pulverised coal	15–25%
Stoker-fired coal	30–50%
Gas turbine combustor (overall)	200–400% (to limit turbine inlet temperature)

- Too little air → incomplete combustion (CO, smoke, unburnt carbon) and wasted fuel; too much air → more heat lost with the large volume of hot flue gas and lower flame temperature. The optimum is the minimum excess air that gives complete combustion.
- Adiabatic flame temperature: temperature of products when combustion occurs without heat loss and without work; maximum near stoichiometric (slightly rich) mixture, ≈ 2000–2300 °C for hydrocarbons in air; falls with excess air, dissociation and preheating absent. Preheating combustion air raises it.

★ PYQ-TREND: 1 kg of carbon needs 8/3 kg O_2 and hydrogen 8 kg O_2 (very frequent); air contains 23% O_2 by mass and 21% by volume; stoichiometric A/F of petrol ≈ 15; excess air effects; adiabatic flame temperature.

26B.5 Flue Gas Analysis

- Orsat apparatus: measures CO_2, O_2 and CO in dry flue gas by volume, by successive absorption: CO_2 in caustic potash (KOH) solution, O_2 in alkaline pyrogallol, CO in ammoniacal cuprous chloride. The order must be $CO_2 \rightarrow O_2 \rightarrow CO$ because pyrogallol also absorbs CO_2 and cuprous chloride also absorbs O_2. N_2 is found by difference. Water vapour is not shown, as it condenses at room temperature – hence 'dry' analysis.
- Modern instruments: NDIR analysers (CO, CO_2), zirconia probe (O_2), electrochemical cells, flame ionisation detector (HC), chemiluminescence (NOx).
- High CO_2 (≈ 12–15% for coal) with low CO indicates good combustion; CO presence means insufficient air or poor mixing; high O_2 means excess air.

Mass of dry flue gas per kg fuel (from volumetric analysis): $m_{dfg} = [11CO_2 + 8O_2 + 7(CO + N_2)]/[3(CO_2 + CO)] \times C$ (C = mass of carbon burnt per kg fuel)

Conversion of volumetric (molar) to mass analysis: multiply each volume fraction by molar mass and divide by the total

★ **PYQ-TREND:** Orsat apparatus absorbents and order (KOH – CO_2, pyrogallol – O_2, cuprous chloride – CO) – very frequent; Orsat gives dry analysis; high CO_2 indicates good combustion.

26B.6 Gaseous and Liquid Fuels

Gaseous fuel	Production	Main constituents
Natural gas	From oil/gas wells	CH_4 85–95%, C_2H_6
CNG	Natural gas compressed to 200–250 bar	CH_4
LPG	By-product of refining / natural gas	Propane + butane (liquefied ≈ 5–8 bar)
Producer gas	Partial combustion of coal/coke with air (and steam)	CO ≈ 25–30%, H_2 ≈ 10–15%, N_2 ≈ 50–55% – low CV
Water gas	Steam over red-hot coke ($C + H_2O \rightarrow CO + H_2$)	CO ≈ 40–45%, H_2 ≈ 50%
Coal gas	Carbonisation of coal	H_2 ≈ 50%, CH_4 ≈ 30%
Biogas (gobar gas)	Anaerobic digestion of biomass/dung	CH_4 55–65%, CO_2 30–40%
Blast-furnace gas	By-product of iron making	CO ≈ 25%, N_2 ≈ 60% – very low CV

Property (liquid fuels)	Meaning
Flash point	Lowest temperature at which vapours ignite momentarily with a flame (safety in storage)
Fire point	Temperature at which vapours burn continuously for at least 5 s (a few degrees above flash point)
Cloud point	Temperature at which wax crystals make the oil cloudy
Pour point	Lowest temperature at which oil still flows (below cloud point)
Viscosity	Affects atomisation and pumping (Redwood, Saybolt viscometers)
Volatility	Ease of vaporisation (distillation curve) – important for SI fuels

Property (liquid fuels)	Meaning
Specific gravity / API gravity	API = 141.5/SG − 131.5

- Flash point order: petrol (≈ −43 °C) < kerosene (≈ 38 °C) < diesel (≈ 52–96 °C) < lubricating oil (≈ 200 °C+). Fire point > flash point; pour point < cloud point.
- Alternative fuels: ethanol (high octane, E20 blending in India), methanol, biodiesel (transesterified vegetable oils – high cetane, more NOx), hydrogen (zero carbon, wide flammability, high flame speed, backfire risk), CNG/LNG (high octane ≈ 120+), DME.

★ **PYQ-TREND:** Producer gas vs water gas composition (frequent); biogas mainly methane; LPG = propane + butane; flash point < fire point; pour point < cloud point; CNG high octane.

✎ **DESCRIPTIVE Q:** Distinguish between proximate and ultimate analysis of coal and explain their uses.

Proximate analysis determines, on a mass basis, the moisture, volatile matter, ash and fixed carbon in coal by simple heating tests: moisture by heating a sample at 105–110 °C, volatile matter by heating in a covered crucible at about 950 °C for seven minutes, ash by burning in an open crucible, and fixed carbon by difference. It indicates how the coal will behave in a furnace – ease of ignition (volatile matter), heat content (fixed carbon), clinkering and disposal problems (ash) – and is used for commercial grading and furnace selection. Ultimate analysis determines the elemental composition – carbon, hydrogen, nitrogen, sulphur and oxygen (by difference) – by chemical methods. It is needed for combustion calculations: theoretical air, flue-gas composition and quantity, and calculation of calorific value by Dulong's formula; it also indicates pollution potential through sulphur and nitrogen content. Proximate analysis is quick and cheap; ultimate analysis requires laboratory equipment but is more fundamental.

CHAPTER 26C

Thermal Measurements and Instruments

26C.1 Temperature Measurement

Instrument	Principle	Range / remarks
Liquid-in-glass (mercury)	Thermal expansion of liquid	−39 to ≈ 357 °C (mercury); alcohol to −110 °C
Bimetallic strip	Unequal expansion of two bonded metals (e.g. invar + brass)	Thermostats, dial thermometers; −30 to 500 °C
Pressure (filled-system) thermometer	Gas/vapour/liquid pressure rises with temperature	Remote dial reading; vapour type has non-linear scale
Thermocouple	Seebeck effect – emf at junction of dissimilar metals depends on temperature difference	−200 to 1800 °C; fast, cheap, small; needs cold-junction compensation
Resistance temperature detector (RTD)	Resistance of metal increases with temperature (positive coefficient)	Platinum Pt-100 (100 Ω at 0 °C, α ≈ 0.00385 /°C); −200 to 850 °C; most accurate and stable
Thermistor	Semiconductor resistance – usually decreases sharply with temperature (NTC)	Very sensitive, non-linear; −50 to 300 °C
Total radiation pyrometer	Stefan–Boltzmann law – total radiation focused on thermopile	High temperatures, non-contact
Optical (disappearing-filament) pyrometer	Brightness of filament matched with target at one wavelength (red, 0.65 μm)	700–3000 °C, furnace and molten metal temperatures
Infrared thermometer / thermal camera	Infrared radiation intensity	Non-contact, needs emissivity setting

Thermocouple type	Materials	Feature
K	Chromel–Alumel	Most common general purpose, −200 to 1250 °C
J	Iron–Constantan	0 to 750 °C; reducing atmospheres
T	Copper–Constantan	Low temperatures (−200 to 350 °C), cryogenic
E	Chromel–Constantan	Highest emf per degree (sensitivity)
R, S	Platinum–Rhodium/Platinum	Up to ≈ 1600 °C, standards, high accuracy
B	Pt-30%Rh / Pt-6%Rh	Up to ≈ 1800 °C

- Laws of thermocouples: (1) law of homogeneous circuit – emf depends only on junction temperatures, not on temperature distribution along wires; (2) law of intermediate metals –

a third metal inserted at uniform temperature does not change the emf (allows meters and leads); (3) law of intermediate temperatures – $E(T_1,T_3) = E(T_1,T_2) + E(T_2,T_3)$ (allows cold-junction correction).

- Thermopile: several thermocouples in series to increase output (radiation pyrometers). Peltier effect (heat absorbed/released at a junction when current flows) and Thomson effect (heat in a single conductor with temperature gradient and current) are related thermoelectric effects.

★ **PYQ-TREND:** Thermocouple works on Seebeck effect (very frequent); RTD platinum; thermistor NTC; optical pyrometer for very high temperature (non-contact); laws of thermocouples; type T for low temperature.

26C.2 Pressure Measurement

Instrument	Principle / use
Manometers (U-tube, inclined, well type)	Liquid column; low pressures and differentials; calibration
Bourdon tube gauge	Elastic C, spiral or helical tube straightens under pressure; most common industrial gauge (boilers)
Diaphragm gauge	Elastic deflection of diaphragm; corrosive fluids, low pressures
Bellows gauge	Expansion of bellows; low pressures/differentials
Dead-weight tester	Known weights on a piston of known area – calibration of gauges
Piezoelectric transducer	Charge generated by crystal under stress – rapidly varying (dynamic) pressure such as engine cylinder pressure
Strain-gauge transducer	Diaphragm strain changes resistance
McLeod gauge	Compression of a known gas volume – vacuum down to $\approx 10^{-4}$ mm Hg; used to calibrate other vacuum gauges
Pirani gauge / thermocouple gauge	Thermal conductivity of gas varies with pressure (vacuum)
Ionisation gauge	Ion current – very high vacuum (to 10^{-10} mm Hg)

★ **PYQ-TREND:** Bourdon tube for boiler pressure (mounting); dead-weight tester for calibration; McLeod gauge for vacuum; piezoelectric for dynamic pressure; Pirani gauge works on thermal conductivity.

26C.3 Flow, Speed and Power Measurement

Quantity	Instruments
Flow rate	Venturimeter, orifice meter, flow nozzle (differential pressure); rotameter (variable area, vertical); turbine meter; electromagnetic meter (conducting liquids, Faraday's law); ultrasonic meter; positive-displacement meters (fuel, water)
Air flow into engine	Air box with orifice (damps pulsations), hot-wire/hot-film mass airflow sensor, viscous flow meter

Quantity	Instruments
Fuel consumption	Burette and stop-watch (volumetric), gravimetric balance, flow meters
Velocity	Pitot tube, hot-wire anemometer (turbulence), cup and vane anemometers (wind, ducts), laser Doppler anemometer
Speed	Mechanical/electrical tachometers, magnetic pick-ups, optical encoders, stroboscope (non-contact)

Dynamometers (brake power)

Type	Examples	Remarks
Absorption	Prony brake, rope brake	Friction; simple, low cost, small engines; heat dissipated by water cooling
Absorption	Hydraulic (Froude) dynamometer	Water churned between rotor and casing; high power, large engines
Absorption	Eddy-current dynamometer	Eddy currents in a rotating disc in magnetic field; easy control, wide speed range
Transmission	Torsion dynamometer, belt dynamometer	Measure torque transmitted without absorbing power
Driving	Swinging-field DC dynamometer	Can absorb power or motor the engine (motoring test for friction power)

- Engine indicator: mechanical indicator (piston, spring and stylus) for slow-speed engines gives the indicator diagram; for high-speed engines, piezoelectric pressure transducer with crank-angle encoder and data acquisition. Area of indicator diagram measured by planimeter → imep → IP.
- Exhaust measurement: NDIR (CO, CO_2), FID (unburnt hydrocarbons), chemiluminescence (NOx), Orsat apparatus (CO_2, O_2, CO); smoke meters – Bosch (filter paper blackening) and Hartridge (light absorption) for diesel smoke.

★ **PYQ-TREND:** Rope brake and hydraulic dynamometers measure BP; eddy-current dynamometer; swinging-field dynamometer for motoring (friction power); indicator diagram gives IP; rotameter variable area; Bosch/Hartridge smoke meters; FID for HC, chemiluminescence for NOx.

26C.4 Humidity, Heat and Property Measurement

Quantity	Instruments
Humidity	Sling psychrometer (DBT and WBT by whirling), Assmann aspiration psychrometer (fan-driven), hair hygrometer (hair length changes with humidity), dew-point hygrometer (cooled mirror), electronic capacitive sensors
Calorific value	Bomb calorimeter (solids, liquids), Junkers/Boys gas calorimeter
Dryness fraction	Barrel, separating, throttling, combined separating–throttling, electrical calorimeters

Quantity	Instruments
Thermal conductivity	Guarded hot-plate (insulators, flat specimens), Searle's apparatus (good conductors – metals bar), Lee's disc (poor conductors), transient hot-wire method (fluids)
Heat flux	Heat flux sensors (thermopile across thin layer), Gardon gauge (radiation)
Emissivity	Comparison with black body at same temperature
Viscosity of oils	Redwood, Saybolt, Engler viscometers
Flash and fire point	Pensky–Martens closed cup, Abel apparatus, Cleveland open cup

- Boiler instruments: water-level gauge glass, pressure gauge (Bourdon), safety valves, fusible plug, feed check valve; steam flow meters; flue gas analysers (O_2, CO_2) for combustion control; draught gauges (inclined manometers measuring mm of water).

★ **PYQ-TREND:** Sling psychrometer measures DBT and WBT; guarded hot plate for thermal conductivity of insulators; Pensky–Martens apparatus for flash point; Redwood viscometer; hair hygrometer.

✔ **TOPPER TRICK:** Matching rule for instruments: 'Seebeck – thermocouple; Peltier – thermoelectric cooler; Stefan–Boltzmann – total radiation pyrometer; brightness at one wavelength – optical pyrometer'.

✎ **DESCRIPTIVE Q:** Explain the working of a thermocouple and state its advantages and limitations.

A thermocouple consists of two wires of dissimilar metals joined at two junctions. When the junctions are at different temperatures, an electromotive force proportional (approximately) to the temperature difference is produced – the Seebeck effect. The measuring (hot) junction is placed at the point whose temperature is required, while the reference (cold) junction is kept at a known temperature (ice bath at 0 °C) or compensated electronically. The emf, of the order of microvolts per degree (about 41 μV/°C for type K), is measured by a millivoltmeter or potentiometer and converted to temperature using standard tables. Advantages: wide range (−200 to 1800 °C depending on type), small size and low thermal mass (fast response), robust and cheap, output is electrical and suitable for remote reading, recording and control; several can be connected in series as a thermopile. Limitations: low output requiring amplification, non-linear characteristics, need for cold-junction compensation, lower accuracy than RTDs, susceptibility to electrical noise and to contamination or ageing at high temperatures.

CHAPTER 26D

Heat Transfer IV – Mass Transfer, Heat Pipes, Insulation and Enhancement

26D.1 Basics of Mass Transfer

- Mass transfer is the movement of a species from a region of high concentration to low concentration – by molecular diffusion (analogous to conduction) and by convective mass transfer (analogous to convection). Examples: evaporation of water into air, drying, humidification, dissolution of sugar, diffusion of hydrogen through metals, absorption in VAR systems.

Fick's law: $J_A = -D_{AB} \frac{dC_A}{dx}$ (kg/m²·s or kmol/m²·s); D_{AB} = mass diffusivity (m²/s)
Convective mass transfer: $\dot{m}_A = h_m A (C_{A,s} - C_{A,\infty})$ (h_m in m/s)

- Diffusivity order: gases ≈ 10^{-5} m²/s; liquids ≈ 10^{-9} m²/s; solids ≈ 10^{-10} to 10^{-14} m²/s. In gases D increases with temperature (≈ T^1.5) and decreases with pressure (∝ 1/p).

Heat transfer	Mass transfer analogue
Temperature T	Concentration C
Thermal conductivity k / diffusivity α	Mass diffusivity D_{AB}
Fourier's law	Fick's law
h	h_m
Nusselt number Nu = hL/k	Sherwood number Sh = h_m L/D_{AB}
Prandtl number Pr = ν/α	Schmidt number Sc = ν/D_{AB}
Nu = f(Re, Pr)	Sh = f(Re, Sc)
Grashof (thermal)	Grashof (concentration)

Lewis number Le = α/D_{AB} = Sc/Pr; Chilton–Colburn analogy: $h/h_m = \rho\, c_p\, Le^{(2/3)}$
Lewis relation for air–water vapour (Le ≈ 1): $h/(h_m \rho\, c_p) \approx 1$ → wet-bulb temperature ≈ adiabatic saturation temperature

- Evaporation from a water surface increases with surface temperature, air velocity and dryness of air (lower vapour pressure of air); it cools the surface – basis of evaporative coolers, cooling towers, sweating and wet-bulb thermometers.

★ **PYQ-TREND:** Fick's law analogy to Fourier's law (frequent); Sherwood and Schmidt numbers; Lewis number ≈ 1 for air–water (WBT = adiabatic saturation temperature); diffusivity of gases ∝ T^1.5/p.

26D.2 Heat Pipes

- A heat pipe is a sealed tube containing a small quantity of working fluid and a wick lining the inner wall. At the hot end (evaporator) the liquid evaporates, absorbing latent heat; vapour flows through the core to the cold end (condenser), condenses releasing latent heat, and the

liquid returns to the evaporator through the wick by capillary action (or by gravity in a thermosiphon, when the condenser is above the evaporator). Between them lies an adiabatic section.

- Because heat is transported as latent heat with a very small temperature drop, its effective thermal conductivity can be hundreds of times that of copper.

Temperature range	Working fluid
Cryogenic (< −150 °C)	Helium, nitrogen
Low (−70 to 200 °C)	Ammonia, methanol, acetone, water (common: water 30–200 °C)
High (> 400 °C)	Sodium, potassium, lithium

- Applications: cooling of laptop and server processors, LED lighting, spacecraft thermal control, evacuated-tube solar water heaters, heat recovery in air-conditioning, stabilisation of permafrost under pipelines.
- Limits: capillary (wicking) limit, boiling limit, sonic limit, entrainment limit, viscous limit.

★ **PYQ-TREND:** Heat pipe principle – evaporation, condensation, capillary return through wick; very high effective conductivity; uses in electronics cooling.

26D.3 Thermal Insulation

- Insulators have low thermal conductivity, mostly because they trap still air or gas in small cells (convection suppressed). Desirable properties: low k, low density, low moisture absorption (wet insulation conducts well), fire resistance, chemical stability, adequate strength, vermin resistance, low cost and suitable temperature limit.

Material	Approx. maximum temperature / use
Polyurethane foam (PUF), expanded polystyrene (EPS, thermocole)	Up to ≈ 80–100 °C; cold storage, refrigerators, buildings (k ≈ 0.02–0.035)
Glass wool	Up to ≈ 250–450 °C; ducts, buildings, ovens
Mineral (rock/slag) wool	Up to ≈ 650–750 °C; pipes, boilers, turbines
Calcium silicate	Up to ≈ 650–1000 °C; high-temperature pipes
Magnesia (85% magnesia)	Up to ≈ 315 °C; steam pipes (older practice)
Ceramic fibre	Up to ≈ 1260–1400 °C; furnaces, kilns
Refractory bricks (fireclay, silica, magnesite)	1500–1800 °C; furnace linings
Aerogel	Super-insulation, k ≈ 0.015 or less
Multilayer insulation (MLI) / vacuum panels	Cryogenic tanks, spacecraft – radiation shields in vacuum

- Cold insulation must have a vapour barrier on the warm (outer) side; otherwise water vapour diffuses into the insulation, condenses or freezes and destroys its insulating value.
- Economic thickness of insulation: the thickness at which the sum of insulation cost (rises with thickness) and the cost of heat lost (falls with thickness) is minimum.
- Thermal conductivity of insulation increases with temperature, density beyond an optimum, and moisture content.

★ **PYQ-TREND:** Insulation works by trapping air; vapour barrier on the warm side of cold insulation (frequent); economic thickness; PUF for cold storage; ceramic fibre for furnaces; moisture increases k.

26D.4 Heat Transfer Enhancement and Electronics Cooling

Passive techniques	Active techniques
Extended surfaces (fins, finned tubes, microfins)	Mechanical stirring, surface scraping
Rough or treated surfaces (promote turbulence or nucleation)	Surface or fluid vibration
Twisted tapes and coil inserts (swirl flow)	Jet impingement, sprays
Corrugated or dimpled tubes, louvered fins	Electrostatic fields
Additives (nanofluids, surfactants in boiling)	Suction/injection

- Enhancement increases h or area but also increases pressure drop; the gain must outweigh extra pumping power.
- Electronics cooling: heat sinks with fins and fans, heat pipes and vapour chambers, thermal interface materials (thermal grease/pads to reduce contact resistance), liquid cold plates, immersion cooling in dielectric fluids for data centres, thermoelectric coolers for spot cooling. Junction temperature limits (≈ 85–125 °C) govern design.
- Thermal contact resistance falls with higher contact pressure, smoother and softer surfaces, and interstitial materials of high conductivity.

★ **PYQ-TREND:** Twisted tapes and rough surfaces increase h and pressure drop; thermal grease reduces contact resistance; heat sinks with fins.

✎ **DESCRIPTIVE Q:** Explain the working of a heat pipe and list its applications.

A heat pipe consists of a sealed, evacuated metal tube whose inner wall is lined with a porous wick saturated with a working fluid such as water, ammonia or methanol. When one end (the evaporator) is heated, the fluid in the wick evaporates and absorbs its latent heat; the vapour pressure rises locally and drives vapour along the hollow core to the cooler end (the condenser). There the vapour condenses on the wick, releasing the latent heat to a heat sink. The condensed liquid is drawn back to the evaporator through the wick by capillary pressure, completing the cycle without any pump; in a gravity-assisted thermosiphon the condensate returns by gravity. Because heat is carried as latent heat and the vapour flows almost isothermally, the heat pipe transfers large amounts of heat with a very small temperature difference – its effective conductivity is many times that of copper – and it is light, silent and has no moving parts. Its capacity is limited by capillary, boiling, sonic and entrainment limits. Applications include CPU cooling in laptops and servers, LED and power-electronics cooling, spacecraft thermal control, heat-recovery units in HVAC, evacuated-tube solar collectors, and keeping permafrost frozen under oil pipelines.

CHAPTER 27

Solved Numericals – Set 1 (Thermodynamics and Gas Cycles)

SSC-style problems solved step by step (no options). Air: R = 0.287 kJ/kg·K, c_p = 1.005, c_v = 0.718 kJ/kg·K, γ = 1.4 unless stated. Temperatures in kelvin inside formulas.

27.1 Ideal Gas and Work Transfer

Problem 1 – Mass of air in a tank

A 2 m³ tank contains air at 5 bar and 27 °C. Find the mass of air.

$m = pV/(RT) = (5 \times 10^5 \times 2)/(287 \times 300) = 10^6/86{,}100 = 11.61$ kg

Problem 2 – Isothermal expansion work

0.1 m³ of gas at 1 bar expands isothermally to 0.3 m³. Find the work done.

$W = p_1V_1 \ln(V_2/V_1) = 10^5 \times 0.1 \times \ln 3 = 10{,}000 \times 1.0986 = 10.99$ kJ (done by the gas)

Problem 3 – Constant pressure process

1 kg of air at 1 bar is heated at constant pressure from 300 K to 500 K. Find Q, W and ΔU.

$Q = m\,c_p\,\Delta T = 1.005 \times 200 = 201$ kJ; $\Delta U = m\,c_v\,\Delta T = 0.718 \times 200 = 143.6$ kJ; $W = mR\Delta T = 0.287 \times 200 = 57.4$ kJ

- Check: Q = ΔU + W = 143.6 + 57.4 = 201 kJ. Fraction of heat converted to work = W/Q = (γ − 1)/γ = 0.286.

Problem 4 – Polytropic compression

1 kg of air at 1 bar, 27 °C is compressed to 8 bar following pV^1.3 = C. Find the final temperature, work and heat transfer.

$T_2 = T_1(p_2/p_1)^{((n-1)/n)} = 300 \times 8^{0.2308} = 300 \times 1.616 = 484.8$ K

$W = mR(T_1 - T_2)/(n - 1) = 0.287 \times (300 - 484.8)/0.3 = -176.8$ kJ (work done on the air)

$\Delta U = m\,c_v\,(T_2 - T_1) = 0.718 \times 184.8 = 132.7$ kJ; $Q = \Delta U + W = 132.7 - 176.8 = -44.1$ kJ (heat rejected)

- Shortcut: Q = W × (γ − n)/(γ − 1) = −176.8 × 0.1/0.4 = −44.2 kJ.

Problem 5 – Adiabatic compression

Air at 1 bar, 300 K is compressed reversibly and adiabatically to 10 bar. Find T_2 and work per kg.

$T_2 = 300 \times 10^{(0.4/1.4)} = 300 \times 1.931 = 579.2$ K

$W = c_v\,(T_2 - T_1) = 0.718 \times 279.2 = 200.5$ kJ/kg (input); steady-flow work = $c_p\,\Delta T$ = 280.6 kJ/kg

- Non-flow work uses $c_v\,\Delta T$, steady-flow (shaft) work uses $c_p\,\Delta T = \gamma \times$ non-flow work.

27.2 First Law – Steady Flow

Problem 6 – Gas turbine

Air enters a turbine at 1000 K with 50 m/s and leaves at 600 K with 150 m/s. The turbine is adiabatic. Find work per kg.

$w = c_p(T_1 - T_2) + (V_1^2 - V_2^2)/2000 = 1.005 \times 400 + (2500 - 22{,}500)/2000 = 402 - 10 = 392$ kJ/kg

Problem 7 – Steam nozzle

Steam enters a nozzle with negligible velocity and $h_1 = 3000$ kJ/kg; $h_2 = 2762$ kJ/kg. Find exit velocity.

$V_2 = 44.72\sqrt{(h_1 - h_2)} = 44.72 \times \sqrt{238} = 44.72 \times 15.43 = 690$ m/s

Problem 8 – Throttling

Steam at 10 bar ($h_f = 762.8$, $h_{fg} = 2015.3$ kJ/kg) is throttled to 1 bar and 110 °C ($h = 2696$ kJ/kg). Find the initial dryness fraction.

$h_1 = h_2 \rightarrow 762.8 + x \times 2015.3 = 2696 \rightarrow x = 1933.2/2015.3 = 0.959$

Problem 9 – Compressor with heat loss

An air compressor takes in air at 100 kPa, 300 K and delivers it at 420 K. Heat lost to the cooling jacket = 20 kJ/kg. Neglect KE. Find work input per kg.

$-w = (h_2 - h_1) + q_{loss} = 1.005 \times 120 + 20 = 120.6 + 20 = 140.6$ kJ/kg (input)

Problem 10 – Mixing chamber

Hot water at 80 °C (2 kg/s) mixes with cold water at 20 °C (3 kg/s) adiabatically. Find the outlet temperature.

$T_3 = (2 \times 80 + 3 \times 20)/5 = 220/5 = 44$ °C

27.3 Second Law, Engines, Refrigerators

Problem 11 – Carnot engine

A Carnot engine works between 527 °C and 27 °C and receives 1000 kJ. Find η, work and heat rejected.

$\eta = 1 - 300/800 = 0.625$; $W = 625$ kJ; $Q_2 = 375$ kJ

Problem 12 – Refrigerator and heat pump

A reversible machine works between −13 °C and 27 °C. Find COP as refrigerator and as heat pump.

$COP_R = T_L/(T_H - T_L) = 260/40 = 6.5$; $COP_{HP} = 300/40 = 7.5 = COP_R + 1$

Problem 13 – Clausius inequality test

An engine receives 1000 kJ at 500 K and rejects heat at 300 K. Check feasibility if it rejects (a) 700 kJ (b) 500 kJ.

(a) $\oint \delta Q/T = 1000/500 - 700/300 = 2 - 2.333 = -0.333 < 0 \rightarrow$ possible (irreversible)

(b) $2 - 500/300 = 2 - 1.667 = +0.333 > 0 \rightarrow$ impossible

- Quick check: Carnot limit = 1 − 300/500 = 40%; (a) η = 30% possible; (b) η = 50% > 40% impossible.

Problem 14 – Two engines in series

Two reversible engines in series work between 1000 K and 300 K. Find the intermediate temperature for (a) equal work (b) equal efficiency.

(a) $T = (1000 + 300)/2 = 650$ K; (b) $T = \sqrt{(1000 \times 300)} = 547.7$ K

Problem 15 – Engine driving a refrigerator

A reversible engine (1000 K to 300 K) drives a reversible refrigerator (250 K to 300 K). Find heat taken from the 250 K space per kJ supplied to the engine.

$W = 1 \times (1 - 0.3) = 0.7$ kJ; $COP_R = 250/50 = 5$; $Q_L = 5 \times 0.7 = 3.5$ kJ

27.4 Entropy and Availability

Problem 16 – Heating water

2 kg of water is heated from 20 °C to 80 °C. Find the change of entropy (c = 4.187 kJ/kg·K).

$\Delta S = mc \ln(T_2/T_1) = 2 \times 4.187 \times \ln(353/293) = 8.374 \times 0.1863 = 1.560$ kJ/K

Problem 17 – Entropy change of air

1 kg of air goes from 1 bar, 300 K to 5 bar, 500 K. Find Δs.

$\Delta s = c_p \ln(T_2/T_1) - R \ln(p_2/p_1) = 1.005 \times 0.5108 - 0.287 \times 1.6094 = 0.5134 - 0.4619 = 0.0515$ kJ/kg·K

Problem 18 – Melting of ice

1 kg of ice at 0 °C melts to water at 0 °C (latent heat 335 kJ/kg). Find ΔS.

$\Delta S = Q/T = 335/273 = 1.227$ kJ/K

Problem 19 – Mixing of water (entropy generation)

1 kg of water at 80 °C is mixed adiabatically with 2 kg at 20 °C. Find the entropy generated.

Final $T = (80 + 40)/3 = 40$ °C = 313 K

$\Delta S = 4.187[1 \times \ln(313/353) + 2 \times \ln(313/293)] = 4.187[-0.1203 + 0.1321] = 0.0494$ kJ/K > 0 (irreversible)

Problem 20 – Heat reservoir entropy (universe)

1000 kJ of heat flows from a reservoir at 800 K to one at 400 K. Find $\Delta S_{universe}$ and the loss of available energy (T_0 = 300 K).

$\Delta S_{univ} = -1000/800 + 1000/400 = -1.25 + 2.5 = 1.25$ kJ/K

Loss of available energy = $T_0 \Delta S_{univ} = 300 \times 1.25 = 375$ kJ

Problem 21 – Available energy

1000 kJ of heat is available at 600 K; surroundings at 300 K. Find the available and unavailable energy.

$AE = Q(1 - T_0/T) = 1000 \times 0.5 = 500$ kJ; UE = 500 kJ

Problem 22 – Second-law efficiency

An engine works between 900 K and 300 K with η = 40%. Find the second-law efficiency.

$\eta_{Carnot} = 1 - 300/900 = 0.667$; $\eta_{II} = 0.40/0.667 = 0.60 = 60\%$

27.5 Properties of Steam

Problem 23 – Wet steam enthalpy

Find the enthalpy and specific volume of steam at 10 bar, x = 0.9 (h_f = 762.8, h_{fg} = 2015.3 kJ/kg, v_g = 0.1944 m³/kg).

$h = 762.8 + 0.9 \times 2015.3 = 762.8 + 1813.8 = 2576.6$ kJ/kg; $v \approx x\, v_g = 0.175$ m³/kg

Problem 24 – Superheated steam (approximate)

Steam at 10 bar (T_sat = 179.9 °C, h_g = 2778.1 kJ/kg) is superheated to 250 °C. Take c_ps = 2.1 kJ/kg·K. Find h and degree of superheat.

Degree of superheat = 250 − 179.9 = 70.1 °C; h = 2778.1 + 2.1 × 70.1 = 2925.3 kJ/kg (tables: ≈ 2943)

Problem 25 – Separating and throttling calorimeter

In a combined calorimeter, the separator collects 0.2 kg water; 3.0 kg steam condensed after throttling with $x_2 = 0.97$. Find the overall dryness.

x_1 (separator) = 3.0/3.2 = 0.9375; x = x_1 × x_2 = 0.9375 × 0.97 = 0.909

27.6 Air-Standard Cycles

Problem 26 – Otto cycle efficiency

An Otto cycle has compression ratio 8. Find air-standard efficiency.

η = 1 − 1/r^(γ−1) = 1 − 1/8^0.4 = 1 − 1/2.297 = 0.565 = 56.5%

Problem 27 – Otto cycle temperatures and work

In Problem 26, $T_1 = 300$ K and heat supplied = 1800 kJ/kg. Find T_2, T_3, T_4 and net work.

T_2 = 300 × 2.297 = 689.2 K; T_3 = 689.2 + 1800/0.718 = 3196 K; T_4 = 3196/2.297 = 1391 K

Q_rejected = 0.718 × (1391 − 300) = 783.4 kJ/kg; W = 1800 − 783.4 = 1016.6 kJ/kg (= 0.565 × 1800 ✓)

Problem 28 – Diesel cycle

A Diesel cycle has r = 16 and cut-off ratio ρ = 2. Find η.

η = 1 − (1/r^(γ−1)) × (ρ^γ − 1)/(γ(ρ − 1)) = 1 − (1/3.031) × (2.639 − 1)/1.4 = 1 − 0.330 × 1.171 = 0.614 = 61.4%

Problem 29 – Cut-off from expansion ratio

A Diesel engine has compression ratio 15 and expansion ratio 7.5. Find the cut-off ratio and percentage cut-off (of stroke).

ρ = r/r_e = 15/7.5 = 2; % cut-off = (ρ − 1)/(r − 1) = 1/14 = 7.14% of stroke

Problem 30 – Brayton cycle

A Brayton cycle has pressure ratio 10. Find η. For T_min = 300 K, T_max = 1200 K, find the optimum pressure ratio and maximum work.

η = 1 − 1/10^0.2857 = 1 − 1/1.931 = 0.482 = 48.2%

r_p,opt = (T_max/T_min)^(γ/(2(γ−1))) = 4^1.75 = 11.31

W_max = c_p(√T_max − √T_min)² = 1.005 × (34.64 − 17.32)² = 1.005 × 300 = 301.5 kJ/kg

Problem 31 – Carnot vs Otto

A Carnot engine and an Otto engine both work between T_max = 1500 K and T_min = 300 K (Otto r = 8). Compare efficiencies.

η_Carnot = 1 − 300/1500 = 80%; η_Otto = 56.5% → Carnot is the upper limit

Problem 32 – Compression ratio from clearance

The swept volume is 450 cc and the clearance volume 50 cc. Find r and Otto efficiency.

r = (450 + 50)/50 = 10; η = 1 − 1/10^0.4 = 1 − 0.398 = 60.2%

CHAPTER 27B

Solved Numericals – Set 2 (IC Engines, Steam Plant, Compressors, Gas Turbines)

SSC-style problems solved step by step (no options).

27B.1 IC Engine Performance

Problem 1 – Indicated power

A single-cylinder four-stroke engine has bore 100 mm, stroke 120 mm, imep 6 bar and speed 1500 rpm. Find IP.

$A = \pi/4 \times 0.1^2 = 7.854 \times 10^{-3}$ m²; $n = N/2 = 750$ power strokes/min

$IP = p_m\, L\, A\, n/60 = 6 \times 10^5 \times 0.12 \times 7.854 \times 10^{-3} \times 750/60 = 7069$ W ≈ 7.07 kW

- Two-stroke engine: n = N, so IP would double to 14.1 kW for the same data.

Problem 2 – Brake power from a rope brake

Net brake load 300 N, effective drum radius 0.5 m, speed 1200 rpm. Find BP. If IP = 23 kW, find η_m.

$T = 300 \times 0.5 = 150$ N·m; $BP = 2\pi NT/60 = 2\pi \times 1200 \times 150/60 = 18{,}850$ W = 18.85 kW

$\eta_m = BP/IP = 18.85/23 = 0.82 = 82\%$; $FP = 23 - 18.85 = 4.15$ kW

Problem 3 – Thermal efficiency and bsfc

The engine of Problem 2 uses 5 kg/h of fuel with CV 42,000 kJ/kg. Find brake thermal efficiency and bsfc.

Fuel power $= 5 \times 42{,}000/3600 = 58.33$ kW; $\eta_{bth} = 18.85/58.33 = 32.3\%$

bsfc $= 5/18.85 = 0.265$ kg/kWh; $\eta_{ith} = 23/58.33 = 39.4\%$

Problem 4 – Volumetric efficiency

A four-cylinder four-stroke engine has swept volume 0.5 L per cylinder and runs at 3000 rpm. Actual air intake (at ambient conditions) is 0.042 m³/s. Find η_v.

Theoretical $= 4 \times 0.5 \times 10^{-3} \times (3000/2)/60 = 0.05$ m³/s; $\eta_v = 0.042/0.05 = 84\%$

Problem 5 – Morse test

A four-cylinder petrol engine gives 40 kW BP with all cylinders firing. With each cylinder cut out in turn, BP = 28, 29, 28.5 and 29.5 kW. Find IP and η_m.

IP of cylinders $= 40 - 28 = 12, 11, 11.5, 10.5$ kW → total IP = 45 kW; $\eta_m = 40/45 = 88.9\%$

Problem 6 – Heat balance

Fuel energy 60 kW; BP 18 kW; cooling water 16 kW; exhaust 18 kW. Draw up the heat balance.

Item	kW	%
Brake power	18	30.0
Cooling water	16	26.7

Item	kW	%
Exhaust gases	18	30.0
Unaccounted (radiation, friction etc.)	8	13.3
Total	60	100

Problem 7 – Mean piston speed and displacement

Bore 80 mm, stroke 90 mm, clearance volume 50 cc, 1500 rpm. Find swept volume, r and mean piston speed.

$V_s = \pi/4 \times 8^2 \times 9 = 452.4$ cc; $r = (452.4 + 50)/50 = 10.05$; $V_p = 2LN/60 = 2 \times 0.09 \times 1500/60 = 4.5$ m/s

Problem 8 – Stoichiometric A/F ratio

Find the stoichiometric air–fuel ratio of octane C_8H_{18} (air contains 23.3% O_2 by mass).

$C_8H_{18} + 12.5\ O_2 \rightarrow 8CO_2 + 9H_2O$; O_2 needed $= 12.5 \times 32 = 400$ kg per 114 kg fuel

Air $= 400/0.233 = 1717$ kg $\rightarrow$ A/F $= 1717/114 \approx 15.1$

Problem 9 – Air required for a fuel by mass analysis

A coal contains C 80%, H 5%, S 1%, O 4% (rest ash). Find the minimum air per kg.

$O_2 = 2.67C + 8H + S - O = 2.67 \times 0.8 + 8 \times 0.05 + 0.01 - 0.04 = 2.136 + 0.4 + 0.01 - 0.04 = 2.506$ kg

Air $= 2.506/0.233 = 10.76$ kg/kg of coal

Problem 10 – Excess air from dry flue gas

If the theoretical air is 10.76 kg/kg and actual air supplied is 15 kg/kg, find the percentage excess air.

Excess air $= (15 - 10.76)/10.76 = 39.4\%$

27B.2 Rankine Cycle and Boilers

Problem 11 – Simple Rankine cycle

Turbine inlet $h_1 = 3230$ kJ/kg; turbine exit $h_2 = 2160$ kJ/kg; condensate $h_3 = 191.8$ kJ/kg ($v = 0.00101$ m³/kg); boiler pressure 40 bar, condenser 0.1 bar. Find net work, η and specific steam consumption.

$W_p = v\,\Delta p = 0.00101 \times (4000 - 10) = 4.03$ kJ/kg; $h_4 = 195.8$ kJ/kg

$W_t = 3230 - 2160 = 1070$ kJ/kg; $W_{net} = 1066$ kJ/kg; $Q_1 = 3230 - 195.8 = 3034.2$ kJ/kg

$\eta = 1066/3034.2 = 35.1\%$; SSC $= 3600/1066 = 3.38$ kg/kWh; work ratio $= 1066/1070 = 0.996$

- Pump work is tiny (≈ 0.4% of turbine work) – the reason the Rankine work ratio is close to 1.

Problem 12 – Turbine isentropic efficiency

In Problem 11, the actual turbine exit enthalpy is 2320 kJ/kg. Find $\eta_{turbine}$.

$\eta_t = (3230 - 2320)/(3230 - 2160) = 910/1070 = 85\%$

Problem 13 – Equivalent evaporation

A boiler produces 10 kg of steam per kg of coal at h = 2800 kJ/kg from feed water at 40 °C (h = 167.5 kJ/kg). CV of coal = 30,000 kJ/kg. Find equivalent evaporation and boiler efficiency.

$Fe = m_a(h - h_{fw})/2257 = 10 \times 2632.5/2257 = 11.66$ kg per kg coal

$\eta_{boiler} = 10 \times 2632.5/30{,}000 = 87.8\%$

Problem 14 – Boiler heat input from fuel rate

A boiler generates 5000 kg/h of steam with h − h_fw = 2600 kJ/kg using 600 kg/h of coal of CV 28,000 kJ/kg. Find efficiency.

```
η = (5000 × 2600)/(600 × 28,000) = 13 × 10⁶/16.8 × 10⁶ = 77.4%
```

Problem 15 – Chimney draught

A chimney 40 m high carries flue gas at 300 °C; ambient 27 °C. Find the theoretical draught in mm of water, taking ρ_air = 1.177 kg/m³ at 300 K and gas density ≈ air density scaled by temperature.

```
ρ_gas ≈ 1.177 × 300/573 = 0.616 kg/m³; Δp = gH(ρ_a − ρ_g) = 9.81 × 40 × 0.561 = 220 Pa
h_w = 220/9810 = 0.0224 m ≈ 22.4 mm of water
```

Problem 16 – Regenerative heating (mass bled)

In a regenerative cycle, bled steam (h = 2800 kJ/kg) heats feed water from h = 200 to h = 640 kJ/kg in an open heater. Find the fraction bled per kg of steam leaving the heater.

```
m(2800) + (1 − m)(200) = 640 → m(2800 − 200) = 440 → m = 0.169 kg
```

27B.3 Nozzles, Turbines, Condensers

Problem 17 – Critical pressure

Find the critical pressure for (a) superheated steam (b) saturated steam (c) air, each entering at 10 bar.

```
(a) 0.546 × 10 = 5.46 bar; (b) 0.577 × 10 = 5.77 bar; (c) 0.528 × 10 = 5.28 bar
```

Problem 18 – Nozzle with friction

Isentropic enthalpy drop 200 kJ/kg, nozzle efficiency 90%, initial velocity negligible. Find exit velocity.

```
V = 44.72 √(0.9 × 200) = 44.72 × 13.42 = 600 m/s
```

Problem 19 – Optimum blade speed (impulse)

Steam leaves the nozzle at 600 m/s at α = 20°. Find the optimum blade speed and maximum blade efficiency (symmetrical blades, no friction).

```
u_opt = V₁ cosα/2 = 600 × 0.9397/2 = 281.9 m/s; η_max = cos²α = 0.883 = 88.3%
```

Problem 20 – Parsons reaction turbine

For α = 20°, find the maximum blade efficiency and optimum blade speed ratio of a 50% reaction stage.

```
ρ_opt = u/V₁ = cosα = 0.94; η_max = 2cos²α/(1 + cos²α) = 1.766/1.883 = 93.8%
```

Problem 21 – Velocity compounding

Why does a two-row Curtis stage use a lower blade speed than a single-row stage for the same nozzle velocity? Find the optimum ratio.

```
Single row: u/V₁ = cosα/2; n rows: u/V₁ = cosα/(2n) → two rows: cosα/4 (blade speed
halved)
```

Problem 22 – Condenser vacuum efficiency

Barometer 760 mm Hg; vacuum gauge on condenser 700 mm Hg; condensate temperature 35 °C (p_sat = 0.0563 bar = 42.2 mm Hg). Find the vacuum efficiency and partial pressure of air.

Absolute pressure = 760 − 700 = 60 mm Hg; ideal vacuum = 760 − 42.2 = 717.8 mm Hg

η_{vacuum} = 700/717.8 = 97.5%; p_{air} = 60 − 42.2 = 17.8 mm Hg

Problem 23 – Cooling water quantity

A condenser condenses 10,000 kg/h of steam, each kg rejecting 2300 kJ. Cooling water rises 10 °C. Find the cooling water flow.

$\dot{m}_w$ = 10,000 × 2300/(4.187 × 10) = 549,300 kg/h ≈ 152.6 kg/s (≈ 55 kg of water per kg of steam)

27B.4 Air Compressors

Problem 24 – Single-stage compressor power

A single-stage compressor takes 1 m³/min of free air at 1 bar and compresses it to 7 bar with n = 1.3. Neglect clearance. Find power and isothermal efficiency.

$(p_2/p_1)^{((n-1)/n)} = 7^{0.2308} = 1.567$

$P = (n/(n-1))\, p_1 \dot{V}_1 [(p_2/p_1)^{((n-1)/n)} - 1] = 4.333 \times 10^5 \times (1/60) \times 0.567 = 4094$ W = 4.09 kW

$P_{iso} = p_1 \dot{V}_1 \ln 7 = 1666.7 \times 1.946 = 3243$ W; η_{iso} = 3243/4094 = 79.2%

Problem 25 – Volumetric efficiency with clearance

Clearance ratio 5%, pressure ratio 7, n = 1.3. Find volumetric efficiency.

$\eta_v = 1 + C - C\,(p_2/p_1)^{(1/n)} = 1.05 - 0.05 \times 7^{0.769} = 1.05 - 0.05 \times 4.468 = 0.827$ = 82.7%

Maximum pressure ratio (η_v = 0): (1 + C)/C = 21 → $r_{max} = 21^{1.3} \approx 52.4$

Problem 26 – Two-stage compression

Air is compressed from 1 bar to 16 bar in two stages with perfect intercooling. Find the optimum intermediate pressure and the stage pressure ratio.

$p_2 = \sqrt{(1 \times 16)} = 4$ bar; each stage ratio = 4; for three stages to 27 bar: ratio = 3 each (3, 9 bar)

Problem 27 – Saving by two-stage compression

For Problem 26 with n = 1.3, compare specific work (per unit p_1V_1) of single and two-stage compression.

Single: $4.333 \times (16^{0.2308} - 1) = 4.333 \times (1.896 - 1) = 3.882$; two stage: $2 \times 4.333 \times (4^{0.2308} - 1) = 8.667 \times 0.377 = 3.268$

Saving = (3.882 − 3.268)/3.882 = 15.8%

27B.5 Gas Turbines

Problem 28 – Actual Brayton cycle

T_1 = 300 K, r_p = 6, T_3 = 1100 K, η_c = 0.85, η_t = 0.90. Find net work, efficiency and work ratio.

$r_p^{0.2857} = 1.6685$; $T_{2s} = 500.6$ K → $T_2 = 300 + 200.6/0.85 = 536.0$ K

$T_{4s} = 1100/1.6685 = 659.3$ K → $T_4 = 1100 - 0.9 \times 440.7 = 703.4$ K

$W_t = 1.005 \times 396.6 = 398.6$; $W_c = 1.005 \times 236 = 237.2$; $W_{net} = 161.4$ kJ/kg

$Q_1 = 1.005 \times (1100 - 536) = 566.8$ kJ/kg; $\eta = 161.4/566.8 = 28.5\%$; work ratio = 161.4/398.6 = 0.405

- Compare ideal $\eta = 1 - 1/1.6685 = 40.1\%$: component inefficiencies lower it sharply because the work ratio is small.

Problem 29 – Effect of regeneration

In Problem 28, a regenerator with effectiveness 100% is added. Is regeneration useful? Find the new heat input.

T_4 (703.4 K) > T_2 (536 K) → regeneration is useful; ideal regenerator heats air to 703.4 K

$Q_1 = 1.005 \times (1100 - 703.4) = 398.6$ kJ/kg; $\eta = 161.4/398.6 = 40.5\%$

Problem 30 – Jet propulsion

An aircraft flies at 250 m/s and the jet leaves at 600 m/s relative to the aircraft; air flow 50 kg/s (neglect fuel mass). Find thrust, thrust power and propulsive efficiency.

Thrust = $\dot{m}(V_j - V_a) = 50 \times 350 = 17{,}500$ N; thrust power = $17{,}500 \times 250 = 4.375$ MW

$\eta_p = 2V_a/(V_j + V_a) = 500/850 = 58.8\%$

CHAPTER 27C

Solved Numericals – Set 3 (Refrigeration, Psychrometry, Heat Transfer)

SSC-style problems solved step by step (no options). 1 TR = 3.5167 kW = 211 kJ/min.

27C.1 Refrigeration

Problem 1 – Tonnage

A refrigerator extracts 63,300 kJ/h from a cold store. Find its capacity in TR.

```
63,300/60 = 1055 kJ/min; TR = 1055/211 = 5 TR
```

Problem 2 – Simple VCR cycle

Enthalpies: evaporator exit (compressor inlet) 1440, compressor exit 1650, condenser exit 320 kJ/kg. Find refrigerating effect, COP, mass flow and power for 10 TR.

```
RE = 1440 − 320 = 1120 kJ/kg; W = 1650 − 1440 = 210 kJ/kg; COP = 1120/210 = 5.33
ṁ = 35.17/1120 = 0.0314 kg/s; P = 0.0314 × 210 = 6.59 kW; heat rejected = 0.0314 × 1330 = 41.8 kW
```

Problem 3 – Carnot COP and relative COP

A refrigerator works between −10 °C and 40 °C with an actual COP of 3.5. Find the Carnot COP and relative COP.

```
COP_Carnot = 263/50 = 5.26; relative COP = 3.5/5.26 = 0.665
```

Problem 4 – Power per TR

A plant has COP 4. Find power per TR.

```
P = 3.5167/4 = 0.879 kW/TR (≈ 1.18 hp/TR)
```

Problem 5 – Bell–Coleman (reversed Brayton) cycle

Find the COP of an ideal air refrigeration cycle with pressure ratio 5.

```
COP = 1/(r_p^((γ−1)/γ) − 1) = 1/(5^0.2857 − 1) = 1/(1.584 − 1) = 1.71
```

Problem 6 – Ideal COP of vapour absorption system

Generator 100 °C, condenser/ambient 30 °C, evaporator −10 °C. Find the maximum COP.

```
COP = [(T_G − T_0)/T_G] × [T_E/(T_0 − T_E)] = (70/373) × (263/40) = 0.1877 × 6.575 = 1.23
```

Problem 7 – Ice plant

Find the ice produced per day by a 10 TR plant making ice at 0 °C from water at 25 °C (c = 4.187, latent heat 335 kJ/kg). Neglect losses.

```
Heat per kg = 4.187 × 25 + 335 = 104.7 + 335 = 439.7 kJ/kg
Capacity = 10 × 211 × 1440 = 3,038,400 kJ/day → ice = 3,038,400/439.7 ≈ 6910 kg/day ≈ 6.9 tonnes
```

Problem 8 – Heat pump heating

A heat pump with COP 3.5 supplies 14 kW to a house. Find the power input and heat taken from outside.

```
W = 14/3.5 = 4 kW; Q_L = 14 − 4 = 10 kW
```

27C.2 Psychrometry

Problem 9 – Specific humidity and RH

Air at 30 °C, 1.01325 bar has vapour pressure 0.025 bar (p_s at 30 °C = 0.04246 bar). Find RH, ω and enthalpy.

```
φ = 0.025/0.04246 = 58.9%
ω = 0.622 × 0.025/(1.01325 − 0.025) = 0.01555/0.98825 = 0.01574 kg/kg dry air
h = 1.005 × 30 + 0.01574 (2500 + 1.88 × 30) = 30.15 + 40.24 = 70.4 kJ/kg dry air
```

Problem 10 – Dew point

In Problem 9, find the dew point (p_sat = 0.025 bar at about 21 °C).

```
DPT = T_sat at p_v = 0.025 bar ≈ 21 °C
```

Problem 11 – Bypass factor

Air at 30 °C passes over a cooling coil with ADP 10 °C and leaves at 15 °C. Find BPF and coil efficiency.

```
BPF = (15 − 10)/(30 − 10) = 0.25; contact factor = 0.75
```

Problem 12 – Adiabatic mixing

2 kg/s of air at 35 °C, ω = 0.020 mixes with 6 kg/s at 25 °C, ω = 0.010. Find the mixed state.

```
ω₃ = (2 × 0.020 + 6 × 0.010)/8 = 0.0125 kg/kg; t₃ ≈ (2 × 35 + 6 × 25)/8 = 27.5 °C
```

Problem 13 – Sensible heating load

1 kg/s of air is heated from 15 °C to 35 °C with no moisture change. Find the heating load.

```
Q = ṁ c_pa ΔT = 1 × 1.022 × 20 = 20.4 kW
```

Problem 14 – Sensible heat factor

A room has sensible heat gain 30 kW and latent heat gain 10 kW. Find SHF.

```
SHF = 30/(30 + 10) = 0.75
```

Problem 15 – Dehumidification load

2 kg/s of air is cooled from h = 75 to h = 45 kJ/kg while its humidity falls from 0.018 to 0.010 kg/kg. Find total cooling load in TR and condensate rate.

```
Q = 2 × (75 − 45) = 60 kW = 60/3.517 = 17.1 TR; condensate = 2 × 0.008 = 0.016 kg/s = 57.6 kg/h
```

Problem 16 – Evaporative cooler

Air at 40 °C DBT, 22 °C WBT passes through an evaporative cooler with humidifying efficiency 80%. Find the exit DBT.

```
t₂ = t₁ − η(t₁ − t_wb) = 40 − 0.8 × 18 = 25.6 °C (WBT stays ≈ 22 °C)
```

27C.3 Conduction

Problem 17 – Plane wall

A brick wall 0.2 m thick (k = 0.7 W/m·K), area 10 m², has faces at 30 °C and 0 °C. Find heat flow.

```
Q = kAΔT/L = 0.7 × 10 × 30/0.2 = 1050 W
```

Problem 18 – Composite wall with convection

Inside air 25 °C (h_1 = 10), brick 0.2 m (k = 0.7), insulation 0.05 m (k = 0.04), outside air −15 °C (h_2 = 25). Find U and heat flux.

```
1/U = 1/10 + 0.2/0.7 + 0.05/0.04 + 1/25 = 0.1 + 0.286 + 1.25 + 0.04 = 1.676 m²K/W → U = 0.597 W/m²K
q = U ΔT = 0.597 × 40 = 23.9 W/m²
Inner surface temperature = 25 − 23.9 × 0.1 = 22.6 °C
```

Problem 19 – Insulated pipe

A steel pipe (r_1 = 50 mm, r_2 = 55 mm, k = 50) carries steam; it is insulated to r_3 = 105 mm (k = 0.06). Inner surface 200 °C, outer insulation surface 40 °C. Find the heat loss per metre.

```
R_steel = ln(1.1)/(2π × 50) = 0.0953/314.2 = 3.03 × 10⁻⁴ K/W
R_ins = ln(105/55)/(2π × 0.06) = 0.6466/0.377 = 1.715 K/W
Q = 160/1.7155 = 93.3 W/m (steel resistance is negligible)
```

Problem 20 – Critical radius

Insulation with k = 0.2 W/m·K is exposed to air with h = 10 W/m²·K. Find the critical radius for a cylinder and a sphere.

```
r_c (cylinder) = k/h = 0.02 m = 20 mm; r_c (sphere) = 2k/h = 40 mm
```

Problem 21 – Heat generation in a wire

A wire of radius 1 mm (k = 20) generates 10^8 W/m³; surface temperature 100 °C. Find centre temperature.

```
T_max = T_s + q_g R²/(4k) = 100 + 10⁸ × 10⁻⁶/80 = 100 + 1.25 = 101.25 °C
```

Problem 22 – Lumped cooling of a steel ball

A steel ball (D = 10 mm, k = 40, ρ = 7800, c = 460) at 400 °C is cooled in air at 30 °C with h = 100 W/m²K. Find the time to reach 100 °C.

```
L_c = R/3 = 1.667 × 10⁻³ m; Bi = 100 × 1.667 × 10⁻³/40 = 0.0042 < 0.1 → lumped valid
τ = ρcL_c/h = 7800 × 460 × 1.667 × 10⁻³/100 = 59.8 s
(100 − 30)/(400 − 30) = 0.189 = e^(−t/59.8) → t = 59.8 × 1.665 = 99.6 s ≈ 100 s
```

Problem 23 – Long pin fin

A long aluminium pin fin (D = 10 mm, k = 200) with base excess temperature 80 °C is exposed to h = 20. Find m, heat loss and effectiveness.

```
P = πD = 0.03142 m; A_c = 7.854 × 10⁻⁵ m²; m = √(hP/(kA_c)) = √(0.6283/0.01571) = √40 = 6.32 m⁻¹
Q = √(hPkA_c) θ₀ = √(9.87 × 10⁻³) × 80 = 0.0994 × 80 = 7.95 W
ε = √(kP/(hA_c)) = √4000 = 63.2 (fin is highly worthwhile)
```

27C.4 Convection, Radiation, Heat Exchangers

Problem 24 – Dittus–Boelter

Water is heated in a 25 mm tube with Re = 20,000, Pr = 5, k = 0.6 W/m·K. Find h.

Nu = 0.023 × 20,000^0.8 × 5^0.4 = 0.023 × 2760 × 1.904 = 120.8

h = Nu k/D = 120.8 × 0.6/0.025 = 2900 W/m²K

Problem 25 – Laminar tube flow

Oil flows in a 20 mm tube in fully developed laminar flow with constant wall temperature (k = 0.14). Find h.

Nu = 3.66 → h = 3.66 × 0.14/0.02 = 25.6 W/m²K

Problem 26 – Black body emission

Find the total emissive power and the wavelength of maximum emission of a black body at 1000 K.

$E_b = 5.67 \times 10^{-8} \times 1000^4 = 56.7$ kW/m²; $\lambda_{max} = 2898/1000 = 2.898$ μm

Problem 27 – Effect of doubling temperature

If the absolute temperature of a black body is doubled, by what factor does its emissive power increase?

$E \propto T^4 \rightarrow 2^4 = 16$ times

Problem 28 – Parallel plates and a shield

Two large parallel plates at 800 K and 500 K have ε = 0.8 each. Find the net radiation, and the value with one shield of ε = 0.8.

$\sigma(T_1^4 - T_2^4) = 5.67 \times 10^{-8} \times (4.096 \times 10^{11} - 6.25 \times 10^{10}) = 19{,}680$ W/m²

q = 19,680/(1/0.8 + 1/0.8 − 1) = 19,680/1.5 = 13.1 kW/m²; with one shield: 13.1/2 = 6.56 kW/m²

Problem 29 – View factors

A sphere of diameter 1 m is inside a cubical box of side 2 m. Find F_{12}, F_{21} and F_{22} (1 = sphere, 2 = box).

$F_{12} = 1$; $F_{21} = A_1/A_2 = \pi/24 = 0.131$; $F_{22} = 1 - 0.131 = 0.869$

Problem 30 – LMTD: parallel vs counter flow

Hot fluid 150 → 100 °C; cold fluid 30 → 80 °C. Find LMTD for parallel and counter flow and compare the areas.

Counter: $\Delta T_1 = 150 - 80 = 70$, $\Delta T_2 = 100 - 30 = 70$ → LMTD = 70 °C

Parallel: $\Delta T_1 = 150 - 30 = 120$, $\Delta T_2 = 100 - 80 = 20$ → LMTD = 100/ln 6 = 55.8 °C

A_counter/A_parallel = 55.8/70 = 0.80 (counter flow needs 20% less area)

Problem 31 – NTU of a condenser

A steam condenser has NTU = 2. Find the effectiveness.

ε = 1 − e^(−NTU) = 1 − e^(−2) = 1 − 0.135 = 0.865

Problem 32 – Heat exchanger area

A cooler transfers 50 kW with U = 500 W/m²K and LMTD = 25 °C. Find the area.

A = Q/(U ΔT_lm) = 50,000/(500 × 25) = 4 m²

CHAPTER 27D

Solved Numericals – Set 4 (Mixed Practice with Steam Tables)

Short step-by-step problems that test one concept each (no options). Steam data from the abridged table at the end of the book.

27D.1 Thermodynamics Quick Problems

Problem 1 – New temperature scale

On a new scale X, ice point = 100 °X and steam point = 300 °X. Find the reading for 50 °C.

```
X = 100 + (300 − 100) × 50/100 = 200 °X
```

Problem 2 – Heating in a rigid tank

1 kg of air in a rigid tank is heated from 300 K to 600 K. Find the heat supplied.

```
Q = m c_v ΔT = 0.718 × 300 = 215.4 kJ (W = 0 because volume is constant)
```

Problem 3 – Paddle work in an insulated tank

A paddle wheel does 50 kJ of work on gas in a rigid insulated tank. Find ΔU.

```
Q = 0, W = −50 kJ → ΔU = Q − W = +50 kJ (paddle work is always work input)
```

Problem 4 – Inventor's claim

An inventor claims an engine with η = 65% between 1000 K and 400 K. Comment.

```
η_Carnot = 1 − 400/1000 = 60% < 65% → claim violates the second law (impossible)
```

Problem 5 – Refrigerator power

A refrigerator removes 2 kW from the cold space with COP 4. Find the power and the heat rejected.

```
W = 2/4 = 0.5 kW; Q_H = 2 + 0.5 = 2.5 kW
```

Problem 6 – Heat pump vs electric heater

A heat pump with COP 3 replaces a 6 kW electric heater. Find the electricity saved.

```
Heat pump input = 6/3 = 2 kW → saving = 4 kW (67%)
```

Problem 7 – Entropy change of a reservoir

A reservoir at 500 K loses 1000 kJ of heat. Find its entropy change.

```
ΔS = −1000/500 = −2 kJ/K (reservoirs: ΔS = Q/T)
```

27D.2 Steam and Rankine with Tables

Problem 8 – Dryness from specific volume

Steam at 10 bar has v = 0.15 m³/kg (v_g = 0.1944 m³/kg). Find x.

```
x ≈ v/v_g = 0.15/0.1944 = 0.772
```

Problem 9 – Entropy of wet steam

Find s for steam at 10 bar with x = 0.9 (s_f = 2.1387, s_{fg} = 4.4478 kJ/kg·K).

$s = 2.1387 + 0.9 \times 4.4478 = 6.1417$ kJ/kg·K

Problem 10 – Isentropic expansion and Rankine efficiency

Dry saturated steam at 10 bar (h_g = 2778.1, s_g = 6.5865) expands isentropically to 0.1 bar (s_f = 0.6493, s_{fg} = 7.5009, h_f = 191.8, h_{fg} = 2392.8). Find exit dryness, turbine work and Rankine efficiency.

$x_2 = (6.5865 - 0.6493)/7.5009 = 0.7915$; $h_2 = 191.8 + 0.7915 \times 2392.8 = 2085.7$ kJ/kg

$W_t = 2778.1 - 2085.7 = 692.4$ kJ/kg; $W_p \approx 0.00101 \times (1000 - 10) = 1.0$ kJ/kg

$Q_1 = 2778.1 - (191.8 + 1.0) = 2585.3$ kJ/kg; $\eta = (692.4 - 1.0)/2585.3 = 26.7\%$

- Exit dryness 0.79 is too wet for a real turbine – which is why superheat and reheat are used.

Problem 11 – Carnot vapour cycle between the same pressures

Find the Carnot efficiency between 10 bar (179.91 °C) and 0.1 bar (45.81 °C).

$\eta = 1 - 318.96/453.06 = 29.6\%$ (Rankine 26.7% is lower but practical)

Problem 12 – Heat to make steam

Find the heat needed to convert 1 kg of water at 25 °C (h = 104.9 kJ/kg) into dry saturated steam at 10 bar.

$Q = h_g - h_w = 2778.1 - 104.9 = 2673.2$ kJ/kg

27D.3 IC Engine Quick Problems

Problem 13 – bsfc to efficiency

An engine has bsfc 0.25 kg/kWh with fuel CV 42,000 kJ/kg. Find brake thermal efficiency.

$\eta_{bth} = 3600/(bsfc \times CV) = 3600/(0.25 \times 42{,}000) = 34.3\%$

Problem 14 – Relative efficiency

An SI engine with r = 7 has indicated thermal efficiency 30%. Find air-standard and relative efficiency.

$\eta_{Otto} = 1 - 1/7^{0.4} = 1 - 1/2.178 = 54.1\%$; relative efficiency = 30/54.1 = 55.5%

Problem 15 – Fuel consumption from power

A 100 kW diesel generator runs at 35% brake thermal efficiency with fuel CV 42,500 kJ/kg. Find fuel per hour.

Fuel energy = 100/0.35 = 285.7 kW; $\dot{m}_f = 285.7 \times 3600/42{,}500 = 24.2$ kg/h

Problem 16 – Two-stroke vs four-stroke

Find IP of a two-stroke single-cylinder engine with imep 5 bar, bore 80 mm, stroke 100 mm, 2000 rpm.

$A = \pi/4 \times 0.08^2 = 5.027 \times 10^{-3}$ m²; $IP = 5 \times 10^5 \times 0.1 \times 5.027 \times 10^{-3} \times 2000/60 = 8.38$ kW

27D.4 Fuels and Power Plant

Problem 17 – Air for methane

Find the stoichiometric air–fuel ratio of methane.

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O \rightarrow O_2 = 64/16 = 4$ kg/kg; air = 4/0.232 = 17.2 kg/kg

Problem 18 – Dulong HCV and LCV

Coal: C 80%, H 5%, O 4%, S 1%. Find HCV and LCV (h_fg = 2442 kJ/kg).

```
HCV = 33,800 × 0.8 + 144,000 × (0.05 − 0.005) + 9,270 × 0.01 = 27,040 + 6,480 + 93 = 33,613 kJ/kg
LCV = 33,613 − 9 × 0.05 × 2442 = 33,613 − 1,099 = 32,514 kJ/kg
```

Problem 19 – Excess air from flue gas

Dry flue gas contains 6% O_2 (complete combustion). Estimate excess air.

```
Excess air ≈ O₂/(21 − O₂) = 6/15 = 40%
```

Problem 20 – Load factor and capacity factor

A plant of 120 MW capacity has maximum demand 100 MW and generates 438,000 MWh in a year. Find load factor and capacity factor.

```
Average load = 438,000/8760 = 50 MW; load factor = 50/100 = 50%; capacity factor = 50/120 = 41.7%
Utilisation factor = 100/120 = 83.3%
```

Problem 21 – Diversity factor

The sum of individual maximum demands of consumers is 150 MW while the station maximum demand is 100 MW. Find diversity factor.

```
Diversity factor = 150/100 = 1.5
```

Problem 22 – Cooling tower

Water is cooled from 40 °C to 30 °C; ambient WBT = 25 °C. Find range, approach and efficiency.

```
Range = 10 °C; approach = 30 − 25 = 5 °C; efficiency = range/(range + approach) = 10/15 = 66.7%
```

Problem 23 – Nozzle exit area

Steam flows at 1 kg/s through a nozzle with exit velocity 600 m/s and specific volume 1.5 m^3/kg. Find the exit area.

```
A = ṁ v/V = 1 × 1.5/600 = 2.5 × 10⁻³ m² = 25 cm²
```

27D.5 RAC and Heat Transfer Quick Problems

Problem 24 – Chiller tonnage

A chiller cools 0.5 kg/s of water from 12 °C to 7 °C. Find its capacity in TR.

```
Q = 0.5 × 4.187 × 5 = 10.47 kW = 10.47/3.517 = 2.98 TR
```

Problem 25 – RH from dew point

DBT 30 °C (p_s = 4.246 kPa); DPT 20 °C (p_s = 2.339 kPa). Find RH and ω at 101.325 kPa.

```
φ = 2.339/4.246 = 55.1%; ω = 0.622 × 2.339/(101.325 − 2.339) = 0.0147 kg/kg
```

Problem 26 – Single vs double glazing

Window area 2 m^2, inside 20 °C (h = 10), outside 0 °C (h = 25), glass 5 mm (k = 0.8). Find heat loss; then with double glazing having a 10 mm still-air gap (k = 0.026).

```
Single: R = 1/20 + 0.005/1.6 + 1/50 = 0.05 + 0.0031 + 0.02 = 0.0731 K/W → Q = 20/0.0731 = 274 W
```

Double: $R = 0.0731 + 0.0031 + 0.01/(0.026 \times 2) = 0.0731 + 0.0031 + 0.1923 = 0.2686$ K/W → $Q = 74.5$ W

Problem 27 – Radiation from a person

Skin at 34 °C ($\varepsilon = 0.95$, $A = 1.6$ m²) in a room with walls at 20 °C. Find the radiation loss.

$Q = \varepsilon\sigma A(T_1^4 - T_2^4) = 0.95 \times 5.67 \times 10^{-8} \times 1.6 \times (307.15^4 - 293.15^4) = 0.95 \times 5.67 \times 10^{-8} \times 1.6 \times 1.515 \times 10^9 \approx 131$ W

Problem 28 – Thermocouple time constant

A thermocouple bead (sphere, D = 1 mm, ρ = 8500, c = 400) is exposed to gas with h = 200 W/m²K. Find its time constant.

$\tau = \rho cV/(hA) = \rho cD/(6h) = 8500 \times 400 \times 0.001/1200 = 2.83$ s

Problem 29 – Laminar flat plate h

Air (k = 0.026, Pr = 0.7) flows over a 1 m plate with $Re_L = 2 \times 10^5$. Find average h.

$Nu = 0.664 \times (2 \times 10^5)^{0.5} \times 0.7^{(1/3)} = 0.664 \times 447.2 \times 0.888 = 263.7$; $h = 263.7 \times 0.026/1 = 6.9$ W/m²K

Problem 30 – Heat exchanger effectiveness from temperatures

Hot fluid ($C_h = 2$ kW/K) cools from 120 °C to 60 °C; cold fluid ($C_c = 4$ kW/K) enters at 20 °C. Find the cold outlet temperature and effectiveness.

$Q = 2 \times 60 = 120$ kW; $T_{c,out} = 20 + 120/4 = 50$ °C

$Q_{max} = C_{min}(120 - 20) = 2 \times 100 = 200$ kW; $\varepsilon = 120/200 = 0.6$

Problem 31 – Doubling insulation thickness

A plane wall loses 400 W. If its insulation thickness is doubled (insulation resistance dominant), find the new loss.

$Q \propto 1/L \rightarrow Q \approx 200$ W (halved)

Problem 32 – Shape factor of a long duct

For a long equilateral triangular duct, find the view factor from one side to another.

By symmetry and summation: $F_{12} + F_{13} = 1$ ($F_{11} = 0$) and $F_{12} = F_{13} \rightarrow F_{12} = 0.5$

CHAPTER 27E

Solved Numericals – Set 5 (Advanced Mixed Problems)

Slightly longer problems combining two or three concepts (no options).

27E.1 Thermodynamics and Cycles

Problem 1 – Reversible engine with three reservoirs

A reversible engine receives 1200 kJ from a reservoir at 600 K and rejects heat to reservoirs at 400 K and 300 K, producing 400 kJ of work. Find the heat rejected to each.

Energy: $Q_2 + Q_3 = 1200 - 400 = 800$ kJ; entropy (reversible): $1200/600 = Q_2/400 + Q_3/300 \rightarrow 2 = Q_2/400 + Q_3/300$

Substituting $Q_3 = 800 - Q_2$: $2 = Q_2/400 + (800 - Q_2)/300 \rightarrow 2 = 2.667 - Q_2/1200 \rightarrow Q_2 = 800$ kJ, $Q_3 = 0$

Problem 2 – Maximum work from two bodies

Two identical blocks (mc = 1 kJ/K) at 600 K and 300 K are used as source and sink for a reversible engine until they reach a common temperature. Find T_f and the work.

Reversible: $\Delta S = 0 \rightarrow T_f = \sqrt{(600 \times 300)} = 424.3$ K

$W = mc(T_1 + T_2 - 2T_f) = 1 \times (900 - 848.5) = 51.5$ kJ (direct contact would give $T_f = 450$ K and $W = 0$)

Problem 3 – Air compressed in a cylinder with heat loss

0.5 kg of air is compressed from 1 bar, 300 K to 8 bar polytropically (n = 1.25). Find the work, the heat transferred and the entropy change.

$T_2 = 300 \times 8^{(0.25/1.25)} = 300 \times 8^{0.2} = 300 \times 1.516 = 454.8$ K

$W = mR(T_1 - T_2)/(n - 1) = 0.5 \times 0.287 \times (-154.8)/0.25 = -88.9$ kJ

$Q = W(\gamma - n)/(\gamma - 1) = -88.9 \times 0.15/0.4 = -33.3$ kJ

$\Delta S = m[c_p \ln(T_2/T_1) - R \ln(p_2/p_1)] = 0.5 \times [1.005 \times 0.4160 - 0.287 \times 2.0794] = 0.5 \times (0.4181 - 0.5968) = -0.0894$ kJ/K

Problem 4 – Dual cycle efficiency

A dual cycle has r = 14, pressure ratio in constant-volume heating α = 1.5 and cut-off ratio ρ = 1.6. Find η.

$\eta = 1 - (1/r^{(\gamma-1)}) \times (\alpha\rho^{\gamma} - 1)/[(\alpha - 1) + \gamma\alpha(\rho - 1)]$

$r^{0.4} = 14^{0.4} = 2.874$; $\rho^{1.4} = 1.6^{1.4} = 1.930$; numerator $= 1.5 \times 1.930 - 1 = 1.895$

denominator $= 0.5 + 1.4 \times 1.5 \times 0.6 = 0.5 + 1.26 = 1.76$; $\eta = 1 - (1/2.874) \times 1.077 = 1 - 0.375 = 62.5\%$

Problem 5 – Mean effective pressure of an Otto cycle

An Otto cycle (r = 8) has swept volume 0.0005 m³ and net work 0.4 kJ per cycle. Find the mep.

$mep = W/V_s = 400/0.0005 = 8 \times 10^5$ Pa = 8 bar

Problem 6 – Stirling engine with ideal regenerator

A Stirling engine works between 800 K and 300 K. Find its efficiency with a perfect regenerator.

```
η = 1 − 300/800 = 62.5% (equal to Carnot)
```

27E.2 Steam Plant

Problem 7 – Reheat cycle

Steam enters the HP turbine at h_1 = 3450 kJ/kg and expands to h_2 = 3000 kJ/kg; it is reheated to h_3 = 3550 kJ/kg and expands in the LP turbine to h_4 = 2400 kJ/kg. Feed enthalpy after pump h_6 = 200 kJ/kg; pump work 10 kJ/kg. Find η.

```
W_t = (3450 − 3000) + (3550 − 2400) = 450 + 1150 = 1600 kJ/kg; W_net = 1590 kJ/kg
Q = (3450 − 200) + (3550 − 3000) = 3250 + 550 = 3800 kJ/kg; η = 1590/3800 = 41.8%
```

Problem 8 – Steam rate and plant size

A 100 MW plant has a net work of 1000 kJ/kg of steam. Find the steam flow rate.

```
ṁ = 100,000/1000 = 100 kg/s = 360 t/h; SSC = 3600/1000 = 3.6 kg/kWh
```

Problem 9 – Heat rate

A power plant has an overall efficiency of 36%. Find its heat rate in kJ/kWh.

```
Heat rate = 3600/η = 3600/0.36 = 10,000 kJ/kWh
```

Problem 10 – Coal consumption

For Problem 9, with coal CV 20,000 kJ/kg and output 500 MW, find coal per hour.

```
Energy input = 500,000 kW × 10,000 kJ/kWh = 5 × 10⁹ kJ/h → coal = 5 × 10⁹/20,000 = 2.5 × 10⁵ kg/h = 250 t/h
```

Problem 11 – Blade work from velocity triangles

In an impulse stage, V_w1 = 550 m/s and V_w2 = −100 m/s (opposite direction), blade speed u = 250 m/s, steam flow 2 kg/s. Find power and blade efficiency if V_1 = 600 m/s.

```
Work per kg = u(V_w1 + |V_w2|) = 250 × 650 = 162,500 J/kg; power = 2 × 162.5 = 325 kW
η_b = 2u(ΣV_w)/V₁² = 2 × 162,500/360,000 = 90.3%
```

Problem 12 – Boiler efficiency with losses

A boiler's heat losses are: dry flue gas 12%, moisture and hydrogen 5%, unburnt carbon 2%, radiation and others 3%. Find efficiency (indirect method).

```
η = 100 − (12 + 5 + 2 + 3) = 78%
```

27E.3 Compressors and Gas Turbines

Problem 13 – FAD and mass of air

A compressor delivers 10 m³/min of free air at 1.013 bar and 27 °C. Find the mass flow.

```
ṁ = pV̇/(RT) = 101.3 × (10/60)/(0.287 × 300) = 16.88/86.1 = 0.196 kg/s
```

Problem 14 – Swept volume from volumetric efficiency

A single-acting compressor running at 600 rpm must deliver 3 m³/min of free air with η_v = 0.8. Find the swept volume.

```
V_s = 3/(0.8 × 600) = 6.25 × 10⁻³ m³ = 6.25 litres
```

Problem 15 – Three-stage compression

Air is compressed from 1 bar to 64 bar in three stages with perfect intercooling. Find the stage pressures.

```
Stage ratio = 64^(1/3) = 4 → 1 → 4 → 16 → 64 bar
```

Problem 16 – Gas turbine air–fuel ratio

In a gas turbine, air leaves the compressor at 550 K and gases leave the combustor at 1200 K (c_p = 1.005 for both, combustion efficiency 100%, CV = 42,000 kJ/kg). Find the air–fuel ratio.

```
(m_a + m_f) c_p T₃ − m_a c_p T₂ = m_f CV → per kg fuel: (A + 1) × 1.005 × 1200 − A × 1.005 × 550 = 42,000
1206(A + 1) − 552.75 A = 42,000 → 653.25 A = 40,794 → A/F ≈ 62.4
```

- Very lean overall (stoichiometric ≈ 15) – excess air limits turbine inlet temperature.

Problem 17 – Thrust from a turbojet with fuel

Air 60 kg/s enters at 200 m/s; fuel 1.2 kg/s; exit jet velocity 700 m/s (pressure thrust zero). Find thrust.

```
F = (ṁ_a + ṁ_f)V_j − ṁ_a V_a = 61.2 × 700 − 60 × 200 = 42,840 − 12,000 = 30,840 N ≈ 30.8 kN
```

27E.4 Refrigeration and Air Conditioning

Problem 18 – VCR with subcooling

In a VCR plant, h_1 = 1450, h_2 = 1650 kJ/kg; condensate is subcooled so that h_3 = 300 instead of 360 kJ/kg. Find the percentage increase in COP.

```
Without subcooling: COP = (1450 − 360)/200 = 5.45; with: (1450 − 300)/200 = 5.75
Increase = (5.75 − 5.45)/5.45 = 5.5%
```

Problem 19 – Compressor displacement

A 5 TR ammonia plant has RE = 1100 kJ/kg and specific volume at suction 0.5 m³/kg; volumetric efficiency 0.8. Find the compressor swept volume per minute.

```
ṁ = 5 × 211/1100 = 0.959 kg/min; actual suction volume = 0.959 × 0.5 = 0.480 m³/min
Swept volume = 0.480/0.8 = 0.60 m³/min
```

Problem 20 – Cooling load from people and lights

An office has 30 people (sensible 75 W, latent 55 W each), lights 2 kW (fluorescent, factor 1.25) and equipment 3 kW. Find the sensible, latent and total load and the room SHF.

```
Sensible = 30 × 75 + 2000 × 1.25 + 3000 = 2250 + 2500 + 3000 = 7750 W; latent = 30 × 55 = 1650 W
Total = 9.4 kW = 2.67 TR; RSHF = 7750/9400 = 0.82
```

Problem 21 – Supply air quantity

The room in Problem 20 is kept at 25 °C; supply air enters at 14 °C. Find the supply air mass flow.

```
ṁ = Q_s/(c_pa ΔT) = 7.75/(1.022 × 11) = 0.689 kg/s (≈ 0.6 m³/s)
```

Problem 22 – Ventilation load

0.5 kg/s of outdoor air at h = 85 kJ/kg replaces room air at h = 50 kJ/kg. Find the fresh-air load.

```
Q = 0.5 × (85 − 50) = 17.5 kW ≈ 5 TR
```

Problem 23 – Product load in cold storage

2000 kg of fish (c above freezing 3.2, below freezing 1.7 kJ/kg·K, latent heat 245 kJ/kg, freezing point −2 °C) is cooled from 18 °C to −18 °C in 24 h. Find the average load in TR.

```
Q = 2000[3.2 × 20 + 245 + 1.7 × 16] = 2000[64 + 245 + 27.2] = 672,400 kJ
Rate = 672,400/(24 × 60) = 467 kJ/min = 2.21 TR
```

27E.5 Heat Transfer

Problem 24 – Interface temperature

A wall has two layers: A (L = 0.1 m, k = 1) and B (L = 0.05 m, k = 0.05). Inner surface 300 °C, outer 30 °C. Find the heat flux and interface temperature.

```
R = 0.1/1 + 0.05/0.05 = 0.1 + 1.0 = 1.1 m²K/W; q = 270/1.1 = 245.5 W/m²
T_interface = 300 − 245.5 × 0.1 = 275.5 °C
```

Problem 25 – Insulated wire (critical radius check)

A 2 mm diameter wire is insulated with plastic (k = 0.15) in air with h = 10 W/m²K. Does insulation increase or decrease heat loss? What thickness gives maximum heat loss?

```
r_c = k/h = 0.015 m = 15 mm > wire radius 1 mm → adding insulation up to 15 mm radius increases heat loss
Thickness for maximum loss = 15 − 1 = 14 mm
```

Problem 26 – Fin with insulated tip

A rectangular aluminium fin (k = 200) has m = 10 m^{-1} and length 0.1 m. Find fin efficiency.

```
mL = 1; η = tanh(1)/1 = 0.762 = 76.2%
```

Problem 27 – Natural convection order of magnitude

A vertical plate has Gr = 10^8 and Pr = 0.7. Is the flow laminar or turbulent?

```
Ra = Gr × Pr = 7 × 10⁷ < 10⁹ → laminar (use Nu ∝ Ra^(1/4))
```

Problem 28 – Grey body exchange in enclosure

A pipe (D = 0.1 m, ε = 0.8) at 500 K runs in a large room at 300 K. Find radiation loss per metre.

```
A = π × 0.1 × 1 = 0.3142 m²; Q = 0.8 × 5.67 × 10⁻⁸ × 0.3142 × (500⁴ − 300⁴)
500⁴ − 300⁴ = 6.25 × 10¹⁰ − 0.81 × 10¹⁰ = 5.44 × 10¹⁰ → Q = 1.425 × 10⁻⁸ × 5.44 × 10¹⁰ = 775 W/m
```

Problem 29 – Overall U with fouling

Clean U = 1000 W/m²K; fouling resistance 0.0004 m²K/W is added. Find the dirty U.

```
1/U_d = 1/1000 + 0.0004 = 0.0014 → U_d = 714 W/m²K (29% reduction)
```

Problem 30 – NTU and area for a counter-flow exchanger

Counter-flow exchanger with C_min = C_max = 2 kW/K needs ε = 0.75 and U = 500 W/m²K. Find NTU and area.

```
C_r = 1: ε = NTU/(1 + NTU) → 0.75 = NTU/(1 + NTU) → NTU = 3
A = NTU × C_min/U = 3 × 2000/500 = 12 m²
```

CHAPTER 28

Concept Recall Bank

Short question → crisp answer. Cover the right column and test yourself (active recall, not MCQs).

R1. Basic Thermodynamics and First Law

Question	Answer
System exchanging neither mass nor energy?	Isolated system
System exchanging energy but not mass?	Closed system (control mass)
Turbine, compressor, nozzle are which systems?	Open systems (control volume)
Intensive property examples?	Pressure, temperature, density, specific volume
Extensive property examples?	Mass, volume, energy, enthalpy, entropy
Is the ratio of two extensive properties intensive?	Yes (e.g. specific volume = V/m)
Basis of temperature measurement?	Zeroth law
Heat and work are path or point functions?	Path functions (inexact differentials)
Thermodynamic properties are?	Point (state) functions – exact differentials
Quasi-static process means?	Infinitely slow, passes through equilibrium states
Work in free expansion?	Zero
Work in a constant-volume process?	Zero ($\int p\,dV = 0$)
Area under p–V curve represents?	Work (non-flow)
Area under T–s curve represents?	Heat (reversible)
First law for a cycle?	$\oint \delta Q = \oint \delta W$
First law for a process?	$Q = \Delta U + W$
Internal energy of an ideal gas depends on?	Temperature only (Joule's law)
PMM-1 violates?	First law
Enthalpy definition?	$H = U + pV$
$c_p - c_v$ for ideal gas?	R (Mayer's relation)
γ for monatomic / diatomic gas?	1.67 / 1.4
Universal gas constant?	8.314 kJ/kmol·K
R for air?	0.287 kJ/kg·K
c_p and c_v of air?	1.005 and 0.718 kJ/kg·K
Isothermal process for ideal gas: ΔU?	Zero; $Q = W$
Adiabatic process: Q?	Zero; $W = -\Delta U$
Polytropic index n = 0, 1, γ, ∞?	Isobaric, isothermal, adiabatic, isochoric
Heat transfer in polytropic process?	$Q = W\,(\gamma - n)/(\gamma - 1)$
Slope of adiabatic vs isothermal on p–V?	Adiabatic is γ times steeper
Steady-flow energy equation for nozzle?	$V_2 = \sqrt{(2(h_1 - h_2))} = 44.72\sqrt{\Delta h}$ (Δh in kJ/kg)
Throttling process?	Isenthalpic ($h_1 = h_2$), irreversible
Joule–Thomson coefficient?	$\mu = (\partial T/\partial p)_h$; zero for ideal gas

Question	Answer
Inversion temperature?	Above it, throttling heats the gas
Flow work?	pv – energy to push fluid across boundary
Steady-flow work of reversible process?	$-\int v\,dp$

R2. Second Law, Entropy, Availability

Question	Answer
Kelvin–Planck statement?	No engine can produce work while exchanging heat with a single reservoir
Clausius statement?	Heat cannot flow from cold to hot without external work
PMM-2 violates?	Second law (Kelvin–Planck)
Carnot efficiency?	$1 - T_L/T_H$
Carnot efficiency depends on?	Only reservoir temperatures, not working fluid
Carnot cycle processes?	Two isothermal + two reversible adiabatic
Why Carnot cycle impractical?	Isothermal heat transfer too slow, adiabatic too fast; low work ratio
COP of Carnot refrigerator?	$T_L/(T_H - T_L)$
COP of heat pump vs refrigerator?	$COP_{HP} = COP_R + 1$
Clausius inequality?	$\oint \delta Q/T \le 0$
Entropy change of reversible adiabatic process?	Zero (isentropic)
Entropy of isolated system?	Never decreases (increase of entropy principle)
Entropy of universe in real process?	Increases
Third law?	Entropy of a perfect crystal at 0 K is zero
Entropy unit?	kJ/K (specific: kJ/kg·K)
Entropy change of ideal gas?	$c_p \ln(T_2/T_1) - R \ln(p_2/p_1)$
Δs in isothermal ideal gas expansion?	$R \ln(v_2/v_1)$
Available energy?	$Q(1 - T_0/T)$
Irreversibility (Gouy–Stodola)?	$I = T_0 S_{gen}$
Dead state?	Equilibrium with surroundings (T_0, p_0)
Gibbs function?	$G = H - TS$
Helmholtz function?	$F = U - TS$
Exergy of heat at T_0?	Zero
Two engines in series, equal efficiency – T?	$\sqrt{(T_1 T_2)}$
Two engines in series, equal work – T?	$(T_1 + T_2)/2$
Maxwell relations derive from?	Exactness of dU, dH, dF, dG
Clausius–Clapeyron equation?	$dp/dT = h_{fg}/(T\,v_{fg})$

R3. Pure Substances and Steam

Question	Answer
Triple point of water?	0.01 °C, 0.6117 kPa
Critical point of water?	374.14 °C, 22.09 MPa
Latent heat at critical point?	Zero

Question	Answer
Dryness fraction x?	Mass of vapour/total mass
Enthalpy of wet steam?	h_f + x h_fg
Latent heat of water at 1 atm?	2257 kJ/kg
Best calorimeter for very wet steam?	Combined separating and throttling
Throttling calorimeter requires?	Steam to become superheated after throttling
Mollier diagram axes?	h (y) vs s (x)
Constant pressure lines in wet region on h–s?	Straight lines
Sublimation occurs below?	Triple point pressure
Dry ice sublimation temperature?	−78.5 °C at 1 atm
Degree of superheat?	T_sup − T_sat
Specific heat of superheated steam (approx.)?	≈ 2.1 kJ/kg·K

R4. Gas Cycles and IC Engines

Question	Answer
Otto efficiency?	1 − 1/r^(γ−1)
Otto efficiency depends on?	Only compression ratio (and γ)
Diesel efficiency vs cut-off ratio?	Decreases as cut-off increases
For same r and heat input, efficiency order?	Otto > Dual > Diesel
For same max pressure and temperature?	Diesel > Dual > Otto
Dual cycle also called?	Limited-pressure / mixed cycle
Brayton efficiency?	1 − 1/r_p^((γ−1)/γ)
Optimum r_p for max Brayton work?	(T_max/T_min)^(γ/2(γ−1))
Stirling cycle processes?	Two isothermal + two constant volume with regeneration
Ericsson cycle processes?	Two isothermal + two constant pressure with regeneration
Atkinson cycle feature?	Expansion ratio > compression ratio
SI engine works on?	Otto cycle
CI engine works on?	Diesel/dual cycle
Typical compression ratio SI / CI?	6–11 / 14–22
Power stroke per cycle in four-stroke?	One per two revolutions
Power stroke in two-stroke?	One per revolution
Scavenging is in?	Two-stroke engines
Port timing in two-stroke uses?	Piston covers/uncovers ports
Inlet valve opens?	≈ 10–20° before TDC
Exhaust valve opens?	≈ 35–45° before BDC
Valve overlap?	Both valves open near TDC
Detonation occurs in?	SI engine (end-gas autoignition)
Knock in CI engine due to?	Long ignition delay
Octane number reference fuels?	Iso-octane (100) and n-heptane (0)
Cetane number reference fuels?	Cetane (100) and α-methyl naphthalene (0)

Question	Answer
Anti-knock additive (old)?	Tetraethyl lead (TEL) – banned
SI knock reduced by?	Lower r, retarded spark, high octane, cooling, turbulence
CI knock reduced by?	High cetane, high r, high intake temperature, reduced delay
Fuel for SI requires?	High octane (resists autoignition)
Fuel for CI requires?	High cetane (easy autoignition)
Carburettor works on?	Venturi principle
Fuel injection in CI?	Injector/nozzle with high-pressure pump (MPFI/CRDI)
Governing in SI?	Quantity governing (throttle)
Governing in CI?	Quality governing (fuel quantity)
Firing order of 4-cyl engine?	1-3-4-2 (or 1-2-4-3)
Mean effective pressure?	Net work/swept volume
Mechanical efficiency?	BP/IP
Morse test is for?	Multi-cylinder engines – IP
Willan's line method?	Friction power of CI engine (fuel vs BP extrapolation)
Volumetric efficiency?	Actual air/swept volume air
Supercharging?	Increasing intake air density (more power)
Turbocharger driven by?	Exhaust gas turbine
Catalytic converter reduces?	CO, HC, NOx (three-way)
EGR reduces?	NOx
Rich mixture A/F (petrol)?	Less than 14.7
bsfc unit?	kg/kWh
Best power mixture?	Slightly rich
Best economy mixture?	Slightly lean

R5. Steam Power Plant

Question	Answer
Rankine cycle processes?	Pump, boiler, turbine, condenser
Rankine efficiency increased by?	Higher boiler p, superheat, lower condenser p, reheat, regeneration
Reheating mainly increases?	Dryness at turbine exit (and work)
Regeneration mainly increases?	Efficiency (mean temperature of heat addition)
Open feed water heater also called?	Deaerator / direct contact heater
Work ratio of Rankine cycle?	Close to 1 (pump work small)
Fire-tube boilers?	Cochran, Lancashire, Cornish, locomotive, Scotch marine
Water-tube boilers?	Babcock & Wilcox, Stirling, La Mont, Benson, Loeffler
Boiler without drum?	Benson (supercritical)
Loeffler boiler feature?	Steam used to evaporate water

Question	Answer
La Mont boiler?	Forced circulation
Fusible plug function?	Melts to put out fire when water level low
Economiser heats?	Feed water (using flue gas)
Air preheater heats?	Combustion air
Superheater?	Converts saturated steam to superheated
Equivalent evaporation reference?	From and at 100 °C (2257 kJ/kg)
Natural draught depends on?	Chimney height and gas/air density difference
Induced draught fan location?	Near chimney base (handles hot gases)
Forced draught fan location?	Before furnace (cold air)
Balanced draught?	FD + ID fans
Critical pressure ratio – superheated steam?	0.546
Critical pressure ratio – saturated steam?	0.577
Critical pressure ratio – air?	0.528
Supersaturated flow in nozzle?	Metastable – steam expands without condensing
Convergent nozzle max exit Mach?	1 (choked)
De Laval turbine?	Simple single-stage impulse
Curtis turbine?	Velocity compounded impulse
Rateau turbine?	Pressure compounded impulse
Parsons turbine?	50% reaction
Optimum u/V_1 impulse?	$\cos\alpha/2$
Optimum u/V_1 Parsons?	$\cos\alpha$
Max blade efficiency – impulse?	$\cos^2\alpha$
Degree of reaction?	Enthalpy drop in moving blades/stage drop
Nozzle governing?	Controls number of nozzles open
Throttle governing?	Throttles steam pressure – Willan's line straight
Condenser improves efficiency by?	Lowering back pressure
Dalton's law in condenser?	Total p = p_steam + p_air
Surface condenser advantage?	Condensate reused as feed
Jet condenser?	Cooling water and steam mix
Cooling tower types?	Natural draught (hyperbolic), mechanical draught
Approach of cooling tower?	Outlet water T − WBT of air
Range of cooling tower?	Inlet − outlet water temperature

R6. Power Plants

Question	Answer
Moderator in thermal reactor?	Heavy water, graphite, light water
Coolant in PWR?	Pressurised light water
Control rods material?	Boron, cadmium, hafnium
Fast breeder reactor coolant?	Liquid sodium
India's three-stage programme fuel ends with?	Thorium
Load factor?	Average load/maximum demand

Question	Answer
Plant capacity factor?	Actual energy/(capacity × time)
Diversity factor?	Sum of max demands/simultaneous max demand (>1)
Peak load plants?	Gas turbine, pumped storage, diesel
Base load plants?	Nuclear, large coal
Combined cycle?	Gas turbine exhaust → HRSG → steam turbine
ESP removes?	Fly ash (particulates)
FGD removes?	SO_2

R7. Compressors and Gas Turbines

Question	Answer
Minimum work compression?	Isothermal
Effect of clearance on volumetric efficiency?	Decreases it
Clearance effect on work per kg (ideal)?	None
Optimum intermediate pressure (2-stage)?	$\sqrt{(p_1 p_3)}$
Perfect intercooling means?	Air cooled back to initial temperature
Free air delivery (FAD)?	Air delivered referred to intake/ambient conditions
Centrifugal compressor slip?	Whirl at exit less than blade speed
Axial compressor degree of reaction?	Usually 50%
Surging?	Flow instability/reversal at low flow
Choking?	Mass flow maximum (sonic)
Stalling?	Flow separation from blades
Gas turbine regeneration useful when?	Turbine exhaust T > compressor exit T
Turbojet thrust?	$\dot{m}(V_j - V_a)$
Propulsive efficiency?	$2V_a/(V_j + V_a)$
Rocket carries?	Both fuel and oxidiser
Ramjet needs?	Initial forward speed (no compressor)

R8. Refrigeration and Air Conditioning

Question	Answer
1 TR equals?	3.5167 kW = 211 kJ/min
VCR components?	Compressor, condenser, expansion valve, evaporator
Process in expansion valve?	Isenthalpic throttling
Subcooling effect?	Increases RE and COP
Superheating effect?	Increases RE but may change COP; protects compressor
COP falls when?	Evaporator T decreases or condenser T increases
Bell–Coleman cycle used in?	Aircraft cooling
Aircraft refrigeration type?	Air refrigeration (simple, bootstrap)
Electrolux third fluid?	Hydrogen
LiBr–water refrigerant?	Water

Question	Answer
Aqua-ammonia refrigerant?	Ammonia
R-717?	Ammonia
R-744?	CO_2
R-718?	Water
R-134a formula?	CH_2FCF_3 ($C_2H_2F_4$)
Refrigerant used in domestic fridge now?	R-134a, R-600a
Ammonia incompatible with?	Copper and its alloys
Leak test for ammonia?	Sulphur candle
Leak test for Freon?	Halide torch (green flame)
Ozone treaty?	Montreal Protocol (1987)
HFC phase-down?	Kigali Amendment (2016)
Capillary tube used in?	Small domestic units
Thermostatic expansion valve maintains?	Constant superheat
Automatic expansion valve maintains?	Constant evaporator pressure
Peltier effect used in?	Thermoelectric refrigeration
DBT, WBT, DPT for saturated air?	All equal
Specific humidity formula?	0.622 p_v/(p_b − p_v)
Sensible heating on chart?	Horizontal line
Evaporative cooling follows?	Constant WBT line
Chemical dehumidification?	DBT rises, ω falls (constant h)
Winter AC process?	Heating + humidification
Summer AC process?	Cooling + dehumidification
Bypass factor?	$(t_2 - t_ADP)/(t_1 - t_ADP)$
SHF?	Sensible/(sensible + latent)
Comfort condition (summer)?	≈ 24–25 °C DBT, 50% RH
Sling psychrometer measures?	DBT and WBT

R9. Heat Transfer

Question	Answer
Fourier's law?	q = −kA dT/dx
Highest k (metal)?	Silver
k of gases with temperature?	Increases
Thermal diffusivity?	k/(ρc_p)
Critical radius (cylinder)?	k/h
Critical radius (sphere)?	2k/h
Fins placed on?	Side with lower h
Fin justified if effectiveness?	> 2
Lumped analysis valid if?	Bi < 0.1
L_c of sphere?	R/3
Biot number?	hL/k_solid
Nusselt number?	hL/k_fluid

Question	Answer
Prandtl number?	ν/α
Pr of air?	≈ 0.71
Pr of liquid metals?	Very low (0.01)
Grashof number significance?	Buoyancy/viscous – free convection
Stanton number?	Nu/(Re Pr)
Laminar tube Nu (const T_w / const q)?	3.66 / 4.36
Dittus–Boelter exponent on Pr (heating)?	0.4
Maximum h in boiling?	Nucleate boiling
Dropwise vs filmwise h?	Dropwise higher
Burnout point?	Critical heat flux
Stefan–Boltzmann constant?	5.67×10^{-8} W/m²K⁴
Wien's constant?	2898 μm·K
Kirchhoff's law?	ε = α at equilibrium
Grey body?	ε constant < 1 for all λ
F_{11} for flat/convex surface?	0
Reciprocity theorem?	$A_1F_{12} = A_2F_{21}$
n shields with same ε reduce heat to?	1/(n + 1)
LMTD formula?	$(\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$
Better HX arrangement?	Counter flow
ε of condenser/evaporator?	1 − e^(−NTU)
NTU?	UA/C_min
Fouling factor?	1/U_dirty − 1/U_clean

CHAPTER 28B

Concept Recall Bank – Part 2

Quick recall for fuels, instruments, modern engines, RAC applications and numerical shortcuts.

R10. Fuels and Combustion

Question	Answer
Coal with highest carbon content?	Anthracite
Lowest rank coal?	Peat
Proximate analysis gives?	Moisture, VM, ash, fixed carbon
Ultimate analysis gives?	C, H, N, S, O
Fixed carbon found by?	Difference
HCV − LCV equals?	Latent heat of water formed (9H × h_fg)
Calorimeter for coal/oil?	Bomb calorimeter
Calorimeter for gas?	Junkers gas calorimeter
Fuel with highest CV per kg?	Hydrogen
O_2 needed per kg carbon?	8/3 kg
O_2 needed per kg hydrogen?	8 kg
Water formed per kg hydrogen?	9 kg
O_2 in air by mass / volume?	23.2% / 21%
Stoichiometric A/F of methane?	17.2
Orsat absorbent for CO_2?	KOH solution
Orsat absorbent for O_2?	Alkaline pyrogallol
Orsat absorbent for CO?	Ammoniacal cuprous chloride
Producer gas main combustible?	CO
Water gas composition?	CO + H_2
Biogas main constituent?	Methane
LPG constituents?	Propane and butane
Flash point vs fire point?	Flash point lower
Pour point vs cloud point?	Pour point lower
Excess air for pulverised coal?	15–25%

R11. Measurements

Question	Answer
Thermocouple principle?	Seebeck effect
Thermocouple for low temperature?	Type T (copper–constantan)
Most common thermocouple?	Type K (chromel–alumel)
RTD material?	Platinum
Thermistor coefficient?	Mostly negative (NTC)

Question	Answer
Optical pyrometer principle?	Brightness matching at one wavelength
Total radiation pyrometer based on?	Stefan–Boltzmann law
Common boiler pressure gauge?	Bourdon tube
Gauge calibration device?	Dead-weight tester
Vacuum gauge standard?	McLeod gauge
Pirani gauge principle?	Thermal conductivity of gas
Dynamic cylinder pressure measured by?	Piezoelectric transducer
Variable-area flow meter?	Rotameter
Brake power of large engines measured by?	Hydraulic (Froude) dynamometer
Dynamometer that can also motor the engine?	Swinging-field DC dynamometer
HC in exhaust measured by?	Flame ionisation detector (FID)
NOx measured by?	Chemiluminescence analyser
CO/CO_2 measured by?	NDIR analyser
Diesel smoke meters?	Bosch, Hartridge
k of insulating material measured by?	Guarded hot-plate apparatus
Flash point apparatus?	Pensky–Martens (closed cup)
Instrument for DBT and WBT?	Sling psychrometer

R12. Modern Engines and Emissions

Question	Answer
CRDI rail pressure?	≈ 1,600–2,500 bar
Pilot injection reduces?	Ignition delay, noise and knock
MPFI injects fuel at?	Intake port of each cylinder
GDI injects fuel?	Directly into the cylinder
Lambda sensor material?	Zirconia
Three-way catalyst metals?	Platinum, palladium, rhodium
Three-way catalyst needs?	Stoichiometric mixture (λ ≈ 1)
SCR reductant?	Urea (AdBlue) → ammonia
DPF traps?	Soot (particulate matter)
EGR reduces?	NOx
BS-VI implemented from?	1 April 2020
BS-VI fuel sulphur limit?	10 ppm
Knock sensor type?	Piezoelectric
Wankel engine rotor shape?	Triangular (Reuleaux)
Wankel weakness?	Apex seal leakage, high HC
Stirling engine combustion?	External
Engine fully balanced (in-line)?	Six-cylinder
Heaviest flywheel needed?	Single-cylinder four-stroke
Secondary force frequency?	Twice crank speed
Power-split hybrid uses?	Planetary gear set

R13. RAC Applications and Air Distribution

Question	Answer
Frozen food storage temperature?	−18 °C or lower
Quick freezing benefit?	Small ice crystals, less drip loss
Brine in ice plant?	NaCl or $CaCl_2$ solution
Ice plant refrigerant?	Ammonia
Inverter AC compressor?	Variable speed (BLDC)
VRF system?	One outdoor unit, many indoor units
Chilled water supply/return?	≈ 7 °C / 12 °C
EER?	Cooling output/power input
Indian AC star rating basis?	ISEER
Most common duct design method?	Equal friction
Duct design for high-velocity systems?	Static regain
Fan power vs speed?	$P \propto N^3$
Fan pressure vs speed?	$\Delta p \propto N^2$
HEPA efficiency?	99.97% at 0.3 µm
Throw of a grille?	Distance until jet velocity ≈ 0.25 m/s
Liquid nitrogen boiling point?	−196 °C
Lowest boiling point gas?	Helium (4.2 K)
Claude cycle adds?	Expansion engine
Heat pipe liquid return?	Capillary action through wick
Vapour barrier for cold insulation?	On the warm side
Economic insulation thickness?	Minimum total (insulation + heat loss) cost
Fick's law analogue?	Fourier's law
Sherwood number analogue?	Nusselt number
Schmidt number analogue?	Prandtl number

R14. Numerical Shortcuts

Situation	Shortcut
Nozzle velocity	$44.72\sqrt{\Delta h}$ (Δh in kJ/kg)
Brake thermal efficiency from bsfc	3600/(bsfc × CV)
kW per TR	3.517/COP
Isothermal work	$p_1V_1 \ln(p_1/p_2) = mRT \ln(V_2/V_1)$
Polytropic heat	$W(\gamma - n)/(\gamma - 1)$
Two engines in series, equal η	$\sqrt{(T_1T_2)}$
Optimum intermediate pressure	$\sqrt{(p_1p_3)}$
Brayton max work r_p	(T_max/T_min)^(γ/2(γ−1)); for air $(T_3/T_1)^{1.75}$
Radiation doubles T	E becomes 16 times
n equal shields	q/(n + 1)
Counter flow with C_h = C_c	LMTD = constant ΔT
Excess air from O_2%	$O_2/(21 - O_2)$
Fin effectiveness (long)	$\sqrt{(kP/(hA))}$

Situation	Shortcut
Lumped sphere L_c	R/3 = D/6
Cooling tower efficiency	Range/(range + approach)

CHAPTER 29

Paper-II Model Answers

Descriptive answers in the length expected for SSC JE Paper-II (conventional) – learn the structure: definition → explanation/sketch → equation → application.

P1. Thermodynamics

✎ **DESCRIPTIVE Q:** State the first law of thermodynamics for a closed system undergoing a cycle and a process, and show that internal energy is a property.

For a closed system executing a cycle, the net heat transfer equals the net work transfer: $\oint \delta Q = \oint \delta W$. For a process, the difference between heat supplied and work done appears as a change in stored energy: $Q - W = \Delta E$ (= ΔU when kinetic and potential energy changes are negligible). To show that U is a property, take a system from state 1 to state 2 by path A and return by path C; then take it from 1 to 2 by path B and return by path C. Applying the first law to both cycles: $Q_A + Q_C = W_A + W_C$ and $Q_B + Q_C = W_B + W_C$. Subtracting, $(Q - W)_A = (Q - W)_B$. Since A and B are arbitrary, $Q - W$ has the same value for every path between the same two states; it therefore depends only on the end states and is a point function – the change of a property called internal energy, $dU = \delta Q - \delta W$. Heat and work individually are path functions, but their difference is not.

✎ **DESCRIPTIVE Q:** Derive the steady-flow energy equation and apply it to a turbine, nozzle and throttling valve.

Consider a control volume with one inlet and one outlet under steady conditions (mass flow, properties and energy content inside do not change with time). Energy entering per unit mass = $h_1 + V_1^2/2 + gz_1$ (enthalpy includes flow work p_1v_1) plus heat q; energy leaving = $h_2 + V_2^2/2 + gz_2$ plus shaft work w. Hence $q - w = (h_2 - h_1) + (V_2^2 - V_1^2)/2 + g(z_2 - z_1)$ per kg. Turbine: adiabatic, KE and PE changes small → $w = h_1 - h_2$, the work output equals the enthalpy drop. Nozzle: no work, adiabatic, PE negligible → $V_2 = \sqrt{(2(h_1 - h_2) + V_1^2)}$; with negligible inlet velocity $V_2 = 44.72\sqrt{(h_1 - h_2)}$ with Δh in kJ/kg. Throttling valve: no work, adiabatic, negligible KE change → $h_1 = h_2$, so throttling is an isenthalpic, irreversible process with a pressure drop (temperature constant for an ideal gas, usually falls for real gases below the inversion temperature).

✎ **DESCRIPTIVE Q:** Explain the Carnot cycle with p–v and T–s diagrams and prove that no engine can be more efficient than a reversible engine between the same reservoirs.

The Carnot cycle consists of four reversible processes: isothermal heat addition at T_1 (1–2), reversible adiabatic expansion from T_1 to T_2 (2–3), isothermal heat rejection at T_2 (3–4) and reversible adiabatic compression back to T_1 (4–1). On the T–s diagram it is a rectangle; heat supplied = $T_1\Delta s$, heat rejected = $T_2\Delta s$, so $\eta = 1 - T_2/T_1$. Carnot's theorem: suppose an engine E is more efficient than a reversible engine R working between the same reservoirs. Let R be reversed as a refrigerator driven by part of the work of E, with both handling the same heat from the hot reservoir. Since E is more efficient it produces more work than R needs, so the net effect is that the combination produces net work while exchanging heat only with the cold reservoir – a violation of the Kelvin–Planck statement. Hence no engine can be more efficient than a reversible one, and all reversible engines between the same two reservoirs have the

same efficiency, depending only on the temperatures. This leads to the thermodynamic (Kelvin) temperature scale.

✎ **DESCRIPTIVE Q:** What is entropy? Explain the principle of increase of entropy.

Entropy is a property whose change in a reversible process is $dS = \delta Q_{rev}/T$; it measures molecular disorder and the degradation of energy. It follows from the Clausius inequality $\oint \delta Q/T \le 0$, which shows that $\int \delta Q_{rev}/T$ between two states is independent of path. For any process, $dS \ge \delta Q/T$, the equality holding for reversible processes; the difference is entropy generation $S_{gen} \ge 0$, caused by irreversibilities such as friction, heat transfer across finite temperature differences, unrestrained expansion and mixing. For an isolated system $\delta Q = 0$, so $dS_{isolated} \ge 0$ – the entropy of an isolated system can never decrease. The universe (system + surroundings) is isolated, so every real process increases the entropy of the universe. This principle determines the direction of natural processes, gives the condition of equilibrium (maximum entropy) and quantifies lost work as $T_0 S_{gen}$ (Gouy–Stodola theorem).

✎ **DESCRIPTIVE Q:** Define available energy, unavailable energy and irreversibility, and explain the loss of available energy due to heat transfer through a finite temperature difference.

Available energy (exergy) is the maximum useful work obtainable from a system or heat source as it comes to equilibrium with the surroundings at T_0. For heat Q at constant temperature T, $AE = Q(1 - T_0/T)$; the rest, $T_0Q/T = T_0\Delta S$, is unavailable energy that must be rejected to the surroundings. Irreversibility is the difference between the maximum (reversible) work and actual work, $I = T_0 S_{gen}$. When heat Q flows from a body at T_1 to one at $T_2 < T_1$, its available energy falls from $Q(1 - T_0/T_1)$ to $Q(1 - T_0/T_2)$. The loss is $T_0Q(1/T_2 - 1/T_1) = T_0\Delta S_{universe}$. So although energy is conserved, its quality is degraded; the larger the temperature difference, the larger the loss. This is why heat exchangers and boilers are designed to keep temperature differences small, and why regenerative feed heating improves steam-plant efficiency.

P2. IC Engines

✎ **DESCRIPTIVE Q:** Compare four-stroke and two-stroke engines.

A four-stroke engine completes its cycle in four strokes (suction, compression, power, exhaust) – two crankshaft revolutions – so there is one power stroke every two revolutions; a two-stroke engine completes the cycle in two strokes (one revolution) by using ports uncovered by the piston and crankcase or blower scavenging, giving one power stroke per revolution. Consequently the two-stroke engine theoretically develops twice the power for the same size and speed (actually 1.6–1.8 times), has a lighter flywheel and more uniform torque, a simpler mechanism (no valves or camshaft) and lower cost. However, scavenging losses (fresh charge escaping with exhaust) and incomplete exhaust give higher fuel consumption, higher HC emissions, lower thermal and volumetric efficiency, greater heating (more frequent power strokes) and higher lubricating-oil consumption (oil mixed with petrol in small engines). Four-stroke engines are used in cars, trucks and most motorcycles; two-stroke engines in small mopeds, chainsaws, outboard motors and very large marine diesel engines (where uniflow scavenging gives high efficiency).

✎ **DESCRIPTIVE Q:** Explain detonation in SI engines and the factors affecting it.

Detonation (knock) in an SI engine is the spontaneous autoignition of the end gas – the last part of the unburnt charge – ahead of the flame front. As the flame advances from the spark plug, it compresses and heats the remaining charge; if that charge reaches its self-ignition temperature and its ignition delay ends before the flame arrives, it ignites almost

instantaneously, producing a sharp pressure rise and pressure waves that cause the characteristic metallic pinging sound. Effects include noise, mechanical stress, overheating of the piston crown and spark plug, erosion, loss of power and efficiency, and in severe cases piston damage. Detonation is favoured by factors that raise the end-gas temperature and pressure or give it more time: high compression ratio, advanced spark timing, high intake temperature and pressure (supercharging), low engine speed, large bore (long flame travel), high wall temperature, lean or slightly rich mixtures at certain conditions and low-octane fuel. It is reduced by using high-octane fuel, lowering compression ratio, retarding the spark, increasing turbulence (shorter combustion time), central spark plug location, compact combustion chambers and good cooling.

✎ **DESCRIPTIVE Q:** Describe the methods used to measure the frictional power of an engine.

(1) Willan's line method (CI engines): fuel consumption is plotted against brake power at constant speed; the nearly straight line is extrapolated back to zero fuel consumption, and the negative intercept on the BP axis gives the friction power. It is not suitable for SI engines because throttling makes the line non-linear. (2) Morse test (multi-cylinder engines): BP is measured with all cylinders firing, then with each cylinder cut out in turn at the same speed (by shorting the spark plug or cutting the fuel injector); the drop in BP equals the IP of the cut-out cylinder, and $\Sigma IP - BP = FP$. (3) Motoring test: the engine is run at the test speed by an electric dynamometer with no combustion; the power absorbed approximates FP, though it overestimates pumping losses at lower temperature. (4) Indicator diagram: IP is obtained from the indicated mean effective pressure and $FP = IP - BP$. (5) Retardation test: the engine is cut off and its deceleration measured to find friction torque.

P3. Steam Power Plant

✎ **DESCRIPTIVE Q:** Explain the Rankine cycle and the methods to improve its efficiency.

The Rankine cycle is the ideal cycle of a steam power plant: (1–2) isentropic expansion of steam in the turbine; (2–3) constant-pressure heat rejection in the condenser until the steam is saturated liquid; (3–4) isentropic pumping of water to boiler pressure; (4–1) constant-pressure heating in the boiler, through sensible heating, evaporation and superheating. Efficiency = $(W_t - W_p)/Q_1 = [(h_1 - h_2) - (h_4 - h_3)]/(h_1 - h_4)$. Unlike Carnot, condensation is complete, so pumping a liquid needs very little work and the work ratio is close to 1. Efficiency is improved by raising the mean temperature of heat addition or lowering that of heat rejection: (a) lowering condenser pressure (limited by cooling-water temperature and turbine-exit wetness); (b) superheating, which increases both work and exit dryness; (c) increasing boiler pressure, which raises efficiency but reduces exit dryness – used with superheat and reheat; (d) reheating the steam after partial expansion, mainly to keep exit dryness above about 88%; (e) regenerative feed heating with bled steam, which raises feed-water temperature and the mean temperature of heat addition; (f) supercritical pressures, combined cycles and binary vapour cycles.

✎ **DESCRIPTIVE Q:** Distinguish between fire-tube and water-tube boilers.

In fire-tube boilers hot flue gases flow inside tubes surrounded by water (Cochran, Lancashire, locomotive); in water-tube boilers water circulates inside tubes heated externally by gases (Babcock & Wilcox, Stirling, La Mont, Benson). Fire-tube boilers are limited to about 25 bar and modest capacities (up to about 9,000 kg/h) because the large shell must withstand the pressure; water-tube boilers handle high pressures (above 100 bar, even supercritical) and capacities of hundreds of tonnes per hour since the small tubes are strong. Water-tube boilers

raise steam quickly due to small water content and good circulation, respond rapidly to load changes, and are safer because a tube burst releases little water; fire-tube boilers hold a large volume of water, so steam raising is slow but they cope with sudden demand from their stored energy. Fire-tube boilers are compact, cheaper, simple to operate and tolerate poorer water; water-tube boilers need treated water and skilled operation but are used in power stations, while fire-tube boilers suit small industries, process heating and locomotives.

✎ **DESCRIPTIVE Q:** Explain the principle of compounding of steam turbines and its types.

If the whole enthalpy drop is expanded in one row of nozzles, steam velocity becomes very high (above 1,000 m/s); for good efficiency the blade speed must be about half this, leading to excessive rotor speeds (De Laval turbines ran up to 30,000 rpm), large centrifugal stresses and high carry-over losses. Compounding uses several rows of blades to absorb the energy in stages and reduce the rotor speed. (1) Pressure compounding (Rateau): the total pressure drop is divided among several stages, each consisting of a nozzle ring and a row of moving blades; each stage acts as a simple impulse turbine with moderate velocity. (2) Velocity compounding (Curtis): the pressure drops entirely in one set of nozzles, and the high-velocity steam passes through two or three rows of moving blades separated by fixed guide blades that redirect it; this gives large work per stage at low blade speed but lower efficiency. (3) Pressure–velocity compounding combines both – a Curtis stage at the high-pressure end followed by Rateau stages. Reaction turbines (Parsons) are inherently pressure compounded because pressure drops in both fixed and moving blades in every stage.

P4. Compressors and Gas Turbines

✎ **DESCRIPTIVE Q:** Explain multistage compression with intercooling and its advantages.

In multistage compression air is compressed in two or more cylinders in series, and is cooled at constant pressure in an intercooler between stages. With perfect intercooling the air returns to its initial temperature before each stage, so the overall process approaches isothermal compression, which requires the minimum work. On the p–V diagram the shaded area between the single-stage polytropic curve and the stepped multistage curve represents the work saved. For minimum work with perfect intercooling the pressure ratio per stage is equal, $p_2/p_1 = p_3/p_2$ = (p_final/p_1)^(1/N), and each stage does equal work. Advantages: reduced total work and power; lower delivery temperatures, which help lubrication and reduce the risk of oil ignition; higher volumetric efficiency since each stage has a small pressure ratio and the effect of clearance is reduced; lighter cylinders for high-pressure stages; better mechanical balance and more uniform torque; moisture can be removed in intercoolers. Disadvantages are higher initial cost and complexity.

✎ **DESCRIPTIVE Q:** Explain the methods to improve the efficiency and work output of a simple gas turbine plant.

The simple open-cycle gas turbine has low efficiency (around 20–30%) because the compressor consumes a large part of the turbine work and hot exhaust is discharged. (1) Regeneration: exhaust gases heat the compressed air in a heat exchanger before the combustion chamber, reducing fuel consumption; it is effective only when exhaust temperature exceeds compressor delivery temperature, i.e. at low pressure ratios. (2) Intercooling between compressor stages reduces compressor work and increases net work, though it lowers the temperature at which heat is added; with regeneration it improves efficiency. (3) Reheating between turbine stages raises turbine work; combined with regeneration it increases efficiency. (4) Increasing turbine inlet temperature through better materials (nickel superalloys, ceramic coatings) and blade

cooling. (5) Improving compressor and turbine efficiencies. (6) Combined cycles: the 500–600 °C exhaust raises steam in a heat-recovery steam generator for a steam turbine, giving overall efficiencies above 55–60%.

P5. Refrigeration and Air Conditioning

✎ **DESCRIPTIVE Q:** Explain a simple vapour compression refrigeration system with p–h and T–s diagrams, and the effect of subcooling and superheating.

A VCR system has four components. Low-pressure dry saturated vapour from the evaporator is compressed isentropically (1–2) to the condenser pressure, becoming superheated. In the condenser (2–3) it rejects heat to cooling air or water at constant pressure, desuperheating and condensing to saturated liquid. The liquid is throttled in the expansion valve (3–4) at constant enthalpy to evaporator pressure, partly flashing to vapour. In the evaporator (4–1) the liquid boils at low temperature, absorbing heat from the space. Refrigerating effect = $h_1 - h_4$, compressor work = $h_2 - h_1$, COP = $(h_1 - h_4)/(h_2 - h_1)$. On the p–h chart the cycle is a near-rectangle with the expansion as a vertical line. Subcooling the condensate below saturation moves point 3 to the left, lowering h_4; the refrigerating effect increases without extra work, so COP rises – achieved with extra condenser surface or a liquid–suction heat exchanger. Superheating the vapour at the evaporator exit, if it is done in the refrigerated space, increases the refrigerating effect, but compressor work and specific volume also increase; the effect on COP depends on the refrigerant. Superheating is still useful because it prevents liquid from entering the compressor.

✎ **DESCRIPTIVE Q:** Explain the psychrometric processes involved in winter air conditioning.

In winter the outdoor air is cold and, although its relative humidity may be high, its specific humidity is low; heating it without adding moisture would give very dry air. Winter air conditioning therefore involves heating with humidification. Fresh and recirculated air are mixed and filtered, then heated in a preheater coil (sensible heating, horizontal line on the chart). The air then passes through a humidifier – a spray of water (adiabatic, along the WBT line, which cools the air and raises its humidity) or steam injection (nearly vertical, slight temperature rise). Finally a reheater coil raises the temperature to the required supply condition (sensible heating). The conditioned air, slightly warmer than the room, is supplied to offset heat loss through walls and infiltration, maintaining about 20–22 °C and 40–50% RH. Controls include a humidistat for the humidifier and thermostats for the heating coils.

P6. Heat Transfer

✎ **DESCRIPTIVE Q:** Define fin efficiency and fin effectiveness. When are fins justified?

Fins (extended surfaces) increase the area for convective heat transfer. Fin efficiency is the ratio of the actual heat dissipated by the fin to the heat it would dissipate if its entire surface were at the base temperature; because temperature falls along the fin, efficiency is less than one. For a fin with an insulated tip $\eta_f = \tanh(mL)/(mL)$, where $m = \sqrt{(hP/(kA_c))}$. Fin effectiveness is the ratio of heat transfer with the fin to that from the base area without the fin; for a long fin $\varepsilon = \sqrt{(kP/(hA_c))}$. A fin is worthwhile only if effectiveness is well above unity – a practical criterion is $\varepsilon > 2$. High effectiveness requires high thermal conductivity, a large perimeter-to-area ratio (thin fins) and a low heat-transfer coefficient; therefore fins are used on the gas or air side of heat exchangers (car radiators, air-cooled engine cylinders, condensers, electronic heat sinks) and not where h is already high (boiling or condensing). Increasing the length beyond $mL \approx 3$ gives little benefit.

✎ **DESCRIPTIVE Q:** Derive the expression for LMTD of a parallel-flow heat exchanger.

Consider an element of area dA where the hot fluid temperature is T_h and the cold T_c, with local difference θ = T_h − T_c. The heat transferred is dQ = U dA θ. The hot fluid cools: dQ = −C_h dT_h; the cold fluid heats: dQ = C_c dT_c. Hence dθ = dT_h − dT_c = −dQ(1/C_h + 1/C_c). Substituting dQ = U dA θ: dθ/θ = −U(1/C_h + 1/C_c) dA. Integrating from inlet (θ_1) to outlet (θ_2): $\ln(\theta_2/\theta_1) = -UA(1/C_h + 1/C_c)$. Over the whole exchanger Q = C_h(T_h1 − T_h2) = C_c(T_c2 − T_c1), so 1/C_h + 1/C_c = [(T_h1 − T_h2) + (T_c2 − T_c1)]/Q = ($\theta_1 - \theta_2$)/Q. Therefore $\ln(\theta_1/\theta_2) = UA(\theta_1 - \theta_2)/Q$, or $Q = UA(\theta_1 - \theta_2)/\ln(\theta_1/\theta_2) = UA \times LMTD$. For parallel flow θ_1 = T_h1 − T_c1 and θ_2 = T_h2 − T_c2. The same form applies to counter flow with θ_1 = T_h1 − T_c2 and θ_2 = T_h2 − T_c1.

✎ **DESCRIPTIVE Q:** Explain the regimes of pool boiling with a boiling curve.

When a heated surface is immersed in a pool of saturated liquid, the heat flux varies with the excess temperature ΔT_e = T_s − T_sat as shown by Nukiyama's boiling curve. (1) Natural-convection boiling (ΔT_e below about 5 °C for water): the liquid is slightly superheated and moves by free convection; evaporation occurs at the free surface. (2) Nucleate boiling (about 5–30 °C): bubbles form at nucleation sites (cavities) on the surface; at first they are isolated and collapse in the liquid, then they form jets and columns. Intense agitation gives very high heat-transfer coefficients, and heat flux rises steeply to a maximum – the critical (peak) heat flux, about 1 MW/m² for water at atmospheric pressure. (3) Transition boiling (about 30–120 °C): an unstable vapour film partly covers the surface; because vapour conducts poorly, heat flux decreases as ΔT_e rises, down to the minimum heat flux at the Leidenfrost point. (4) Film boiling (above about 120 °C): a stable vapour film blankets the surface; heat passes by conduction and radiation through the film, and flux rises again only at very high temperatures. In heat-flux-controlled equipment, exceeding the critical flux causes a jump from nucleate to film boiling with a huge temperature rise (burnout), so boilers and nuclear reactors are designed to operate in nucleate boiling with a safety margin below CHF.

CHAPTER 29B

Paper-II Model Answers – Set 2

More conventional-style answers covering topics that frequently appear in descriptive papers.

P7. Thermodynamics and Cycles

✎ **DESCRIPTIVE Q:** Prove the equivalence of the Kelvin–Planck and Clausius statements.

The two statements are equivalent if violation of one implies violation of the other. (a) Suppose the Clausius statement is violated: a device transfers heat Q_2 from a cold reservoir to a hot reservoir with no work input. Let an ordinary heat engine operate between the same reservoirs, taking Q_1 from the hot reservoir, producing $W = Q_1 - Q_2$ and rejecting Q_2 to the cold reservoir. Combining the two, the cold reservoir has no net heat exchange; the hot reservoir supplies net heat $Q_1 - Q_2$ and the combination produces equal work. This is a cyclic device that converts heat from a single reservoir completely into work – a violation of the Kelvin–Planck statement. (b) Suppose the Kelvin–Planck statement is violated: an engine converts Q_1 from the hot reservoir fully into work $W = Q_1$. Let this work drive a refrigerator that takes Q_2 from the cold reservoir and delivers $Q_1 + Q_2$ to the hot reservoir. The net effect is transfer of Q_2 from the cold to the hot reservoir with no external work – a violation of the Clausius statement. Since each violation implies the other, the statements are equivalent.

✎ **DESCRIPTIVE Q:** Compare Otto, Diesel and Dual cycles for the same compression ratio and heat input, and for the same maximum pressure and temperature.

For the same compression ratio and the same heat input, the three cycles share the same compression process. On the T–s diagram they reject heat along the same constant-volume line; the Otto cycle adds all heat at constant volume, reaching the highest temperature, and therefore rejects the least heat; the Diesel cycle adds heat at constant pressure and rejects the most heat; the dual cycle lies between them. Since $\eta = 1 - Q_2/Q_1$, the order is $\eta_{Otto} > \eta_{Dual} > \eta_{Diesel}$. For the same maximum pressure and maximum temperature (a better comparison of real engines, because the limits are set by strength and materials), the Diesel cycle must have a much higher compression ratio; the heat rejected is the same for all, but the Diesel cycle receives the most heat, so $\eta_{Diesel} > \eta_{Dual} > \eta_{Otto}$. This explains why actual CI engines are more efficient than SI engines: they work at higher compression ratios, which SI engines cannot use because of detonation.

P8. Boilers, Nozzles and Turbines

✎ **DESCRIPTIVE Q:** Explain boiler mountings and accessories with the function of each.

Mountings are fittings mounted on the boiler for its safe operation and control; accessories are auxiliary devices that improve efficiency. Mountings: (1) Water-level indicator (gauge glass) – shows the water level in the drum; (2) Pressure gauge (Bourdon tube) – indicates steam pressure; (3) Safety valves (dead-weight, lever, spring-loaded, high-steam low-water) – release steam when pressure exceeds the safe limit; (4) Fusible plug – a plug with low-melting alloy above the furnace crown that melts when the water level falls dangerously low, letting steam and water extinguish the fire; (5) Steam stop valve – regulates steam flow from the boiler to the

steam pipe; (6) Feed check valve – non-return valve that admits feed water and prevents back-flow; (7) Blow-off cock – removes sludge and sediment from the bottom and empties the boiler; (8) Manholes and mud holes for inspection and cleaning. Accessories: (1) Economiser – heats feed water using flue gas; (2) Air preheater – heats combustion air using flue gas; (3) Superheater – converts wet/saturated steam to superheated; (4) Feed pump and injector – supply water to the boiler against its pressure; (5) Steam separator and steam trap – remove water from steam and drain condensate; (6) Draught fans and chimney; (7) Soot blowers. The order of flue gas passage is superheater → economiser → air preheater.

✎ **DESCRIPTIVE Q:** Derive the expression for critical pressure ratio in a nozzle.

For isentropic flow of a gas or superheated steam obeying $pv^n = C$, the steady-flow energy equation with negligible inlet velocity gives $V_2^2/2 = h_1 - h_2 = [n/(n-1)]\, p_1v_1[1 - (p_2/p_1)^{((n-1)/n)}]$. The mass flow per unit area is $\dot{m}/A = V_2/v_2$, with $v_2 = v_1(p_1/p_2)^{(1/n)}$. Substituting, $\dot{m}/A = \sqrt{\{[2n/(n-1)](p_1/v_1)[(p_2/p_1)^{(2/n)} - (p_2/p_1)^{((n+1)/n)}]\}}$. For a given inlet state, $\dot{m}/A$ is maximum (throat condition) when the bracketed term is maximum. Differentiating with respect to $r = p_2/p_1$ and equating to zero: $(2/n)\, r^{((2-n)/n)} = ((n+1)/n)\, r^{(1/n)}$, which gives $r_c = (2/(n+1))^{(n/(n-1))}$. For air ($n = \gamma = 1.4$) $r_c = 0.528$; for superheated steam ($n = 1.3$) $r_c = 0.546$; for dry saturated steam ($n = 1.135$) $r_c = 0.577$. At this pressure ratio the velocity at the throat equals the local sonic velocity; further lowering of back pressure does not increase the mass flow – the nozzle is choked.

✎ **DESCRIPTIVE Q:** Differentiate between impulse and reaction turbines.

In an impulse turbine the entire pressure drop of a stage occurs in the fixed nozzles; steam enters the moving blades at high velocity and constant pressure, and the blades, being symmetrical, only change the direction of the jet – the force comes from the change of momentum. In a reaction turbine the pressure drops in both fixed and moving blades (the blade passages are converging nozzles), so the steam accelerates relative to the moving blades and exerts a reaction force in addition to the impulse. Consequences: impulse blades are symmetrical with constant-area passages, reaction blades are aerofoil-shaped with converging passages; in impulse turbines pressure is constant across the rotor, so tip leakage is small and partial admission is possible; reaction turbines need full admission, have pressure difference across the moving blades, require balance pistons/dummy pistons for axial thrust and close tip clearances; impulse stages handle a larger enthalpy drop per stage so fewer stages are needed, while reaction turbines have more stages and are more efficient at their design point. Optimum blade speed ratio is $\cos\alpha/2$ for impulse and $\cos\alpha$ for 50% reaction stages. Examples: De Laval, Curtis and Rateau (impulse); Parsons (reaction).

P9. IC Engines, Power Plants, Compressors

✎ **DESCRIPTIVE Q:** Explain the necessity of cooling in IC engines and compare air and water cooling.

During combustion gas temperatures reach about 2000–2500 °C; without cooling, the cylinder walls, piston and valves would overheat, losing strength and causing thermal stresses, distortion and seizure of the piston; the lubricating oil film would break down and burn; overheated surfaces and spark plugs would cause pre-ignition and detonation in SI engines; and volumetric efficiency would fall because the incoming charge is heated. About 25–35% of fuel energy is removed by the cooling system. Over-cooling must also be avoided, as it increases heat loss, fuel consumption, friction (viscous oil) and corrosion from condensed acids. Air cooling uses fins on the cylinder and head, with natural airflow or a fan (motorcycles, small

engines, aircraft piston engines): it is light, simple, cheap, needs no radiator or coolant and warms up quickly, but cooling is less uniform, noisier, and limited to small engines. Water (liquid) cooling circulates coolant through jackets around the cylinders by a pump (forced circulation) or thermosiphon, with a thermostat to control temperature and a radiator with fan to reject heat; it gives uniform and effective cooling for large and high-output engines, quieter operation and allows higher compression ratios, but adds weight, cost, maintenance and the risk of leaks and freezing (antifreeze such as ethylene glycol is added).

✎ **DESCRIPTIVE Q:** Explain the working of a nuclear power plant with its main components.

A nuclear power plant generates heat by controlled fission of U-235 (or Pu-239) and uses it to raise steam for a conventional turbine-generator. Main components of the reactor: (1) Fuel – natural or enriched uranium as UO_2 pellets in zircaloy tubes; (2) Moderator – slows fast neutrons to thermal energies (heavy water, light water, graphite); (3) Control rods – absorb neutrons to regulate or shut down the chain reaction (boron, cadmium, hafnium); (4) Coolant – removes heat from the core (pressurised water, heavy water, CO_2, liquid sodium); (5) Reflector – returns escaping neutrons to the core; (6) Reactor vessel and biological shield (thick concrete) to protect personnel from radiation. In a pressurised water reactor (PWR) the primary coolant is kept at about 155 bar so it does not boil; it transfers heat to a secondary circuit in a steam generator, and the steam drives the turbine. In a boiling water reactor (BWR) steam is generated in the reactor itself. Indian PHWRs use natural uranium with heavy water as moderator and coolant. Advantages include very low fuel requirement and no CO_2 emission; disadvantages are high capital cost, radioactive waste disposal, safety concerns and the need for skilled operation.

✎ **DESCRIPTIVE Q:** What is supercharging? Explain its effects on SI and CI engines.

Supercharging is supplying air (or mixture) to an engine at a pressure above atmospheric, so that a greater mass of charge is inducted per cycle and more fuel can be burned, increasing power output for the same engine size. The supercharger may be mechanically driven by the crankshaft (Roots blower, vane or centrifugal compressor) or driven by an exhaust-gas turbine (turbocharger), which uses otherwise wasted exhaust energy. Objectives: to increase power-to-weight ratio, to compensate for loss of power at high altitude (aircraft), and to reduce engine size for a given power. In CI engines supercharging is very beneficial: the higher pressure and temperature of the charge reduce ignition delay, giving smoother combustion and reduced diesel knock, and excess air improves combustion; it is limited mainly by thermal and mechanical loads. In SI engines, supercharging increases the tendency to detonate because the end gas is at a higher pressure and temperature; it is therefore used with lower compression ratios, intercooling and high-octane fuels. Intercooling (charge-air cooling) after the supercharger increases charge density further and reduces thermal load and NOx.

P10. Refrigeration, AC and Heat Transfer

✎ **DESCRIPTIVE Q:** Explain the desirable properties of a refrigerant.

A good refrigerant should have (1) a low boiling point at atmospheric pressure, so that low temperatures are obtained at pressures above atmospheric, preventing air leakage into the system; (2) a moderate condensing pressure to keep equipment light; (3) high latent heat of vaporisation to give a large refrigerating effect per kg and low mass flow; (4) low specific volume of vapour so that the compressor is small; (5) a high critical temperature, well above the condensing temperature, and a low freezing point; (6) high COP and low power per TR; (7) high thermal conductivity and low viscosity for good heat transfer and low pressure drop; (8)

chemical stability, non-toxicity, non-flammability and non-corrosiveness; (9) miscibility or compatibility with lubricating oil, and high dielectric strength for hermetic compressors; (10) ease of leak detection; (11) zero ozone depletion potential and low global-warming potential; (12) low cost and easy availability. No refrigerant has all these properties – ammonia has excellent thermodynamic properties but is toxic and attacks copper; halocarbons are safe but have environmental drawbacks – so the choice depends on the application.

✎ **DESCRIPTIVE Q:** Explain the electrical analogy of heat conduction through a composite wall.

Steady heat conduction obeys Q = ΔT/R, analogous to Ohm's law I = ΔV/R, where heat flow corresponds to current, temperature difference to potential difference, and thermal resistance to electrical resistance. A plane layer has conduction resistance L/(kA), a convective surface film 1/(hA), and a cylindrical layer $\ln(r_2/r_1)/(2\pi kL)$. For a composite wall of layers in series – for example inside film, plaster, brick, insulation, outside film – the same heat flows through each layer, so the resistances add: $R_{total} = 1/(h_1A) + \Sigma L_i/(k_iA) + 1/(h_2A)$, and $Q = (T_i - T_o)/R_{total}$. The temperature drop across each layer is proportional to its resistance, which allows interface temperatures to be calculated. When different materials are placed side by side (parallel paths, such as a wall with a window or studs), the conductances add: $1/R_{eq} = \Sigma 1/R_i$. The overall heat transfer coefficient U is defined by 1/(UA) = R_{total}. The analogy simplifies the analysis of insulated pipes, building walls and heat exchanger tubes, and extends to radiation networks using surface and space resistances.

CHAPTER 29C

Paper-II Model Answers – Set 3

Further descriptive answers; practise writing each in 10–12 minutes with a neat sketch.

P11. Thermodynamics and Steam

✎ **DESCRIPTIVE Q:** Explain the measurement of dryness fraction of steam by a combined separating and throttling calorimeter.

A throttling calorimeter alone works only if steam becomes superheated after throttling, which requires the initial dryness to be high (about 0.95 or more). For wetter steam a separating calorimeter is placed before it. Steam from the main first enters the separating calorimeter, where a sudden change in direction makes the heavier water particles separate by inertia and collect in a chamber (mass m_1 measured from a gauge glass). The partly dried steam (mass m_2, measured by condensing it after the throttling calorimeter) then passes through a narrow orifice into the throttling calorimeter, where it is throttled to nearly atmospheric pressure and becomes superheated; its pressure and temperature are measured. Since throttling is isenthalpic, $h_{f1} + x_2 h_{fg1} = h_{sup2}$ at the calorimeter pressure and temperature, giving x_2. The dryness fraction indicated by the separator is $x_1 = m_2/(m_1 + m_2)$. The overall dryness fraction of the steam is $x = x_1 \times x_2$. The combined arrangement measures dryness of very wet steam accurately, the separator removing most of the water and the throttling calorimeter measuring the remaining wetness.

✎ **DESCRIPTIVE Q:** Explain the Mollier (h–s) diagram and its uses.

The Mollier chart plots specific enthalpy on the vertical axis against specific entropy on the horizontal axis for steam. It shows the saturation line (separating wet and superheated regions), constant-pressure lines (straight in the wet region, curving upward in the superheated region), constant-temperature lines (coinciding with constant-pressure lines in the wet region and becoming nearly horizontal at low pressures in the superheated region), constant-dryness lines below the saturation curve and constant-volume lines. Its main advantage is that isentropic processes are vertical lines and enthalpy changes are read directly as vertical distances. It is therefore used for quick solution of steam-turbine and nozzle problems: finding the isentropic enthalpy drop and exit dryness, actual exit state using isentropic efficiency, reheat and regenerative cycle states, throttling (horizontal lines), and calculation of nozzle velocities. It avoids interpolation in steam tables and gives a clear visual picture of the processes.

P12. IC Engines

✎ **DESCRIPTIVE Q:** Explain the working of a simple carburettor and its limitations.

A simple carburettor consists of a float chamber, a fuel jet (nozzle) with metering orifice, a venturi, and a throttle valve. The float and needle valve maintain a constant fuel level slightly below the jet tip. During the suction stroke air flows through the venturi; its velocity increases and pressure decreases at the throat. The pressure difference between the float chamber (atmospheric) and the throat forces fuel out of the jet, where it is atomised by the high-velocity

air and partly vaporised, forming the air–fuel mixture. The throttle valve controls the quantity of mixture entering the engine, and hence the power (quantity governing). A choke valve upstream of the venturi enriches the mixture for cold starting. Limitations of the simple carburettor: as air flow increases, the mixture becomes progressively richer because fuel flow rises faster than air flow (air density falls with velocity, and fuel is incompressible); it cannot give the rich mixture needed at idle and for acceleration and full power, nor the lean mixture needed for economy; it is affected by altitude. Practical carburettors therefore add compensating jets, idle systems, accelerating pumps, power (economiser) systems and choke; modern engines have replaced carburettors with electronic fuel injection.

✎ **DESCRIPTIVE Q:** Explain the different lubrication systems used in IC engines.

Lubrication reduces friction and wear, cools moving parts, seals the piston rings, removes contaminants and prevents corrosion. (1) Petroil (mist) lubrication: used in small two-stroke petrol engines – lubricating oil mixed with petrol (2–3%) enters with the charge through the crankcase and lubricates bearings and cylinder walls; simple but causes carbon deposits, smoke and oil waste. Modern two-strokes use separate oil injection. (2) Splash lubrication: a dipper on the connecting rod dips into the oil sump and splashes oil onto cylinder walls and bearings; used in small single-cylinder engines; uneven supply. (3) Pressure (forced-feed) lubrication: a gear or rotor pump draws oil from the sump through a strainer and delivers it under pressure (2–4 bar) through a filter to main bearings, crankpin bearings (via drilled crankshaft), camshaft and gudgeon pins; cylinder walls are lubricated by oil thrown from bearings; used in all modern multi-cylinder engines. A pressure-relief valve, oil cooler and pressure gauge are included. (4) Dry-sump system: oil is scavenged from a shallow sump by a separate pump to an external tank; used in racing cars, aircraft and off-road vehicles to prevent starvation during manoeuvres. Lubricating oil properties: suitable viscosity and viscosity index, oiliness, high flash and fire point, low pour point, oxidation stability and detergency.

P13. Power Plant and Compressors

✎ **DESCRIPTIVE Q:** Describe the layout of a modern coal-fired thermal power plant.

A coal-fired plant consists of several circuits. (1) Coal and ash circuit: coal received by rail is unloaded, crushed, stored, and fed through bunkers to pulverising mills that grind it to fine powder; it is blown with hot primary air into the furnace burners. Bottom ash collects below the furnace and fly ash is captured by electrostatic precipitators; ash is disposed of by hydraulic or pneumatic systems and increasingly used in cement and bricks. (2) Air and flue gas circuit: forced-draught fans push air through the air preheater into the furnace; flue gases pass over the superheater, reheater, economiser and air preheater, then through the ESP and flue-gas desulphurisation unit, and are drawn by induced-draught fans to the chimney. (3) Feed water and steam circuit: water from the condenser is pumped through LP heaters, deaerator and HP heaters to the economiser and boiler drum (or once-through supercritical boiler); steam is superheated, expanded in HP turbine, reheated, and expanded through IP and LP turbines, then condensed in the surface condenser. (4) Cooling water circuit: cooling water from a river or cooling tower circulates through the condenser tubes. (5) The turbine drives the alternator, and power is stepped up by transformers to the grid. Overall efficiencies are about 33–38% for subcritical and up to 40–42% for supercritical and ultra-supercritical units.

✎ **DESCRIPTIVE Q:** Compare reciprocating and rotary (centrifugal and axial) compressors.

Reciprocating compressors compress air by a piston in a cylinder in a positive-displacement manner. They give high pressure ratios (up to about 1000 bar with multistaging), but deliver

small quantities of air intermittently, are heavy, need careful balancing and foundations, have many wearing parts (valves, rings) and the air may be contaminated with lubricating oil; their efficiency is good at high pressure ratios. Rotary positive-displacement compressors (vane, Roots, screw) give steady flow at moderate pressures. Dynamic (rotary) compressors – centrifugal and axial – add kinetic energy to the air by a rotating impeller or blades and convert it to pressure in diffusers. They handle large volume flows continuously and smoothly, are compact and light for their capacity, have few wearing parts, deliver oil-free air and can be directly coupled to turbines and motors at high speed; but the pressure ratio per stage is low (centrifugal about 4:1 per stage, axial about 1.2–1.3 per stage), and they suffer from surging and choking and lose efficiency away from the design point. Centrifugal compressors are used in turbochargers, small gas turbines and refrigeration chillers; axial compressors, with many stages and high efficiency, are used in large gas turbines and jet engines.

P14. RAC and Heat Transfer

✎ **DESCRIPTIVE Q:** Explain the working of a vapour absorption refrigeration system (aqua-ammonia) with its components.

In the aqua-ammonia system, ammonia is the refrigerant and water the absorbent. Low-pressure ammonia vapour from the evaporator enters the absorber, where it is absorbed by the weak aqua-ammonia solution; since absorption releases heat, the absorber is cooled by water to keep its capacity high. The resulting strong solution is pumped to the generator at condenser pressure, passing through a heat exchanger where it is preheated by the hot weak solution returning from the generator. In the generator, heat from steam, gas or waste heat drives ammonia vapour out of the strong solution. The vapour, which carries some water vapour, passes through an analyser (where it meets the incoming strong solution and some water condenses) and a rectifier (a water-cooled dephlegmator that condenses the remaining water and returns it to the generator), so nearly pure ammonia reaches the condenser. It is condensed, passes through the expansion valve and evaporates in the evaporator, producing refrigeration, and returns to the absorber. The weak solution from the generator returns to the absorber through the heat exchanger and a pressure-reducing valve. Since the pump handles liquid, its work is very small; the main energy input is heat. The system can use low-grade heat and has few moving parts, but its COP is low (0.5–0.7).

✎ **DESCRIPTIVE Q:** Explain Kirchhoff's law of radiation and the concept of a grey body.

Consider a small body placed inside an isothermal enclosure and in thermal equilibrium with it. Radiation inside the enclosure is black-body radiation with irradiation $G = \sigma T^4$. The body absorbs αG and emits E; for equilibrium $E = \alpha G$, so $E/\alpha = \sigma T^4 = E_b$. Since this holds for any body, the ratio of emissive power to absorptivity is the same for all bodies at a given temperature and equals the emissive power of a black body. Defining emissivity $\varepsilon = E/E_b$ gives $\varepsilon = \alpha$: at thermal equilibrium the emissivity of a surface equals its absorptivity. The law holds strictly for spectral, directional properties; for total properties it holds when the incident radiation comes from a source at the same temperature as the surface. A grey body is an idealised surface whose spectral emissivity (and absorptivity) is the same at all wavelengths. For a grey body, the total emissivity equals the total absorptivity regardless of the source temperature, so Kirchhoff's law can be applied freely in engineering calculations of radiation exchange between real surfaces, which are approximated as grey and diffuse.

CHAPTER 29D

Abridged Property Tables and Data

Standard textbook values (rounded) for quick numericals. Exam papers supply the data they need – use these for practice only.

T1. Saturated Water and Steam (Pressure Table)

p (bar)	T_sat (°C)	v_g (m³/kg)	h_f (kJ/kg)	h_fg (kJ/kg)	h_g (kJ/kg)	s_f (kJ/kg·K)	s_fg (kJ/kg·K)	s_g (kJ/kg·K)
0.05	32.88	28.19	137.8	2423.7	2561.5	0.4764	7.9187	8.3951
0.1	45.81	14.67	191.8	2392.8	2584.6	0.6493	7.5009	8.1502
0.5	81.33	3.240	340.5	2305.4	2645.9	1.0910	6.5029	7.5939
1.0	99.63	1.694	417.5	2258.0	2675.5	1.3026	6.0568	7.3594
2.0	120.23	0.8857	504.7	2201.9	2706.7	1.5301	5.5970	7.1271
5.0	151.86	0.3749	640.2	2108.5	2748.7	1.8607	4.9606	6.8213
10	179.91	0.1944	762.8	2015.3	2778.1	2.1387	4.4478	6.5865
20	212.42	0.0996	908.8	1890.7	2799.5	2.4474	3.8935	6.3409
40	250.40	0.0498	1087.3	1714.1	2801.4	2.7964	3.2737	6.0701
60	275.64	0.0324	1213.4	1571.0	2784.3	3.0267	2.8625	5.8892
100	311.06	0.0180	1407.6	1317.1	2724.7	3.3596	2.2544	5.6141
220.9	374.14	0.00316	2099.3	0	2099.3	4.4298	0	4.4298

- Trends: as pressure rises, T_sat, h_f and s_f rise; h_fg falls to zero at the critical point; h_g peaks at about 30 bar (≈ 2804 kJ/kg) and then falls.
- Specific volume of saturated liquid v_f ≈ 0.001 m³/kg at low pressures (0.00101 at 0.1 bar, 0.00113 at 10 bar).

T2. Saturation Pressure of Water (Temperature Table, for Psychrometry)

T (°C)	p_sat (kPa)	T (°C)	p_sat (kPa)
0.01	0.6117	25	3.169
5	0.8725	30	4.246
10	1.228	35	5.628
15	1.705	40	7.384
20	2.339	50	12.35
21	2.487	100	101.42

T3. Gas Properties (≈ 300 K)

Gas	M (kg/kmol)	R (kJ/kg·K)	c_p (kJ/kg·K)	c_v (kJ/kg·K)	γ
Air	28.97	0.287	1.005	0.718	1.40
Nitrogen	28.01	0.297	1.039	0.743	1.40

Gas	M (kg/kmol)	R (kJ/kg·K)	c_p (kJ/kg·K)	c_v (kJ/kg·K)	γ
Oxygen	32.00	0.260	0.918	0.658	1.40
Hydrogen	2.016	4.124	14.31	10.18	1.41
Helium	4.003	2.077	5.193	3.116	1.67
Argon	39.95	0.208	0.520	0.312	1.67
Carbon dioxide	44.01	0.189	0.846	0.657	1.29
Steam (low pressure)	18.02	0.461	1.87	1.41	1.33
Methane	16.04	0.518	2.25	1.73	1.30

T4. Thermal Conductivity and Emissivity (≈ room temperature)

Material	k (W/m·K)	Typical ε
Silver	429	0.02 (polished)
Copper	401	0.03 polished / 0.78 oxidised
Aluminium	237	0.05 polished / 0.2–0.3 anodised
Carbon steel	45–60	0.8 (oxidised)
Stainless steel	15	0.2–0.6
Concrete / brick	1.4 / 0.7	0.9
Glass	0.8–1.4	0.9
Water	0.6	0.96
Wood	0.12–0.17	0.9
Glass wool / mineral wool	0.04	–
Polyurethane foam	0.02–0.03	–
Air	0.026	–
Human skin	–	0.95
White paint / black paint	–	0.9 / 0.97 (infrared)

T5. Unit Conversions

Conversion	Value
1 kWh	3600 kJ
1 hp (metric)	735.5 W
1 kcal	4.187 kJ
1 TR	3.5167 kW = 211 kJ/min
1 bar	10^5 Pa = 0.9869 atm
1 atm	1.01325 bar = 760 mm Hg
1 mm Hg (torr)	133.32 Pa
1 mm water	9.81 Pa
1 Btu	1.055 kJ
T (K)	T (°C) + 273.15
T (°F)	1.8 T (°C) + 32
T (°R)	T (°F) + 459.67

CHAPTER 29E

High-Yield Topic Map

A chapter-by-chapter checklist of what SSC JE papers repeatedly test. Tick each line only when you can explain it without notes.

M1. Thermodynamics (Chapters 1–7)

Concept	What is usually asked
Systems and properties	Identify open/closed/isolated systems; intensive vs extensive; point vs path functions
Zeroth law	Basis of thermometry; temperature scale conversions
Work and heat	Work in different processes; sign convention; free expansion work = 0
First law	$Q = \Delta U + W$; internal energy a property; PMM-1
Ideal gas processes	n values for processes; work and heat formulas; $c_p - c_v = R$; γ values
SFEE	Nozzle velocity; turbine/compressor work; throttling isenthalpic
Second law	Kelvin–Planck and Clausius statements; PMM-2; equivalence
Carnot	Efficiency and COP formulas; dependence only on temperatures; engines in series
Entropy	Clausius inequality; entropy change formulas; increase of entropy principle; T–s areas
Availability	$AE = Q(1 - T_0/T)$; irreversibility; second-law efficiency; Gibbs/Helmholtz functions
Pure substances	Triple and critical points; dryness fraction; steam enthalpy; calorimeters; Mollier chart

M2. Gas Cycles and IC Engines (Chapters 8–12B)

Concept	What is usually asked
Air-standard cycles	Otto, Diesel, Dual, Brayton, Stirling, Ericsson – processes and efficiency formulas
Cycle comparison	Same r & heat input vs same max p & T orderings
Engine terminology	Compression ratio, swept volume, mean piston speed, mep
Four- vs two-stroke	Power strokes per revolution, scavenging, merits and demerits
Valve timing	IVO, IVC, EVO, EVC typical angles; valve overlap
SI combustion	Stages (Ricardo), flame propagation, detonation causes and remedies
CI combustion	Ignition delay, diesel knock, remedies opposite to SI

Concept	What is usually asked
Fuel ratings	Octane and cetane numbers, reference fuels, additives
Fuel systems	Carburettor and its circuits; MPFI, GDI, CRDI; injectors
Ignition, cooling, lubrication	Battery/magneto/electronic ignition; air vs water cooling; splash/pressure lubrication
Performance	IP, BP, FP, efficiencies, bsfc, volumetric efficiency, heat balance
Testing	Morse test, Willan's line, motoring test, dynamometers
Supercharging	Effect on SI and CI; turbocharger; intercooler
Emissions	CO, HC, NOx, PM sources; catalytic converter, EGR, DPF, SCR; BS-VI
Special engines	Wankel, Stirling, dual-fuel, hybrids

M3. Steam Engineering and Power Plants (Chapters 13–18)

Concept	What is usually asked
Rankine cycle	Processes, efficiency, work ratio, effect of p, T and condenser pressure
Reheat and regeneration	Purpose of each; open vs closed feed-water heaters; deaerator
Boilers	Fire-tube vs water-tube; named boilers; high-pressure boilers (Benson, La Mont, Loeffler)
Mountings and accessories	Function of each; fusible plug; economiser and air preheater
Draught	Natural vs artificial; chimney height formula; FD, ID, balanced draught
Boiler performance	Equivalent evaporation; boiler efficiency; heat losses
Nozzles	Velocity, critical pressure ratio values, choking, supersaturation
Steam turbines	Impulse vs reaction; velocity triangles; optimum blade speed ratios; compounding types
Turbine governing	Throttle, nozzle, bypass governing
Condensers	Surface vs jet; vacuum efficiency; Dalton's law; air extraction
Cooling towers	Natural vs mechanical draught; range, approach
Power plants	Load curve factors; nuclear reactor components and types; combined cycle; renewable sources

M4. Compressors, Gas Turbines, Jet Propulsion (Chapters 19–20)

Concept	What is usually asked
Reciprocating compressor	Work for isothermal/polytropic/adiabatic; isothermal efficiency; FAD
Clearance	Volumetric efficiency formula; effect on work and capacity

Concept	What is usually asked
Multistaging	Optimum intermediate pressure; perfect intercooling; work saved
Rotary compressors	Vane, Roots, screw, centrifugal (slip, power input factor), axial (degree of reaction)
Instabilities	Surging, choking, stalling
Gas turbines	Brayton cycle; optimum pressure ratio; work ratio; regeneration, reheating, intercooling
Jet propulsion	Turbojet, turbofan, turboprop, ramjet, rocket; thrust and propulsive efficiency

M5. Refrigeration and Air Conditioning (Chapters 21–23B)

Concept	What is usually asked
Basics	TR, COP of refrigerator and heat pump; reversed Carnot cycle
Air refrigeration	Bell–Coleman cycle COP; aircraft refrigeration systems
VCR cycle	p–h and T–s diagrams; RE, work, COP; subcooling and superheating effects
VCR components	Compressor types; condensers; expansion devices (capillary, TEV, AEV); evaporators
Refrigerants	Numbering; properties; ammonia vs halocarbons; ODP/GWP; Montreal and Kigali
VAR	Aqua-ammonia, LiBr–water, Electrolux; ideal COP
Other systems	Steam jet, thermoelectric, vortex tube, cryogenics
Psychrometry	DBT, WBT, DPT, ω, RH, enthalpy; chart processes; SHF; BPF; mixing
AC systems	Comfort conditions; summer/winter AC; cooling load components; split, central AC, VRF
Applications	Cold storage, ice plant, duct design methods, fan laws, filters

M6. Heat Transfer (Chapters 24–26D)

Concept	What is usually asked
Conduction	Fourier's law; thermal conductivity values and trends; diffusivity
Resistances	Plane, cylinder, sphere formulas; composite walls; overall U
Critical radius	k/h and 2k/h; effect on heat loss
Heat generation	Maximum temperature formulas
Fins	m, efficiency, effectiveness; when fins are useful
Transient	Biot number, lumped analysis, time constant, characteristic length
Convection	Dimensionless numbers and their meanings; laminar/turbulent correlations; Nu = 3.66/4.36
Boiling/condensation	Boiling curve regimes; CHF; dropwise vs filmwise

Concept	What is usually asked
Radiation	Black/grey bodies; Stefan–Boltzmann, Wien, Kirchhoff, Planck; view factors; shields
Heat exchangers	Types; LMTD; ε–NTU; fouling; parallel vs counter flow
Others	Mass transfer analogy, heat pipes, insulation

M7. Fuels and Measurements (Chapters 26B–26C)

Concept	What is usually asked
Coal	Ranks; proximate and ultimate analysis
Calorific value	HCV vs LCV; Dulong's formula; bomb and Junkers calorimeters
Combustion	Oxygen and air requirement; A/F ratios; excess air
Flue gas	Orsat apparatus absorbents and order
Gaseous and liquid fuels	Compositions; flash, fire, cloud and pour points
Instruments	Thermocouples, RTDs, pyrometers; Bourdon gauge; vacuum gauges; dynamometers; gas analysers

M8. How to Use This Map in the Last Two Weeks

- Week 1: for each row above, revise the chapter section, then answer the matching rows of the Concept Recall Bank without looking.
- Week 2: solve one Solved Numericals set per day under time pressure (about 1 minute per short problem), and write two Paper-II answers per day.
- Final three days: Formula Appendix, PYQ-Trend Appendix and the Last-Minute Revision Sheet only.
- Mark any row you cannot explain in 30 seconds and revisit it the next morning – spaced repetition beats re-reading.

CHAPTER 30

Last-Minute Revision Sheet

30.1 Must-Know Values

Quantity	Value
Universal gas constant R	8.314 kJ/kmol·K
R / c_p / c_v / γ of air	0.287 / 1.005 / 0.718 kJ/kg·K / 1.4
1 atm	101.325 kPa = 760 mm Hg = 10.33 m water
1 bar	10^5 Pa = 100 kPa
Triple point of water	0.01 °C, 0.6117 kPa
Critical point of water	374.14 °C, 22.09 MPa (220.9 bar)
Latent heat of water (1 atm)	2257 kJ/kg
Latent heat of ice	335 kJ/kg
Specific heat of water	4.187 kJ/kg·K
1 TR	3.5167 kW = 211 kJ/min = 12,660 kJ/h
Stefan–Boltzmann σ	5.67×10^{-8} W/m²·K⁴
Wien's constant	2898 μm·K
Critical pressure ratio: superheated / saturated steam / air	0.546 / 0.577 / 0.528
Laminar tube Nu (T_w const / q const)	3.66 / 4.36
Lumped analysis limit	Bi < 0.1
Pr of air / water (20 °C)	0.71 / ≈ 7
Mechanical equivalent of heat	4.187 J/cal
Octane / cetane reference fuels	Iso-octane & n-heptane / cetane & α-methyl naphthalene
Stoichiometric A/F petrol / diesel	≈ 14.7 / ≈ 14.5
Typical r of SI / CI	6–11 / 14–22
Calorific value: petrol / diesel	≈ 44–47 / ≈ 42–45 MJ/kg
Air composition by mass / volume O_2	23.2% / 21%
Comfort condition (summer)	24–25 °C DBT, 50% RH

30.2 Formula Flash Card

Polytropic work $W = (p_1V_1 - p_2V_2)/(n - 1)$; isothermal $W = p_1V_1 \ln(V_2/V_1)$

$T_2/T_1 = (p_2/p_1)^{((n-1)/n)} = (V_1/V_2)^{(n-1)}$

$Q_{polytropic} = W(\gamma - n)/(\gamma - 1)$

Nozzle velocity $V = 44.72 \sqrt{\Delta h}$ (Δh in kJ/kg)

$\eta_{Carnot} = 1 - T_L/T_H$; $COP_R = T_L/(T_H - T_L)$; $COP_{HP} = T_H/(T_H - T_L)$

Δs (ideal gas) $= c_p \ln(T_2/T_1) - R \ln(p_2/p_1) = c_v \ln(T_2/T_1) + R \ln(v_2/v_1)$

Available energy $= Q(1 - T_0/T)$; $I = T_0 S_{gen}$

Otto $\eta = 1 - 1/r^{(\gamma-1)}$; Brayton $\eta = 1 - 1/r_p^{((\gamma-1)/\gamma)}$

Diesel $\eta = 1 - (1/r^{(\gamma-1)})\ (\rho^{\gamma} - 1)/(\gamma(\rho - 1))$

IP = $p_m\ L\ A\ n\ k/60$ (n = N/2 four-stroke, N two-stroke; k cylinders); BP = $2\pi NT/60$

Equivalent evaporation = $m(h - h_{fw})/2257$

Impulse turbine: $u_{opt} = V_1\cos\alpha/2$, $\eta_{max} = \cos^2\alpha$; Parsons: $u_{opt} = V_1\cos\alpha$, $\eta_{max} = 2\cos^2\alpha/(1 + \cos^2\alpha)$

Compressor work = $(n/(n-1))\ p_1V_1[(p_2/p_1)^{((n-1)/n)} - 1]$; $\eta_v = 1 + C - C(p_2/p_1)^{(1/n)}$

Two-stage optimum intermediate pressure $p_2 = \sqrt{(p_1p_3)}$

Brayton optimum $r_p = (T_{max}/T_{min})^{(\gamma/(2(\gamma-1)))}$; $W_{max} = c_p(\sqrt{T_{max}} - \sqrt{T_{min}})^2$

Propulsive efficiency = $2V_a/(V_j + V_a)$

VCR COP = $(h_1 - h_4)/(h_2 - h_1)$; Bell–Coleman COP = $1/(r_p^{((\gamma-1)/\gamma)} - 1)$

VAR ideal COP = $[(T_G - T_0)/T_G] \times [T_E/(T_0 - T_E)]$

$\omega = 0.622\ p_v/(p_b - p_v)$; $h = 1.005t + \omega(2500 + 1.88t)$; BPF = $(t_2 - t_{ADP})/(t_1 - t_{ADP})$

$r_c = k/h$ (cylinder), $2k/h$ (sphere); $Q_{cyl} = 2\pi kL\ \Delta T/\ln(r_2/r_1)$

Fin: $m = \sqrt{(hP/kA)}$; long fin $Q = \sqrt{(hPkA)}\ \theta_0$; $\varepsilon = \sqrt{(kP/hA)}$

Lumped: $\theta/\theta_0 = \exp(-hAt/\rho Vc)$; $\tau = \rho Vc/(hA)$

LMTD = $(\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$; NTU = UA/C_{min}; ε (phase change) = $1 - e^{(-NTU)}$

Parallel plates radiation $q = \sigma(T_1^4 - T_2^4)/(1/\varepsilon_1 + 1/\varepsilon_2 - 1)$; n shields → $1/(n+1)$

30.3 Comparison Quick-Hits

Topic	Remember
Same r, same heat input	η: Otto > Dual > Diesel
Same max p and T	η: Diesel > Dual > Otto
Minimum compression work	Isothermal (polytropic n smaller = less work)
Carnot vs Rankine (vapour)	Rankine practical: complete condensation, small pump work
Reheat vs regeneration	Reheat → dryness; regeneration → efficiency
Fire-tube vs water-tube	Low p, small vs high p, large; water-tube for power plants
Impulse vs reaction	Pressure drop only in nozzles vs in both fixed and moving blades
SI vs CI knock	Late autoignition of end gas vs long ignition delay at start
Remedies for SI knock	High octane, low r, retard spark – these cause knock in CI
VCR vs VAR	Work-driven, high COP vs heat-driven, low COP
Dropwise vs filmwise	Dropwise higher h; filmwise used for design
Parallel vs counter flow	Counter flow: higher LMTD, smaller area, higher effectiveness
Biot vs Nusselt	k of solid vs k of fluid
Free vs forced convection	Gr vs Re governs

30.4 Traps Toppers Avoid

- Always convert temperatures to kelvin in efficiency, COP, entropy and radiation formulas.
- Work done by a system is positive in Q = ΔU + W; work done on it is negative.
- Steady-flow work uses $c_p\ \Delta T$; non-flow (closed) adiabatic work uses $c_v\ \Delta T$.

- Four-stroke IP uses N/2 power strokes per minute; two-stroke uses N.
- Equivalent evaporation is referred to 'from and at 100 °C' – divide by 2257, not 2500.
- COP of a heat pump is one more than COP of a refrigerator between the same temperatures – not double.
- Clearance reduces volumetric efficiency but does not change the ideal work per kg of air delivered.
- In LMTD, define ΔT_1 and ΔT_2 according to flow arrangement.
- Critical radius: insulation increases heat loss until r = r_c – check whether the pipe radius is above or below r_c.
- Lumped analysis: L_c = V/A (sphere R/3), not the radius itself.
- In refrigerant numbering, add 90 to get C-H-F atoms; R-7XX = 700 + molecular mass.
- Wet-bulb temperature lies between DBT and DPT; all three are equal only at 100% RH.

30.5 Final 10-Minute Checklist

- Laws: zeroth (temperature), first (energy), second (direction/quality), third (zero entropy at 0 K).
- Cycles: Carnot, Otto, Diesel, Dual, Brayton, Stirling, Ericsson, Rankine, Bell–Coleman, VCR – know diagrams and efficiencies.
- IC engines: valve timing, knock, octane/cetane, Morse test, Willan's line, performance formulae.
- Steam: boiler types and mountings/accessories, nozzles and critical pressure, turbine compounding and blade efficiency, condensers and cooling towers.
- Power plants: nuclear components, load curve factors, combined cycle.
- Compressors: multistage, intercooling, volumetric efficiency; rotary compressors, surging and choking.
- RAC: TR, VCR cycle and effects, refrigerant numbering, VAR (Electrolux), psychrometric processes, SHF and BPF.
- Heat transfer: conduction resistances, critical radius, fins, lumped analysis, dimensionless numbers, boiling, radiation laws, LMTD/NTU.

APPENDIX

Appendix A – One-Place Formula Sheet

Every formula box from the theory chapters, collected chapter by chapter for the final revision. Read it with a pen and rewrite any formula you hesitate on.

Basic Concepts of Thermodynamics

Property test: ∮dX = 0 for any cycle; path function: ∮δQ ≠ 0 in general

Kelvin: T(K) = t(°C) + 273.15; Fahrenheit: t(°F) = 1.8 t(°C) + 32; Rankine: T(R) = t(°F) + 459.67 = 1.8 T(K)

°C and °F read the same at −40; °F and K read the same at 574.59

Linear thermometer scale: (t − ice point)/(steam point − ice point) is the same on all linear scales

1 bar = 10^5 Pa; 1 atm = 1.01325 bar = 760 mm Hg = 10.33 m H_2O; 1 torr = 1 mm Hg

p_abs = p_atm + p_gauge; vacuum = p_atm − p_abs

1 kJ = 1000 N·m; 1 kWh = 3600 kJ; 1 cal = 4.187 J; 1 hp (metric) = 735.5 W

Work, Heat and the First Law (Closed Systems)

Work done in different quasi-static processes (closed system, ideal gas)

Constant volume (isochoric): W = 0

Constant pressure (isobaric): W = $p(V_2 - V_1) = mR(T_2 - T_1)$

Isothermal (pV = C): W = $p_1V_1 \ln(V_2/V_1) = mRT \ln(p_1/p_2)$

Adiabatic reversible (pV^γ = C): W = $(p_1V_1 - p_2V_2)/(\gamma - 1) = mR(T_1 - T_2)/(\gamma - 1)$

Polytropic (pV^n = C): W = $(p_1V_1 - p_2V_2)/(n - 1)$

Isentropic relations: $T_2/T_1 = (V_1/V_2)^{(\gamma-1)} = (p_2/p_1)^{((\gamma-1)/\gamma)}$

Polytropic: $T_2/T_1 = (V_1/V_2)^{(n-1)} = (p_2/p_1)^{((n-1)/n)}$

Cycle: ∮δQ = ∮δW

Process: $Q_{1-2} = (U_2 - U_1) + W_{1-2}$; per unit mass q = Δu + w

Enthalpy H = U + pV; h = u + pv

Ideal gas: du = cv dT, dh = cp dT, cp − cv = R, γ = cp/cv

cv = R/(γ − 1); cp = γR/(γ − 1); molar: $\bar{C}p - \bar{C}v = \bar{R} = 8.314$ kJ/kmol·K

Polytropic heat: Q = W (γ − n)/(γ − 1); polytropic specific heat $c_n = cv\ (n - \gamma)/(n - 1)$ (negative for 1 < n < γ)

Ideal gas: pv = RT; pV = mRT = n $\bar{R}$ T; R = $\bar{R}$/M

Compressibility factor Z = pv/RT (Z = 1 ideal; Z < 1 attractive forces dominate; Z > 1 repulsive)

Van der Waals: $(p + a/\bar{v}^2)(\bar{v} - b) = \bar{R}T$; a – intermolecular attraction, b – molecular volume

Critical constants (vdW): $a = 27\bar{R}^2Tc^2/(64pc)$; $b = \bar{R}Tc/(8pc)$; Zc = 3/8 = 0.375 (actual gases ≈ 0.27–0.29)

Incompressible substances: cp ≈ cv = c; du ≈ dh = c dT

First Law for Open Systems (Control Volume)

Mass balance: $\dot{m} = \rho_1 A_1 V_1 = \rho_2 A_2 V_2 = A\,V/v$

Steady Flow Energy Equation (per kg): $h_1 + V_1^2/2 + g\,z_1 + q = h_2 + V_2^2/2 + g\,z_2 + w$

Rate form: $\dot{m}(h_1 + V_1^2/2 + gz_1) + \dot{Q} = \dot{m}(h_2 + V_2^2/2 + gz_2) + \dot{W}$

Units check: $V^2/2$ in J/kg – divide by 1000 to add to h in kJ/kg

Shaft work (reversible, negligible KE/PE): $w = -\int v\,dp$

Nozzle exit velocity (V_1 negligible): $V_2 = 44.72\,\sqrt{(h_1 - h_2)}$ m/s with Δh in kJ/kg

Joule–Thomson coefficient $\mu_{JT} = (\partial T/\partial p)_h$

Filling an evacuated rigid tank from a line at constant h_line (adiabatic): $u_2 = h_{line} \rightarrow T_2 = \gamma\,T_{line}$ (ideal gas)

General: $m_2u_2 - m_1u_1 = (m_2 - m_1)h_{line} + Q - W$

Second Law of Thermodynamics

Heat engine efficiency: $\eta = W/Q_1 = 1 - Q_2/Q_1$

Refrigerator: $COP_R = Q_2/W = Q_2/(Q_1 - Q_2)$

Heat pump: $COP_{HP} = Q_1/W = Q_1/(Q_1 - Q_2)$

$COP_{HP} = COP_R + 1$ (for the same Q_1, Q_2)

Carnot efficiency: $\eta_C = 1 - T_2/T_1$ (temperatures in kelvin)

Carnot refrigerator: $COP_R = T_2/(T_1 - T_2)$; Carnot heat pump: $COP_{HP} = T_1/(T_1 - T_2)$

Engines in series (same heat flow through): equal work → $T = (T_1 + T_2)/2$; equal efficiency → $T = \sqrt{(T_1T_2)}$

$Q_1/Q_2 = T_1/T_2$ (reversible engine)

Entropy

Clausius inequality: $\oint \delta Q/T \le 0$ (= 0 for reversible cycles, < 0 for irreversible cycles)

Definition: $dS = (\delta Q/T)_{rev}$; $S_2 - S_1 = \int(\delta Q/T)_{rev}$ (kJ/K; specific s in kJ/kg·K)

For any process: $dS \ge \delta Q/T$; $S_2 - S_1 = \int \delta Q/T + S_{gen}$ ($S_{gen} \ge 0$)

Isolated system: $\Delta S_{isolated} \ge 0$; $\Delta S_{universe} = \Delta S_{system} + \Delta S_{surroundings} \ge 0$

Heat transfer Q from T_H to T_L: $\Delta S_{gen} = Q(1/T_L - 1/T_H) > 0$

Tds relations: $T\,ds = du + p\,dv$; $T\,ds = dh - v\,dp$

Ideal gas: $s_2 - s_1 = cv\,\ln(T_2/T_1) + R\,\ln(v_2/v_1) = cp\,\ln(T_2/T_1) - R\,\ln(p_2/p_1) = cv\,\ln(p_2/p_1) + cp\,\ln(v_2/v_1)$

Isothermal: $\Delta s = R\,\ln(v_2/v_1) = R\,\ln(p_1/p_2)$

Isochoric: $\Delta s = cv\,\ln(T_2/T_1)$; Isobaric: $\Delta s = cp\,\ln(T_2/T_1)$

Polytropic: $\Delta s = cv\,(n - \gamma)/(n - 1)\,\ln(T_2/T_1)$

Reversible adiabatic: $\Delta s = 0$ (isentropic)

Solids/liquids: $\Delta s = c\,\ln(T_2/T_1)$

Phase change at constant T: $\Delta s = h_{fg}/T_{sat}$ (water at 100 °C: 2257/373.15 = 6.05 kJ/kg·K)

Mixing of two equal masses at T_1 and T_2: $\Delta s = 2mc\,\ln[(T_1 + T_2)/(2\sqrt{(T_1T_2)})] > 0$ (AM > GM)

Turbine: η_T = actual work/isentropic work = $(h_1 - h_2)/(h_1 - h_{2s})$

Compressor/pump: η_C = isentropic work/actual work = $(h_{2s} - h_1)/(h_2 - h_1)$

Nozzle: η_N = actual KE at exit / isentropic KE = $(h_1 - h_2)/(h_1 - h_{2s})$

Availability, Thermodynamic Relations and Mixtures

Available energy from heat Q at constant T: $AE = Q(1 - T_0/T)$; unavailable energy $UE = T_0\,\Delta S$

Heat from a body cooled from T_1 to T_0 (variable T): $AE = Q - T_0\ \Delta s = mc[(T_1 - T_0) - T_0 \ln(T_1/T_0)]$

Loss of available energy due to heat transfer through finite ΔT: $\Delta AE = T_0\ \Delta S_{gen}$

Closed-system availability: $\varphi = (u - u_0) + p_0(v - v_0) - T_0(s - s_0) + V^2/2 + gz$

Flow availability (exergy): $\psi = (h - h_0) - T_0(s - s_0) + V^2/2 + gz$

Irreversibility (Gouy–Stodola theorem): $I = W_{rev} - W_{actual} = T_0\ S_{gen}$

Second-law efficiency: $\eta_{II} = \eta_{th}/\eta_{rev}$ (engines) = COP/COP_rev (refrigerators)

Helmholtz function $F = U - TS$; Gibbs function $G = H - TS$

$dU = T\,dS - p\,dV$; $dH = T\,dS + V\,dp$; $dF = -S\,dT - p\,dV$; $dG = -S\,dT + V\,dp$

Maxwell: $(\partial T/\partial v)_s = -(\partial p/\partial s)_v$; $(\partial T/\partial p)_s = (\partial v/\partial s)_p$; $(\partial p/\partial T)_v = (\partial s/\partial v)_T$; $(\partial v/\partial T)_p = -(\partial s/\partial p)_T$

Clausius–Clapeyron: $dp/dT = h_{fg}/(T\ v_{fg})$ ($\approx h_{fg}\ p/(R\ T^2)$ for vapour as ideal gas, $v_f \ll v_g$)

$c_p - c_v = -T\ (\partial v/\partial T)_p^2\ (\partial p/\partial v)_T = T\ v\ \beta^2/\kappa_T \geq 0$

Joule–Thomson coefficient: $\mu = (\partial T/\partial p)_h = [T(\partial v/\partial T)_p - v]/c_p$

Dalton's law: $p = \Sigma p_i$; $p_i = y_i\ p$ (y_i = mole fraction)

Amagat's law: $V = \Sigma V_i$; volume fraction = mole fraction

Mixture properties: $R_{mix} = \bar{R}/M_{mix}$; $M_{mix} = \Sigma y_i M_i$; $c_{p,mix} = \Sigma$(mass fraction × c_{pi})

Entropy of mixing ideal gases: $\Delta S = -\bar{R}\ \Sigma\ n_i\ \ln\ y_i > 0$

Properties of Pure Substances and Steam

Gibbs phase rule: $F = C - P + 2$ (pure substance at triple point: $F = 1 - 3 + 2 = 0$)

Dryness fraction $x = m_{vapour}/(m_{vapour} + m_{liquid})$ $(0 \leq x \leq 1)$

Wetness $= 1 - x$; quality as percentage $= 100x$

Wet steam: $v = v_f + x\ v_{fg} \approx x\ v_g$; $h = h_f + x\ h_{fg}$; $s = s_f + x\ s_{fg}$; $u = u_f + x\ u_{fg}$

Superheated: $h = h_g + c_{ps}(T_{sup} - T_{sat})$ ($c_{ps} \approx 2.1$ kJ/kg·K); degree of superheat = $T_{sup} - T_{sat}$

Internal energy: $u = h - pv$

Throttling calorimeter: $h_{f1} + x\ h_{fg1} = h_{g2} + c_{ps}(T_2 - T_{sat2}) \rightarrow x$

Combined: $x = x_{separating} \times x_{throttling}$

Gas Power Cycles (Air-Standard Cycles)

Compression ratio $r = V_1/V_2 = (V_s + V_c)/V_c$; clearance ratio $c = V_c/V_s \rightarrow r = 1 + 1/c$

Cut-off ratio $\rho = V_3/V_2$ (Diesel); pressure (explosion) ratio $\alpha = p_3/p_2$ (Otto, Dual)

Mean effective pressure: MEP = net work per cycle / swept volume

Work ratio = net work / gross (positive) work

$\eta_{Otto} = 1 - 1/r^{(\gamma-1)}$

Heat added $Q_1 = c_v(T_3 - T_2)$; rejected $Q_2 = c_v(T_4 - T_1)$; $T_2/T_1 = T_3/T_4 = r^{(\gamma-1)}$

$MEP_{Otto} = p_1\ r\ [(\alpha - 1)(r^{(\gamma-1)} - 1)]\ /\ [(\gamma - 1)(r - 1)]$

For maximum work between T_{min} and T_{max}: optimum $r^{(\gamma-1)} = \sqrt{(T_3/T_1)} \rightarrow T_2 = T_4 = \sqrt{(T_1 T_3)}$

$\eta_{Diesel} = 1 - (1/r^{(\gamma-1)}) \times [(\rho^\gamma - 1)/(\gamma(\rho - 1))]$

$Q_1 = c_p(T_3 - T_2)$; $Q_2 = c_v(T_4 - T_1)$; $T_3/T_2 = \rho$; $T_4/T_1 = \rho^\gamma$

$MEP_{Diesel} = p_1\ r^\gamma\ [\gamma(\rho - 1) - r^{(1-\gamma)}(\rho^\gamma - 1)]\ /\ [(\gamma - 1)(r - 1)]$

Cut-off as % of stroke: $(V_3 - V_2)/(V_1 - V_2) = (\rho - 1)/(r - 1)$

$\eta_{Dual} = 1 - (1/r^{(\gamma-1)}) \times [(\alpha\rho^\gamma - 1)/((\alpha - 1) + \gamma\alpha(\rho - 1))]$

$\eta_{Brayton} = 1 - 1/r_p^{((\gamma-1)/\gamma)}$ (r_p = pressure ratio)

Net work $w = c_p[(T_3 - T_4) - (T_2 - T_1)]$

Optimum pressure ratio for maximum net work: $r_{p,opt} = (T_3/T_1)^{\gamma/(2(\gamma-1))} \rightarrow T_2 = T_4 = \sqrt{(T_1 T_3)}$

Maximum net work: $w_{max} = c_p(\sqrt{T_3} - \sqrt{T_1})^2$

Back-work ratio = compressor work / turbine work (40–80% in gas turbines – very high)

With actual efficiencies: $\eta_c = (T_2s - T_1)/(T_2 - T_1)$; $\eta_t = (T_3 - T_4)/(T_3 - T_4s)$

Regenerator effectiveness $\varepsilon = (T_x - T_2)/(T_4 - T_2)$

IC Engines – Classification, Working and Components

Bore D; stroke L = 2 × crank radius; swept (displacement) volume $V_s = (\pi/4)D^2L$

Total cylinder volume $V = V_s + V_c$; compression ratio $r = (V_s + V_c)/V_c$

Mean piston speed $V_p = 2LN/60$ (m/s)

Engine displacement = V_s × number of cylinders

Stroke-to-bore ratio L/D: square engine L = D; over-square (short stroke) L < D – high-speed engines; under-square (long stroke) L > D

Scavenging ratio = mass of delivered air / (swept volume × ambient density); scavenging efficiency = mass of fresh charge retained / total charge in cylinder

IC Engines – Combustion, Knocking and Fuels

Equivalence ratio $\varphi = (F/A)_{actual} / (F/A)_{stoich}$; $\varphi > 1$ rich, $\varphi < 1$ lean; $\lambda = 1/\varphi$

Stoichiometric A/F for a hydrocarbon C_xH_y: $(A/F)_s = 34.56\,(4x + y)/(12x + y) \rightarrow$ octane $C_8H_{18} \approx 15{:}1$

IC Engines – Fuel Supply, Ignition, Cooling, Lubrication

Air velocity at throat: $V = \sqrt{(2\,\Delta p/\rho_a)}$; fuel flow $\dot{m}_f = C_{df} A_f \sqrt{(2\rho_f(\Delta p - \rho_f g z))}$

IC Engines – Performance, Testing, Supercharging, Emissions

Indicated power $IP = p_{mi} L A n k / 60$ (W, with p_{mi} in Pa) – $n = N/2$ (four-stroke), $n = N$ (two-stroke); k = number of cylinders

Brake power $BP = 2\pi NT/60$ = torque × angular speed

Friction power FP = IP − BP

Mechanical efficiency $\eta_m = BP/IP$ (typically 75–90%)

Indicated thermal efficiency $\eta_{ith} = IP/(\dot{m}_f \times CV)$; brake thermal efficiency $\eta_{bth} = BP/(\dot{m}_f \times CV) = \eta_{ith} \times \eta_m$

Relative efficiency $\eta_{rel} = \eta_{ith}$ / air-standard efficiency

Brake mean effective pressure: $bmep = BP \times 60/(L A n k)$

Specific fuel consumption: $bsfc = \dot{m}_f/BP$ (kg/kWh); $bsfc = 3600/(\eta_{bth} \times CV)$ with CV in kJ/kg

Volumetric efficiency η_v = actual mass (or volume at intake conditions) of air inducted / swept volume × ρ_{air}

Air–fuel ratio = $\dot{m}_{air}/\dot{m}_{fuel}$

Torque and bmep: $T = bmep \times V_s \times k/(2\pi \times n_R)$ ($n_R = 2$ for four-stroke, 1 for two-stroke)

Imep from indicator card: p_{mi} = (area of diagram / length of diagram) × spring stiffness

Heat to coolant = $\dot{m}_w c_{pw} (t_{out} - t_{in})$; heat to exhaust = $\dot{m}_g c_{pg} (t_g - t_{ambient})$

IC Engines – Modern Technologies, Emission Norms and Special Engines

$\Delta E = I\,\omega^2$ $C_s = m\,k^2\,\omega^2$ C_s; $C_s = (\omega_{max} - \omega_{min})/\omega_{mean}$

Vapour Power Cycles (Rankine and Modifications)

Pump work $w_p = v_f\,(p_2 - p_1) = h_2 - h_1$ (≈ 0.001 m³/kg × Δp)

Turbine work $w_t = h_3 - h_4$

Heat supplied $q_1 = h_3 - h_2$; heat rejected $q_2 = h_4 - h_1$

$\eta_{Rankine} = (w_t - w_p)/q_1 = [(h_3 - h_4) - (h_2 - h_1)]/(h_3 - h_2) \approx (h_3 - h_4)/(h_3 - h_1)$ if pump work is neglected

Specific steam consumption SSC = 3600/w_net (kg/kWh, w_net in kJ/kg)

Heat rate = 3600/η (kJ/kWh)

Work ratio = w_net/w_t

Mean temperature of heat addition $T_{m1} = (h_3 - h_2)/(s_3 - s_2)$; $\eta = 1 - T_2/T_{m1}$ (T_2 = condenser temperature)

Turbine work = $(h_3 - h_4) + (h_5 - h_6)$ (HP + LP turbines); heat supplied = $(h_3 - h_2) + (h_5 - h_4)$

$\eta_{reheat} = [(h_3 - h_4) + (h_5 - h_6) - w_p] / [(h_3 - h_2) + (h_5 - h_4)]$

Open heater mass balance: $m(h_{bled}) + (1 - m)h_{feed} = h_{sat,out} \rightarrow m = (h_{out} - h_{feed})/(h_{bled} - h_{feed})$

Combined efficiency (ideal coupling): $\eta_{cc} = \eta_{GT} + \eta_{ST} - \eta_{GT}\,\eta_{ST}$

Utilisation factor (cogeneration) = (net work + process heat)/heat supplied

Steam Generators (Boilers)

Natural draught (mm of water): $h_w = 353\,H\,[1/T_a - ((m+1)/m)(1/T_g)]$ (H = chimney height in m, m = air–fuel ratio by mass, temperatures in K)

Draught as column of hot gas: $H_1 = H[(m/(m+1))(T_g/T_a) - 1]$

Condition for maximum discharge of gases through a chimney: $T_g/T_a = 2(m+1)/m$ ($\approx 2 \rightarrow$ flue gases about twice ambient absolute temperature)

Chimney efficiency = (useful work equivalent of draught) / (extra heat carried by flue gases) – very low, ≈ 0.25–0.5%

Equivalent evaporation (from and at 100 °C): $m_e = m_a\,(h - h_{fw})/2257$ kg of steam per kg of fuel

Boiler efficiency $\eta_b = m_a\,(h - h_{fw})/CV$ (m_a = actual steam per kg fuel)

Factor of evaporation $F_e = (h - h_{fw})/2257$

Boiler horsepower (old unit) = equivalent evaporation of 15.65 kg/h from and at 100 °C

Steam Nozzles and Diffusers

Exit velocity (adiabatic, inlet velocity negligible): $V_2 = \sqrt{(2(h_1 - h_2))} = 44.72\,\sqrt{(\Delta h,\ kJ/kg)}$

With inlet velocity: $V_2 = \sqrt{(2000\,\Delta h + V_1^2)}$

Isentropic (pv^n = const): $V_2 = \sqrt{[(2n/(n-1))\,p_1v_1\,(1 - (p_2/p_1)^{((n-1)/n)})]}$

Mass flow per unit area: $\dot{m}/A = V/v$; continuity $A_1V_1/v_1 = A_2V_2/v_2$

Area–velocity relation: $dA/A = (M^2 - 1)\,dV/V \rightarrow$ subsonic: area decreases as velocity increases; supersonic: area increases

Critical pressure ratio: $p_c/p_1 = (2/(n + 1))^{(n/(n-1))}$

Critical (throat) velocity: $V_c = \sqrt{(2n/(n+1) \times p_1 v_1)} = \sqrt{(n\ p_c\ v_c)}$ = speed of sound at throat

Maximum mass flow (choked): $\dot{m}_{max}/A_t = \sqrt{[n\ (p_1/v_1)\ (2/(n+1))^{((n+1)/(n-1))}]}$

Nozzle efficiency η_n = actual enthalpy drop / isentropic enthalpy drop (0.90–0.98)

Velocity coefficient $C_v = \sqrt{\eta_n}$; coefficient of discharge = actual mass flow / theoretical mass flow

Steam Turbines

Whirl components: $V_{w1} = V_1 \cos\alpha$; $V_{w2} = V_2 \cos\beta$

Work per kg = $u\ (V_{w1} \pm V_{w2})$ (+ when V_{w2} is opposite to V_{w1} – usual)

Tangential force (driving thrust) per kg/s = $V_{w1} \pm V_{w2}$; axial thrust per kg/s = $V_{f1} - V_{f2}$

Blade (diagram) efficiency $\eta_b = 2u(V_{w1} \pm V_{w2})/V_1^2$

Stage efficiency = blade efficiency × nozzle efficiency = work/(enthalpy drop in stage)

Impulse blade efficiency (symmetrical blades, no friction, $k = 1$): $\eta_b = 4\rho(\cos\alpha - \rho)$, where $\rho = u/V_1$

Optimum blade speed ratio (single-stage impulse): $u/V_1 = \cos\alpha/2$; $\eta_{b,max} = \cos^2\alpha$

With blade friction ($k = V_{r2}/V_{r1}$) and blade angles: $\eta_{b,max} = (\cos^2\alpha/2)(1 + k\ \cos\varphi/\cos\theta)$

Velocity-compounded (2 rows): optimum $u/V_1 = \cos\alpha/4$; work in rows ratio 3:1 (first row does 3/4 of total work)

For n rows: optimum $u/V_1 = \cos\alpha/(2n)$; work ratio of rows: $(2n-1):(2n-3):\ldots:1$

Degree of reaction R = enthalpy drop in moving blades / enthalpy drop in the stage = $\Delta h_{moving}/(\Delta h_{fixed} + \Delta h_{moving})$

Parsons (50% reaction): optimum $u/V_1 = \cos\alpha$; $\eta_{b,max} = 2\cos^2\alpha/(1 + \cos^2\alpha)$

Height of blades from continuity: $\dot{m}\ v = \pi\ D_m\ h\ V_f$ (D_m = mean diameter, h = blade height)

Reheat factor RF = $\Sigma(\Delta h_{stage,isentropic}) / (\Delta h_{overall,isentropic})$; internal efficiency $\eta_{int} = \eta_{stage} \times RF$

Willans line: steam consumption varies linearly with load for throttle-governed turbines

Condensers and Cooling Towers

Vacuum = barometric reading – absolute condenser pressure (both in mm Hg)

Vacuum efficiency = actual vacuum / ideal vacuum; ideal vacuum = barometer reading – saturation pressure of steam at condensate temperature

Condenser efficiency = $(t_{w,out} - t_{w,in}) / (t_{sat} - t_{w,in})$ (actual rise of cooling water / maximum possible rise)

Cooling water required: $\dot{m}_w\ c_{pw}\ (t_{w,out} - t_{w,in}) = \dot{m}_s\ (h_{exhaust} - h_{condensate})$

Dalton's law in condenser: $p_{condenser} = p_s + p_a$; mass of air $\dot{m}_a = p_a\ V/(R\ T)$

Range = water inlet temperature – water outlet temperature

Approach = water outlet temperature – wet bulb temperature of entering air (the outlet water can never be cooled below WBT)

Cooling tower efficiency = range/(range + approach) = $(t_{in} - t_{out})/(t_{in} - t_{wb})$

Power Plant Engineering

Load factor = average load / maximum demand (over the period)

Demand factor = maximum demand / connected load (≤ 1)

Diversity factor = sum of individual maximum demands / maximum demand of the whole system (≥ 1)

Plant capacity factor = average load / rated plant capacity = actual energy produced / (capacity × hours)

Plant use factor = actual energy produced / (capacity × hours of actual operation)

Reserve capacity = plant capacity − maximum demand

Power $P = \eta \rho g Q H$ (W); energy = P × time

Air Compressors

Work per cycle (no clearance), polytropic $pV^n = C$: $W = (n/(n-1))\, p_1V_1\, [(p_2/p_1)^{((n-1)/n)} - 1]$

Isentropic ($n = \gamma$): $W = (\gamma/(\gamma-1))\, p_1V_1\, [(p_2/p_1)^{((\gamma-1)/\gamma)} - 1] = \dot{m}\, c_p\, (T_2 - T_1)$

Isothermal ($n = 1$): $W = p_1V_1 \ln(p_2/p_1)$ — minimum work

With clearance, the work formula is the same with V_1 replaced by the effective swept (induced) volume $V_1 - V_4$

Isothermal efficiency η_{iso} = isothermal work / actual work

Delivery temperature: $T_2 = T_1\, (p_2/p_1)^{((n-1)/n)}$

Power = W × N_working strokes per second (single-acting: N/60; double-acting: 2N/60)

Clearance ratio C = V_c/V_s (typically 4–10%)

Volumetric efficiency $\eta_v = (V_1 - V_4)/V_s = 1 + C - C\, (p_2/p_1)^{(1/n)}$

Maximum pressure ratio ($\eta_v = 0$): $(p_2/p_1)_{max} = ((1 + C)/C)^n$

Free air delivery (FAD): volume of air delivered reduced to intake (free-air) conditions — $p_a V_a/T_a = p_1V_1/T_1$

Overall volumetric efficiency referred to FAD: η_v' = (FAD per cycle)/V_s

Two-stage, perfect intercooling, equal n: optimum intermediate pressure $p_2 = \sqrt{(p_1p_3)}$

Pressure ratio per stage for z stages: $(p_{final}/p_1)^{(1/z)}$ (equal in each stage)

Minimum total work (z stages): $W = z\, (n/(n-1))\, p_1V_1\, [(p_{z+1}/p_1)^{((n-1)/(zn))} - 1]$

Under these conditions: work in each stage is equal, delivery temperature from each stage is equal, heat rejected in each intercooler is equal

Heat rejected in intercooler = $\dot{m}\, c_p\, (T_2 - T_1)$; heat rejected during polytropic compression = $W\, (\gamma - n)/(\gamma - 1)$

Roots blower work = $(p_2 - p_1)V$; Roots efficiency = isentropic work / actual work

Work input per kg (radial vanes, no prewhirl) = $\sigma\, \psi\, u_2^2$ (σ = slip factor ≈ 0.9, ψ = power input factor ≈ 1.03–1.05)

Pressure coefficient = isentropic work / u_2^2; temperature rise of air $\Delta T = \sigma\psi u_2^2/c_p$

Gas Turbines and Jet Propulsion

Simple cycle with component efficiencies: $w_{net} = c_p[\eta_t T_3(1 - 1/r_p^k) - T_1(r_p^k - 1)/\eta_c]$, $k = (\gamma-1)/\gamma$

Thermal efficiency $\eta = w_{net} / [c_p(T_3 - T_2)]$

Thrust (air-breathing): $F = \dot{m}_a[(1 + f)V_j - V_a] + (p_e - p_a)A_e \approx \dot{m}_a(V_j - V_a)$

Thrust power = F × V_a

Propulsive (Froude) efficiency η_p = thrust power / KE supplied to jet = $2V_a/(V_a + V_j) = 2/(1 + V_j/V_a)$

Thermal efficiency = increase in KE of gases / heat supplied by fuel; overall efficiency = $\eta_p \times \eta_{th} = F\, V_a/(\dot{m}_f\, CV)$

Thrust specific fuel consumption TSFC = $\dot{m}_f / F$

Rocket thrust $F = \dot{m}\, V_e + (p_e - p_a)A_e$; specific impulse $I_{sp} = F/(\dot{m}\, g)$ (seconds)

Refrigeration – Fundamentals, Air Refrigeration and VCR

1 tonne of refrigeration (TR) = 211 kJ/min = 3.5167 kW (≈ 3.5 kW) ≈ 50 kcal/min ≈ 12,000 BTU/h

COP = refrigerating effect / work input; relative COP = actual COP / theoretical COP

Carnot (reversed) COP_R = T_L/(T_H − T_L); COP_HP = COP_R + 1

Energy efficiency ratio EER = COP × 3.412 (BTU/Wh)

COP = $T_1/(T_2 - T_1)$ = 1/(r_p^((γ−1)/γ) − 1) (r_p = pressure ratio; T_1 = compressor inlet, T_2 = compressor outlet)

Refrigerating effect = cp($T_1 - T_4$); work = cp[($T_2 - T_3$) − ($T_1 - T_4$)]

Refrigerating effect RE = $h_1 - h_4 = h_1 - h_3$ (kJ/kg)

Compressor work W = $h_2 - h_1$; heat rejected Q_c = $h_2 - h_3$ = RE + W

COP = $(h_1 - h_4)/(h_2 - h_1)$

Refrigerant mass flow $\dot{m}$ = 211 × TR / RE (kg/min) or 3.5167 × TR / RE (kg/s)

Compressor power = $\dot{m}$ ($h_2 - h_1$); power per TR = 3.5167/COP kW

Volumetric efficiency of compressor: η_v = 1 + C − C (p_d/p_s)^(1/n); piston displacement = $\dot{m}$ v_1/η_v

Refrigerants, Vapour Absorption and Other Systems

Halocarbons $C_mH_nF_pCl_q$: R-(m−1)(n+1)(p) — hundreds digit = C − 1, tens = H + 1, units = F

Inorganic: R-7XX where XX = molecular mass (NH_3 = 17 → R-717; H_2O = 18 → R-718; CO_2 = 44 → R-744; air = 29 → R-729)

Blends: zeotropic R-4XX (R-404A, R-407C, R-410A); azeotropic R-5XX (R-502); hydrocarbons R-600 (n-butane), R-600a (isobutane), R-290 (propane)

Ideal (maximum) COP of VAR: COP = [(T_G − T_0)/T_G] × [T_E/(T_0 − T_E)] (T_G generator, T_0 ambient/condenser, T_E evaporator)

Practical COP: single-effect LiBr ≈ 0.6–0.8; double-effect ≈ 1.0–1.3; aqua-ammonia ≈ 0.5–0.7

Psychrometry and Air Conditioning

ω = 0.622 p_v/(p_b − p_v) kg/kg of dry air

φ = p_v/p_s; μ = ω/ω_s = φ (p_b − p_s)/(p_b − p_v) (μ ≈ φ)

Enthalpy of moist air: h = 1.005 t + ω(2500 + 1.88 t) kJ/kg of dry air (t in °C)

Humid specific heat c_pa = 1.005 + 1.88ω ≈ 1.022 kJ/kg·K

Specific volume v = R_a T/p_a = 0.287 T/(p_b − p_v) m^3/kg dry air

Carrier's equation: p_v = p_vs' − (p_b − p_vs')(t_db − t_wb)/(1544 − 1.44 t_wb) (p_vs' = saturation pressure at WBT)

Sensible heat: Q_s = $\dot{m}$_a c_pa ΔT ≈ 1.022 $\dot{m}$_a ΔT; latent heat: Q_L = $\dot{m}$_a Δω h_fg ≈ 2500 $\dot{m}$_a Δω

Total heat = $\dot{m}$_a ($h_2 - h_1$) = Q_s + Q_L

Sensible heat factor SHF = Q_s/(Q_s + Q_L) (slope of the condition line)

Bypass factor BPF = (t_2 − t_ADP)/(t_1 − t_ADP); contact (coil) efficiency = 1 − BPF

Mixing: $\dot{m}_1\omega_1 + \dot{m}_2\omega_2 = (\dot{m}_1 + \dot{m}_2)\omega_3$; $\dot{m}_1h_1 + \dot{m}_2h_2 = (\dot{m}_1 + \dot{m}_2)h_3$; t_3 ≈ weighted average

Humidifying efficiency (air washer) = ($t_1 - t_2$)/(t_1 − t_wb)

Refrigeration and AC – Applications and Equipment

Product load (freezing): $Q = m[c_1(T_1 - T_f) + L + c_2(T_f - T_2)]$ (c_1 above freezing, c_2 below freezing, L latent heat of product)

Energy efficiency ratio EER = cooling capacity (W)/power input (W) (= COP for cooling)

Chiller performance often quoted in kW/TR = 3.517/COP (efficient water-cooled centrifugal ≈ 0.5–0.65 kW/TR)

Equivalent diameter of rectangular duct (same friction and flow): D_eq = 1.3(ab)^0.625/(a + b)^0.25; for same velocity and friction: D_h = 2ab/(a + b)

Fan laws (same fan, density constant): $Q \propto N$; $\Delta p \propto N^2$; $P \propto N^3$

Geometrically similar fans: $Q \propto N D^3$; $\Delta p \propto N^2 D^2$; $P \propto N^3 D^5$

Heat Transfer I – Conduction, Fins, Transient Conduction

Thermal diffusivity $\alpha = k/(\rho c_p)$ (m^2/s) – how fast temperature changes propagate in transient conduction

Heat conduction equation (Cartesian): $\partial^2T/\partial x^2 + \partial^2T/\partial y^2 + \partial^2T/\partial z^2 + q_g/k = (1/\alpha)\,\partial T/\partial t$

Steady 1-D, no generation: $d^2T/dx^2 = 0 \rightarrow$ linear temperature profile (Laplace equation)

Poisson equation (steady with generation): $\nabla^2T + q_g/k = 0$; Fourier equation (transient, no generation): $\nabla^2T = (1/\alpha)\partial T/\partial t$

Plane wall: $Q = kA(T_1 - T_2)/L$; thermal resistance $R = L/(kA)$

Convection resistance $R = 1/(hA)$; radiation resistance $R = 1/(h_r A)$, $h_r = \varepsilon\sigma(T_1^2 + T_2^2)(T_1 + T_2)$

Composite wall (series): $Q = (T_{\infty 1} - T_{\infty 2})/(1/(h_1A) + \Sigma L_i/(k_iA) + 1/(h_2A))$

Parallel layers: $1/R_{eq} = \Sigma 1/R_i$

Overall heat transfer coefficient (plane): $1/U = 1/h_1 + \Sigma L/k + 1/h_2$; $Q = U A \Delta T$

Hollow cylinder: $Q = 2\pi kL(T_1 - T_2)/\ln(r_2/r_1)$; $R = \ln(r_2/r_1)/(2\pi kL)$

Logarithmic mean area (cylinder): $A_m = (A_2 - A_1)/\ln(A_2/A_1) \rightarrow Q = kA_m \Delta T/(r_2 - r_1)$

Hollow sphere: $Q = 4\pi k\, r_1r_2(T_1 - T_2)/(r_2 - r_1)$; $R = (r_2 - r_1)/(4\pi k\, r_1r_2)$; geometric mean area $A_m = \sqrt{(A_1A_2)}$

Variable conductivity $k = k_0(1 + \beta T)$: use $k_m = k_0(1 + \beta\, T_{mean})$ in the constant-k formula (plane wall)

Cylinder: $r_c = k_{ins}/h_o$; sphere: $r_c = 2k_{ins}/h_o$

Plane wall (thickness 2L, both surfaces at T_s): $T_{max} = T_s + q_g L^2/(2k)$ at the centre

Solid cylinder (radius R): $T_{max} = T_s + q_g R^2/(4k)$; $T(r) = T_s + q_g(R^2 - r^2)/(4k)$

Solid sphere: $T_{max} = T_s + q_g R^2/(6k)$

Surface temperature from convection: $T_s = T_\infty + q_g V/(hA)$ (cylinder: $T_\infty + q_g R/(2h)$)

Fin parameter $m = \sqrt{(hP/(kA_c))}$ (P = perimeter, A_c = cross-section area)

Infinitely long fin: $Q = \sqrt{(hPkA_c)}\,\theta_0$; $\theta/\theta_0 = e^{(-mx)}$

Fin with insulated tip: $Q = \sqrt{(hPkA_c)}\,\theta_0 \tanh(mL)$; $\theta/\theta_0 = \cosh m(L - x)/\cosh mL$

Fin with convective tip: $Q = \sqrt{(hPkA_c)}\,\theta_0$ [sinh mL + (h/mk) cosh mL]/[cosh mL + (h/mk) sinh mL]; approximate with corrected length $L_c = L + t/2$ (rectangular), $L + D/4$ (pin)

Fin efficiency η_f = actual heat transfer / heat transfer if the whole fin were at base temperature; insulated tip: $\eta_f = \tanh(mL)/(mL)$; long fin $\eta \approx 1/(mL)$

Fin effectiveness ε_f = heat transfer with fin / without fin = $Q_{fin}/(h A_c \theta_0) = \sqrt{(kP/(hA_c))}$ (long fin)

Biot number $Bi = h L_c / k_{solid}$ (ratio of internal conduction resistance to surface convection resistance)

Characteristic length $L_c = V/A_s$: plane wall (thickness 2L) = L; long cylinder = R/2; sphere = R/3; cube of side a = a/6

Lumped-capacitance analysis (valid if Bi < 0.1): (T − T_∞)/(T_i − T_∞) = exp(−hA_s t/(ρVc)) = exp(−Bi·Fo)

Fourier number Fo = α t/L_c² (dimensionless time)

Time constant τ = ρVc/(hA_s) — time to reach 63.2% of the total temperature change

Instantaneous heat transfer rate q(t) = hA_s(T_i − T_∞) e^(−t/τ); total Q = ρVc(T_i − T_∞)(1 − e^(−t/τ))

Heat Transfer II – Convection, Boiling and Condensation

Flat plate, laminar (Re_x < 5 × 10⁵): local Nu_x = 0.332 Re_x^0.5 Pr^(1/3); average Nu_L = 0.664 Re_L^0.5 Pr^(1/3)

Flat plate, turbulent: Nu_x = 0.0296 Re_x^0.8 Pr^(1/3); average Nu_L = 0.037 Re_L^0.8 Pr^(1/3)

Laminar fully developed tube flow: Nu = 3.66 (constant wall temperature), Nu = 4.36 (constant heat flux)

Turbulent tube flow (Dittus–Boelter): Nu = 0.023 Re^0.8 Pr^n; n = 0.4 (fluid being heated), 0.3 (cooled)

Sieder–Tate (large viscosity variation): Nu = 0.027 Re^0.8 Pr^(1/3) (μ/μ_s)^0.14

Reynolds analogy (Pr = 1): St = C_f/2; Colburn analogy: St Pr^(2/3) = C_f/2 (j-factor)

Hydraulic diameter for non-circular ducts: D_h = 4A/P

Bulk (mixing-cup) temperature and LMTD for tubes: q = h A ΔT_lm

Nu = C (Gr Pr)^n = C Ra^n; laminar (10⁴ < Ra < 10⁹): n = 1/4; turbulent (Ra > 10⁹): n = 1/3 (h independent of length)

Volume expansion coefficient for ideal gas: β = 1/T (absolute film temperature)

Film temperature T_f = (T_s + T_∞)/2 — properties evaluated at it

Nusselt: h_film (vertical plate) = 0.943 [ρ_l(ρ_l − ρ_v)g h_fg k_l³/(μ_l L ΔT)]^(1/4); horizontal tube uses 0.725 with D in place of L

Heat Transfer III – Radiation and Heat Exchangers

Absorptivity α + reflectivity ρ + transmissivity τ = 1; opaque body τ = 0 → α + ρ = 1

Stefan–Boltzmann: $E_b = \sigma T^4$; σ = 5.67 × 10⁻⁸ W/m²·K⁴; real surface $E = \varepsilon\sigma T^4$

Wien's displacement law: $\lambda_{max} T = 2898$ μm·K (≈ 2900)

Planck's law: $E_{b\lambda} = C_1\lambda^{-5}/(e^{(C_2/\lambda T)} - 1)$; $C_1 = 3.742 \times 10^8$ W·μm⁴/m², $C_2 = 1.4388 \times 10^4$ μm·K

Kirchhoff's law: at thermal equilibrium ε = α (for a grey body at all temperatures)

Lambert's cosine law: intensity of a diffuse emitter I is independent of direction; emissive power E = πI

Shape (view/configuration) factor F_{12}: fraction of radiation leaving surface 1 that reaches surface 2 directly

Reciprocity: $A_1F_{12} = A_2F_{21}$; summation: $\Sigma_j F_{1j} = 1$ (enclosure)

Flat or convex surface: $F_{11} = 0$; concave surface: $F_{11} > 0$

Small body (1) completely enclosed by a large enclosure (2): $F_{12} = 1$; $F_{21} = A_1/A_2$

Two large parallel plates: $q = \sigma(T_1^4 - T_2^4)/(1/\varepsilon_1 + 1/\varepsilon_2 - 1)$ per unit area

Concentric cylinders/spheres: $Q = \sigma A_1(T_1^4 - T_2^4)/[1/\varepsilon_1 + (A_1/A_2)(1/\varepsilon_2 - 1)]$

Small body in large enclosure: $Q = \varepsilon_1\sigma A_1(T_1^4 - T_2^4)$

Black surfaces: $Q_{12} = A_1F_{12}\sigma(T_1^4 - T_2^4)$

Surface resistance $(1 - \varepsilon)/(\varepsilon A)$; space (geometric) resistance $1/(A_1F_{12})$ — radiation network

Hemisphere over its base: F(base → hemisphere) = 1; F(hemisphere → base) = 0.5; F(hemisphere → itself) = 0.5

n shields between two parallel plates (all emissivities equal): heat transfer reduced to 1/(n + 1) of the unshielded value

One shield with ε_s between plates with ε_1, ε_2: resistance = $(1/\varepsilon_1 + 1/\varepsilon_2 - 1) + (2/\varepsilon_s - 1)$ per unit area

Heat transfer: $Q = U A \Delta T_{lm}$

LMTD = $(\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$

Parallel flow: $\Delta T_1 = T_{h,in} - T_{c,in}$, $\Delta T_2 = T_{h,out} - T_{c,out}$; counter flow: $\Delta T_1 = T_{h,in} - T_{c,out}$, $\Delta T_2 = T_{h,out} - T_{c,in}$

When $\Delta T_1 = \Delta T_2$ (counter flow, equal capacity rates): LMTD = ΔT (constant along length)

Cross-flow and multipass: $Q = U A F \Delta T_{lm,counter}$ (F = correction factor ≤ 1)

Overall U (tube, based on outer area): $1/U_o = 1/h_o + R_{fo} + (r_o \ln(r_o/r_i))/k + (r_o/r_i) R_{fi} + r_o/(r_i h_i)$

Fouling factor (scale resistance) $R_f = 1/U_{dirty} - 1/U_{clean}$

Capacity rate $C = \dot{m} c_p$; C_{min}, C_{max}; capacity ratio $C_r = C_{min}/C_{max}$

Effectiveness $\varepsilon = Q_{actual}/Q_{max}$; $Q_{max} = C_{min} (T_{h,in} - T_{c,in})$

Number of transfer units NTU = $U A / C_{min}$

Parallel flow: $\varepsilon = [1 - \exp(-NTU(1 + C_r))]/(1 + C_r)$

Counter flow: $\varepsilon = [1 - \exp(-NTU(1 - C_r))]/[1 - C_r \exp(-NTU(1 - C_r))]$; $C_r = 1$: $\varepsilon = NTU/(1 + NTU)$

Phase change (condenser/evaporator, $C_r = 0$): $\varepsilon = 1 - \exp(-NTU)$ (same for all arrangements)

Fuels and Combustion

LCV = HCV − $m_w \times h_{fg}$; m_w = 9H (+ moisture M) kg per kg fuel; $h_{fg} \approx 2442$ kJ/kg (at 25 °C)

Dulong's formula: HCV ≈ 33,800 C + 144,000 (H − O/8) + 9,270 S kJ/kg (C, H, O, S as mass fractions)

Minimum O_2 per kg of fuel = (8/3)C + 8H + S − O kg

Theoretical (stoichiometric) air = O_2 required/0.232 ≈ 4.31 × O_2 (mass; air has 23.2% O_2 by mass, 21% by volume)

Hydrocarbon C_xH_y: $C_xH_y + (x + y/4)(O_2 + 3.76\ N_2) \rightarrow x\ CO_2 + (y/2)\ H_2O + 3.76(x + y/4)\ N_2$

Equivalence ratio $\varphi = (F/A)_{actual}/(F/A)_{stoich}$; $\varphi > 1$ rich, $\varphi < 1$ lean; λ (excess-air ratio) = $1/\varphi$

Excess air % = (actual air − theoretical air)/theoretical air × 100 ≈ $O_2\%/(21 - O_2\%)$ × 100 (from dry flue gas, complete combustion)

Mass of dry flue gas per kg fuel (from volumetric analysis): $m_{dfg} = [11CO_2 + 8O_2 + 7(CO + N_2)]/[3(CO_2 + CO)] \times C$ (C = mass of carbon burnt per kg fuel)

Conversion of volumetric (molar) to mass analysis: multiply each volume fraction by molar mass and divide by the total

Heat Transfer IV – Mass Transfer, Heat Pipes, Insulation and Enhancement

Fick's law: $J_A = -D_{AB}\, dC_A/dx$ (kg/m²·s or kmol/m²·s); D_{AB} = mass diffusivity (m²/s)

Convective mass transfer: $\dot{m}_A = h_m A (C_{A,s} - C_{A,\infty})$ (h_m in m/s)

Lewis number Le = α/D_{AB} = Sc/Pr; Chilton–Colburn analogy: $h/h_m = \rho\, c_p\, Le^{(2/3)}$

Lewis relation for air–water vapour (Le ≈ 1): h/(h_m ρ c_p) ≈ 1 → wet-bulb temperature ≈ adiabatic saturation temperature

APPENDIX

Appendix B – PYQ-Trend Concept Map

All PYQ-Trend boxes in one place, chapter-wise. These are the concepts SSC JE has tested repeatedly; revise this list in the last week.

Basic Concepts of Thermodynamics

★ Classification of systems with examples (turbine – open; piston–cylinder – closed; universe – isolated). Adiabatic vs isolated. Control mass vs control volume.

★ Intensive vs extensive identification (very frequent); heat and work as path functions; quasi-static process definition; thermodynamic equilibrium components; state postulate (two independent properties).

★ Zeroth law is the basis of temperature measurement (very frequent); temperature equal on °C and °F (−40); triple point of water 273.16 K; constant-volume gas thermometer as standard; thermocouple works on Seebeck effect.

★ Unit conversions; absolute vs gauge pressure; internal energy is a property; cp > cv for gases.

Work, Heat and the First Law (Closed Systems)

★ Work in free expansion = 0 (very frequent); n values for each process; adiabatic steeper than isothermal on p–V; work formulas; area under p–V = work.

★ First law for cycle and process; internal energy as a property; PMM-1; U of ideal gas depends only on T (Joule's law); cp − cv = R; γ values; heat in each process.

★ Ideal behaviour at low p and high T; meaning of vdW constants a and b; Zc = 0.375 from vdW; Dalton's and Amagat's laws; R = 8.314 kJ/kmol·K and R_air = 0.287.

★ cp = cv for incompressible substances; specific heat of water.

First Law for Open Systems (Control Volume)

★ Flow work = pv; SFEE terms; reversible steady-flow work $-\int v\,dp$ vs closed-system $\int p\,dv$; enthalpy as the natural property for flow processes.

★ Throttling is isenthalpic (very frequent); nozzle velocity $44.72\sqrt{\Delta h}$; turbine work = $h_1 - h_2$; boiler heat = $h_2 - h_1$; diffuser converts KE to pressure.

★ Throttling isenthalpic and irreversible; $\mu_{JT} = 0$ for ideal gas; cooling only below inversion temperature; hydrogen heats on throttling at room temperature.

★ Final temperature when filling an evacuated tank = γ T_line; filling and emptying processes are unsteady.

Second Law of Thermodynamics

★ Direction of heat flow; reservoirs (source and sink).

★ COP_HP − COP_R = 1 (very frequent); COP can exceed unity; definitions of efficiency and COP.

★ Kelvin–Planck vs Clausius statements (very frequent); PMM-2 violates Kelvin–Planck; equivalence of the two statements.

★ Causes of irreversibility (friction, finite ΔT heat transfer, free expansion, mixing, throttling) – frequent; reversible process is an idealisation.

★ Carnot efficiency formula and dependence only on temperatures (very frequent); Carnot cycle processes; Carnot COP formulas; series engines ($\sqrt{T_1T_2}$ for equal efficiency); impracticality of Carnot cycle.

★ Thermodynamic temperature scale is independent of the working substance.

Entropy

★ Clausius inequality; dS = δQ_rev/T; entropy as a property; area under T–s = heat; work does not transfer entropy.

★ Entropy of isolated system never decreases (very frequent); entropy generated in heat transfer across finite ΔT; entropy of a system may decrease.

★ Entropy change formulas for each process; reversible adiabatic = isentropic (very frequent); isochore steeper than isobar on T–s; entropy change in free expansion; entropy change of vaporisation.

★ Third law statement; unattainability of absolute zero.

★ Definitions of isentropic efficiency of turbine and compressor (note the inverse ratio for compressors).

Availability, Thermodynamic Relations and Mixtures

★ Unavailable energy = $T_0\Delta S$ (frequent); Gouy–Stodola theorem; available energy decreases when heat flows to a lower temperature; second-law efficiency.

★ Gibbs and Helmholtz definitions; Maxwell relations; Clausius–Clapeyron relates saturation pressure slope to latent heat; cp ≥ cv always; cp = cv for water at 4 °C; G minimum at equilibrium at constant T, p.

★ Dalton's law of partial pressures; mole fraction = volume fraction; composition of air; entropy of mixing positive.

Properties of Pure Substances and Steam

★ Critical point of water (221.2 bar, 374 °C) and latent heat zero there (very frequent); triple point 0.01 °C, 0.611 kPa; degrees of freedom at triple point = 0; negative slope of fusion curve for water; pure substance definition; sublimation of dry ice.

★ Dryness fraction definition; h = h_f + x h_fg; degree of superheat; reference state of steam tables.

★ Throttling calorimeter uses isenthalpic throttling and needs the steam to become superheated; combined calorimeter for very wet steam; $x = x_1x_2$ (very frequent).

★ Mollier diagram is h–s (very frequent); isentropic = vertical on h–s; throttling = horizontal on h–s, vertical on p–h; isotherms nearly horizontal in superheated region far from saturation.

Gas Power Cycles (Air-Standard Cycles)

★ Air-standard assumptions; compression ratio and clearance ratio relation; MEP definition (very frequent).

★ Otto efficiency formula and dependence only on r (very frequent); limit of r by knocking; $T_2 = T_4 = \sqrt{(T_1T_3)}$ for maximum work.

★ Diesel efficiency formula; efficiency decreases with increasing cut-off ratio; Diesel → Otto when ρ = 1; cut-off percentage of stroke.

★ Comparison orders (asked almost every year). Remember: same r → Otto best; same peak pressure → Diesel best.

★ Stirling and Ericsson efficiencies equal Carnot (with regenerator); Atkinson cycle efficiency higher than Otto; Lenoir cycle has no compression; Bell–Coleman for air refrigeration.

★ Brayton efficiency depends only on r_p (very frequent); optimum r_p for maximum work; back-work ratio high; regeneration increases efficiency (not work); intercooling and reheating increase work; closed vs open cycle; combined cycle efficiency.

IC Engines – Classification, Working and Components

★ EC vs IC engine examples; classification bases; radial engines in aircraft; quantity vs quality governing.

★ Swept volume and compression ratio calculations; mean piston speed; definition of TDC/BDC; square and over-square engines.

★ Valve overlap definition and purpose (very frequent); camshaft speed half of crankshaft; EVO before BDC (blow-down); IVC after BDC (ram effect); spark advance increases with speed.

★ Two-stroke completes cycle in one revolution; uniflow scavenging most efficient; lighter flywheel for two-stroke; higher HC and lube-oil consumption; comparison table points.

★ Number and function of piston rings (compression vs oil rings); connecting rod I-section; camshaft speed; materials of piston and liner; sodium-cooled exhaust valves.

IC Engines – Combustion, Knocking and Fuels

★ Stages of SI combustion; stoichiometric A/F 14.7; rich mixture for maximum power and cold start; turbulence increases flame speed.

★ Detonation due to auto-ignition of end gas (very frequent); factors reducing SI knock (lower r, retard spark, higher octane, more turbulence, higher speed); pre-ignition vs detonation.

★ Diesel knock due to long ignition delay; CI knock at start vs SI knock at end of combustion; opposite effect of factors in SI and CI engines (very frequent).

★ I-head most efficient SI chamber; open chamber vs swirl/pre-combustion chambers; glow plugs in IDI engines; squish and swirl.

★ Octane and cetane reference fuels (very frequent); good SI fuel = high ON, good CI fuel = high CN; TEL banned; aromatics high ON; paraffins high CN; HCV vs LCV; bomb calorimeter; biodiesel by transesterification; CNG main constituent methane.

IC Engines – Fuel Supply, Ignition, Cooling, Lubrication

★ Functions of float chamber, venturi, throttle and choke (very frequent); simple carburettor richens mixture at high speeds; compensating jet; MPFI advantages; choke gives rich mixture.

★ CRDI full form and features; jerk pump with helical plunger; multi-hole nozzle for DI, pintle for IDI; air injection obsolete; unit injector.

★ Function of condenser in battery ignition (very frequent); magneto better at high speed; cold plug for high-compression engines; centrifugal and vacuum advance; firing order 1-3-4-2; secondary voltage range.

★ Function of thermostat (very frequent); thermosyphon has no pump; pressurised radiator cap raises boiling point; antifreeze (ethylene glycol); percentage of heat to coolant.

★ Petroil lubrication in two-strokes; splash vs pressure-feed; dry sump in racing/aircraft; viscosity index; SAE 20W-40 meaning; flash and fire points; PCV valve.

★ Quantity governing in SI and quality governing in CI (very frequent); starter is a DC series motor; Bendix drive.

IC Engines – Performance, Testing, Supercharging, Emissions

★ IP formula with n = N/2 for four-stroke (very frequent); BP = 2πNT/60; mechanical efficiency; bsfc units; volumetric efficiency definition; relation $\eta_{bth} = \eta_{ith} \times \eta_m$.

★ Willan's line for CI engines, Morse test for multi-cylinder SI engines (very frequent); heat balance components and percentages; Orsat apparatus; Prony brake; motoring test.

★ Purpose of supercharging; turbocharger driven by exhaust gases (very frequent); Roots blower; intercooler; supercharging better for CI engines; altitude compensation in aircraft.

★ EGR reduces NOx (very frequent); three-way catalyst metals and $\lambda = 1$ requirement; CO from rich mixture; HC from quenching; DPF and SCR for diesels; BS-VI from 2020; NOx–PM trade-off.

IC Engines – Modern Technologies, Emission Norms and Special Engines

★ MPFI vs carburettor; CRDI common rail pressure and multiple injections (very frequent); lambda sensor for closed-loop control; knock sensor piezoelectric; GDI.

★ Three-way catalyst needs stoichiometric mixture; SCR uses urea (very frequent); DPF traps soot; EGR reduces NOx; BS-VI from 2020 skipping BS-V; NOx–PM trade-off; PCV and charcoal canister.

★ Wankel engine rotor shape and apex seals (frequent); Stirling engine is external combustion with regenerator; dual-fuel engine with pilot diesel.

★ Six-cylinder in-line engine fully balanced; four-stroke single-cylinder engine needs the heaviest flywheel; secondary force at twice crank speed.

Vapour Power Cycles (Rankine and Modifications)

★ Reasons the Carnot cycle is not used in steam plants; Rankine work ratio close to 1.

★ Rankine efficiency expression; pump work small due to low specific volume of water (very frequent); specific steam consumption 3600/w_{net}; mean temperature of heat addition concept; heat rate.

★ Effect of boiler pressure, superheat, condenser pressure, reheat and regeneration (very frequent). Main purpose of reheat = dryness; regeneration raises efficiency by raising feedwater temperature; exhaust dryness ≥ 0.88.

★ Reheat mainly improves exhaust dryness; optimum reheat pressure ≈ 1/4 of boiler pressure.

★ Regeneration increases efficiency and reduces condenser size; open heater = deaerator; closed heater = no mixing; bleeding fraction calculation.

★ Combined cycle uses HRSG and gas turbine as topping cycle; $\eta = \eta_1 + \eta_2 - \eta_1\eta_2$; mercury in binary vapour cycles; cogeneration definition and back-pressure turbines.

Steam Generators (Boilers)

★ Fire-tube vs water-tube classification with examples (very frequent); Lancashire has two flue tubes; Cornish single flue; water-tube for high pressure; Indian Boilers Act.

★ Boiler–feature matching (Benson supercritical drumless; Loeffler avoids salt deposits; Velox supersonic gases; La Mont forced circulation; B&W externally fired water-tube; Cochran vertical fire-tube; locomotive draught by exhaust steam) – very frequent.

★ Mountings vs accessories (economiser is an accessory; safety valve and fusible plug are mountings) – very frequent; function of fusible plug; economiser heats feedwater; air preheater heats air; order of heat-recovery equipment in gas path; injector principle; number of safety valves.

★ Natural draught formula and dependence on chimney height and temperature; condition for maximum discharge ($T_g \approx 2T_a$); FD vs ID vs balanced draught (very frequent); steam-jet draught in locomotives; chimney efficiency very low.

★ Equivalent evaporation from and at 100 °C with 2257 kJ/kg (very frequent); boiler efficiency formula; largest loss is dry flue gas; scale, priming, foaming, caustic embrittlement; deaeration.

Steam Nozzles and Diffusers

★ Nozzle converts enthalpy to KE; exit velocity $44.72\sqrt{\Delta h}$ (very frequent); convergent vs convergent–divergent nozzles; area–velocity relation.

★ Critical pressure ratio values (0.546 superheated, 0.577 saturated, 0.528 air) – very frequent; M = 1 at the throat; choked flow and maximum discharge.

★ Supersaturated flow increases discharge (very frequent); Wilson line; nozzle efficiency definition; friction effect.

Steam Turbines

★ Impulse vs reaction turbine differences (very frequent); De Laval simple impulse; Parsons reaction; pressure constant on impulse moving blades; partial admission in impulse; axial thrust balanced by dummy piston in reaction turbines.

★ Optimum speed ratio $\cos\alpha/2$ and $\eta_{max} = \cos^2\alpha$ for impulse turbine (very frequent); work = $u(V_{w1} + V_{w2})$; axial thrust from difference in flow components.

★ Curtis = velocity compounding; Rateau = pressure compounding (very frequent); purpose of compounding (reduce rotor speed); work ratio 3:1 in two-row Curtis wheel; De Laval needs gearing.

★ Degree of reaction definition; Parsons turbine is 50% reaction with identical fixed and moving blades (very frequent); optimum ratio $\cos\alpha$ for Parsons; losses list.

★ Throttle vs nozzle-control governing; Willans line; reheat factor > 1; back-pressure and pass-out turbines for cogeneration; overspeed trip.

Condensers and Cooling Towers

★ Purpose of condenser; jet vs surface condensers (surface used in power plants, condensate reused) – very frequent; effects of air leakage and Dalton's law; vacuum efficiency and condenser efficiency formulas; barometric condenser height ≈ 10.3 m.

★ Approach and range definitions (very frequent); water cannot be cooled below WBT; natural draught towers are hyperbolic; induced vs forced draught towers; losses and make-up.

Power Plant Engineering

★ Load factor, diversity factor (≥ 1), demand factor, capacity factor definitions (very frequent); base vs peak load plants; two-part tariff.

★ Four circuits of thermal plant; ESP removes fly ash; proximate vs ultimate analysis; anthracite highest carbon; FBC and limestone; pulverised coal firing; largest loss in condenser.

★ Surge tank function; pumped storage for peak load; turbine selection by head; $P = \eta\rho gQH$.

★ Moderator and control rod materials (very frequent); heavy water allows natural uranium; FBR has no moderator and uses liquid sodium; PHWR in India; fertile vs fissile; Tarapur first; three-stage programme.

★ Diesel plants for standby and peak loads; gas turbines quick start; combined cycle efficiency.

★ Wind power $\propto V^3$ and Betz limit 59.3% (very frequent); solar constant; PV cell material; biogas composition; OTEC temperature difference; fuel cell principle; MHD generates DC.

Air Compressors

★ Classification; reciprocating for high pressure/low flow; centrifugal and axial for large flow; uses of compressed air; fan/blower/compressor pressure ratios.

★ Isothermal compression requires minimum work (very frequent); work formulas; isothermal efficiency definition; delivery temperature; methods to reduce work.

★ Volumetric efficiency formula and effect of clearance and pressure ratio (very frequent); FAD definition; maximum pressure ratio for zero delivery; work per kg independent of clearance (ideal).

★ Optimum intermediate pressure $\sqrt{(p_1 p_3)}$ (very frequent); equal work and equal pressure ratio per stage; advantages of multistaging; intercooler heat.

★ Roots blower has no internal compression; surging at low flow and choking at maximum flow (very frequent); slip factor; axial compressors for jet engines (high efficiency, large flow, 50% reaction); centrifugal for turbochargers; screw compressor.

Gas Turbines and Jet Propulsion

★ High back-work ratio; overall air–fuel ratio is very lean because of TIT limit; open vs closed cycle; blade cooling; uses of gas turbines.

★ Propulsive efficiency $2/(1 + V_j/V_a)$ (very frequent); ramjet cannot start from rest and has no moving parts; rocket carries oxidiser; turbofan most efficient for airliners; turboprop for low speeds; afterburner increases thrust; specific impulse.

Refrigeration – Fundamentals, Air Refrigeration and VCR

★ 1 TR = 3.5 kW = 211 kJ/min (very frequent); Carnot COP formula and that COP decreases as the temperature difference increases; $COP_{HP} = COP_R + 1$.

★ Why reversed Carnot is not practical; throttle valve replaces expander.

★ Air refrigeration used in aircraft because of low weight per TR (very frequent); Bell–Coleman COP formula; low COP; bootstrap system.

★ Throttling in expansion valve is isenthalpic (very frequent); $COP = (h_1 - h_4)/(h_2 - h_1)$; subcooling increases COP; lowering evaporator temperature decreases COP; mass flow $211 \times TR/RE$; dry vs wet compression; condenser heat = RE + W.

★ Capillary tube in domestic refrigerators (very frequent); TEV maintains constant superheat; hermetic compressors; centrifugal compressors for large capacity; evaporative condenser; purpose of drier.

Refrigerants, Vapour Absorption and Other Systems

★ Refrigerant numbering (R-717 ammonia, R-744 CO_2, R-718 water; R-134a formula) – very frequent; ammonia not used with copper; halide torch for Freon leaks; sulphur candle for ammonia; Montreal (ozone) vs Kigali (HFC) protocols; R-134a has zero ODP; desirable properties; secondary refrigerants (brine).

★ Electrolux uses hydrogen as third fluid with no pump (very frequent); LiBr–water: water is the refrigerant (above 0 °C only); aqua-ammonia: ammonia is refrigerant, needs rectifier and analyser; ideal COP of VAR formula; VAR vs VCR comparison.

★ Peltier effect; vortex tube; Linde–Hampson and Claude cycles; desert coolers in dry climates; dry ice −78.5 °C; liquid nitrogen −196 °C; cascade systems for very low temperature.

Psychrometry and Air Conditioning

★ Order DBT > WBT > DPT and equality at saturation (very frequent); ω = 0.622 p_v/(p_b − p_v); definitions of RH, specific humidity, degree of saturation, dew point; enthalpy of moist air; sling psychrometer.

★ Sensible heating is horizontal (very frequent); evaporative cooling follows constant WBT; chemical dehumidification along constant enthalpy with DBT rising; cooling and dehumidification for summer AC; SHF and BPF definitions; adiabatic mixing on a straight line; ADP.

★ Comfort conditions (24 °C, 50% RH) and effective temperature; sources of sensible and latent load; split AC components; winter AC = heating + humidification; central AC with chillers and AHUs.

Refrigeration and AC – Applications and Equipment

★ Quick freezing preserves quality (small crystals); frozen food at −18 °C; cold storage load components; chilling vs freezing; cryogenic freezing with liquid nitrogen.

★ Split AC components (very frequent); inverter AC variable-speed compressor; chillers and AHUs in central AC; EER/COP definitions; VRF systems; thermal storage for peak shaving.

★ Equal friction method most common (very frequent); static regain for high-velocity systems; fan laws $P \propto N^3$; HEPA filter 0.3 µm; throw definition; forward-curved fans overloading.

★ Liquid nitrogen −196 °C, helium lowest boiling point; Linde vs Claude cycle; J–T precooling of hydrogen and helium; Dewar flask.

Heat Transfer I – Conduction, Fins, Transient Conduction

★ Fourier's law and its sign; order of thermal conductivity (silver highest metal; diamond overall); k of gases increases with temperature; thermal diffusivity definition (very frequent); forms of heat conduction equation.

★ Electrical analogy of heat flow (series/parallel resistances) – frequent; overall U; cylinder formula with $\ln(r_2/r_1)$; log-mean area for cylinders, geometric mean for spheres; temperature distribution shapes.

★ r_c = k/h (cylinder) and 2k/h (sphere) – very frequent; heat loss maximum at critical radius; electric cables use this.

★ Maximum temperature formulas for plane wall, cylinder, sphere with generation; location of maximum temperature at the centre.

★ Fins used where h is low and placed on that side (very frequent); effectiveness formula and > 2 condition; fin efficiency definition; long-fin heat transfer; corrected length; thin closely spaced fins of high-k material.

★ Lumped analysis valid for Bi < 0.1 (very frequent); Biot vs Nusselt (k of solid vs fluid); characteristic lengths (sphere R/3); time constant and 63.2%; Heisler charts; Fourier number.

Heat Transfer II – Convection, Boiling and Condensation

★ Forced vs free convection; boiling/condensation give the highest h; h is lowest for free convection in gases.

★ Meaning of Nu, Pr, Gr, St, Bi (very frequent); Pr of air ≈ 0.7, liquid metals very low; Nu = f(Re, Pr) for forced and f(Gr, Pr) for free convection; Biot uses k of solid, Nusselt uses k of fluid.

★ Nu = 3.66 and 4.36 (very frequent); Dittus–Boelter exponents 0.8 and 0.4/0.3; Reynolds/Colburn analogy; flat plate laminar Nu ∝ Re^0.5; hydraulic diameter 4A/P.

★ Grashof number governs free convection; β = 1/T for gases; turbulent free convection exponent 1/3; film temperature for properties.

★ Nucleate boiling gives maximum h (very frequent); critical heat flux / burnout point; Leidenfrost point marks film boiling minimum; regimes order; film boiling insulates by vapour.

★ Dropwise condensation gives higher h but filmwise is used for design (very frequent); effect of non-condensable gases; horizontal tubes better than vertical.

Heat Transfer III – Radiation and Heat Exchangers

★ Black body α = ε = 1 (very frequent); Kirchhoff's law α = ε; Wien's law constant 2898 μm·K; Stefan–Boltzmann constant; grey body definition; greenhouse effect; N_2/O_2 transparent, CO_2/H_2O participating; α + ρ + τ = 1.

★ Shape factor reciprocity and summation rules (very frequent); F_{11} = 0 for flat/convex surfaces; small body in large enclosure; parallel plates formula; n shields → 1/(n+1); hemisphere view factors.

★ Counter flow more effective than parallel flow (very frequent); LMTD formula; ε = 1 − e^(−NTU) for condensers/evaporators; NTU definition; fouling factor; car radiator is cross-flow compact HX; regenerators vs recuperators; LMTD = ΔT when $\Delta T_1 = \Delta T_2$.

Fuels and Combustion

★ Order of coal rank (anthracite highest carbon/CV; peat lowest); coke made by carbonisation; Indian coal high ash.

★ Proximate vs ultimate analysis (very frequent); fixed carbon found by difference; oxygen by difference in ultimate analysis.

★ HCV vs LCV difference (latent heat of water vapour from hydrogen) – very frequent; bomb calorimeter for solid/liquid, Junkers for gaseous fuels; hydrogen highest CV per kg; Dulong's formula.

★ 1 kg of carbon needs 8/3 kg O_2 and hydrogen 8 kg O_2 (very frequent); air contains 23% O_2 by mass and 21% by volume; stoichiometric A/F of petrol ≈ 15; excess air effects; adiabatic flame temperature.

★ Orsat apparatus absorbents and order (KOH – CO_2, pyrogallol – O_2, cuprous chloride – CO) – very frequent; Orsat gives dry analysis; high CO_2 indicates good combustion.

★ Producer gas vs water gas composition (frequent); biogas mainly methane; LPG = propane + butane; flash point < fire point; pour point < cloud point; CNG high octane.

Thermal Measurements and Instruments

★ Thermocouple works on Seebeck effect (very frequent); RTD platinum; thermistor NTC; optical pyrometer for very high temperature (non-contact); laws of thermocouples; type T for low temperature.

★ Bourdon tube for boiler pressure (mounting); dead-weight tester for calibration; McLeod gauge for vacuum; piezoelectric for dynamic pressure; Pirani gauge works on thermal conductivity.

★ Rope brake and hydraulic dynamometers measure BP; eddy-current dynamometer; swinging-field dynamometer for motoring (friction power); indicator diagram gives IP; rotameter variable area; Bosch/Hartridge smoke meters; FID for HC, chemiluminescence for NOx.

★ Sling psychrometer measures DBT and WBT; guarded hot plate for thermal conductivity of insulators; Pensky–Martens apparatus for flash point; Redwood viscometer; hair hygrometer.

Heat Transfer IV – Mass Transfer, Heat Pipes, Insulation and Enhancement

★ Fick's law analogy to Fourier's law (frequent); Sherwood and Schmidt numbers; Lewis number ≈ 1 for air–water (WBT = adiabatic saturation temperature); diffusivity of gases ∝ T^1.5/p.

★ Heat pipe principle – evaporation, condensation, capillary return through wick; very high effective conductivity; uses in electronics cooling.

★ Insulation works by trapping air; vapour barrier on the warm side of cold insulation (frequent); economic thickness; PUF for cold storage; ceramic fibre for furnaces; moisture increases k.

★ Twisted tapes and rough surfaces increase h and pressure drop; thermal grease reduces contact resistance; heat sinks with fins.